

	Real Time Simulation Oslo ASAP (Oslo Advanced Sectorization and Automation Project)	
--	--	---

EUROCONTROL EXPERIMENTAL CENTRE

VALIDATION REPORT

EEC Technical/Scientific Report No. 2010-002
Final Version



Issue Date: 29 January 2010

© European Organisation for the Safety of Air Navigation EUROCONTROL 2007

This document is published by EUROCONTROL in the interest of the exchange of information. It may be copied in whole or in part providing that the copyright notice and disclaimer are included. The information contained in this document may not be modified without prior written permission from EUROCONTROL.

EUROCONTROL makes no warranty, either implied or express, for the information contained in this document, neither does it assume any legal liability or responsibility for the accuracy, completeness or usefulness of this information.

Page intentionally left blank

REPORT DOCUMENTATION PAGE

Reference: EEC Technical/Scientific Report No. 2010-002	Security Classification: Unclassified					
Originator: EEC - ATC (Air Traffic Control)	Originator (Corporate Author) Name/Location: EUROCONTROL Experimental Centre Centre de Bois des Bordes B.P.15 F - 91222 Brétigny-sur-Orge CEDEX FRANCE Telephone: +33 (0)1 69 88 75 00 Internet : www.eurocontrol.int					
Sponsor:	Sponsor (Contract Authority) Name/Location: EUROCONTROL Agency 96, Rue de la Fusée B - 1130 Brussels BELGIUM Telephone: +32 2 729 90 11 Internet : www.eurocontrol.int					
TITLE: REAL TIME SIMULATION OSLO ASAP (Oslo Advanced Sectorization and Automation Project)						
Author Stefano Tiberia François Vergne	Date 01/2010	Pages xiv + 172	Figures 31	Tables 8	Annexes 14	References 8
	Project Oslo ASAP		Task No. Sponsor		Period 2009	
Distribution Statement: (a) Controlled by: (b) Special Limitations: None (c) Copy to NTIS: YES / NO						
Descriptors (keywords): Point Merge; P-RNAV; Arrival Manager; Continuous Descent Approach (CDA); TMA; Working procedures.						
Abstract: The document presents the results and evaluation of a real-time simulation (RTS) based on future Oslo AoR operations. The RTS was carried out in March 2009 at the EUROCONTROL Experimental Centre (EEC), in the framework of the Oslo Advanced Sectorization and Automation Project (ASAP) project led by AVINOR (Norwegian ANSP). The analysis presented in the document addresses the impact of the new airspace structure, procedures and working methods on controller's roles and human and system performances.						

Page intentionally left blank

EVOLUTION SHEET

Date	Change status	Changes	Version
14/05/09	Draft		0.1
25/06/09	Second Draft	Overall review.	0.2
17/07/09	First Release	Overall review.	0.3
31/07/09	Second Release	Overall review.	0.4
25/09/09	Third Release	Overall review.	0.5
29/10/09	Internal Release	Final review.	1.0
29/01/10	Final	Template and content adaptation	1.1

Page intentionally left blank

EXECUTIVE SUMMARY

This document presents the results of the real time simulation conducted from the 10th to the 18th March 2009 at the EUROCONTROL Experimental Centre, in the framework of the Oslo Advanced Sectorization and Automation (ASAP) project led by AVINOR.

The project objective is to establish a new Air Traffic Management system for Oslo AoR in the timeframe of 2009 – 2011, in order to increase capacity to meet future demand forecast, to integrate environmental constraints and comply with the directives given by the Norwegian CAA.

The solution envisaged to achieve these objectives was to develop new procedures for Oslo TMA and redesign Oslo AoR airspace accordingly. After an evaluation of different existing procedures for the handling of traffic in terminal airspace, AVINOR decided the Point Merge method would better suit their operational needs for Oslo TMA.

The Point Merge is an innovative procedure to merge arrival flows, developed by the EUROCONTROL Experimental Centre to improve the efficiency of terminal airspace operations. It associates a dedicated P-RNAV route structure with a systemised operating method to integrate arrival flows into one sequence while keeping aircraft on FMS lateral navigation mode, thus allowing efficient use of FMS advanced functions and consequent optimisation of vertical profiles.

The new procedure will be combined with the deployment of a new arrival manager, for traffic arriving at Oslo Gardermoen airport.

For the validation of the new airspace organisation a step wise approach was applied. Six prototyping sessions were conducted before the final large-scale real time simulation took place.

The primary objective of the simulation was to assess the operability of the new organisation. The analysis mainly covered human performances aspects linked to operability. The impact that the new organisation and related working methods had on the controllers was assessed in terms of acceptability, workload, situation awareness and job satisfaction. Secondary objective was to provide initial indications on capacity, efficiency and safety.

RESULTS OVERVIEW

The simulation results show that the new organisation is operationally viable. The proposed airspace organisation and the new working methods based on the implementation of Point Merge and AMAN for Oslo arrivals were overall well accepted by the participants of the simulation.

The Point Merge method was assessed to be a suitable and effective solution for Oslo TMA. It relieved the controllers of a significant part of the workload currently experienced and improved their situation awareness because of the standardisation of arrivals trajectories.

Containment of the workload and improvement of situation awareness are deemed as both beneficial for safety.

Moreover the simulation provided a clear indication on how effectively AMAN can work in combination with Point Merge.

In terms of performances, the new organisation has potential to allow a good level of quality of service and flight efficiency. Point Merge procedures led to more standardised operations, limiting radar vectoring to the extent needed for managing non-nominal situations. That along with the optimisation of vertical trajectories (bringing potential for CDA) is expected to result in an increase of flight efficiency.

Finally the new working methods and procedures are believed to have a positive impact on the environmental sustainability. Containment of trajectories dispersion and improved flight profiles have potential for reducing fuel consumption and emissions and for responding to the new noise abatement procedures.

ACKNOWLEDGMENTS

The EUROCONTROL Experimental Centre would like to thank all the people who have contributed to the success of the present real time simulation. In particular, we extend our appreciation to the AVINOR.

We would like to specifically thank the AVINOR Project Manager Geir Gillebo, the ASAP Core Team led by Hans Jacob Hofgaard, Øystein Brøvik and Kristian Pjaaten and ably supported by Trym Fyrileiv, Jan Storøy, Per Christian Aasland, Torgeir Braathen and Hans Christian Erstad. Their excellent input and expertise was invaluable.

During the simulation itself, the ASAP Core Team and the other AVINOR operational participants continuously demonstrated a high level of professionalism and enthusiasm and provided precious input and feedback to the study.

Page intentionally left blank

TABLE OF CONTENTS

EVOLUTION SHEET	V
EXECUTIVE SUMMARY	VII
ACKNOWLEDGMENTS.....	IX
LIST OF ANNEXES.....	XII
LIST OF FIGURES	XIII
LIST OF TABLES.....	XIII
1. INTRODUCTION	1
1.1. PURPOSE OF THIS DOCUMENT	1
1.2. INTENDED AUDIENCE.....	1
1.3. ACRONYMS AND ABBREVIATIONS	2
1.4. REFERENCES	3
2. OSLO ASAP PROJECT AND SIMULATION OBJECTIVES	5
2.1. OSLO ASAP PROJECT	5
2.2. OSLO ASAP SIMULATIONS.....	5
2.3. DESCRIPTION OF NEW ATM ELEMENTS.....	6
2.3.1. Point Merge Method	6
2.3.2. Arrival Manager (AMAN)	7
2.4. MAIN OUTCOMES FROM PHASE A AND B.....	8
2.5. SIMULATION OBJECTIVES	9
3. SIMULATION SETTINGS	11
3.1. SIMULATED ENVIRONMENT	11
3.1.1. Airspace.....	11
3.1.2. Sectors and Positions.....	11
3.1.3. SID/STAR Procedures.....	13
3.1.4. Military Areas.....	18
3.1.5. Traffic.....	19
3.1.6. Meteorological Environment	21
3.2. CONTROLLERS' ROLES.....	22
3.2.1. Oslo ACC.....	22
3.2.2. Oslo TMA.....	23
3.2.3. Farris TMA.....	25
3.3. SPECIAL PROCEDURES	27
3.3.1. ILS Cross Over Procedures.....	27
3.3.2. Missed Approach Procedures	27
3.4. SUPPORTING TOOL (AMAN)	28
4. EXPERIMENTAL DESIGN AND SIMULATION CONDUCT	31
4.1. EXPERIMENTAL VARIABLES AND ORGANISATIONS	31
4.2. OPS ROOM LAYOUT	33
4.3. PARTICIPANTS AND SEATING PLAN.....	33
4.4. SIMULATION SCHEDULE	34

4.5.	MEASUREMENTS	36
4.5.1.	Human Data Collection Methods	36
4.5.2.	Objective Measurements	37
5.	RESULTS.....	39
5.1.	OPERABILITY	39
5.1.1.	Acceptability	39
5.1.2.	Workload	45
5.1.3.	Job Satisfaction	49
5.2.	SAFETY	51
5.3.	CAPACITY.....	54
5.3.1.	Controllers' Availability	54
5.3.2.	Sectors Throughput.....	55
5.3.3.	Throughput at ENGM	57
5.4.	EFFICIENCY	57
5.4.1.	Quality of Service	58
5.4.2.	Vertical Profiles.....	61
6.	CONCLUSIONS	63

LIST OF ANNEXES

ANNEX A:	SIMULATION ROOM LAYOUT	67
ANNEX B:	BLANK QUESTIONNAIRES	71
ANNEX C:	INSTANTANEOUS SELF-ASSESSMENT (ISA)	83
ANNEX D:	WORKLOAD COMPONENTS	91
ANNEX E:	R/T AND PHONE USAGE	95
ANNEX F:	REPARTITION OF INSTRUCTIONS	99
ANNEX G:	AIRCRAFT ON FREQUENCY	107
ANNEX H:	ENGM STARS - ARRIVALS VERTICAL PROFILES	113
ANNEX I:	ENGM SIDS - DEPARTURES VERTICAL PROFILES	125
ANNEX J:	ENGM SPECIAL SIDS - DEPARTURES VERTICAL PROFILES	141
ANNEX K:	ENRY SIDS - DEPARTURES VERTICAL PROFILES	145
ANNEX L:	ENTO SIDS - DEPARTURES VERTICAL PROFILES	147
ANNEX M:	OSLO TMA - 2D FLOWN TRAJECTORIES.....	149
ANNEX N:	FARRIS TMA - 2D FLOWN TRAJECTORIES	165

LIST OF FIGURES

Figure 2-1 – Example of Point Merge System	7
Figure 3-1 – Oslo ACC Sectorization.....	11
Figure 3-2 – Oslo and Farris Terminal Areas.....	13
Figure 3-3 – ENGM STARs - RWYs 01 (left) and RWYs 19 (right)	14
Figure 3-4 – ENGM Sequencing Legs - RWYs 01 (top) and RWYs 19 (bottom).....	15
Figure 3-5 – ENGM SIDs - RWYs 01 (left) and RWYs 19 (right).....	16
Figure 3-6 – ENRY RWY 12 STARs (left) and SIDs (right).....	17
Figure 3-7 – ENRY RWY 30 STARs (left) and SIDs (right).....	17
Figure 3-8 – ENTO RWY 18 STARs (left) and SIDs (right).....	18
Figure 3-9 – ENTO RWY 36 STARs (left) and SIDs (right).....	18
Figure 3-10– Military Areas	19
Figure 3-11 – Missed Approach Procedure at ENGM	28
Figure 3-12 – AMAN Timelines	29
Figure 4-1 – Simulation Schedule	35
Figure 5-1 – Subjective Feedback on Airspace Design and Routes Segregation	40
Figure 5-2– Interaction ACC Sectors ↔ TMAs (ACC Perspective).....	41
Figure 5-3 – Interaction ACC sectors ↔ TMAs (TMA Perspective)	42
Figure 5-4 – Examples of Non P-RNAV Aircraft (red) and Missed Approach (green) trajectories.....	45
Figure 5-5 – ISA: Oslo ACC Sectors.....	46
Figure 5-6 – ISA: Oslo TMA Positions	47
Figure 5-7 – ISA: Oslo TMA Positions (MPO – Non Nominal Scenarios)	48
Figure 5-8 – ISA: Farris TMA	49
Figure 5-9 – Job Satisfaction	50
Figure 5-10 – Situation Awareness.....	52
Figure 5-11 – Frequency Occupancy in Oslo TMA.....	55
Figure 5-12 – Example of 2D Flown Trajectories for ENGM Arrivals (IPA).....	58
Figure 5-13 – Inter Aircraft Spacing (MPO)	60
Figure 5-14 – Inter Aircraft Spacing (IPA).....	60
Figure 5-15 – Example of Average Profile for Arrivals (Higher Side).....	61
Figure 5-16 – Example of Average Profile for Arrivals (Lower Side).....	62
Figure 5-17 – Example of Average Profile for Departures	62

LIST OF TABLES

Table 2-1 – Oslo ASAP Real Time Simulations.....	6
Table 3-1– Simulation Traffic Samples (TS).....	20
Table 3-2 – Meteorological Setting	21
Table 4-1 – Simulation Organisations.....	32
Table 4-2 – ISA Rating Scale.....	37
Table 5-1 – Sector Throughput (Oslo ACC).....	56
Table 5-2 – Position Throughput (Oslo and Farris TMA)	56
Table 5-3 – ENGM Throughput.....	57

Page intentionally left blank

1. INTRODUCTION

1.1. PURPOSE OF THIS DOCUMENT

This document presents the results of the simulation analysis, as the outcome of the EUROCONTROL assessment.

The simulation was conducted in March 2009 at the EUROCONTROL Experimental Centre (EEC), in the framework of the Oslo Advanced Sectorization and Automation Project (ASAP) project led by AVINOR (Norwegian ANSP).

The document aims to back up AVINOR's operational understanding of the simulation. Therefore it will be of use in complementing the AVINOR's assessment.

The analysis presented in the document addresses the impact of the new airspace structure, procedures and working methods on controller's roles and human and system performances.

Apart from this introductory part, the document is structured as follows:

- Section 2 presents the Oslo ASAP project and the objectives of the simulation;
- Section 3 describes the simulation settings and the controllers tasks;
- Section 4 provides details of the experimental design and the conduct of the simulation;
- Section 5 presents the simulation results;
- Conclusions and final recommendations in section 6.

1.2. INTENDED AUDIENCE

This document is intended to the following audience:

- The AVINOR Oslo ASAP core team members and AVINOR management;
- The EUROCONTROL Oslo ASAP project team members and ATC Operations and System.area;
- The ANSP/stakeholder having an interest in PMS and/or new arrival procedures.

1.3. ACRONYMS AND ABBREVIATIONS

Term	Definition
ACC	Area Control Centre
AIP	Aeronautical Information Publication
AMAN	Arrival Manager (Tool)
ANSP	Air Navigation Service Provider
AoR	Area of Responsibility
ASAP	Advanced Sectorization & Automation Project
ATC	Air Traffic Control
ATCC	Air Traffic Control Centre
ATM	Air Traffic Management
ATS	Air Traffic Services
AVINOR	Norwegian ANSP
CAA	Civil Aviation Authority
CCD	Continuous Climb Departures
CDA	Continuous Descent Approach
CFL	Cleared Flight Level
CND	Cooperative Network Design
CRDS	Central European Research, Development and Simulation
EEC	EUROCONTROL Experimental Centre
FIR	Flight Information Region
FL	Flight Level
FMS	Flight Management System
FPB	Flight Progress Board
FPS	Flight Progress Strip
HMI	Human Machine Interface
IAF	Initial Approach Fix
ILS	Instrument Landing System
IPA	Independent Parallel Approaches
ISA	Instantaneous Self-Assessment
LNAV	Lateral Navigation
MPO	Mixed Parallel Operations
MUDPIE	Multiple User Data Processing Interactive Environment
NM	Nautical Miles
OSD	Operational Services and Environment Definition
PMS	Point Merge System
P-RNAV	Precision Area Navigation

Term	Definition
R/T	Radio Telephony
RNAV	Area Navigation
RWY	Runway
SAAM	System for Assignment and Analysis at a Macroscopic level
SID	Standard Instrument Departure (Route)
STAR	Standard Terminal Arrival Route
TMA	Terminal Control Area
TSA	Temporary Segregated Area
UIR	Upper (flight) Information Region
WTC	Wake Turbulence Categories

1.4. REFERENCES

- [1] AVINOR, Controller Handbook ASAP C, 6th March 2009
- [2] EUROCONTROL, Oslo ASAP A1 Validation Report, Version 1.1, 9th September 2008
- [3] EUROCONTROL, Oslo ASAP A2 Validation Report, Version 1, 9th September 2008
- [4] EUROCONTROL, Oslo ASAP A3 Validation Report, Version 1, 18th November 2008
- [5] EUROCONTROL, Oslo ASAP B1 Validation Report, Version 1, 25th May 2009
- [6] EUROCONTROL, Oslo ASAP B2 Validation Report, Version 1, 25th May 2009
- [7] EUROCONTROL, Oslo ASAP Facility Specification, Edition 10.0, 5th March 2009
- [8] EUROCONTROL, Point Merge Integration of Arrival Flows Enabling Extensive RNAV Application and CDA (OSD), Version 1.0, 21st April 2009

Page intentionally left blank

2. OSLO ASAP PROJECT AND SIMULATION OBJECTIVES

2.1. OSLO ASAP PROJECT

The Oslo ASAP project was established in May 2006 and is led by AVINOR.

The project objective is to establish a new Air Traffic Management system for Oslo AoR in the timeframe of 2009 - 2011. The drivers are:

- To increase capacity to meet future demands - full utilisation of today's runway capacity at Gardermoen (ENGM) as well as expected traffic growth at Torp (ENTO) and Rygge (ENRY) airports;
- To integrate environmental constraints - new noise abatement procedures at ENGM;
- To comply with the directives given by the Norwegian CAA - better segregation of flows and reduction of TCAS hot spots.

The solution envisaged to achieve these objectives was to develop new procedures for Oslo TMA and redesign Oslo AoR airspace accordingly.

At a really first stage of the project AVINOR foresaw the implementation of an arrival manager (AMAN) from BarcoTM to support the management of the ENGM arrivals.

At the same time an evaluation of different existing procedures for the handling of traffic in terminal airspace started (e.g. trombone technique as currently implemented in Frankfurt and Munich). Eventually AVINOR decided the Point Merge method would better suit their operational needs for Oslo TMA.

The Point Merge is an innovative procedure developed by the EEC aiming at improving and standardising terminal airspace operations. The method has not been implemented by any provider yet; therefore AVINOR took on the challenge to be a pioneer in implementing it.

2.2. OSLO ASAP SIMULATIONS

As part of the project a series of model-based and real time simulations were conducted to identify and then validate the new airspace organisation for Oslo AoR.

The activity began in February 2007 at EUROCONTROL's CRDS facilities in Budapest with a model-based simulation which identified the operational scenario to be then developed and validated through the real time simulations at the EEC.

From April 2008 to February 2009, six simulation sessions were conducted and finally a large-scale real time simulation of the Oslo area took place from the 10th to the 18th March 2009. The six sessions were part of two phases: Phase A and Phase B. The large-scale simulation constituted the so called Phase C.

Table 2-1 – Oslo ASAP Real Time Simulations

	2008									2009		
	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Sessions phase A	A1		A2		A3							
Sessions phase B							B1		B2		B3	
Simulation phase C												C

2.3. DESCRIPTION OF NEW ATM ELEMENTS

The new Oslo AoR airspace organisation, built up in the previous sessions, relies on:

- SID/STAR for Oslo TMA airport (ENGM) with Point Merge;
- SID/STAR for Farris TMA airports (ENRY and ENTO);
- New sectorization and route structure for the Oslo ATCC;
- New positions configurations for Oslo and Farris TMAs;
- AMAN for ENGM arrivals.

A short description of Point Merge method and AMAN follows hereafter.

2.3.1. Point Merge Method

Point Merge is a P-RNAV application that has been developed by the EEC as an innovative technique aiming at improving and standardising terminal airspace operations.

A Point Merge procedure associates a dedicated route structure with a systemised operating method to integrate arrival flows with extensive use of RNAV while keeping aircraft on Flight Management System (FMS) lateral navigation mode. It thus enables an efficient use of FMS advanced functions and consequent optimisation of vertical profiles, making it possible to apply Continuous Descent Approaches (CDAs) even under high traffic load. Open-loop radar vectoring is limited to the extent needed for recovering from non nominal situations.

The dedicated RNAV route structure relies on the following key elements: merge point and sequencing legs:

- A single point – denoted ‘merge point’, is used for traffic integration;
- Pre-defined legs – denoted ‘sequencing legs’, equidistant from the merge point, are dedicated to path stretching/shortening for each inbound flow.

Point Merge operating method aims at integrating inbound flows, using this route structure, without normally relying on open loop vectors. It comprises the following main steps:

- Create the sequence order and inter-aircraft spacing. This is achieved by iteratively:
 - Leaving each aircraft fly along the sequencing leg as long as necessary for path stretching/shortening , and

- Issuing a 'Direct To' instruction to the merge point when the appropriate spacing is reached with the preceding aircraft in the sequence (already on course to the merge point).
- Maintain the sequence order and inter-aircraft spacing. This is achieved through speed control after leaving the legs.

Upon leaving the sequencing legs following the 'Direct To' instruction, the descent profile can be optimised in the form of a CDA as the distance to go is then known by the FMS.

Considering a simple configuration involving the integration of two inbound flows, Figure 2-1 provides a typical example of Point Merge.

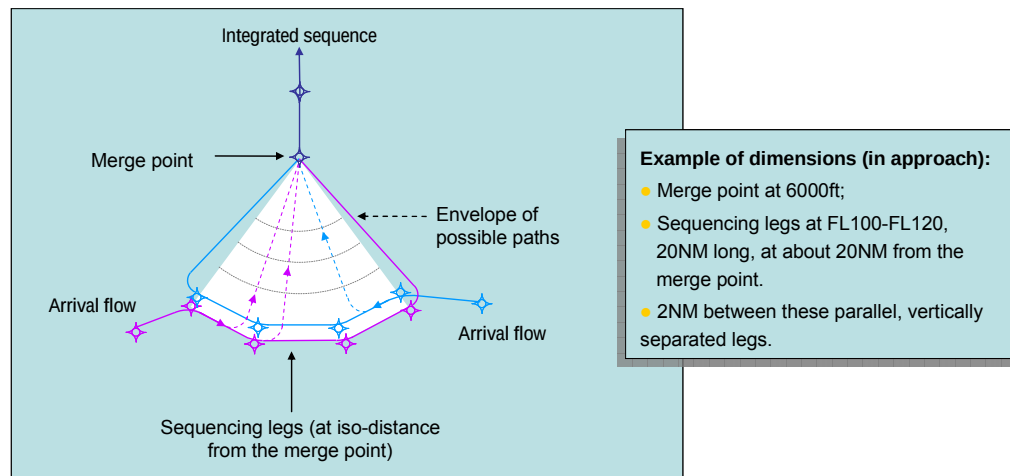


Figure 2-1 – Example of Point Merge System

2.3.2. Arrival Manager (AMAN)

AMAN is a sequence planning and support tool for arriving traffic. The objective of AMAN is to advise controllers in upstream ACC sectors to adjust approaching flights in a manner ensuring a smooth flow of traffic entering the TMA in order to use the airport's capacity in the most efficient way.

The AMAN functionality:

- Establishes the initial arrival sequence, based on the first-come, first-served rule, for the stream of inbound traffic considered, and subsequently optimises it to take into account different factors (e.g. WTC, preferred runway allocation, etc.);
- Generates advisories for the controllers in order to meet and maintain the optimised arrival sequence;
- Presents advisories to controllers through the timeline HMI;
- Automatically adapts the established inbound traffic sequence to the actual traffic evolution as well as to the controller decisions deemed necessary to meet exceptional cases.

It comprises three areas of different functionality:

- Eligibility Horizon: This range includes all flights which are relevant for consideration by the AMAN function. These inbound flights are inserted into a natural sequence based on the first-come, first-served rule. The natural sequence serves the controller as a kind of sector load forecast for the inbound traffic;

- Active Advisory Horizon: For flights within this area, an optimised arrival sequence will be generated and time advisories are provided to the controller. Time to lose or gain or holding advisories are given within the active advisory horizon but outside the common path horizon;
- Common Path Horizon: The common path horizon should be kept as small as possible (e.g. base leg and final approach). Advisories related to the common path ensure that the required spacing between consecutive arrivals is established and maintained.

2.4. MAIN OUTCOMES FROM PHASE A AND B

The simulation activity within the Oslo ASAP project followed a step wise approach. In other words, each simulation session was built on the results obtained in the preceding session and incremented the scope of the operational scenario simulated.

The Phase A extensively explored on the Oslo TMA, and produced feedbacks which were used to refine the new SID/STAR design and the working procedures relying on Point Merge.

In particular, Point Merge and related working method were found to be an effective solution allowing the arrival controllers to work comfortably and safely and enabling a good quality of service. Among the benefits, the following can be mentioned:

- Decrease of workload and radio communications;
- Increase of predictability and situation awareness;
- Increase of lateral navigation mode usage for the aircraft;
- Improvement of descent profiles (potential for CDA);
- Containment of trajectories dispersion.

The Phase A explored as well on the interface between the Oslo and Farris TMAs, and on the new SID/STAR design for Farris TMA airports. In particular the new P-RNAV based STARs, defined for ENRY and ENTO, encountered the favour of the participants, as allowed for an easier handling of arrivals than the current procedures do. The new procedures increased the trajectories predictability and in turn improved the controllers' situation awareness. Much appreciated was then the chosen configuration with two controller positions being active, as it reduced the controllers' workload. Further the layout tested with the side by side positions largely enhanced the situation awareness of the controllers.

The Phase B enlarged the scope of the operational scenario tested adding the Oslo ACC sectorization. Therefore the realism of the simulated scenario was enhanced and allowed to focus also on the interaction between the Oslo ACC and the two TMAs. Moreover AMAN was introduced and the ACC sectors and TMAs positions were supplied with the tool supporting in the management of the arrivals to ENGM.

The sessions within the Phase B confirmed again the positive outcomes from the previous phase concerning the use of Point Merge method. In addition, it was found that the new ACC sectorization was appropriate for a more efficient management of the traffic. Generally, the participants reported their own satisfaction with the new airspace structure which, compared to the today's situation, allowed for a better interaction between the en-route sectors and the TMAs positions. The sessions allowed then for a tuning of AMAN in the Oslo AoR context. Although its configuration could not be fully finalised within the Phase B timeframe, AMAN revealed to be well accepted by the participants who recognised its potential. AMAN enhanced the controllers' situation awareness and decreased the coordination need between the ACC sectors and the Oslo TMA positions.

Finally the two phases aimed also to familiarise the controllers with the new operational scenario and the simulator facilities. The selection of the participants to the sessions was done to allow as much as possible the consistency. That meant the participants to the large-scale simulation had already achieved an appropriate level of knowledge of the concepts and of the simulator.

2.5. SIMULATION OBJECTIVES

The primary objective of the simulation was to assess the operability of the new organisation. The analysis conducted by EUROCONTROL mainly covered human performances aspects linked to operability. The impact that the new organisation and related working methods had on the controllers was assessed in terms of:

- Acceptability;
- Workload;
- Situation Awareness;
- Job satisfaction.

Secondary objective was to provide initial indications on capacity, efficiency and safety (the analysis scope was in that case limited by the absence of a reference scenario being simulated).

Page intentionally left blank

3. SIMULATION SETTINGS

3.1. SIMULATED ENVIRONMENT

3.1.1. Airspace

The airspace simulated was the Oslo AoR within the Norway FIR/UIR, as described in the AIP Norway.

More in detail the simulation was based on the operational scenario formed by two adjacent terminal areas, the Oslo and Farris TMAs, and the Oslo ACC airspace.

The Oslo TMA was set up to deal with operations at ENGM when the two parallel runways are in use.

The Farris TMA controlled inbound and outbound traffic to and from single runway airports, which are ENRY, ENTO and ENSN.

The Oslo ACC airspace was made up of eight blocks of airspace to be combined in order to make a maximum of six sectors responsible for ATS provision to aircraft within their own lateral limits.

3.1.2. Sectors and Positions

3.1.2.1. Oslo ACC Sectors

The Oslo ACC airspace was made up of eight blocks of airspace (S1, S2, S3, S4, S5, S6, S7 and S8) to be combined in order to simulate a maximum of six sectors.

The lateral limits of the various airspace blocks are described hereafter.

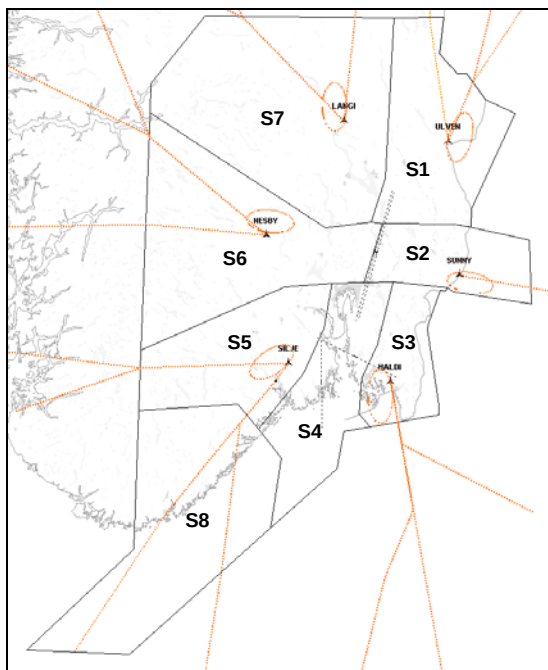


Figure 3-1 – Oslo ACC Sectorization

3.1.2.2. Oslo TMA Positions

The manning configuration for the Oslo TMA was dependent on the runway mode in use at ENGM. Two different runways mode were applied during the simulation which are:

- Mixed Parallel Operations¹ (MPO);
- Independent Parallel Approaches² (IPA).

Both the simulated runway modes represent a change in comparison with the today's situation where stricter rules are applied for the use of runways by departures and arrivals.

The full configuration, being active with IPA, consisted of five positions. They were:

- Approach East (**APPE**) and Approach West (**APPW**);
- Director East (**DIRE**) and Director West (**DIRW**);
- Final (**FIN**).

In case of MPO the two director positions collapsed into one single position (**DIR**).

The area of responsibility of the positions was in relationship with the runways orientation in use at ENGM which could be either 01 or 19.

3.1.2.3. Farris TMA Positions

The Farris TMA manning configuration consisted of two positions, which were:

- Farris East (**FARE**);
- Farris West (**FARW**).

The above refer to the full configuration for Farris TMA. When operationally feasible the two positions could collapse into a single one (**FAR**).

The two diagrams hereafter describe the location of the Oslo and Farris positions' area of responsibility when runways orientation in use at ENGM is respectively 01 and 19.

¹ MPO are operations where radar separation minima between aircraft on adjacent extended runway centre lines apply. Both runways may be used for a mix of departures and arrivals.

² IPA are simultaneous approaches to independent parallel or near-parallel instrument runways where radar separation minima between aircraft on adjacent extended runway centre lines are not prescribed.

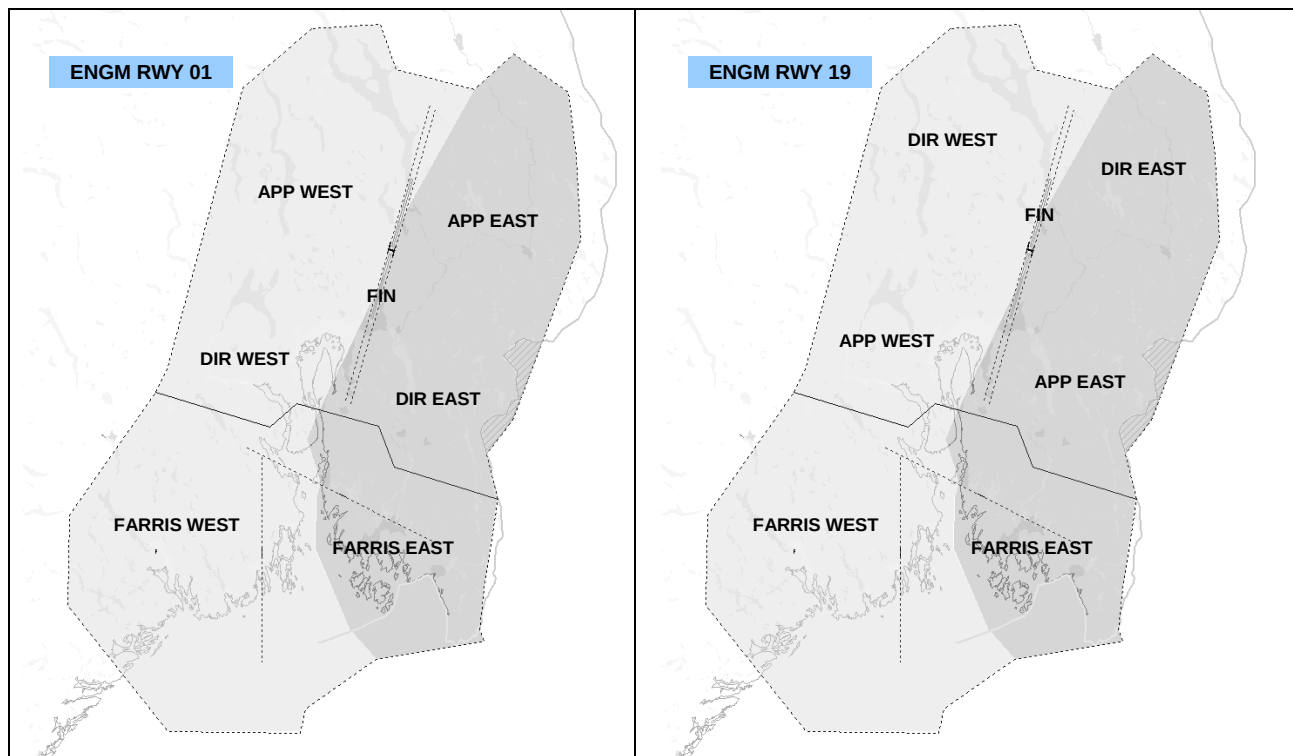


Figure 3-2 – Oslo and Farris Terminal Areas

3.1.3. SID/STAR Procedures

3.1.3.1. Oslo TMA

All the STARs to ENGM tested in the simulation were P-RNAV. Six STARs per runways orientation were designed. The STARs designators took after the waypoints listed hereafter, which from now onwards we refer to as STAR-start-points:

- LANGI, NESBY and SILJE (west side);
- ULVEN, SUNNY and HALDI (east side).

The two diagrams below depict the ENGM STARs when runways orientation in use is either 01 or 19.

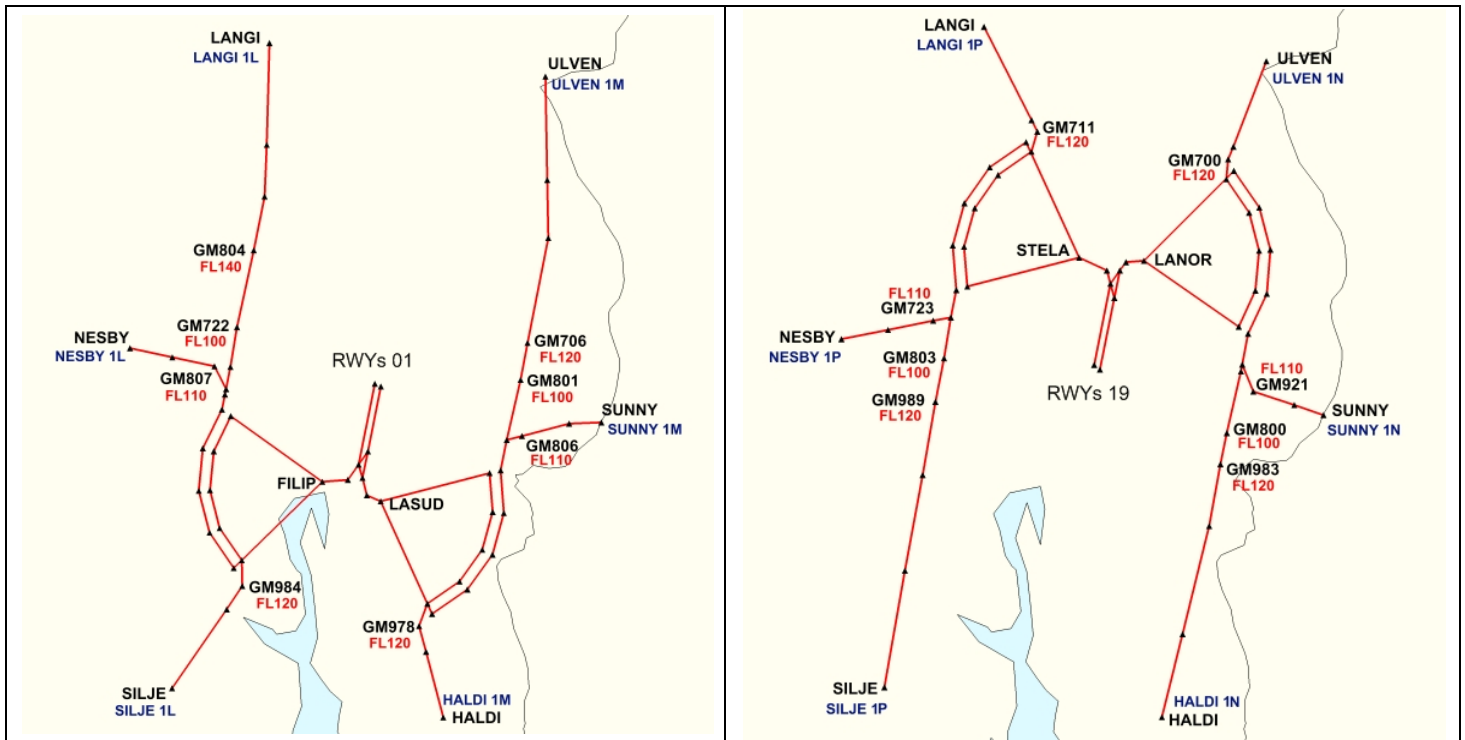


Figure 3-3 – ENGM STARS - RWYs 01 (left) and RWYs 19 (right)

The STARS were based on the Point Merge system. For each runway orientation the six STARS integrated a dual Point Merge system which consisted of:

- Four sequencing legs - curved and parallel sequencing legs of opposite directions with 2NM offset;
- Two merge points which were respectively:
 - FILIP (west side) and LASUD (east side) with RWYs 01 and
 - STELA (west side) and LANOR (east side) with RWYs 19.

Vertical constraints along the procedures were defined as well. In particular FL constraints were defined at the sequencing legs to allow for vertical separation of parallel and of opposite directions flows (traffic on inner legs at higher FL than traffic on outer legs). Vertical constraints were as well defined in order to provide appropriate separation prior joining the ILS axis for close parallel approaches.

The two diagrams below better focus on the sequencing legs when runways orientation in use at ENGM is either 01 or 19.

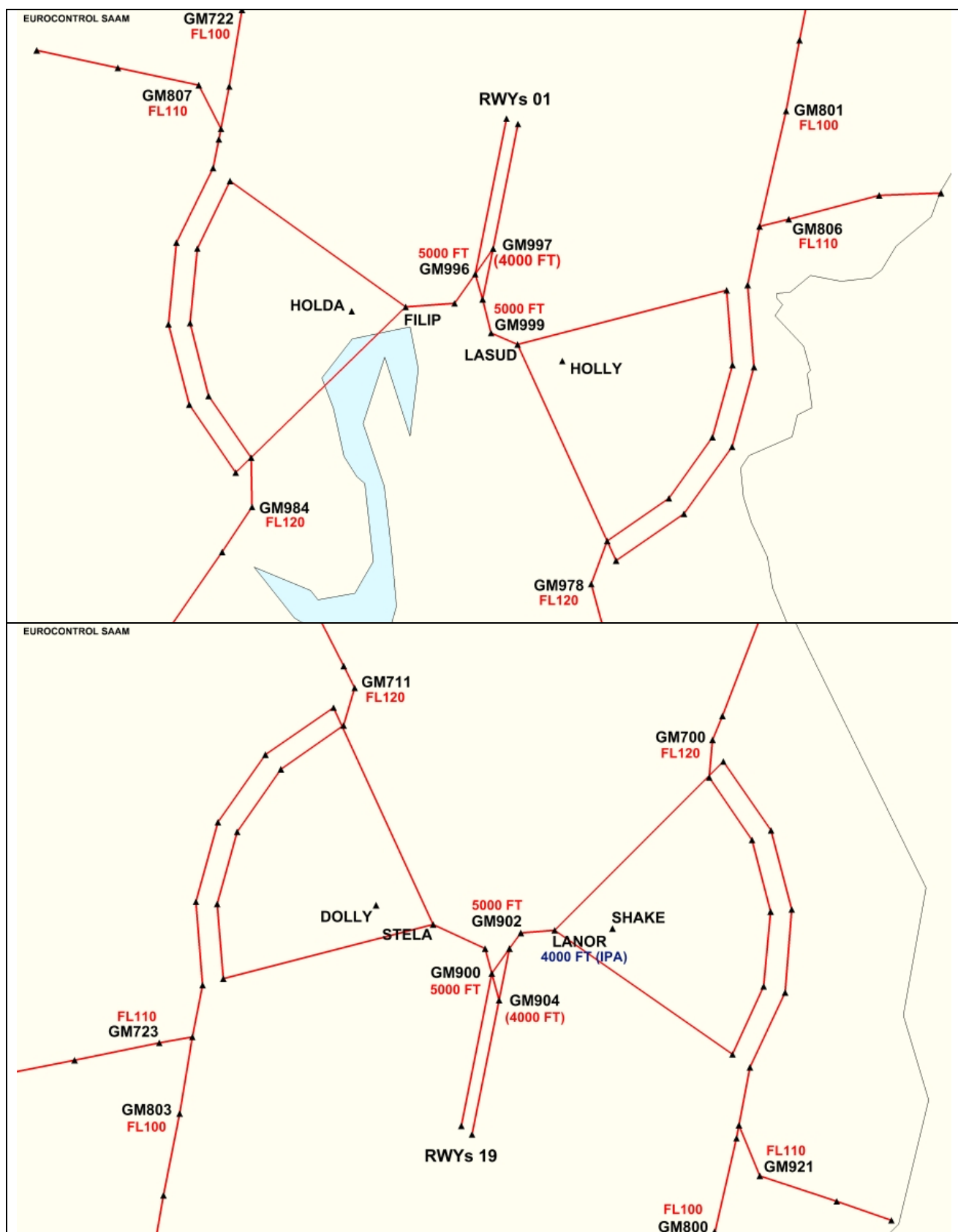


Figure 3-4 – ENGM Sequencing Legs - RWYs 01 (top) and RWYs 19 (bottom)

The RNAV SIDs at ENGM were conceived and designed so that to integrate with the STARs and to segregate arriving and departing traffic provided that defined vertical constraints were respected.

The diagrams hereafter draw the main ENGM SIDs (with the exclusion of the special SIDs) which operated respectively with runways orientation 01 and 19. Special SIDs (not shown in the diagrams) were used for either propellers, general aviation aircraft or aircraft requiring the longest runway.

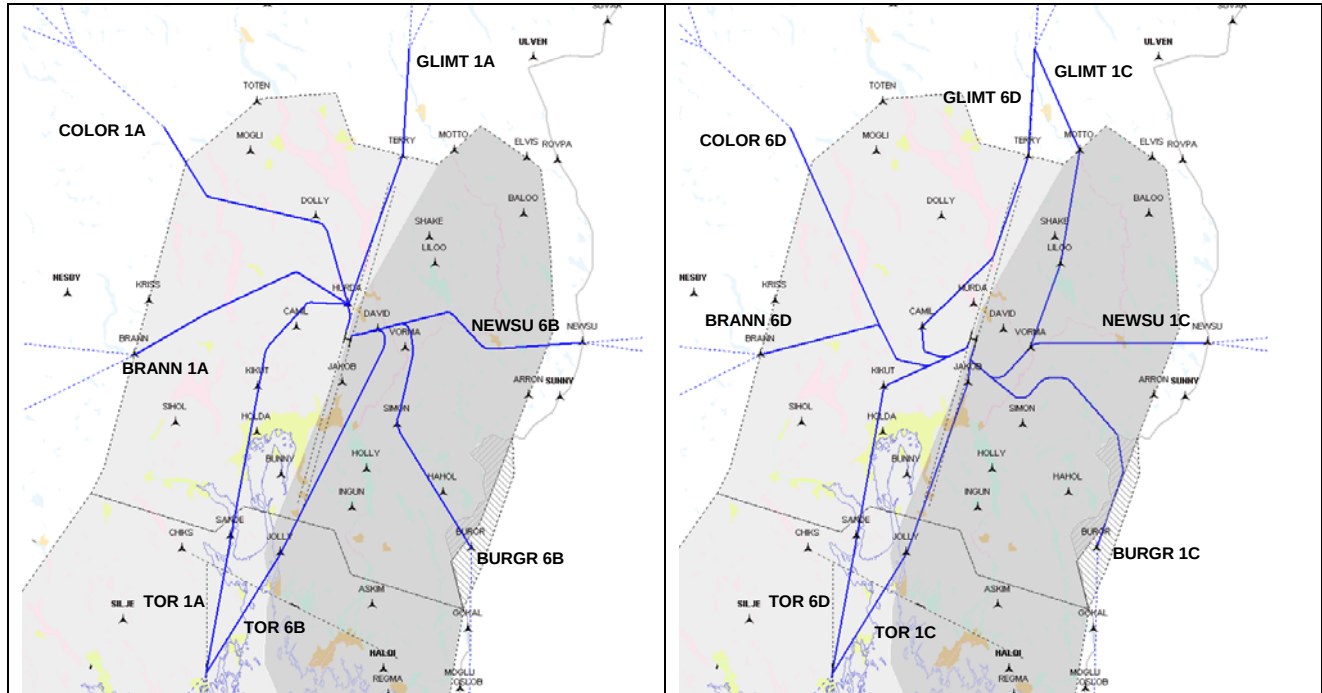


Figure 3-5 – ENGM SIDs - RWYs 01 (left) and RWYs 19 (right)

3.1.3.2. Farris TMA

The new STARs defined for ENRY and ENTO were P-RNAV based.

The arriving and departing procedures concerning ENRY and ENTO were designed to interfere as little as possible with the main traffic flows which head up to ENGM. The STARs and SIDs, as they were defined, are shown in the following diagrams.

No SID/STAR was defined for ENSN (as today's situation).

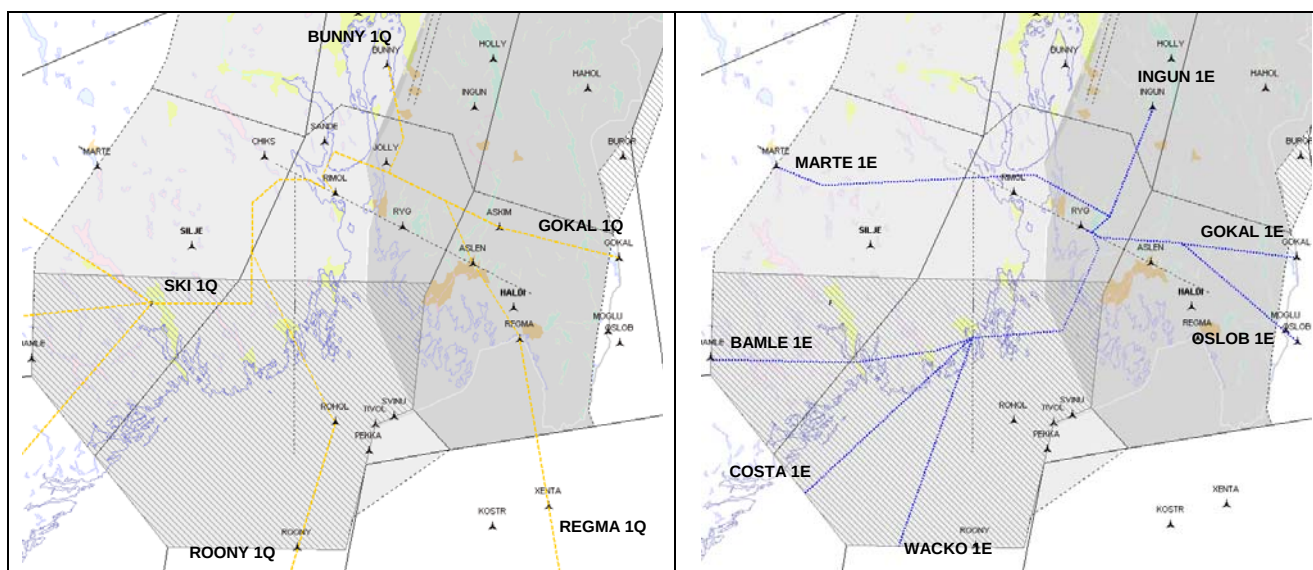


Figure 3-6 – ENRY RWY 12 STARs (left) and SIDs (right)

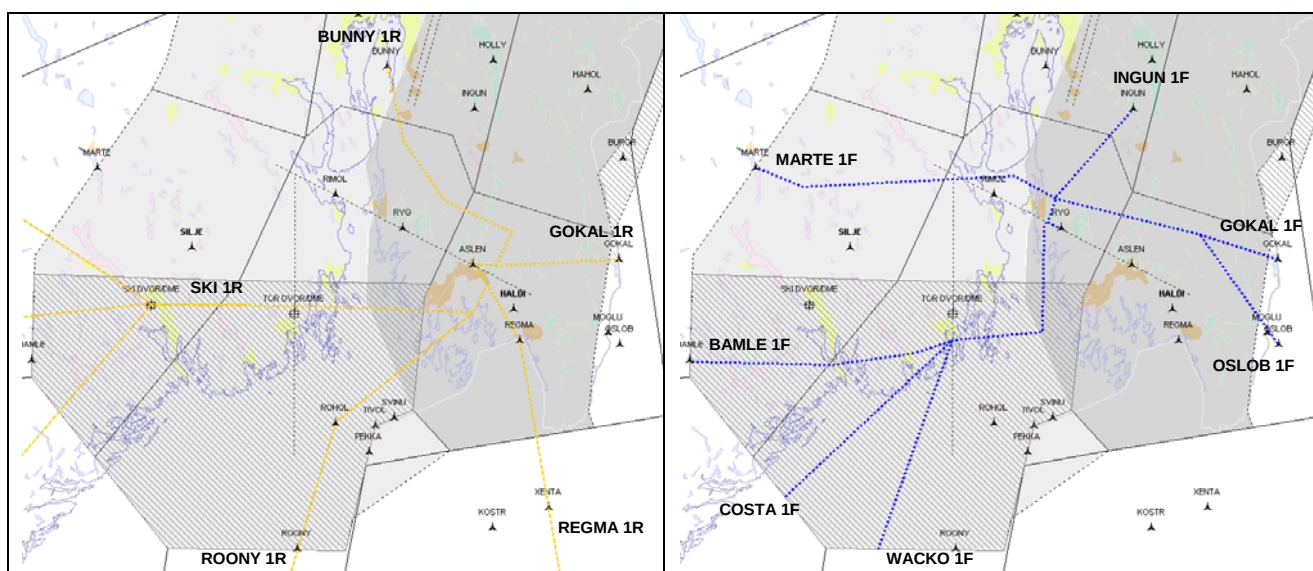


Figure 3-7 – ENRY RWY 30 STARs (left) and SIDs (right)

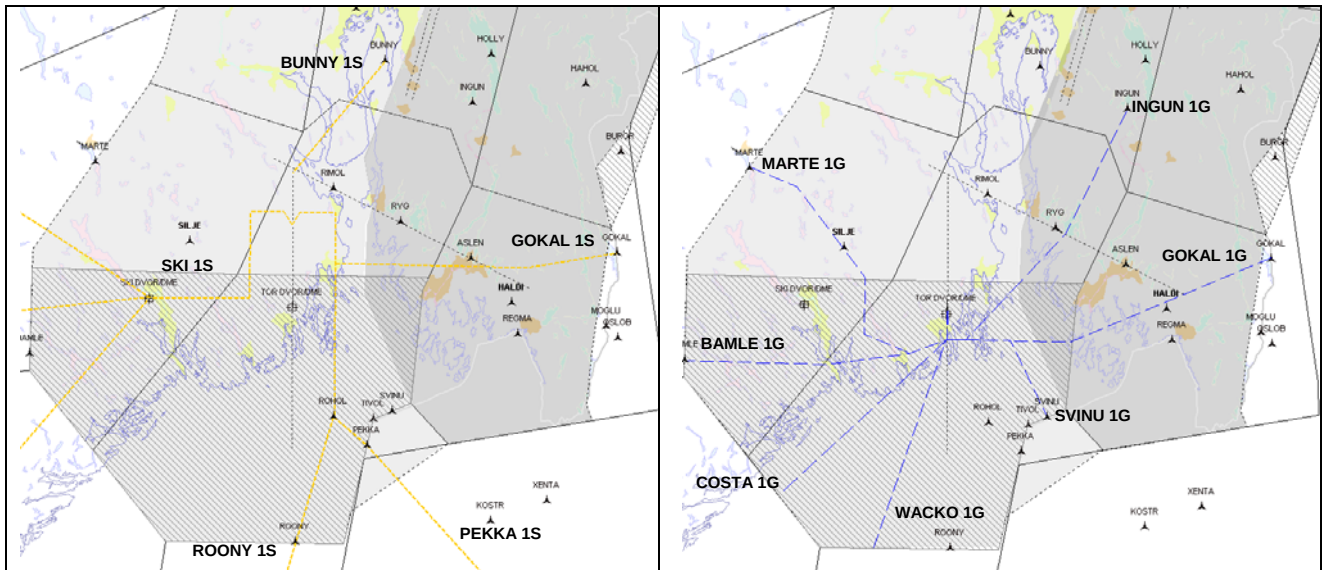


Figure 3-8 – ENTO RWY 18 STARs (left) and SIDs (right)

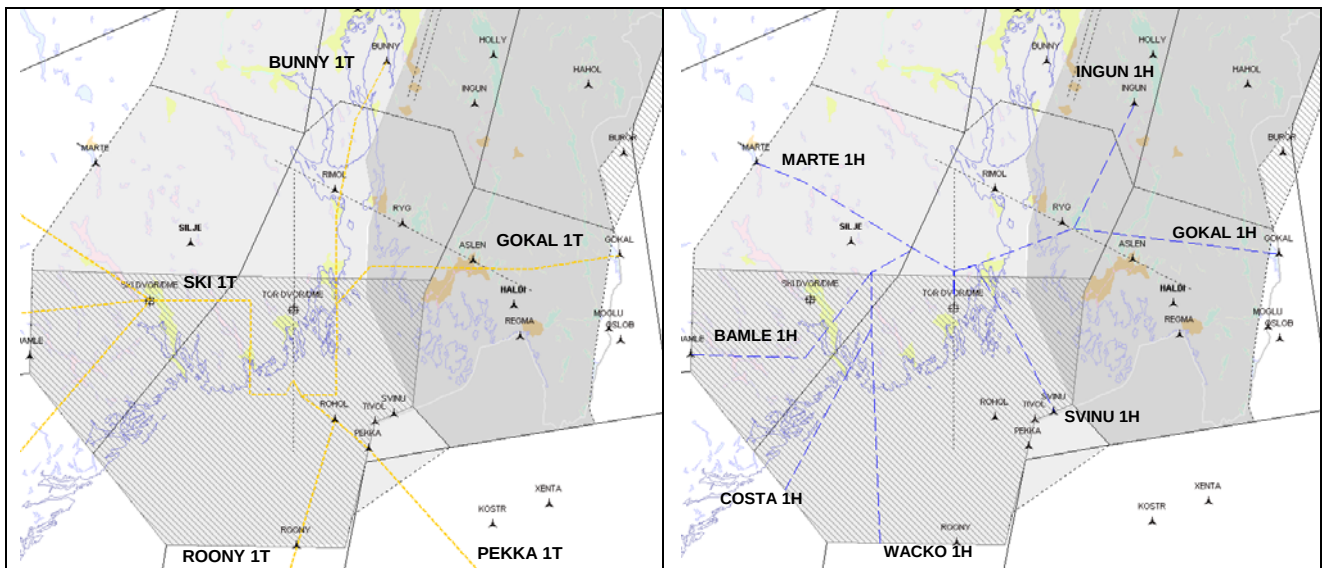


Figure 3-9 – ENTO RWY 36 STARs (left) and SIDs (right)

3.1.4. Military Areas

Two temporary segregated areas were simulated and were Dovre and Rena areas. They are located in the northern part of Oslo AoR. In case of military activity the standard routing of traffic in contact with the Oslo ACC sectors S1 and S7 was impacted (and in particular ENGM inbound traffic via LANGI and ULVEN). A military corridor joining the two areas was simulated as well.

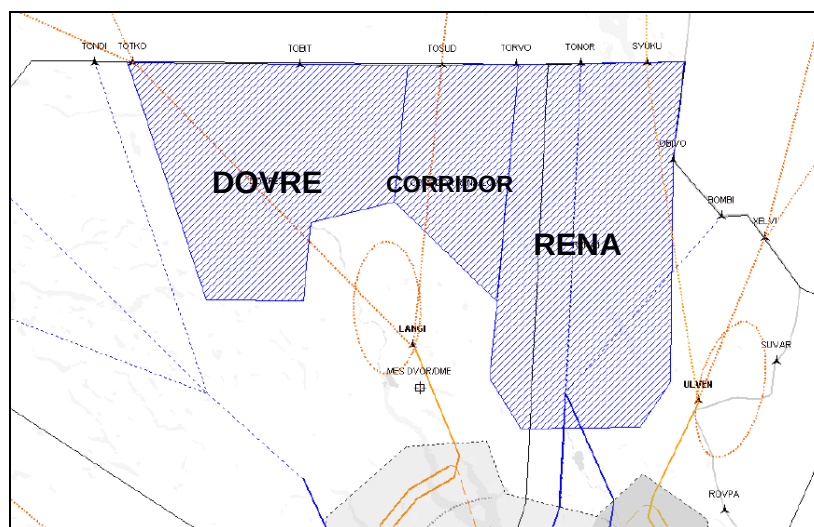


Figure 3-10– Military Areas

The vertical limits of the simulated military areas were as follows:

- DOVRE: FL115/135 – FL285
- RENA: FL095-FL215
- Corridor FL145-FL205

3.1.5. Traffic

3.1.5.1. Characteristic

The traffic scenarios applied in the simulation were based on seven traffic samples. The traffic samples were prepared to better suit the operational requirements (e.g. modes for parallel approaches at ENGM) and present different levels of load and complexity.

Three traffic samples (EXE2, 3 and 7) were designed to go along with the ENGM 01L + 01R runway configuration. Four traffic samples (EXE1, 4, 5 and 6) were instead used to simulate 19L + 19R runways usage. All the traffic samples were constructed to suit MPO at ENGM, with the exception of EXE4 which in fact was designed to reflect the higher volume of traffic at the airport that IPA allows.

The main traffic samples characteristics are reported in the next table.

Table 3-1– Simulation Traffic Samples (TS)

TS	Characteristic		
	ACC traffic	ENGM traffic	Farris traffic
EXE1	Approximately 25-34 flights per hour.	Approximately 80-83 movements per hour. Traffic evenly distributed between West and East side.	Approximately 31-34 movements per hour. Traffic evenly distributed between West and East side.
EXE2	Approximately 21-37 flights per hour.	Approximately 79 movements per hour. Traffic evenly distributed between West and East side.	Approximately 24 movements per hour. Traffic evenly distributed between West and East side.
EXE3	Approximately 22-34 flights per hour.	Approximately 83-84 movements per hour. Traffic evenly distributed between West and East side.	Approximately 31-34 movements per hour. More traffic in West side.
EXE4	Approximately 25-34 flights per hour.	Approximately 95 movements per hour. Traffic based on 16.10.2008, compressed and slightly modified to be more evenly distributed between West and East side.	Approximately 31-34 movements per hour. More traffic in West side.
EXE5	Approximately 21-37 flights per hour.	Approximately 79 movements per hour. Traffic evenly distributed between West and East side.	Approximately 24 movements per hour. Traffic evenly distributed between West and East side.
EXE6	Approximately 25-34 flights per hour.	Approximately 80-83 movements per hour. Traffic evenly distributed between West and East side.	Approximately 31-34 movements per hour. More traffic in West side.
EXE7	Approximately 25-34 flights per hour.	Approximately 80-83 movements per hour. Traffic evenly distributed between West and East side.	Approximately 31-34 movements per hour. More traffic in West side.

In terms of movements, all the exercises mean a significant increase in the departures/arrivals per hour in comparison with the today's traffic. ENGM has currently a practical capacity of 65 movements per runway system, but generally not more than 60 movements are performed.

3.1.5.2. Aircraft Capabilities

All the traffic scenarios were assumed being composed of P-RNAV capable aircraft for a wide majority. A small percentage of arrivals (some 5%) were simulated operating with non P-RNAV capable aircraft.

Currently 85-90% of flights concerning Oslo TMA are certified being P-RNAV capable. It is therefore expected that within a short timeframe the values will be comparable to those simulated.

3.1.6. Meteorological Environment

Two meteorological environments were simulated. The two environments were designed to reflect prevailing wind conditions and consequently the choice of ENGM runways orientation. The algorithm for wind applied changes in speed (knots) and direction (heading) with altitude.

A comprehensive description of the wind characteristic is provided in Table 3-2.

Table 3-2 – Meteorological Setting

RWY 01		RWY 19	
Flight Level / Altitude	Wind (direction / knots)	Flight Level / Altitude	Wind (direction / knots)
FL 310 and above	340° / 30	FL 410 and above	200° / 30
FL 300	340° / 35	FL 350	190° / 30
FL 250	330° / 40	FL 280	190° / 35
FL 200	330° / 35	FL 220	180° / 40
FL 140	320° / 30	FL 180	180° / 35
FL 100	320° / 30	FL 110	170° / 30
6000'	310° / 25	FL 80	160° / 25
5000'	300° / 15	5000'	150° / 15
4000'	300° / 10	3000'	150° / 5
3000'	300° / 5	2000'	No wind
2000'	No wind		

3.2. CONTROLLERS' ROLES

Hereafter follows a brief description of what the tasks and roles of the simulated positions and sectors were, with regard to inbound and outbound traffic to and from Oslo and Farris airports.

3.2.1. Oslo ACC

The Oslo ACC sectors (with the exception of S4 and S8) had to clear arriving traffic to Oslo TMA in accordance with the tables hereafter unless otherwise coordinated. The vertical constraints and/or the route network design made arriving traffic set apart from the departing traffic. Moreover the respect of those constraints prepared the traffic, once in the TMA, to properly engage the sequencing legs so as to allow a vertical separation between flows heading up to the same leg and/or to parallel legs (refer to Figure 3-3).

The non P-RNAV equipped traffic to ENGM was subject to approval request.

ENGM RWY 01

ACC Sector	STAR-start-point	STAR name	CFL
S1	ULVEN	ULVEN 1M	FL120 at GM706
S2	SUNNY	SUNNY 1M	FL110 at GM806
S3	HALDI	HALDI 1M	FL120 at GM798
S5	SILJE	SILJE 1L	FL120 at GM984
S6	NESBY	NESBY 1L	FL110 at GM807
S7	LANGI	LANGI 1L	FL140 at GM804

ENGM RWY 19

ACC Sector	STAR-start-point	STAR name	CFL
S1	ULVEN	ULVEN 1N	FL120 at GM700
S2	SUNNY	SUNNY 1N	FL110 at GM921
S3	HALDI	HALDI 1N	FL120 at GM983
S5	SILJE	SILJE 1P	FL120 at GM989
S6	NESBY	NESBY 1P	FL110 at GM723
S7	LANGI	LANGI 1P	FL120 at GM711

The ACC sectors arranged traffic to pass the STAR-start-points with speed below 250 knots and to respect the time advisories issued by AMAN (see § 3.4).

In case of necessity the Oslo ACC sectors could instruct arriving traffic to hold at published holding patterns at STAR-start-points.

As for arriving traffic to Farris TMA, the concerned Oslo ACC sectors had to deliver traffic in descent using level clearances in accordance with the tables hereafter unless otherwise coordinated.

The ACC sectors arranged traffic on the same STAR to pass the STAR-start-point with a longitudinal separation of minimum 5NM constant or increasing, unless otherwise coordinated.

ENRY RWY 12/30

ACC Sector	STAR-start-point	STAR name	CFL	Handoff to sector
S3	GOKAL	GOKAL 1Q/R	FL120	FARE
	REGMA	REGMA 1Q/R	FL120	FARE
S4	ROONY	ROONY 1Q/R	FL160	FARW
S5	SKI	SKI 1Q/R	FL160/FL120*	FARW
S7	BUNNY	BUNNY 1Q/R	FL100	FARW

ENTO RWY 18/36

ACC Sector	STAR-start-point	STAR name	CFL	Handoff to sector
S3	GOKAL	GOKAL 1S/T	FL120	FARE
S4	PEKKA	PEKKA 1S/T	FL160	FARW
	ROONY	ROONY 1S/T	FL160	FARW
S5	SKI	SKI 1S/T	FL160/FL120*	FARW
S7	BUNNY	BUNNY 1S/T	FL100	FARW

* FL160 for flights from South-West; FL120 for flights from West and North-West

3.2.2. Oslo TMA

3.2.2.1. APP Positions

Oslo APPW was responsible for arriving traffic on LANGI 1L and NESBY 1L STARs with runway orientation 01, and for arriving traffic on NESBY 1P and SILJE 1P STARs with runway orientation 19.

After transfer from S7, APPW had to clear traffic to FL100 at waypoint GM722 on LANGI 1L STAR. Traffic handed off by S5 had to be cleared to FL100 at waypoint GM803 on SILJE 1P STAR.

As soon as the traffic was clear of traffic, and prior to the entrance in the sequencing leg, unless otherwise coordinated, APPW had to transfer it to DIRW (or DIR).

APPW was responsible for departing traffic flying western SIDs and for flights along the transit route in the western side of Oslo TMA between FL 130 and FL210.

Oslo APPE was responsible for arriving traffic on ULVEN 1M and SUNNY 1M STARs while runway orientation was 01, and for arriving traffic on HALDI 1N and SUNNY 1N STARs while runway orientation was 19.

After transfer from S1, APPE had to clear traffic to FL100 at waypoint GM801 on ULVEN 1M STAR. Traffic handed off by S3 had to be cleared to FL100 at waypoint GM800 on HALDI 1N STAR.

As soon as the traffic was clear of traffic, and prior to the entrance in the sequencing leg, unless otherwise coordinated, APPE had to transfer it to DIRE (or DIR).

APPE was responsible for departing traffic flying eastern SIDs and for flights along the transit route in the eastern side of Oslo TMA between FL 130 and FL210.

Oslo APPs had to clear departing traffic according to tables below, climbing to FL210 or cruising level if lower, unless otherwise coordinated.

Oslo TMA had to transfer departing traffic with a longitudinal separation of 5NM constant or increasing, unless otherwise coordinated.

ENGM RWY 01

Towards ACC Sector	SID name	CFL	Cleared direct
S1	GLIMT 1A	FL210	TERRY / MOTTO
S2	NEWSU 6B	FL210	NEWSU
S3	BURGR 6B	FL210	BURGR
S4	TOR 1A/6B	FL210	SANDE / JOLLY
S6	BRANN 1A	FL210	BRANN
S7	COLOR 1A	FL210	COLOR

ENGM RWY 19

Towards ACC Sector	SID name	CFL	Cleared direct
S1	GLIMT 1C/6D	FL210	MOTTO / TERRY
S2	NEWSU 1C	FL210	NEWSU
S3	BURGR 1C	FL210	BURGR
S4	TOR 1C/6D	FL210	JOLLY / SANDE
S6	BRANN 6D	FL210	BRANN
S7	COLOR 6D	FL210	COLOR

3.2.2.2. DIR Positions

Oslo DIRW was responsible for:

- ENGM arriving traffic on SILJE 1L STAR received from S5 with runway orientation 01;
- ENGM arriving traffic on LANGI 1P STAR received from S7 with runway orientation 19;
- ENGM arriving traffic received from APPW;
- Low level flights along the transit route in the western side of Oslo TMA.

Oslo DIRE was responsible for:

- ENGM arriving traffic on HALDI 1M STAR received from S3 with runway orientation 01;
- ENGM arriving traffic on ULVEN 1N STAR received from S1 with runway orientation 19;
- ENGM arriving traffic received from APPE;
- Low level flights along the transit route in the eastern side of Oslo TMA.

Oslo DIR took on both DIRE and DIRW duties when Mixed Parallel Operation worked.

In case of P-RNAV equipped aircraft, arriving traffic engaged the sequencing legs with appropriate flight level. Unless otherwise coordinated, traffic was level at FL120 on the inner sequencing legs, and at FL110/100 on the outer legs. The fact that the flight level along the inner legs was higher than the ones on the outer legs was commanded by safety aspects (in case of immediate descent when leaving the legs).

Before transferring to the FIN position, the DIR issued the following clearances:

- Turn to merge point, descent and ILS approach clearance, or
- Heading instructions and descent (required coordination with FIN);
- Speed reductions/adjustments in both cases.

Altitude restrictions were imposed when intercepting the localizer (because of close parallel approaches).

The descent clearance given by the directors after the turn to the merge point was by means of the following phraseology: "when ready descent to Altitude xxxx to be level at [Point yyyy]". On an airborne side, the procedure allowed the pilots for achieving a continuous descent for that part of the flight.

In case of necessity (e.g. temporary runways closure) the DIR positions could instruct arriving traffic to hold. With that purpose, two holding patterns per runway orientation were defined in correspondence of waypoints HOLDA and HOLLY (with runway orientation 01), and. Waypoints DOLLY and SHAKE (with runway orientation 19).

3.2.2.3. *FIN Position*

Oslo FIN confirmed the clearances given by the DIR positions and issued the instructions required to achieve/refine spacing on final as prescribed.

With runway mode IPA the prescribed spacing between aircraft on the same final was 5.5NM, which would allow for departures between two consecutive arrivals to both the runways.

In case of runway mode MPO, when doing staggered approaches, the minima separation between aircraft to different runways was 3NM and consequently the spacing between two consecutive arrivals to the same runway is in the 6.5/7.5nm range.

3.2.3. *Farris TMA*

The Farris FARE and FARW positions mainly handled traffic to/from ENTO and ENRY. Only a small amount of traffic in Farris TMA originated or terminated at ENSN.

In general FARE covered departing/arriving traffic from/to ENRY, whereas FARW departing/arriving traffic from/to ENTO.

Arriving traffic was released to Farris TMA by Oslo ACC sectors, as described in § 3.2.1. The prescribed aircraft spacing between the landings was 6NM for all Farris airports. As

far as ENSN is concerned, the FARW position had to vector the arriving traffic, as no STAR was designed.

Farris TMA positions had to clear departing traffic according to tables below, climbing to FL110/150 or cruising level if lower, unless otherwise coordinated. For flights along the same route (SID), Farris positions had to transfer departing traffic with a longitudinal separation of 5NM constant or increasing, unless otherwise coordinated.

Low level traffic transiting through Oslo TMA was directly transferred from TMA sectors to Farris after coordination.

FAR took on both FARE and FARW duties when the Farris configuration was set to one single position.

ENRY RWY 12/30

Towards ACC Sector	SID name	CFL	Cleared direct
S3	GOKAL 1E/F	FL110	GOKAL
	OSLOB 1E/F	FL110	OSLOB
S4	COSTA 1E/F	FL150	SF
	BAMLE 1E/F	FL150	SF
	MARTE 1E/F	FL110	MARTE
	INGUN 1E/F	FL110	INGUN
	WACKO 1E/F	FL150	SF

ENTO RWY 18/36

Towards ACC Sector	SID name	CFL	Cleared direct
S3	GOKAL 1G/H	FL110	GOKAL
S4	SVINU 1G/H	FL150	SVINU
	COSTA 1G/H	FL150	COSTA
	MARTE 1G/H	FL110	MARTE
	INGUN 1G/H	FL110	INGUN
	WACKO 1G/H	FL150	WACKO
S5	BAMLE 1G/H	FL150	BAMLE

3.3. SPECIAL PROCEDURES

Throughout the simulation some special procedures were simulated in order to assess their operability. In particular they refer to runway changes (e.g. from 01L to 01R using ILS cross over procedure) and missed approaches.

Hereafter a brief description of those procedures is provided.

3.3.1. ILS Cross Over Procedures

Duty of the DIR position was to initiate, on request of either TWR or PLN controller, ILS cross over procedures. The aim of that procedure is generally to suit operational needs (e.g. ease the sequence building by balancing the traffic load from one RWY to another, create gaps between landing to serve departing traffic, suit airport preferences).

The ILS cross over procedure was used with runway mode MPO only. The runway change was ideally announced to the pilot when the aircraft was still flying along the sequencing legs, so as s/he could update the route to join the localizer. Therefore the DIR controller cleared the left/right turn direct to merge point (as “normal” procedure). The descent clearance could vary from the normal procedure.

3.3.2. Missed Approach Procedures

Different missed approach procedures were defined and tested during the simulation. The procedures were designed so as to allow the re-insertion of the aircraft in the landing sequence without any disruption of the system based on Point Merge method.

The tasks within the Oslo TMA positions were defined as follows (the example is based on runway orientation 01):

- TWR instructs the aircraft to fly a heading (270 / runway heading / 055) and climb to altitude 4000 / 5000 / 4000. Then, TWR performs the appropriate coordination (with PLN and APP sectors, and transfers the flight to the concerned position (APPE or APPW);
- APP issues radar vectors as appropriate to rejoin the sequence and transfers the flight to DIR;
- DIR issues radar vectors as appropriate to rejoin the sequence and transfers the flight to FIN;
- FIN vectors the flight towards the ILS.
- AMAN is updated by PLN during the process.

The diagrams hereafter detail the missed approach procedures in both runway orientations.

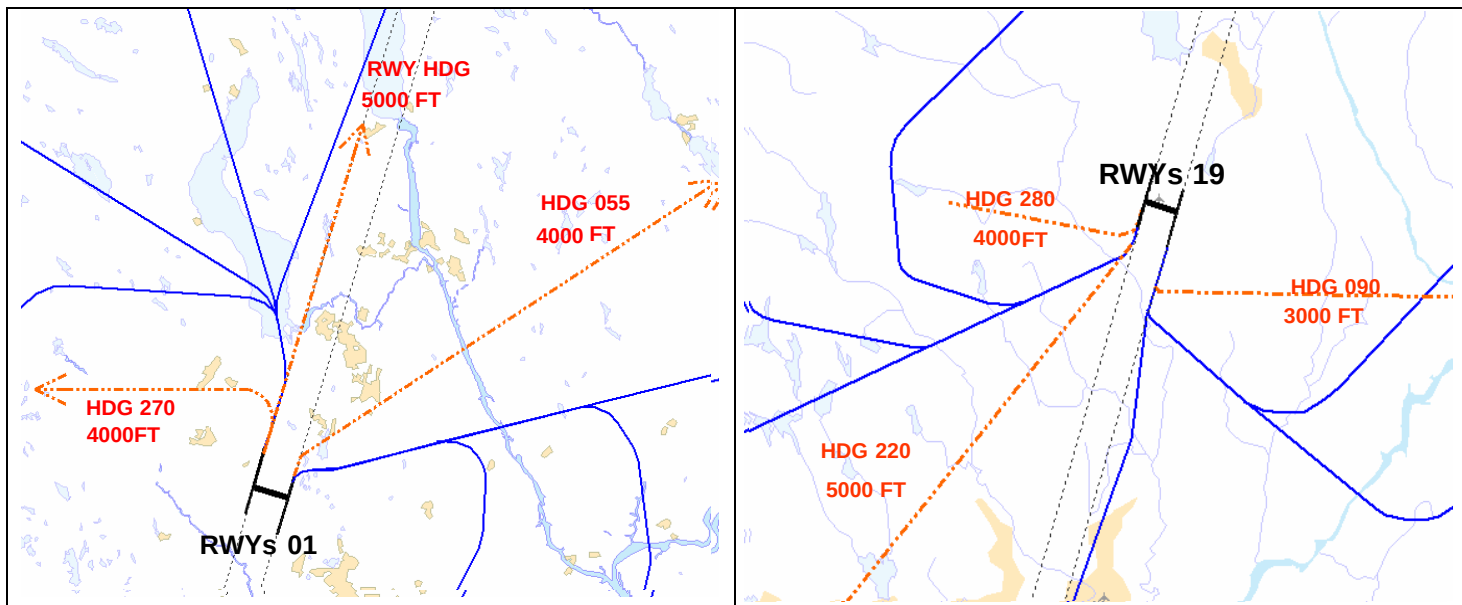


Figure 3-11 – Missed Approach Procedure at ENGM

3.4. SUPPORTING TOOL (AMAN)

Based on the experience from the previous phases, the simulation employed an arrival manager (AMAN) which was configured for the ENGM arriving traffic management.

AMAN proposed optimised sequences both at the STAR-start-points and at the runway thresholds. Along with that, the arrival manager provided time advisories (time to lose/gain at the reference points in order to comply with the sequence proposed).

The planner controller (PLN) was in charge of monitoring AMAN and proposing strategies to the Oslo ACC sectors in order to meet the advisories. The PLN had also the power to intervene in the process changing the sequence order proposed by AMAN or simply updating the time estimations over the reference points (STAR-start-points, runway threshold).

As much as it was possible, the relevant ACC sectors could apply speed control and/or headings for delay absorption. To lose a significant amount of time or when instructed by the PLN, the controllers could hold traffic at published holding patterns at the STAR-start-points.

The guideline given to the ACC controllers for the simulation was to meet the AMAN time advisories as closely as possible³. The compliance with AMAN proposals by the upstream sectors would allow an appropriate delivery of traffic to Oslo TMA positions (APPE/W and DIR). In case of nominal situations the absorption of the delay by means of speed control and/or vectors might also reduce the need to hold traffic, especially if those actions were taken appropriately in advance (before traffic entered the TMA).

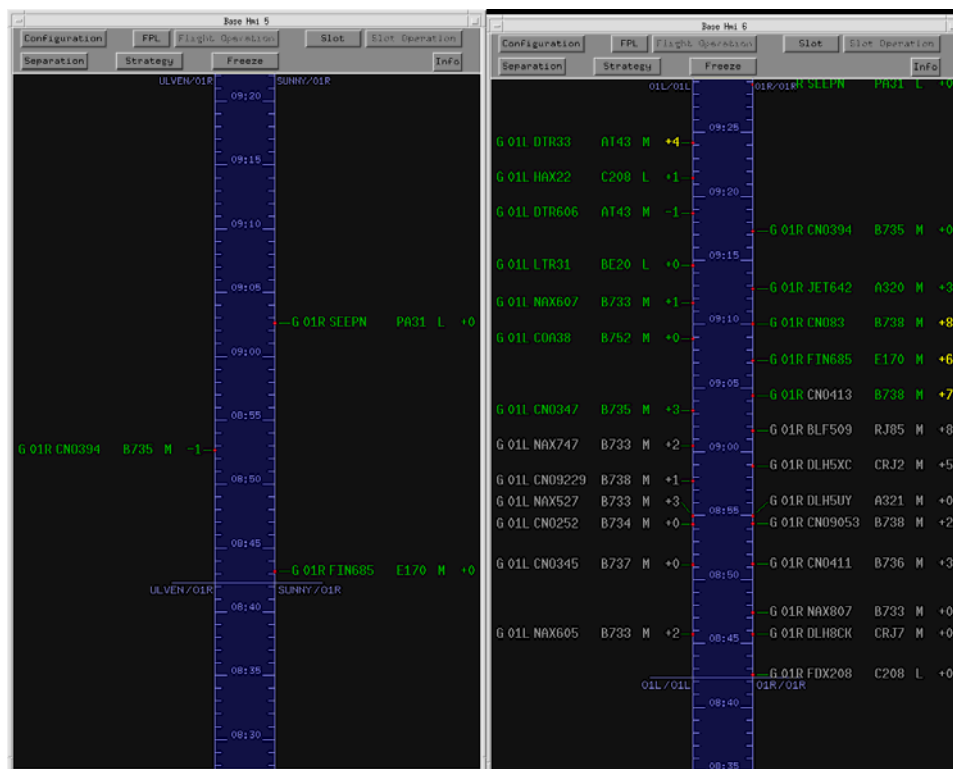


Figure 3-12 – AMAN Timelines

³ In the sessions within Phase B the guideline was to meet AMAN time advisories within a +/- 4 minutes tolerance in case the controllers made use of speed controls or vectors. Otherwise in case of holdings the target was to meet the AMAN time more accurately (normally within a +/- 2 minutes tolerance).

Page intentionally left blank

4. EXPERIMENTAL DESIGN AND SIMULATION CONDUCT

4.1. EXPERIMENTAL VARIABLES AND ORGANISATIONS

The design of the simulation consisted of seven different set of experimental conditions which we refer to as organisations. The experimental conditions were built around the following independent variables. They were:

- V1: The ENGM modes for parallel approaches:
 - Mixed Parallel Operations (MPO)
 - Independent Parallel Approaches (IPA)
- V2: The ENGM runway usage:
 - RWY 01L + 01R
 - RWY 19L + 19R
- V3: The ENRY runway usage:
 - RWY 12
 - RWY 30
- V4: The ENTO runway usage:
 - RWY 18
 - RWY 36
- V5: The ACC and TMAs configurations (measured sectors⁴)
- V6: The activation of temporary military areas:
 - No temporary segregated areas active
 - Activation of Dovre, Rena temporary segregated areas along with the corridor (the activation lasted the whole duration exercise)
- V7: The meteorological environment:
 - No wind
 - Wind conditions (setting as described in § 3.1.6)

Out of all the possible combinations of the independent variables, **seven organisations** were set up to address the simulation objectives. For practical reasons we refer to the organisations as:

- **Org1, Org2, Org3, Org4, Org5, Org6 and Org7**

No organisation reflecting the current operations in Oslo AoR was run in the simulation. All the organisations set up were therefore employing the re-organisation of the airspace and the new procedures.

The seven traffic scenarios (see § 3.1.5.1) were created to be appropriate for the different organisations. Moreover the different characteristics of the traffic scenarios ensured variability in the simulation. This, combined with the rotation of controllers in different sectors/positions, reduced significantly the controllers' learning effect throughout the simulation.

⁴ We refer to the measured sectors as those sectors and positions which were, over the duration of the simulation, the main object of the analysis.

Each traffic scenario was coupled to only one organisation. The next table summarises and adds details on the different organisations employed in the simulation.

Table 4-1 – Simulation Organisations

						Runway Usage					
			Measured Sectors			ENGM		ENRY		ENTO	
Org	Traffic	Mil Activities	Oslo TMA	Farris TMA	Oslo ACC	RWY01	RWY19	RWY12	RWY30	RWY18	RWY36
Org4	EXE 4	-	APPE APPW DIRE DIRW FIN	FARE FARW	S1_2 S3 S4 S5 S6_7 S8		IPA		X	X	
Org1	EXE 1		APPE APPW DIR FIN				MPO	X		X	
Org6	EXE 6						MPO		X	X	
Org3	EXE 3					MPO			X		X
Org7	EXE 7					MPO			X		X
Org2	EXE 2	Dovre Rena Corridor		FAR	S1_2 S3 S4_5 S6 S7 S8	MPO		X			X
Org5	EXE 5						MPO	X		X	

The difference in the experimental conditions between Org3 and Org7 lies in the meteorological environment. The former did not feature any wind. The traffic load in EXE3, as far as ENGM arrivals are concerned (see § 3.1.5.1), had been deemed by the AVINOR Core Team too high for the Oslo TMA positions in case of adverse wind conditions. Based on that rational was the decision to run Org3 exercises with no meteorological environment active.

The Org2 and Org5 were characterised by the traffic scenarios which presented a lower arrivals rate to ENGM (see Table 3-1). Therefore the exercises within those organisations were selected to host temporary runway closures at ENGM.

4.2. OPS ROOM LAYOUT

Three different ops room layouts, varying in accordance to the different measured sector combinations, were prepared and used during the simulation (see in Annex A).

Each measured sector was associated to a single CWP (single man operations manned by an Executive controller). Each CWP was equipped with:

- A BARCOTM monitor, with a multi-window working environment. The HMI used for the simulation was simplified compared to the one which will be available at Oslo ATCC when the new airspace organisation goes live;
- A three-button mouse;
- A Flight Progress Strip (FPS) printer;
- A Flight Progress Board (FPB);
- A digital voice communication system (Audio-LAN) with a headset, a loudspeaker, a footswitch and a panel-mounted push-to-talk facility (to simulate radio and telephone communications);
- An ISA (Instantaneous Self-Assessment) subjective workload input device.

In addition, all the CWPs (with the exception of S4⁵), were accompanied by a display showing the BARCOTM AMAN timelines. The timelines provided the controllers with the proposed sequences both at the IAFs and at the runway thresholds, along with the time advisories.

Apart from the CWPs corresponding to the measured sectors, the layouts consisted of other two manned but not measured positions, devoted to the planner controller (PLN) and to the ENGM tower position (TWR). Finally an automated position for ENRY and ENTO tower was present.

4.3. PARTICIPANTS AND SEATING PLAN

Fifteen controllers in total, committed by AVINOR, participated in the simulation. All of them were already familiar with the simulation setup (and concepts) because of their participation in the sessions in the previous project phases.

The participants adhered to a roster which had been prepared by AVINOR.

Seven controllers were in charge of managing the Oslo and Farris TMAs positions, which, depending on the configuration, varied from five to seven positions. Among these controllers, only two had the opportunity to man positions of both the TMAs.

A separate group of six controllers were devoted to controlling the Oslo ACC sectors.

Finally two controllers, who were also part of the Oslo ASAP core team, were employed to manage the non-measured positions, which were the ENGM TWR and the PLN.

⁵ S4, departure sector, with no route leading to an IAF, was not participating in the management of ENGM arriving traffic.

4.4. SIMULATION SCHEDULE

The simulation consisted of 22 exercises over seven days.

Each traffic scenario (EXEx) was designed so as to allow the execution of two consecutive exercises (EXEx-1 and EXEx-2), and each exercise was about 1h and 15min long. The traffic scenario prepared to suit IPA runway mode at ENGM allowed for the execution of only one exercise (approximately 1h and 30min long).

The schedule allowed for running three exercises per organisation. In addition, for Org2, an additional exercise was run at the end of the simulation.

As already stated, all the controllers involved in the simulation had taken part in the other sessions (and the majority participated in session B3 which took place only two weeks before the simulation). Thus it was assumed the participants had already an appropriate grasp of the HMI and of the new operational elements to assess.

As a consequence, differently to what was done in the previous sessions, no training exercises were planned at the early stage of the simulation. Generally the objectives of the training are to:

- Provide the controllers with a sufficient knowledge of the ATM concepts assessed during the simulation;
- Familiarise the controllers with the airspace settings and with the operational procedures and working methods that will be applied during the simulation;
- Provide the controllers with a sufficient knowledge and practice of the platform functions e.g. Audio-LAN, ISA) and HMI.

At the end of each exercise, the participants were asked to complete a post-exercise questionnaire to get their subjective feedback on the run. Moreover, briefing and debriefing sessions were held before and after the exercises.

The schedule of the simulation is depicted in Figure 4-1.

[illegible]

Figure 4-1 – Simulation Schedule

4.5. MEASUREMENTS

The results shown in the present document are the outcomes of both qualitative and quantitative assessments, which were based on human data and objective measurements.

4.5.1. Human Data Collection Methods

4.5.1.1. Briefings

Before each exercise, the objectives and the general organisation of the exercise were presented to the controllers.

4.5.1.2. Debriefings

Debriefings were conducted at the end of each exercise to enable participants to discuss their feeling regarding the feasibility and the acceptability of the tested concept, and evoke more specifically what they experienced during the exercise, e.g. confirm appropriateness of procedures, describe problems or difficulties encountered.

4.5.1.3. Post-Exercise Questionnaires

The aim of the Post-exercise questionnaire (see in Annex B) was to collect immediate feedback on the run, with a specific focus on:

- Workload;
- Situation Awareness;
- Feasibility and acceptability of the concept and the induced new working method.

4.5.1.4. Instantaneous Self-Assessment (ISA)

In ATM, the objective is to keep controllers' global workload within a range where they are kept stimulated without going to the point where they become overloaded and start to be behind the tasks.

During the simulation exercises, the participants (measured sectors only) could assess and report their perceived workload by means of the ISA technique.

With ISA, the controllers were periodically prompted visually to assess and then report the workload level on a scale from 1 to 5. As done for the past sessions, the period between two interrogations was 3 minutes long.

The scale interpretation is given in Table 5.

Table 4-2 – ISA Rating Scale

Level	Workload	Spare Capacity	Description
5	Very High (VH)	None	Behind on tasks. Losing track of the full picture.
4	High (H)	Very Little	Non essential tasks suffering. Could not work at this level very long.
3	Fair (F)	Busy	All tasks well in hand. Busy but stimulating pace. Could keep going continuously at this level.
2	Low (L)	Ample	More than enough time for all tasks. Active on ATC task less than 50% of the time.
1	Very Low (VL)	Very Much	Nothing to do. Rather boring.

4.5.2. Objective Measurements

4.5.2.1. General requirements

Several aspects were assessed by objective data, collected by means of system recordings during the simulation exercises. The recorded data concerned controller and pilot inputs, communications (R/T and telephone) and aircraft navigation data.

The MUDPIE (Multiple User Data Processing Interactive Environment) analysis tool was used both to retrieve the recorded data from the simulation platform and to deliver them in a format that could be used for data analysis and exploration.

The SAAM (System for air traffic Assignment and Analysis at a Macroscopic level) tool was used to produce charts depicting 2D flown trajectories.

4.5.2.2. Data samples

The data were collected for the full duration of the exercise and comprised also the traffic build-up period (5 up to 15 minutes at the beginning of the exercise).

The analysis was done and results are shown with regard to one hour of each exercise. Therefore the build-up and familiarisation periods were excluded from the analysis.

Page intentionally left blank

5. RESULTS

5.1. OPERABILITY

The literature defines operability as: *“the changes envisioned by the novel concept are] usable by and suitable for those who operate the system, e.g. controllers and pilots. Satisfaction of usability and suitability issues leads to operational acceptability.”*

The objective of the sessions beforehand and the simulation was to provide an assessment of operability aspects of the changes envisioned by the implementation of a new airspace structure and the adoption of new working methods, from a controller perspective, relying on subjective/objective measurements. Finally the assessment aims to answer whether the changes can acceptably be operated by Air Traffic Controllers.

5.1.1. Acceptability

5.1.1.1. Airspace Organisation

The participants generally agreed that the new airspace organisation offered a significant improvement compared to the current situation.

The controllers who manned the ACC sectors reported their level of satisfaction with the proposed airspace design, in terms of sectorization and ATS routes.

More in detail the participants recognised that various configurations in the ACC sectorization are viable solutions. Two different configurations were tested which varied in terms of sectors being collapsed. Subject to appropriate traffic demand level, the controllers found manning the proposed sector collapsed as acceptable. Other alternatives that the simulation could not explore due to lack of time might exist. Beyond the total amount of traffic in a collapsed sector, the recommendation is to take into consideration the range needed to display the airspace on the radar. The underlying issue is to prevent clutter and assure the readability of the information displayed.

The additional value the new airspace design brought is the longitudinal segregation of inbound and outbound routes from and to ENGM. With that regard the feedback obtained was positive and the participants expressed an ample satisfaction.

Minor concerns were expressed concerning S5 because of the adjacency of S4 border with the SILJE 1L STAR. In fact with runway 01 in use the leg between the points CHIKS and GM984 was deemed being too close to the border (2.5NM).

The following diagrams (Figure 5-1) summarise the feedback the participants gave by means of the questionnaires on airspace design and routes segregation. The results refer to the average values obtained grouping all the exercises in the different conditions. Feedback concerning S4_5, S6 and S7 are not presented here as it is supposed the outcome is already captured by the feedback concerning S4, S5 and S6_7.

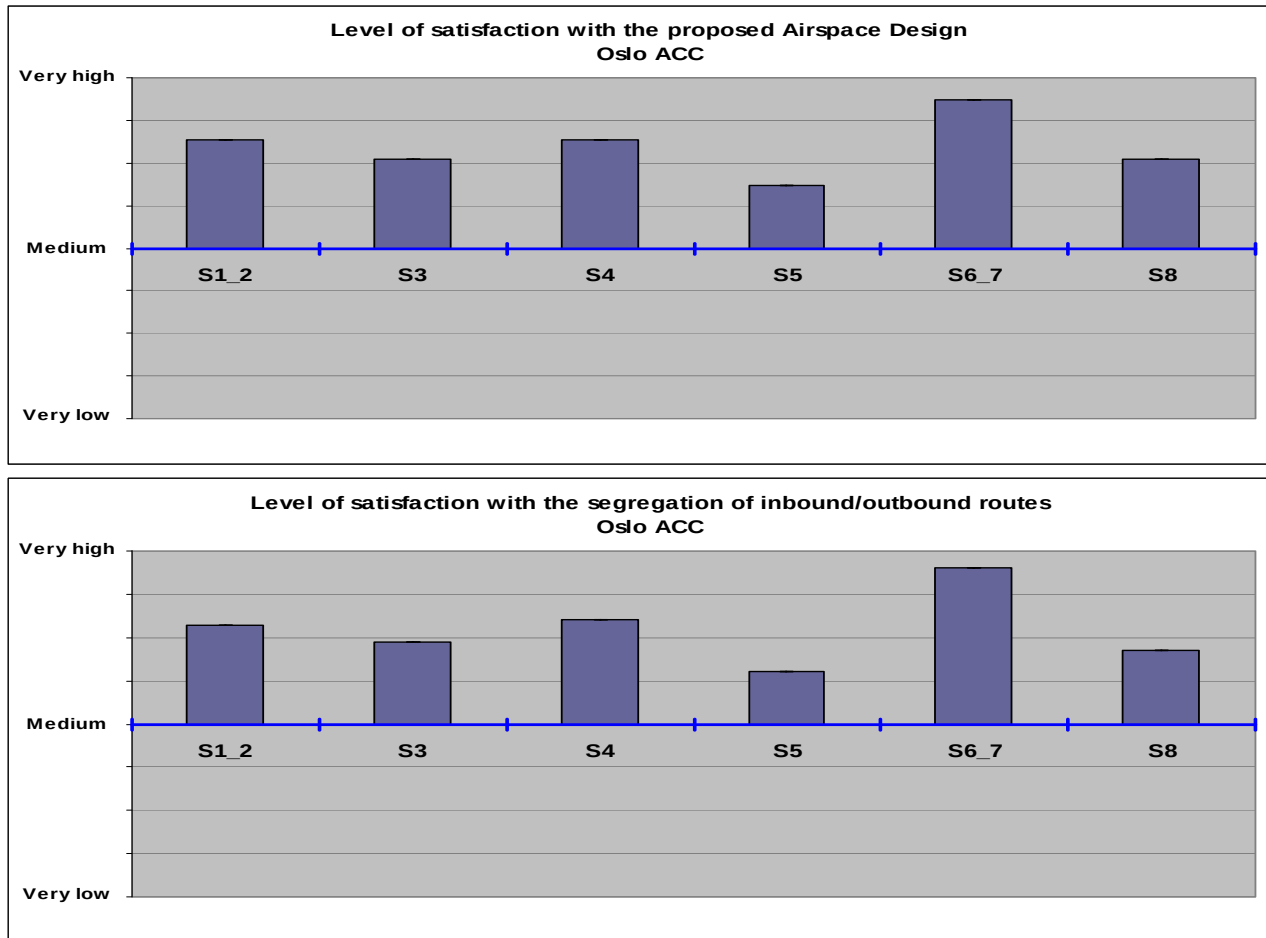


Figure 5-1 – Subjective Feedback on Airspace Design and Routes Segregation

One of the objectives of implementing a new airspace organisation in Oslo AoR is to make the interaction between the ACC and the Oslo and Farris TMAs more efficient. The feedback obtained from the participants was encouraging. Globally the participants agreed that the proposed airspace organisation facilitated the interactions also by reducing the amount of coordination needed in comparison with the today's situation.

The following diagrams (Figure 5-2 and Figure 5-3) summarise the feedback the participants gave by means of the questionnaires. They were asked whether the proposed airspace organisation allowed for a good interaction between the ACC and the TMAs.

The results are presented from both perspectives (ACC vs Oslo/Farris TMA and Oslo/Farris TMA vs ACC) and refer to the average values obtained grouping the exercises according to the different runways orientations in use at ENGM, ENRY and ENTO. The same diagram shows all the concerned positions or sectors which were simulated in the different configurations.

The interaction between Oslo and Farris TMAs was very limited because of the airspace design. Consequently it was not considered as a main objective to look at. However it did not raise any issues.

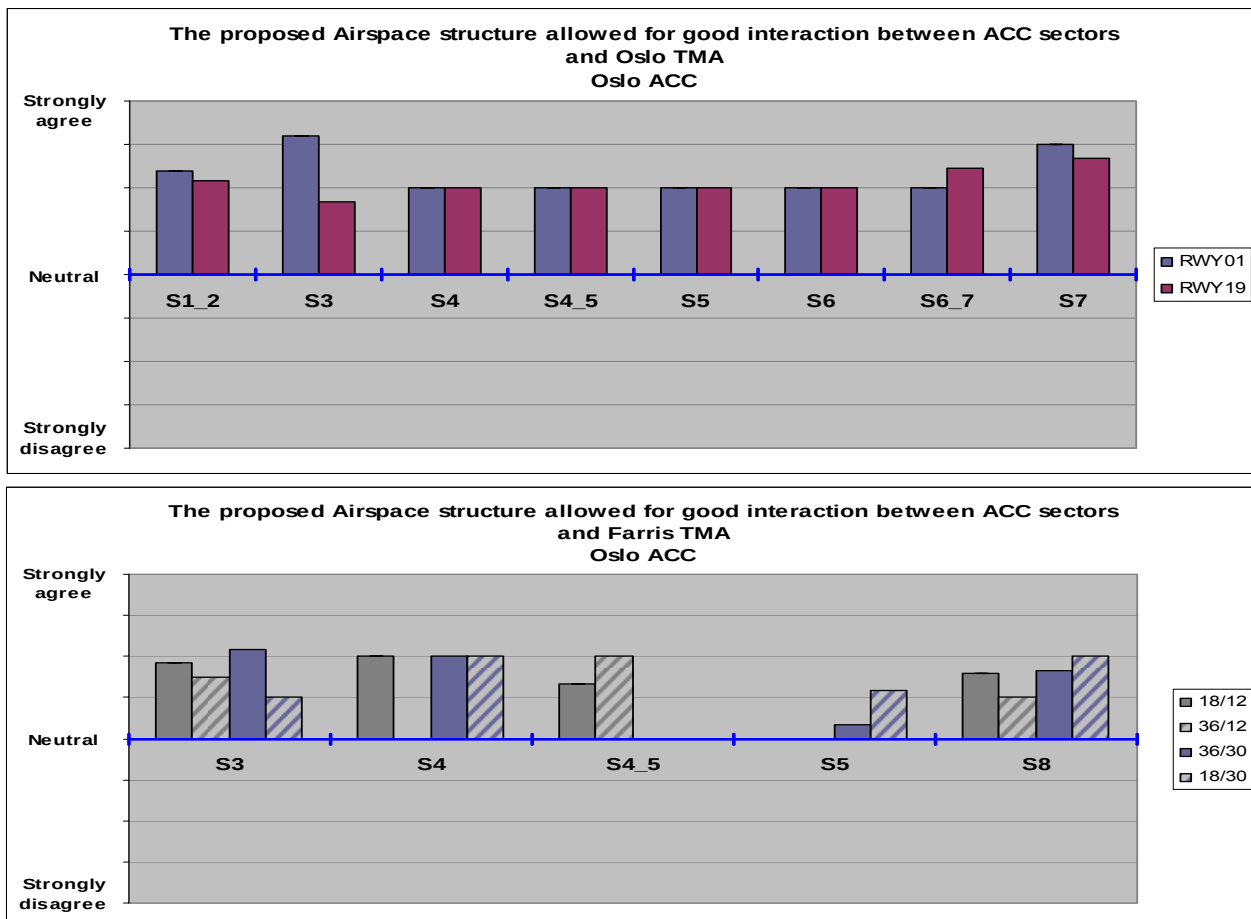


Figure 5-2– Interaction ACC Sectors ↔ TMAs (ACC Perspective)

On the ACC perspective (Figure 5-2), the new airspace organisation was generally deemed as suitable to allow for good interactions between controllers. The result concerning the interaction with Oslo TMA appears being independent from the ENGM runways orientation (with the exception of S3). The interaction between the ACC sectors and the Farris TMA seems instead more influenced by the different airports runway schemes simulated.

On the TMA perspective (Figure 5-3), the controllers who were in charge of the Farris positions largely agreed on the benefits brought by the proposed airspace organisation in terms of good interaction with the ACC (in that case the effect of the airports runway schemes was minimal). So did, but to a minor extent, the controllers responsible for Oslo TMA positions.

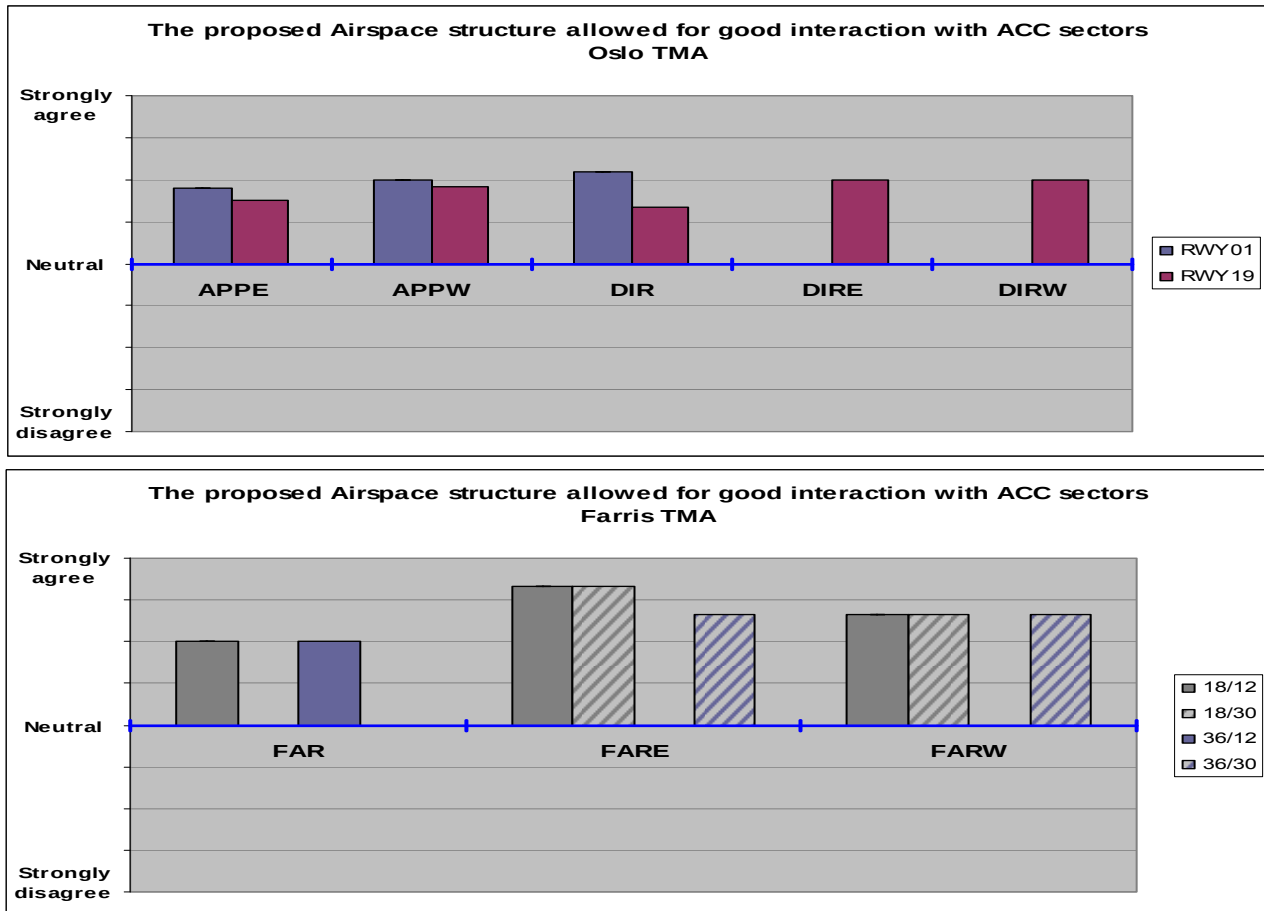


Figure 5-3 – Interaction ACC sectors ↔ TMAs (TMA Perspective)

The SID/STAR organisation in Oslo and Farris TMA had been simulated for several sessions before the simulation. All the controllers had already gained a lot of experience and had witnessed the successive improvements implemented in the design. The simulation “only” confirmed the feedback gathered during the previous phases of the project.

As for Oslo TMA, the SIDs were conceived and designed so as to best integrate with the STARs and to strategically segregate arriving and departing traffic.

Following the procedures as they were designed, the controllers agreed on the effectiveness of the design. The participants who rotated on the APPs positions did not encounter any problems in managing departing and arriving traffic. The respect of the procedures and of the vertical constraints assured the inbound and outbound flows were not in conflict. That led to a reduced amount of radar interventions by the controllers and generally the prevailing task was the monitoring of the traffic. If the situation allowed for it, the departing traffic was instructed to shortcut. The participants reported that it was not difficult to handle traffic on direct routing as most of the SIDs allowed for it with a safe tolerance.

As far as the Farris TMA is concerned, the proposed arrival routes allowed for an easier handling of the traffic than compared to the current system. That response was unanimous and independent from the runways scheme active during the simulation exercise. As well, the management of departing traffic potentially in conflict with arriving traffic was found easy.

5.1.1.2. AMAN

The ACC sectors and the Oslo TMA positions were supported in the management of the ENGM arriving traffic by AMAN. It provided the controllers with sequence proposals (at STAR-start-points and runway thresholds) along with time advisories (time to lose/gain to comply with the sequence proposed)⁶.

As already stated in § 3.4, the guideline given to the ACC controllers for the simulation was to meet the AMAN time advisories as closely as possible.

The compliance with the AMAN proposals was recognised by the participants having potential for more appropriate delivery of traffic at metering points (IAFs). On that purpose AMAN worked effectively in combination with Point Merge. The prevention of traffic bunching, because of the pre-sequencing of the arriving flows upstream, assured the traffic did not overwhelm the delay absorption capability of the sequencing legs. In other words, what the controllers experienced in the simulation was the possibility to share the necessary delay between the ACC sectors and the Oslo TMA positions. The additional work done by the ACC sectors was found to be fully acceptable. In turn it was beneficial for Oslo TMA as the positions were relieved of part of their workload.

The controllers also gave the feedback that AMAN would lead towards a reduction in the amount of coordination concerning the management of the arriving traffic. What the participants experienced in the simulation is that, even when coordination was needed (between ACC sectors and Oslo TMA positions or on request of the PLN), AMAN helped them out in the task, making the coordination process easier and clearer, which was particularly valid for the participants who manned the ACC sectors.

With regard to Oslo TMA positions, the DIR relied on the proposed sequence before deciding when to instruct the aircraft on the sequencing legs in case of simultaneous entries.

AMAN was helpful for DIR and FIN also to predict the evolution of the traffic and work on a plan to successively implement, being aware of the future workload. AMAN facilitated that process presenting relevant information in the timelines. By monitoring the timelines, the controllers could start the planning process even well in advance of receiving the applicable FPSs.

The experience gained over the simulation confirmed again an outcome of the previous sessions, that being the usage of AMAN during the runway closure. In that circumstance both DIR and FIN positions relied on the timelines in order to act efficiently and not waste the runway capacity (ref [6]).

5.1.1.3. Point Merge

The Oslo TMA organisation relied on the concept of Point Merge method to: a) create the sequence order and inter-aircraft spacing; and b) maintain the sequence order and inter-aircraft spacing, keeping the aircraft on FMS lateral navigation mode.

The simulation confirmed again the suitability of Point Merge for the accomplishment of those tasks. Independently from the runway mode and orientation Point Merge method was applied effectively and safely by the participants.

⁶ The AMAN performance was generally suitable for the simulation purposes and the potential benefits that AMAN is going to bring in the operational life were clearly grasped by the participants. Nevertheless the AMAN implementation was subject to some limitations which at times impaired the correct working of the tool. The main one was that the system did not update the timelines when the flights were along the sequencing legs. That was often the cause of large shifts in the time advisories on the ACC sectors.

Creating the sequence with appropriate inter-aircraft spacing and maintaining it were found less demanding tasks if performed with Point Merge method rather than with the current working method based on radar vectoring.

The Point Merge system defined in Oslo TMA was found suitable to accommodate the simulated traffic demands⁷ the scenarios were characterised by. The simulated meteorological conditions were not of impediment to achieve the controllers' tasks. The participants recognised problem might occur in case of strong wind (50 knots or greater around FL100). On the other hand, such meteorological condition would impose some traffic restrictions to/from ENGM and in the surrounding sectors.

All the traffic scenarios were composed of a wide variety of P-RNAV capable aircraft. The composition of the fleet was miscellaneous, therefore the performances could vary. Moreover a small percentage of arrivals (some 5%) were simulated operating with non P-RNAV capable aircraft. Using Point Merge, the controllers found it easy to manage the differences in the aircraft performances. And in case of non P-RNAV capable aircraft, or whenever it was needed, reverting to radar vectoring was not considered a problem.

The participants agreed that Point Merge method had a significant impact on their role.

On the one hand, creating and maintaining the arriving sequence (provided the conditions simulated, e.g. mix of traffic, meteorological conditions, etc.) implied fewer tactical interventions than it would be with the current working method.

On the other hand the monitoring of the traffic still remained an essential task for the controllers. In particular the DIR position required the controller to maintain his/her attention all the time. This was a prerequisite to efficiently deliver the traffic to the FIN position with appropriate characteristics, in terms of spacing and speed. Therefore the main recommendation from the participants was to reduce to a minimum all the non-essential tasks for DIR, especially those that would take the focus away from the radar screen.

The handling of the FPB/FPSs was deemed a major issue. The task was found time-consuming and more importantly diverting the attention of the controller away from the radar screen. The implementation of a stripless system would smooth this situation and therefore was deemed by the participants as recommendable.

Finally the simulation gave the participants the possibility to evaluate different missed approach procedures. The procedures were designed so as to allow for re-inserting the aircraft back into the landing sequence by means of radar vectors. The procedures did not imply any disruption of the system based on Point Merge. The controllers agreed that the Point Merge method is flexible enough to cope with the missed approach procedures.

The following chart (Figure 5-4) depicts the 2D trajectories of inbound flights (EXE3-1 and 3-2 using runways 01L + 01R) and highlights the trajectory flown by a non P-RNAV capable aircraft (red track, with the use of NESBY holding) and the trajectory concerning a missed approach flight (green track). In both circumstances the TMA positions had to revert to radar vectoring.

⁷ All the traffic scenarios simulated meant a significant increase in the number of movements per hour (up to 95 movements) if compared with the today's situation. ENGM has currently a practical capacity of 65 movements per runway system, but generally not more than 60 movements are performed.

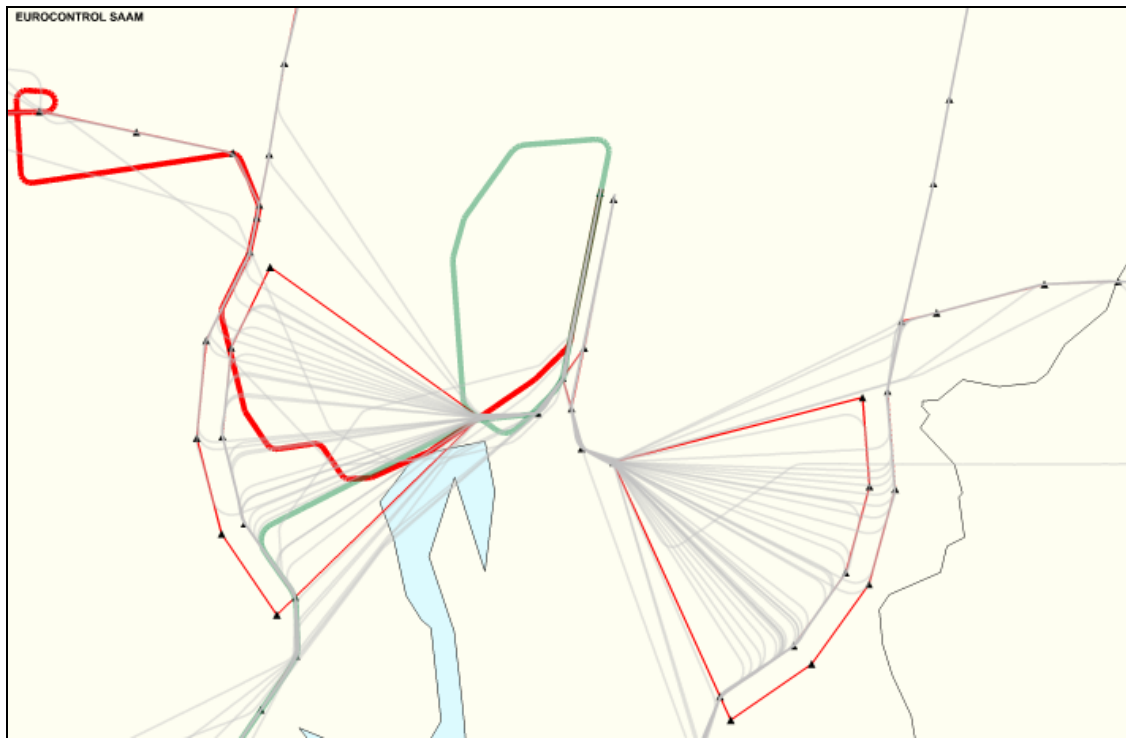


Figure 5-4 – Examples of Non P-RNAV Aircraft (red) and Missed Approach (green) trajectories

5.1.2. Workload

This section is based on subjective feedback collected by means of questionnaires, debriefing and ISA technique. Annex C presents the whole set of ISA diagrams for each exercise.

5.1.2.1. Oslo ACC

The participants never perceived the workload as an issue throughout the simulation. Although the level of workload depended on the sectors or positions, the controllers globally reported after each exercise that they were able to work comfortably.

The ISA analysis confirms that for most of the time the controllers rated their perceived workload as “Low”. The next diagrams show ISA average values for the two ACC configurations simulated. We recall that Configuration 2 came along with the activation of the military areas (Dovre and Rena) and corridor. In addition the traffic scenarios adopted with that configuration were not as busy as the scenarios used in Configuration 1⁸.

⁸ The exercises for Org2 and Org5 were based on a traffic presentation characterised by a lower arrivals rate to ENGM (see Table 3-1) to address specific objectives (see § 4.1).

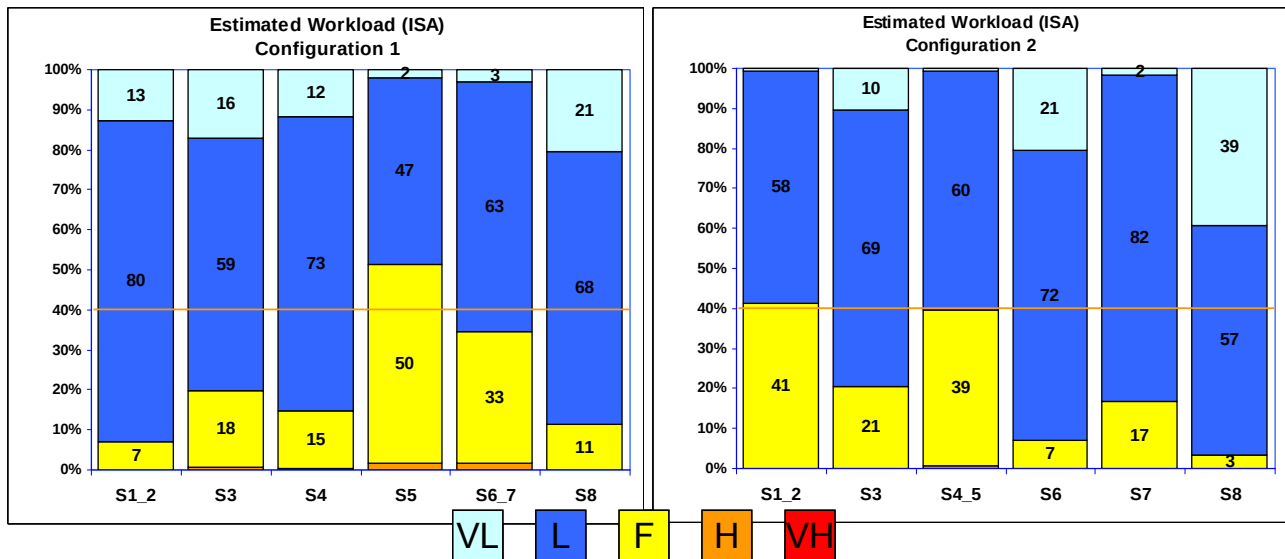


Figure 5-5 – ISA: Oslo ACC Sectors

With regard to Configuration 1, S5 was considered the most demanding sector in terms of workload. And that was particularly true in case of high traffic load (Org3 and Org4). Despite of that, the participants never deemed the workload level as unacceptable.

The combined sectors were not a concern either and were just fairly demanding. Both S1_2 and S6_7 are characterised by low/fair level of workload. Only occasionally (Org4 – IPA runway mode) the perception of workload for S6_7 was higher than the average.

S6_7 was responsible for ENGM arrivals via LANGI and NESBY, and accepted departures from Oslo TMA on different SIDs in the North-Western part of Oslo AoR. In addition, transit flights made S6_7 further busy in terms of traffic load. Nevertheless the controllers agreed that the routes segregation worked really fine in that part of the airspace and mitigated the weight of the traffic amount. Thus, the perceived workload was mainly due to monitoring tasks and frequency occupancy.

With regard to Configuration 2, S1_2 and S4_5 were considered as the most demanding in terms of workload. Nevertheless, even in this case the controllers did not feel uncomfortable with manning collapsed sectors.

The activation of the Rena military area (TSA) had a direct impact on ENGM arriving flow via ULVEN. The response of the participants was that, provided a fair amount of traffic, the collapsed sector S1_2 was still allowing for an acceptable workload. The simultaneity of active TSAs and the use of holding stacks raised some concerns.

As a general comment the participants reported that the proximity of the TSAs and corridor borders to the holding patterns (at LANGI and ULVEN) constrained the area of intervention. The controllers questioned whether in real life it would be feasible to comply with non nominal situations which require holding traffic at LANGI and ULVEN while the military areas are active.

The traffic load which concerned S4_5 was not as high as it would be using traffic scenarios tested with the Configuration 1. Therefore the controllers had to manage a fairly higher amount of traffic and they were not particularly concerned about the workload.

Nevertheless the participants agreed that a main workload component for S4_5 might come from the traffic complexity. The collapsed sector had to interact with both Oslo and Farris TMAs and so handle mix of traffic. Despite the well structured airspace design which allowed routes segregation, a few potential conflict areas still existed.

In order to keep the workload to a fair level, the participants' recommendation was that in real life the configuration with S4_5 should be operative in case of an appropriate traffic load (which might correspond to the one simulated). In addition they proposed that an assistant for S4_5 could mitigate part of the workload due to the FPS and FPB management.

5.1.2.2. Oslo TMA

As far as the Oslo TMA is concerned, the distribution of the workload was clearly influenced by the runway mode for parallel approaches at ENGM. Although the TMA positions considered the tasks to carry out as feasible and within an acceptable level in all the conditions, the way the total workload was shared among the positions varied in accordance with the runway mode.

The next diagrams display ISA average values for the two runway modes (MPO and IPA).

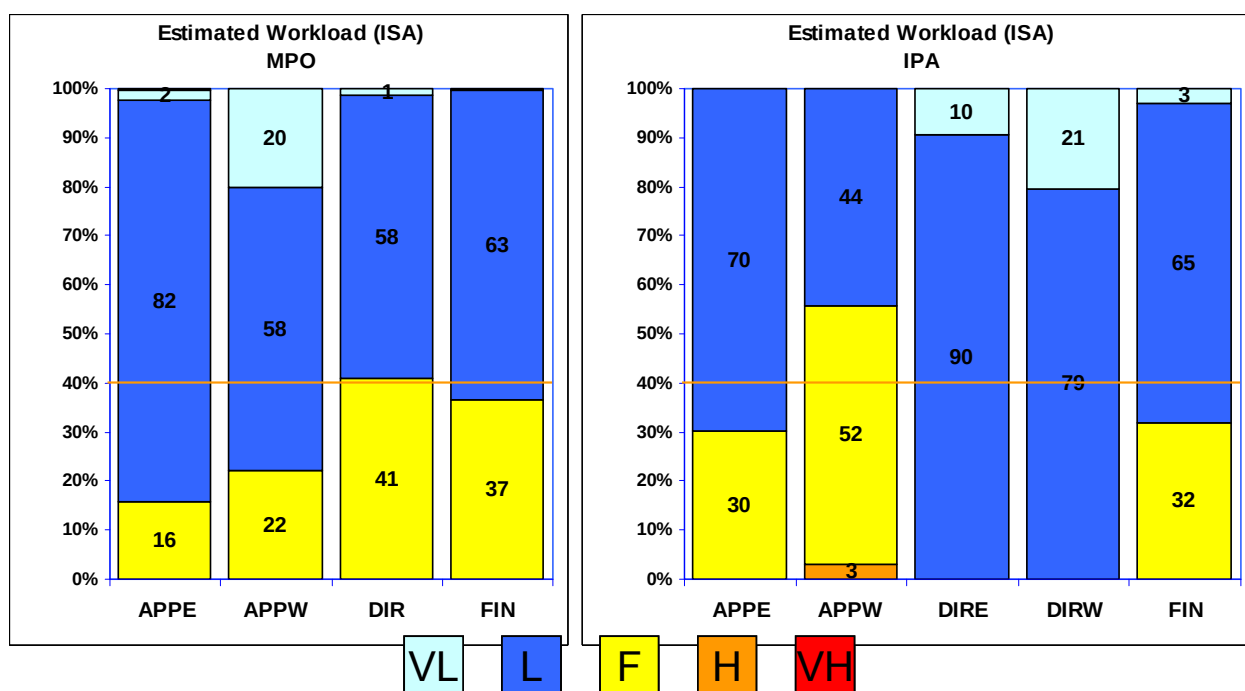


Figure 5-6 – ISA: Oslo TMA Positions

With regard to MPO runway mode, the highest workload was observed within the DIR and FIN positions. Point Merge method, however, made the tasks for those positions less demanding than compared to the current working method based on radar vectoring. The simulation showed that most of the time the DIR could comply with his/her tasks efficiently and that resulted in relieving the FIN position of part of the workload.

Receiving the traffic towards the merge points with the appropriate spacing and speed profile meant for FIN that his/her task was mainly to monitor the aircraft and refine the spacing with further speed control. Eventually FIN always felt comfortable with the amount of traffic and so always experienced a fair level of workload.

According to the participants, with Point Merge the prevailing task for DIR and FIN was radar monitoring.

The APP positions, on the other hand, did not particularly suffer from high workload and most of the time had a low perception of it. In the management of arrivals to ENGM, they felt sometimes having a downsized role. With the pre-regulation of the traffic, which was done by the upstream ACC sectors (accordingly to the AMAN advisories), the APP positions were relieved of part of their tasks.

The unbalanced task sharing within Oslo TMA positions appeared quite clear when runway closures scenarios were tested (Org2 and Org5). The APP positions hardly took an active part during those situations. While the ACC sectors were busy to hold the traffic at the IAFs, and the DIR positions were spending many resources holding traffic at the specified patterns inside the “triangles”⁹, the APPs were not able to lend their spare capacity to relieve the other sectors and positions of the temporary peak of workload (see Figure 5-7). That was also a matter of frustration for the participants.

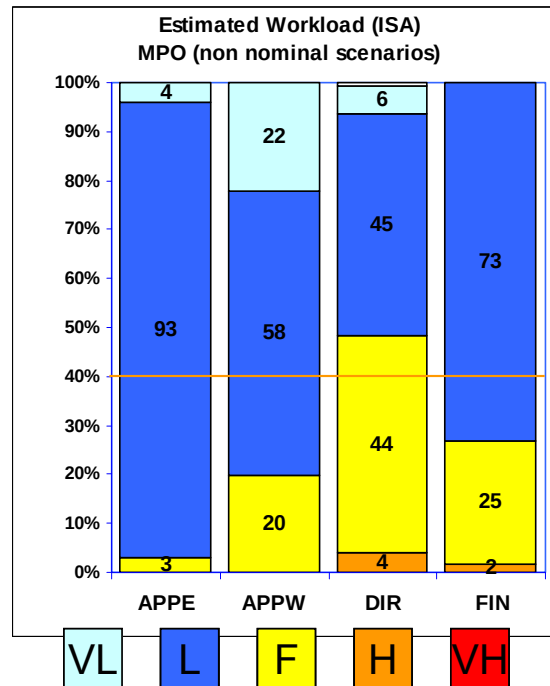


Figure 5-7 – ISA: Oslo TMA Positions (MPO – Non Nominal Scenarios)

With regard to IPA runway mode, the APP positions were instead characterised by a higher level of workload. The increase in traffic load, because of more departures and arrivals, led to a significant increase in monitoring tasks. It was imperative that the ACC sectors delivered arrival traffic in strict compliance with the vertical restrictions defined on the STARs, so as to allow the vertical segregation with the departing flows¹⁰, and ease the tasks for the APP positions.

Some of the participants expressed concerns that the APP positions might need support in FPS handling in case of high/excessive traffic loads.

Splitting the DIR in two positions was clearly beneficial in terms of workload as it counter balanced the increase of the traffic demand. DIRE and DIRW reported a generally low level of workload. The FIN position was barely impacted by the increase of traffic load and substantially kept the same level of workload as in MPO runway mode.

As a general result, regardless the runway mode in use, the implementation of missed approach procedures did produce a fair amount of additional workload on the Oslo TMA positions, which was anyway deemed as acceptable. The procedures required some coordination between PLN and TWR positions, but they were never a concern for the participants.

5.1.2.3. Farris TMA

⁹ Runways closure had a drawback in terms of situation awareness (see § 5.2).

¹⁰ The SIDs/STARs design assured the vertical separation between flows as long as the restrictions were respected.

As far as Farris TMA is concerned, the perceived level of workload varied in accordance with the configurations. The next diagrams display ISA average values for the two configurations tested.

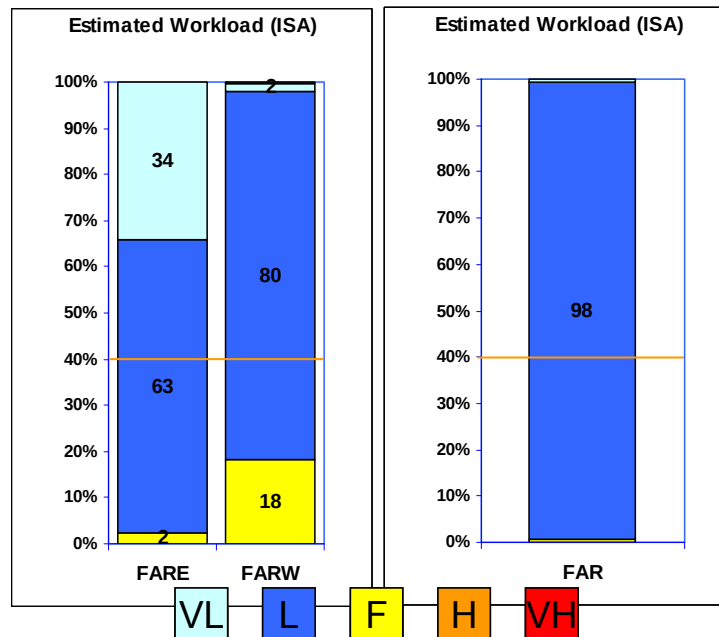


Figure 5-8 – ISA: Farris TMA

With the exception of Org1, all the exercises featured traffic samples with more traffic in west side of Farris TMA. And consistently with that, the perception of workload level was significantly higher for FARW than FARE. But in any case the workload level was never reported as greater than fair.

The configuration with one single position (in Org2 and Org5) did not raise any concerns and was not considered too demanding for only one controller. Nevertheless the participants agreed that in real life they would be keen to have the standard configuration for Farris made up of two positions. The collapse might be accepted only in case of low traffic load.

5.1.3. Job Satisfaction

The main human factors problem concerning controller job satisfaction in air traffic control is not to improve it but to sustain it.

The assessment of controller job satisfaction was carried out through post exercise questionnaires. The results are presented in Figure 5-9. The diagrams show the average values obtained grouping all the exercises in the different conditions. The same diagram displays all the concerned positions or sectors which were simulated in the different configurations.

Globally the level of job satisfaction was high for the ACC sectors (with a few exceptions - e.g. S8) and Farris TMA in the full configuration (with two positions). However, for Oslo TMA positions the result does not follow the same tendency.

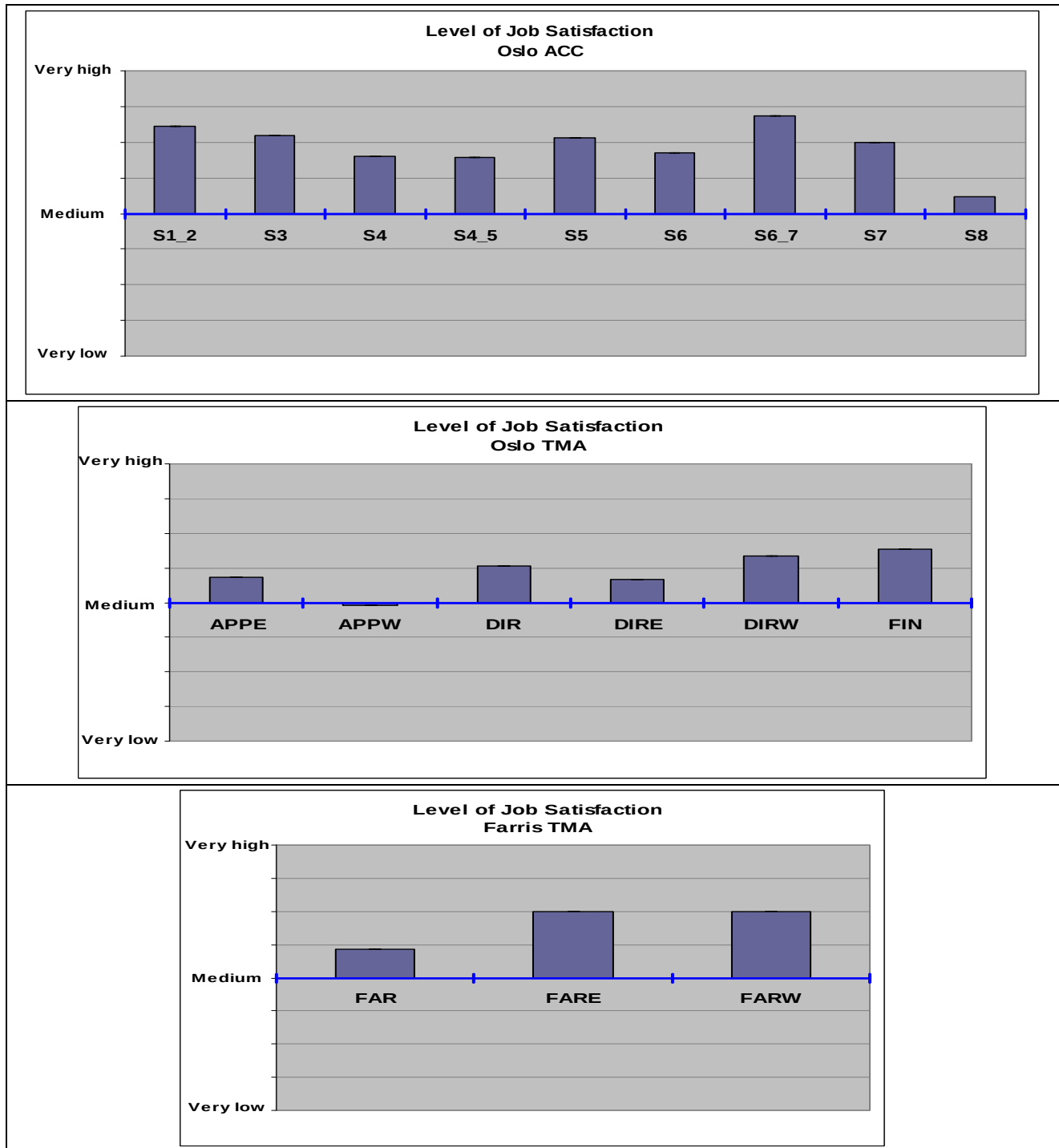


Figure 5-9 – Job Satisfaction

The outcome concerning the Oslo TMA positions is of difficult analysis, although, it is worth mentioning, job satisfaction was never rated below a medium value.

The literature shows that many proposed changes (e.g. new working method, new support tool) seem sometimes to threaten the job satisfaction as perceived by the controllers to reduce opportunities to use their skills and active interventions¹¹.

¹¹ There could be an analogy with introduction of automation. It takes always time to make the actors understand that the satisfaction could eventually be found in the quality of the service that can result from the change.

The APP positions, in situations where the traffic load was not demanding (e.g. decrease of the departures rate), reported slight frustration as were not in the conditions to lend their spare capacity to relieve DIR positions of part of the workload. That had an influence on controllers' job satisfaction which on average was rated as medium for the two positions. The difference between APPE and APPW might be partly explained with the choice done for the seating plan. Two different groups of controllers manned the two positions throughout the simulation.

The job satisfaction for the DIR position was connected to the application of Point Merge method. The easiness of use of Point Merge working method had been a clear outcome of the previous sessions. So was it for the simulation.

To some extent, the simplicity of Point Merge was perceived by the controllers as a "downgrade" of their role, as if the new working method would no longer require their skills (as opposed to the current working method based on radar vectoring allowing them to fully exploit their skills). Therefore the method being less demanding was associated with the possibility to lose part of that satisfaction which, fairly enough, the controllers want to preserve.

The usual skills of the controllers have to be retained because situations in which the abilities are required will always exist. The risk of controllers' de-skilling with regards to open-loop vectoring had been often remarked during the various debriefing sessions. The recommendation was that simulation recurrent training might, in future, be the essential way to keep controllers' skills up to the level required to cope with anomalies (e.g. unexpected/non-nominal situations).

5.2. SAFETY

The present report addresses safety related aspects such as the controllers' workload and situation awareness. The hypothesis underneath is that the containment of the workload and the improvement of situation awareness are both beneficial for safety.

The participants agreed that the new airspace organisation along with the proposed operational elements (Point Merge and AMAN) led to a perceivable decrease of workload compared to today's situation, especially considering that the traffic load of the exercises was generally higher than the current one. The organisation proposed for Oslo AoR, as described in § 5.1.2, allowed for a fair distribution of workload among the sectors and positions, which never experienced situations of unacceptable workload during the simulation.

As far as the controllers' situation awareness is concerned, the simulation showed the new operational scenario enabled a very good level of it. The assessment of the situation awareness was conducted by means of the post exercise questionnaires and the results are presented in Figure 5-10. The diagrams show the average values obtained grouping all the exercises in the different conditions. The same diagram displays all the concerned positions or sectors which were simulated in the different configurations.

The participants generally reported they had constantly a clear picture of the traffic and they were able to foresee the evolution of the traffic.

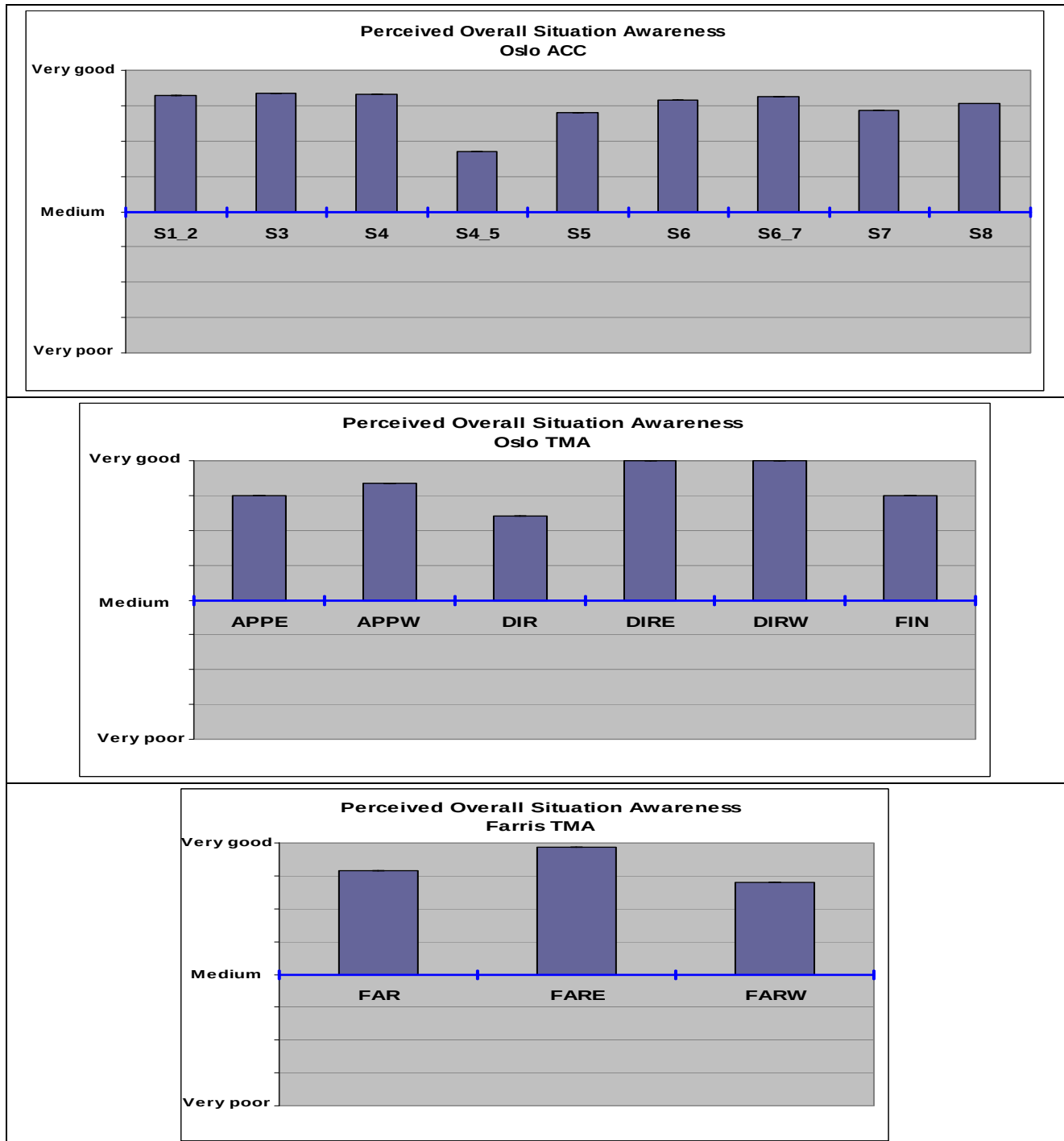


Figure 5-10 – Situation Awareness

As for Oslo ACC, the controllers reported their situation awareness as almost very good in all the sectors, with the exception of S4_5. As already stated in § 5.1.2, S4_5 interacted with a mix of traffic from/to Farris/Oslo. In real life the controllers would organise the FPB so as to display most of the potential conflicts, although they also confirmed the airspace structure itself allowed a good routes segregation. But the limitations of the simulator in terms of FPS management prevented the controllers from accomplishing the task effectively. As a result the perceived situation awareness in S4_5 was not as good as in the other ACC sectors.

The situation awareness perception was really positive for both Oslo and Farris TMAs positions. Only occasionally the participants felt it slightly degraded, typically Oslo DIR in case of runway closures. The feedback is however strictly influenced by the lack of

knowledge/experience the controllers had in managing that situation¹². Apart from that episode the TMAs positions always profited from a clear picture of the traffic.

Three main factors contributed to enhance the controllers' situation awareness compared to today's situation.

a) For most of the participants the new airspace organisation led to an improvement of the situation awareness compared to the current situation.

Asked by means of questionnaires, the controllers agreed the new sectorisation and the organisation of the routes (based on the principle of segregating arriving and departing flows) decreased the number of conflict areas and enabled a good picture of the traffic in the majority of the sectors. The feedback was a little more conservative as far as S5 and S8 are concerned. As S8 is the only ACC sector in the Oslo AoR considered to be an "en-route" sector, its proposed ATS routes and lateral limits are similar to today's organisation. Therefore the result obtained (neutrality) may be expected.

b) Another factor that clearly had an impact on the controllers' situation awareness was the availability of a supporting tool such as AMAN.

The general belief was that AMAN contributed to a good situation awareness. In the upstream sectors, the support offered by the arrival manager was a means to make the controllers more aware of the arriving traffic evolution so as to better understand the strategies dictated by the PLN. This was particularly the case for S3 and S5.

The support tool enhanced the situation awareness of DIR and FIN (refer to § 5.1.1.2).

c) Finally Point Merge method contributed to the high level of situation awareness reported by the DIR and FIN positions. Firstly, the upstream sectors (ACC sectors and APP positions) were more committed to deliver the traffic to the DIR with a more standardised 3D profile in order not to vanish the benefits Point Merge was going to bring. Secondly, the application of the new working method by the DIR assured a stricter adherence to the aircraft lateral navigation. All that contributed to increase the trajectory predictability which in turn meant for the TMA positions the achievement of an enhanced situation awareness.

Similarly in the Farris TMA the new P-RNAV based SID/STAR system meant for the controllers a significant leap in terms of trajectory predictability compared to today's situation. Even regarding the Farris context, the participants agreed that the situation awareness was enhanced by the new procedures and also by the particular layout chosen with side by side positions. That facilitated the coordination required between the controllers and allowed them to be easily aware of adjacent positions traffic situation. The general recommendation of the participants was that the upcoming layout of the operational room shall foresee the two positions to be close to each other.

¹² When the scenario employing the closure of the two runways at ENGM (MPO mode) was simulated for the first time, the controller in charge of the TMA positions had not been properly briefed on the control procedures. More in detail, the scenario was setup to force the implementation of missed approach procedures and the opening of the stacks. That because the runways closure communication had to be passed to the controllers with a short notice (and thus while traffic was still engaging the sequencing legs and other flights were in frequency with the FIN position and already established on the ILS). In the briefing beforehand the exercise, the controllers were not given precise information on the holding procedures defined at points DOLLY and SHAKE (the waypoints are located inside the "triangles" at 5NM from the merge points). The vagueness of the control procedures was a stress-inducing factor for the controller who in fact was not completely aware of what he/she had to proceed in such situation.

5.3. CAPACITY

This Key Performance Area addresses the ability of the ATM System to cope with air traffic demand (in number and distribution through time and space).

The simulation with the series of measured exercises aimed to provide trends on the reduction in controller task load achieved by a reduced requirement for controller tactical interventions.

That reduced workload might provide a potential for capacity increase, but it has to be proven that freed cognitive resources can be used for capacity.

Achieved throughput is also considered as an indicator of capacity.

5.3.1. Controllers' Availability

The simulation demonstrated that the controllers were able to manage a traffic demand higher than the current one with no particular concerns in terms of workload in most of the circumstances (§ 5.1.2).

Airspace design, methods (e.g. Point Merge for Oslo TMA positions) and support tool (AMAN) worked suitably and consequently downsized the requirement for controllers' tactical interventions and the need of coordination between sectors and positions. The reduced workload is then expected to result in an increase in airspace and terminal area capacity.

With regard to Point Merge, the participants stressed that one of the benefits the new procedure brings is the reduction of radio communications. In fact the controllers firmly believed Point Merge method will be crucial to decongest the frequency use in Oslo TMA, allowing therefore an increase in controllers' availability. Creating and maintaining the arriving sequence (provided the conditions simulated, e.g. mix of traffic, meteorological conditions, etc.) implied less tactical interventions than it would with the current working method. Consequently the amount of R/T (transmissions) decreased¹³.

Hereafter (Figure 5-11), in order to provide trends for the R/T task load, diagrams show the mean values and dispersion (standard deviation) of the frequency occupancy in Oslo TMA positions for both runway modes (MPO and IPA). The frequency occupancy corresponds to the percentage of time a controller spends on frequency to handle aircraft, issuing instructions and receiving read-back/requests from the pilots. Detailed results on frequency occupancy are presented in the Annex E.

¹³ However, in absence of a reference scenario, replicating the today's situation we can not measure the magnitude of the reduction in the controllers' task load (number of instructions and clearances, R/T Usage).

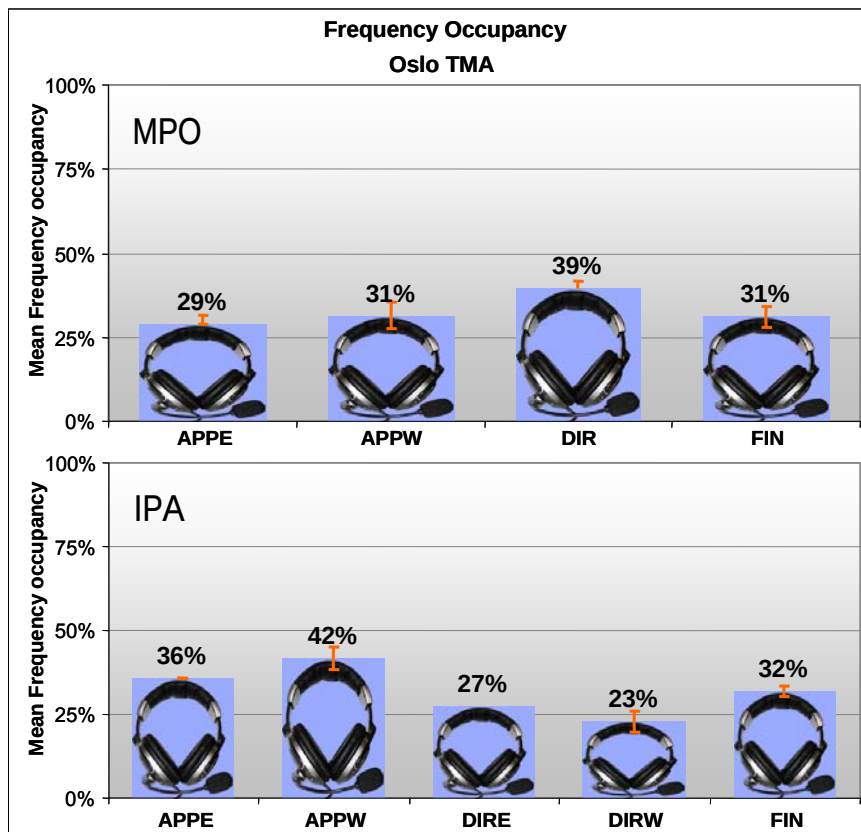


Figure 5-11 – Frequency Occupancy in Oslo TMA

In MPO runway mode the task load is well balanced among the positions. DIR is fairly busier although the average value is still below 40% of the available time. In IPA runway mode the split of the DIR position in East and West sides led to a redistribution of the time spent on frequency which is now on average about 25% for DIRE and DIRW.

Apparently the difference in traffic load between MPO and IPA exercises is not reflected in the figures concerning FIN which show an almost steady value in the frequency occupancy. The difference in traffic load had instead an impact on the APP positions (especially APPW) which increased significantly the time spent on radio (but still to acceptable values).

5.3.2. Sectors Throughput

As already stated, the traffic demand simulated in the different traffic scenarios could be accommodated with no stress for the controllers and with the appropriate quality of service level.

Measurements were taken throughout the simulation to provide trends on sector throughput. That measurement is often used as an indicator of the controller's workload.

For readability purpose the outcomes of the throughput analysis are presented in the Annex G. The diagrams contained in the annex display the number of aircraft handled by a sector in each organisation within a time interval. The information contained in each diagram are:

- Total number of aircraft on frequency (in 60 minutes);
- Mean number of aircraft on frequency at a time;
- Max and min number of aircraft on frequency at a time.

It is worth reminding that the figures presented are purely indicative of what happened during the simulation and do not aim to a comparison with the today's situation. The absence of a reference scenario in the simulation prevented such comparison.

In this section Table 5-1 and Table 5-2 summarise the average value of aircraft handled by ACC sectors and Oslo/Farris TMAs positions in the different configurations.

Table 5-1 – Sector Throughput (Oslo ACC)

Sector	MPO		IPA
	ACC Conf1	ACC Conf2	ACC Conf1
S1_2	28	32	28
S3	29	28	30
S4	32	-	30
S5	31	-	33
S4_5	-	42	-
S6	-	26	-
S7	-	26	-
S6_7	38	-	44
S8	34	28	35

Table 5-2 – Position Throughput (Oslo and Farris TMA)

Oslo TMA		
Positions	MPO	IPA
APPE	39	50
APPW	38	53
DIRE	-	25
DIRW	-	28
DIR	43	-
FIN	41	51
Farris TMA		
Positions	Split Positions	Single Position
FARE	16	-
FARW	28	-
FAR	-	28

Generally the distribution of traffic was fairly balanced among all the ACC sectors in the different configurations. As expected sectors formed by two airspace blocks were characterised by busier traffic (with the exception of S1_2). However, the traffic samples were prepared to keep sectors throughput within acceptable values.

As far as the Oslo TMA is concerned, the throughput is clearly affected by the runway mode. In particular the APP positions saw almost a 30% increase in their throughput moving from MPO to IPA runway mode.

5.3.3. Throughput at ENGM

With focus on ENGM, the runways throughput in one hour period was calculated and the average figures are presented in the Table 5-3 (values concerning the exercises, where non nominal situations such as temporarily runway closure were simulated, were excluded).

Table 5-3 – ENGM Throughput

	MPO		IPA
	RWY 01	RWY 19	RWY 01
Arrivals	38	37	47
Departures	40	41	47
Max movements	82	82	95

The figures above prove again the simulation featured traffic demands significantly higher than ENGM current situation where generally not more than 60 movements are performed. That meant in IPA, for instance, more than a 50% increase in the runways throughput (of course in experimental situations).

5.4. EFFICIENCY

In the present report the efficiency is addressed from both service providers and airlines' perspective in terms of quality of service and flight efficiency, through the optimisation of vertical trajectories as well as containment of trajectory dispersion, expected to result in an increase in fuel efficiency. The assessment concerns mainly inbound and outbound traffic to and from ENGM.

Even though it was not possible within the simulation to carry out detailed study, the new working methods and procedures are believed to have a positive impact on the environmental sustainability.

Containment of trajectories dispersion and improved flight profiles both have potential for reducing fuel consumption and emissions.

5.4.1. Quality of Service

5.4.1.1. Trajectories Flown

For each measured exercise the aircraft trajectories were looked at. In the absence of advanced metrics such as “distance flown on the legs” or “trajectory containment rate”, these trajectories provide a support for a subjective analysis. Typically these trajectories illustrate:

- The usage of the sequencing legs for ENGM arrivals;
- The containment of trajectories (for ENGM arrivals and departures);
- The working methods used (and more specifically the frequency and location of occurrence of heading instructions).

In terms of sequencing legs usage, the analysis of the flown trajectories showed that at all times the designed arcs were suitable to absorb the further delay which could not be absorbed upstream. Usually the DIR used not more than half of the legs. Rarely, and only because of non-nominal situations (e.g. runway closure) sequencing leg run-offs were observed.

At DIR discretion, if the traffic condition allowed, the arriving traffic could be sent by the APP controller direct to the merge point before entering the sequencing leg. The analysis of the trajectories in fact confirmed that in many circumstances the arriving aircraft did not even enter the legs.

As already stated in the paragraph above, heading instructions in Oslo TMA were limited to the extent needed to manage particular situations. Whereas in the ACC sectors the controllers sometimes gave vectors for delaying actions prior to rejoin the STAR tracks.

As far as the containment of trajectories is concerned, definitely the implementation of Point Merge working method meant a reduction of the trajectories dispersion (as an example see Figure 5-12).

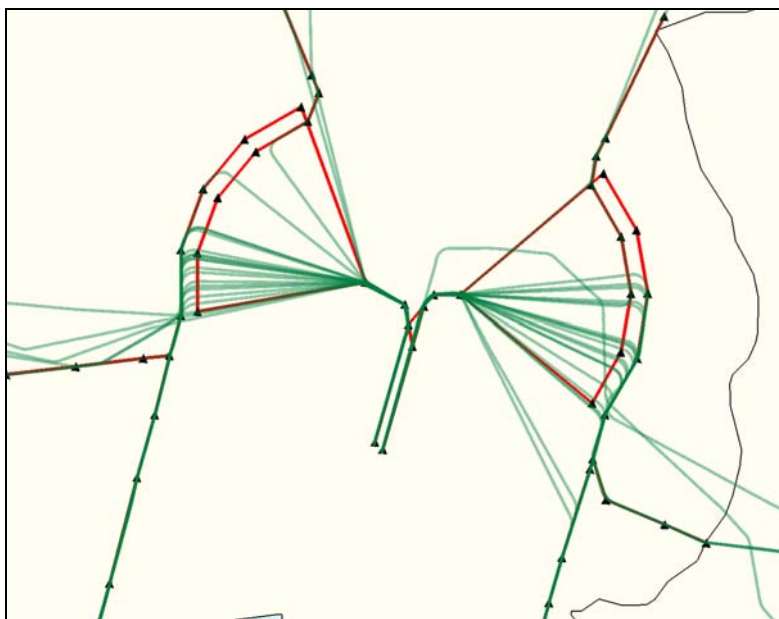


Figure 5-12 – Example of 2D Flown Trajectories for ENGM Arrivals (IPA)

The implementation of the new working method, and especially Point Merge related working method, worked to the advantage of the service provided as it enabled an efficient use of aircraft FMS even under high traffic load.

With specific reference to ENGM arrivals, the measurements taken throughout the simulation showed that generally the controllers did not make the aircraft deviate from the P-RNAV procedure track.

Apart from non P-RNAV capable aircraft which had to be vectored, open-loop instructions were limited to the extent needed to manage particular situations (e.g. sequencing leg run off, go around flight) or sometimes those flights operated with low performance aircraft.

Charts of the flown trajectories are presented in the Annexes M and N. They refer to arrival and departure flows to/from the three airports ENGM, ENTO and ENRY.

The colour convention adopted in the charts of the annexes is the following:

- **ENGM arrivals**
- **ENGM departures**

and

- **ENRY arrivals**
- **ENTO arrivals**
- **ENRY departures**
- **ENTO departures**

5.4.1.2. Spacing on ENGM Final

The quality of service in Oslo TMA was also addressed in terms of delivery conditions to the tower.

In MPO mode the controllers were tasked to ensure at least 3NM separation between arrivals to the parallel finals using staggered approaches and consequently the spacing between two consecutive arrivals to the same runway had to be in the 6.5/7.5nm range.

In IPA mode the controllers were instructed to provide 5.5NM spacing on each final. This would allow spacing for a departure between two consecutive arrivals.

The measurement of inter aircraft spacing was performed taking into account as metering points the runway thresholds¹⁴. The exercises in Org2 and Org5 were excluded from the data sample, as they contained runway closures scenarios.

The result of the analysis is displayed on the following figures (the inter aircraft spacing on the same final is marked on the axis as [01/19][L/R], whilst the separation between successive landings on the two finals is marked as the [01/19][L/R]+ [01/19][L/R]).

¹⁴ The presence of a manned TWR feed sector in the simulation setting allowed for changes in the speed profile of the aircraft on the final at low altitude. This bypassed the technical limitations of the simulator which assigns default speeds to aircraft that in some cases might not be appropriate. In the previous sessions the measurement of inter aircraft spacing was performed taking into account as metering points waypoints located on the finals at a higher altitude so as to prevent the result would be affected by unrealistic aircraft performance.

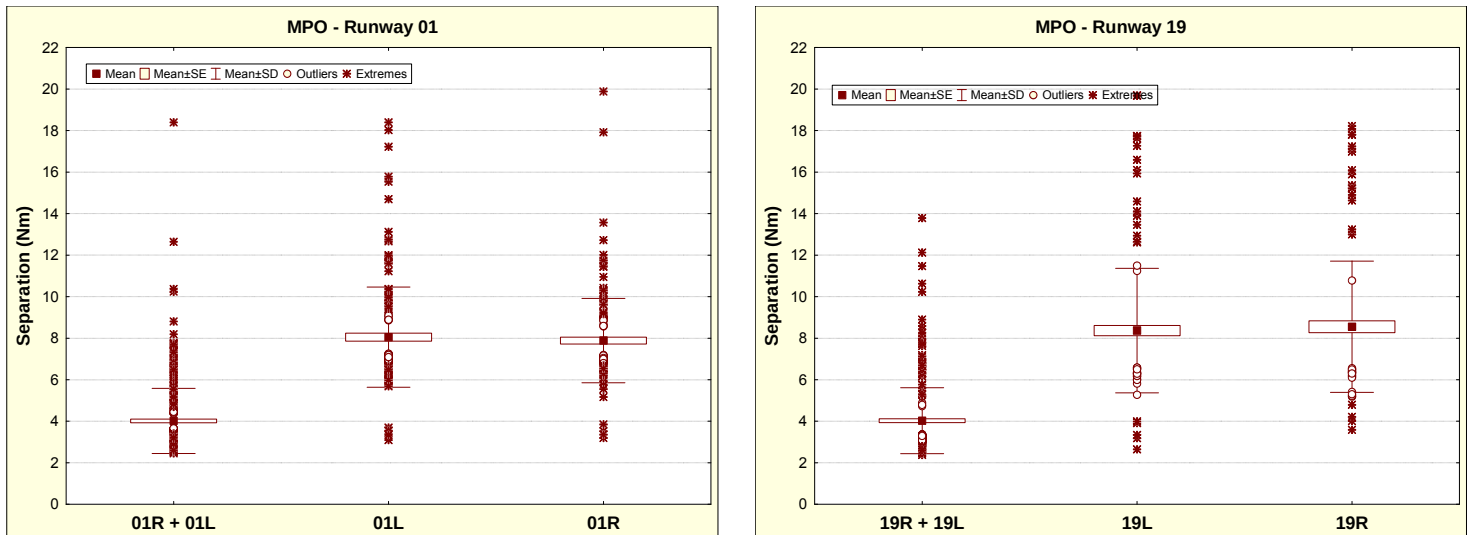


Figure 5-13 – Inter Aircraft Spacing (MPO)

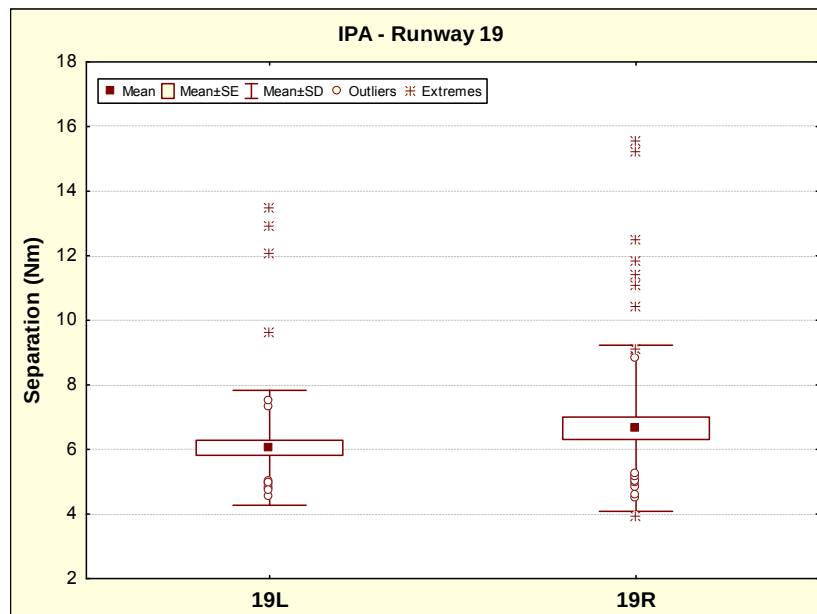


Figure 5-14 – Inter Aircraft Spacing (IPA)

The diagrams show that the inter aircraft spacing was not consistent all the time. Although the average values are not so far from the targets, the distribution of the spacing at the runway thresholds appears to be significantly spread.

On the other hand, the way the traffic samples were designed allowed for peaks and troughs in the arriving flows which consequently made the inter aircraft spacing distribution such a spread one.

In IPA runway mode the spacing measured between arrivals was always well above 3NM, whereas in MPO there were a few cases where the separation went below that threshold as if the operations were temporarily parallel approaches. When it happened it was under coordination with TWR.

5.4.2. Vertical Profiles

The assessment of the flight efficiency was carried out through the analysis of the vertical profiles which concerned arriving and departing aircraft to and from ENGM.

Upon leaving the sequencing legs following the 'Direct To' instruction, the descent profile of the flight can be optimised in the form of a basic Continuous Descent Approach (CDA) as the distance to go is then known by the FMS.

The previous sessions had already provided indications of the potential for CDA enabled by the dedicated RNAV structure and Point Merge method. The simulation allowed to collect further data and confirmed the past outcome.

The aircraft were levelled off along the sequencing legs if they could not be turned to the merge point. After they were given the direct to and cleared to descend, the vertical profile assumed was a continuous descent until either the interception of the localiser or to the merge point according to the restrictions linked to the close parallel approaches. So, there was a "high side" (aircraft coming from APPW) and a "low side" (aircraft coming from APPE) when approaching the ILS axis. In the former case usually the descent continued without the aircraft levelled off (see Figure 5-15), whereas in the latter case generally the aircraft levelled off before continuing the descent once on the runway axis: localiser down to the interception altitude then glide path (see Figure 5-16).

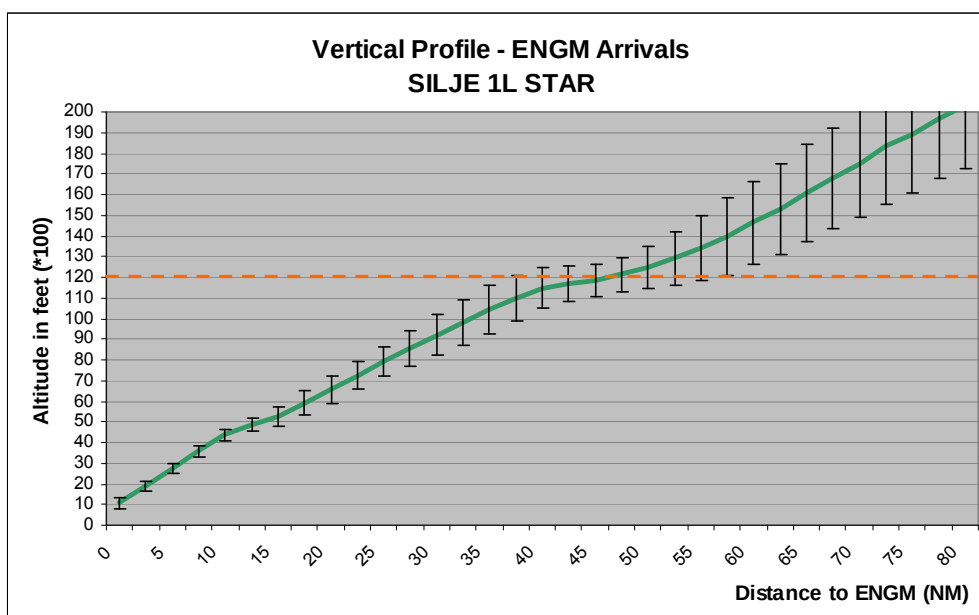


Figure 5-15 – Example of Average Profile for Arrivals (Higher Side)

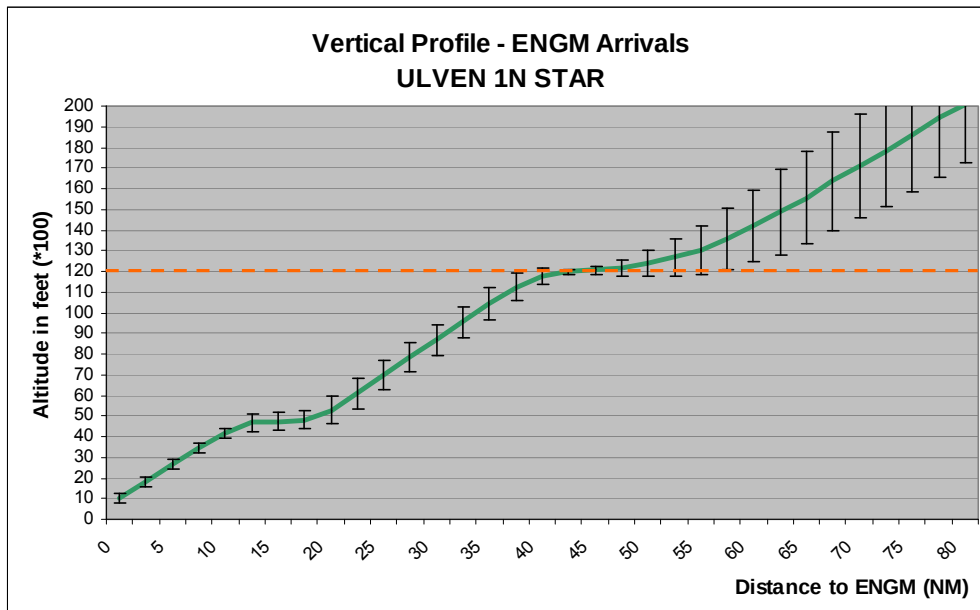


Figure 5-16 – Example of Average Profile for Arrivals (Lower Side)

The SIDs from both runway orientations were conceived and designed so as to best integrate with the STARs and to segregate (at least vertically) arriving and departing traffic. Other principle applied in the design was to enable continuous climb departures (CCD). An assessment was conducted therefore on departures vertical profiles. It showed that the whole SIDs/STARs design was suitable to clear outbound traffic to their exit flight level in most of the circumstances (see an example Figure 5-17).

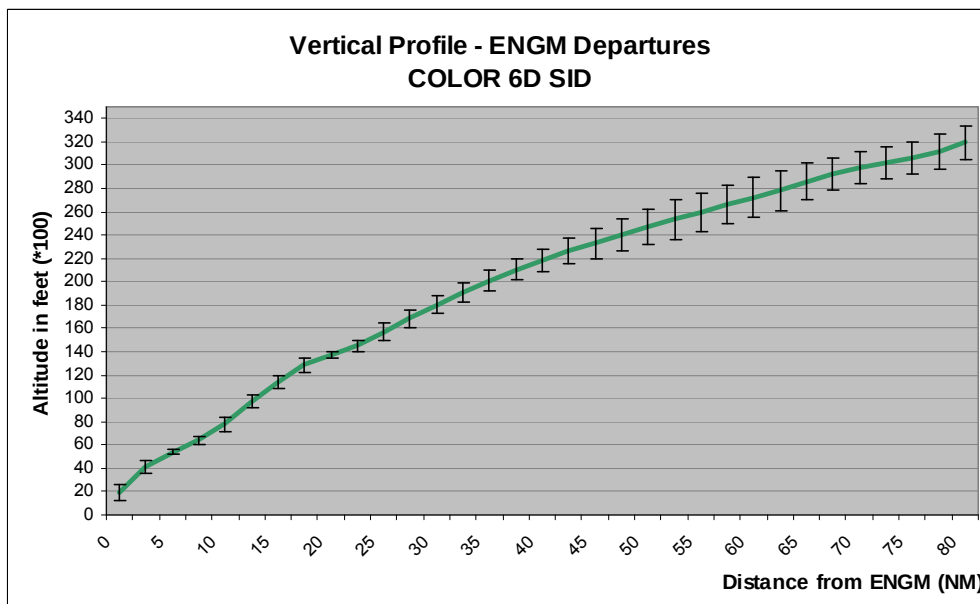


Figure 5-17 – Example of Average Profile for Departures

Diagrams of the vertical profiles are presented from Annex H to Annex L. For ENGM they refer to the different SIDs and STARs. For each procedure two diagrams are presented and they respectively show the average vertical profile and the set of the whole profiles. For ENTO and ENRY the diagrams only concern the departures vertical profiles and show all of them.

6. CONCLUSIONS

The objectives of the simulation were to assess the operability of the new Oslo AoR organisation and to provide initial indications on capacity, efficiency and safety.

The simulation results show that the new organisation is operationally viable. The proposed airspace organisation and the new working methods based essentially on the implementation of Point Merge and AMAN for Oslo arrivals were overall well accepted by the controllers who took part in the simulation.

With different traffic demands and meteorological conditions the controllers were always able to work comfortably. The measurements prove the controllers' workload was constantly at an acceptable level even with a traffic demand in ENGM which was up to 50% higher than the current levels.

The increase of the controllers' availability provided them with more time to monitor the traffic and predict its evolution. That in turn enabled an enhancement of the situation awareness which was further facilitated, as reported, by the new airspace structure (strategic segregation of the flows).

Containment of the workload and improvement of situation awareness are both beneficial for safety.

The Point Merge method was assessed to be an effective solution for Oslo TMA. The participants fully agreed on its suitability and expected benefits. Among these, it is worth listing the following:

- Easy to learn and use;
- Flexible, i.e. operationally viable with different traffic mix and under non nominal situations;
- Relieving the controllers of a significant part of the workload currently experienced;
- Increasing the lateral navigation (FMS LNAV) usage for the aircraft;
- Improving the descent profiles (potential for CDA);
- Containing the trajectories dispersion.

The simulation provided a clear indication on how effectively AMAN worked in combination with Point Merge. The respect of AMAN sequence proposals assured traffic did not overwhelm the delay absorption capability of the sequencing legs. Further it empowered the ACC sectors which felt themselves more in the loop for ENGM arrival management. The AMAN has potential for:

- Optimal delivery of traffic at metering points (IAFs);
- Increase in regularity (prevention of traffic bunching);
- Predictability;
- Reduction in use of airborne holding.

In terms of performances, the new Oslo AoR organisation has potential to allow a good level of quality of service and flight efficiency. The measurements show that the controllers operated in order to keep as much as possible aircraft on their lateral navigation mode. Radar vectoring was limited to the extent needed for managing particular situations. That along with the optimisation of vertical trajectories (potential for CDA) enabled by arrival and departure procedures is expected to result in an increase of the flight efficiency.

Even though it was not possible within the simulation to carry out a comparative study (because of the absence of a baseline), the new working methods and procedures are

believed to have a positive impact on the environmental sustainability. Containment of trajectories dispersion and improved flight profiles have potential for reducing fuel consumption and emissions and for responding to the new noise abatement procedures.

The motivations which have been driving the project were therefore addressed in the simulation and the outcomes are generally positive.

Finally there might be still some room for refinements in the airspace design (e.g. small changes in sectors borders). Further ad hoc activities might take place to address aspects which could not be covered in the simulation due to time or technical constraints.

Therefore the recommendation would be for AVINOR to dedicate resources to conduct additional activities (e.g. small scale simulations) prior to the implementation of the new airspace organisation such as:

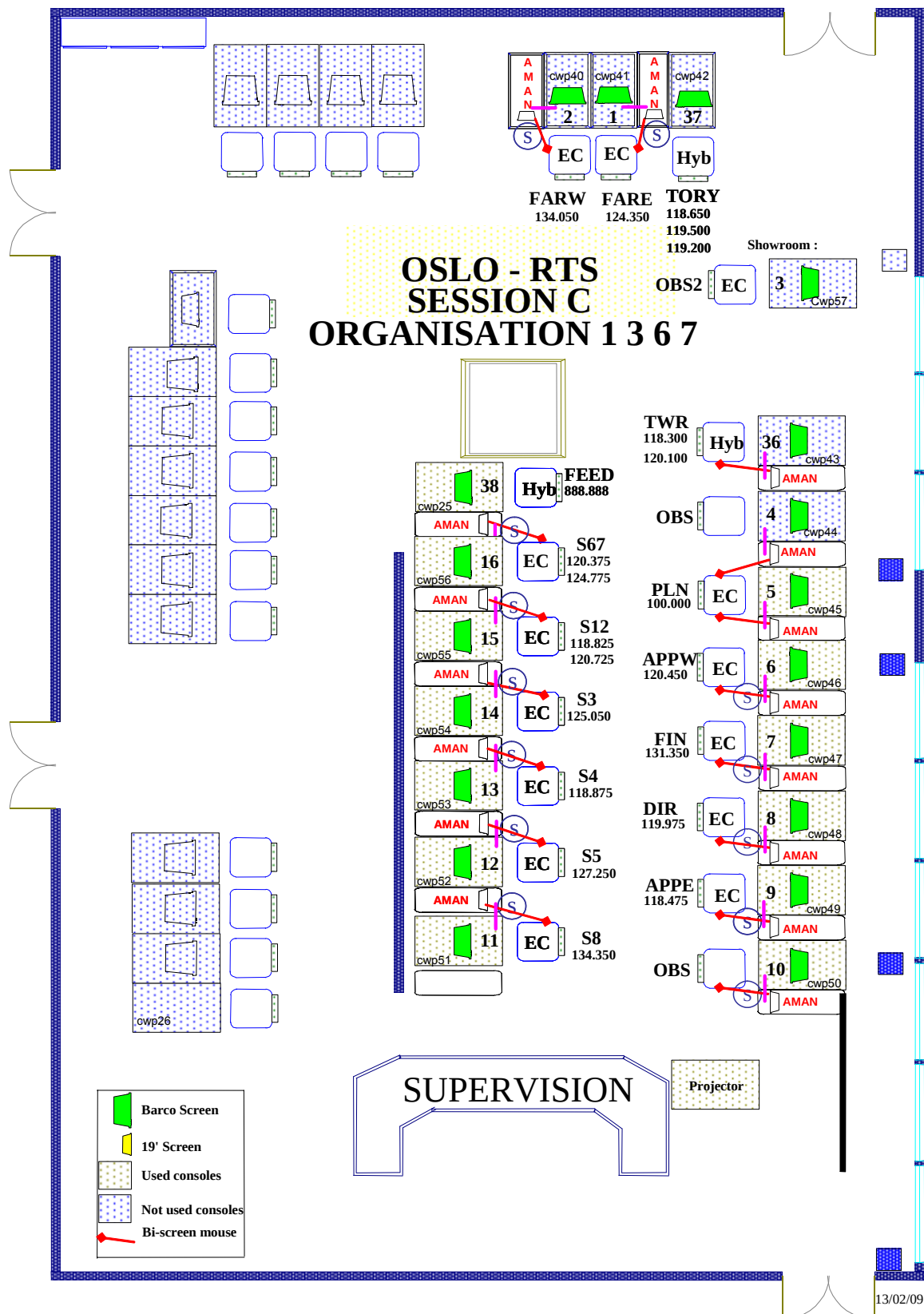
- Investigate other combinations to collapse airspace blocks to form ACC sectors;
- Assess the possibility of having additional holding patterns in Oslo TMA;
- Finalise the design and definition of the missed approach procedures in ENGM;
- Clarify the layout for Flight Progress Board to be operative in ACC sectors;
- Investigate further possibility of stripless system for Oslo TMA positions.

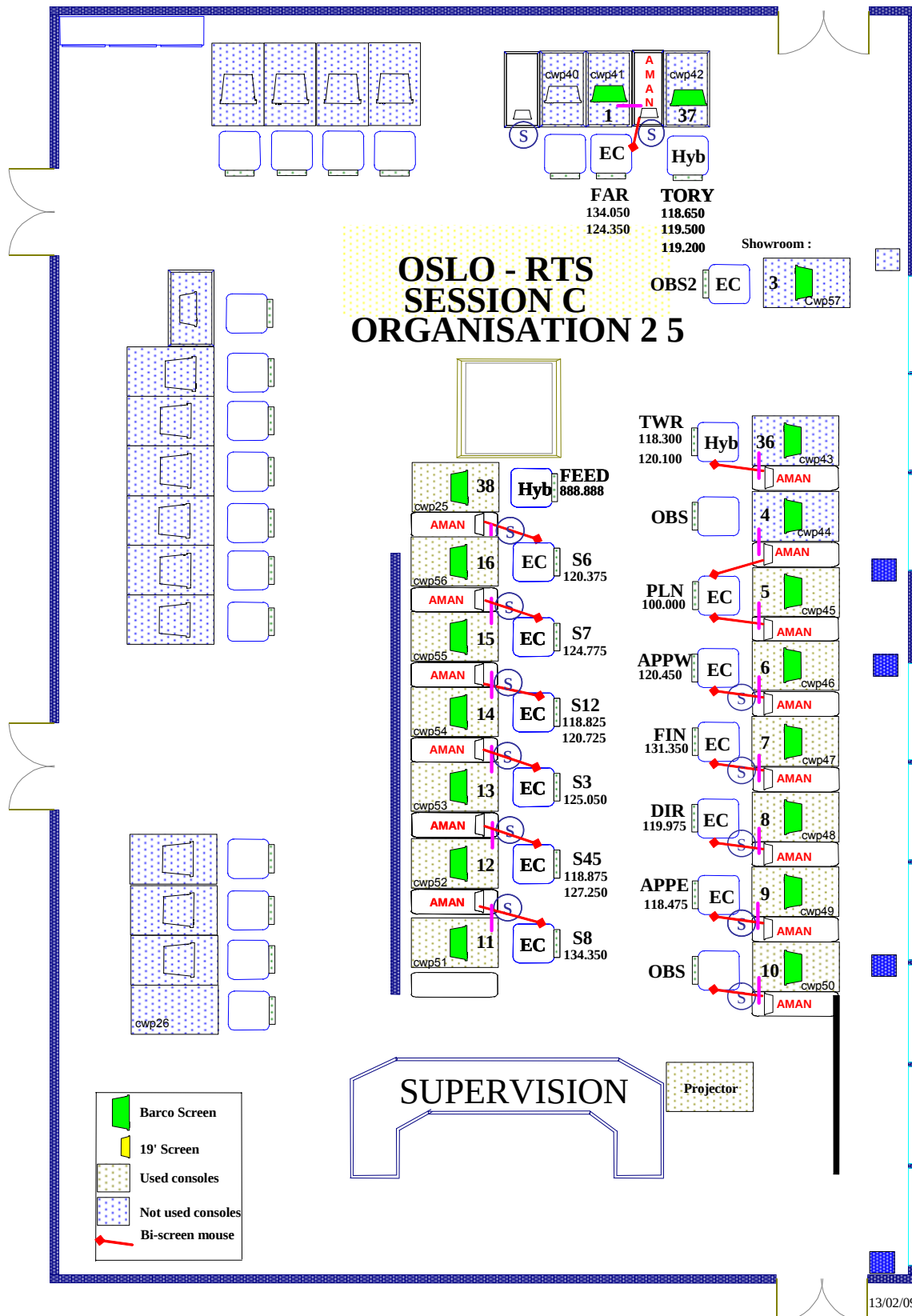
Despite the needs for those limited additional activities, the simulation scope was wide, comprehensive and the results and feedback obtained have definitely paved the way towards the implementation of the new airspace organisation which is planned to go operative on the 7th April 2011.

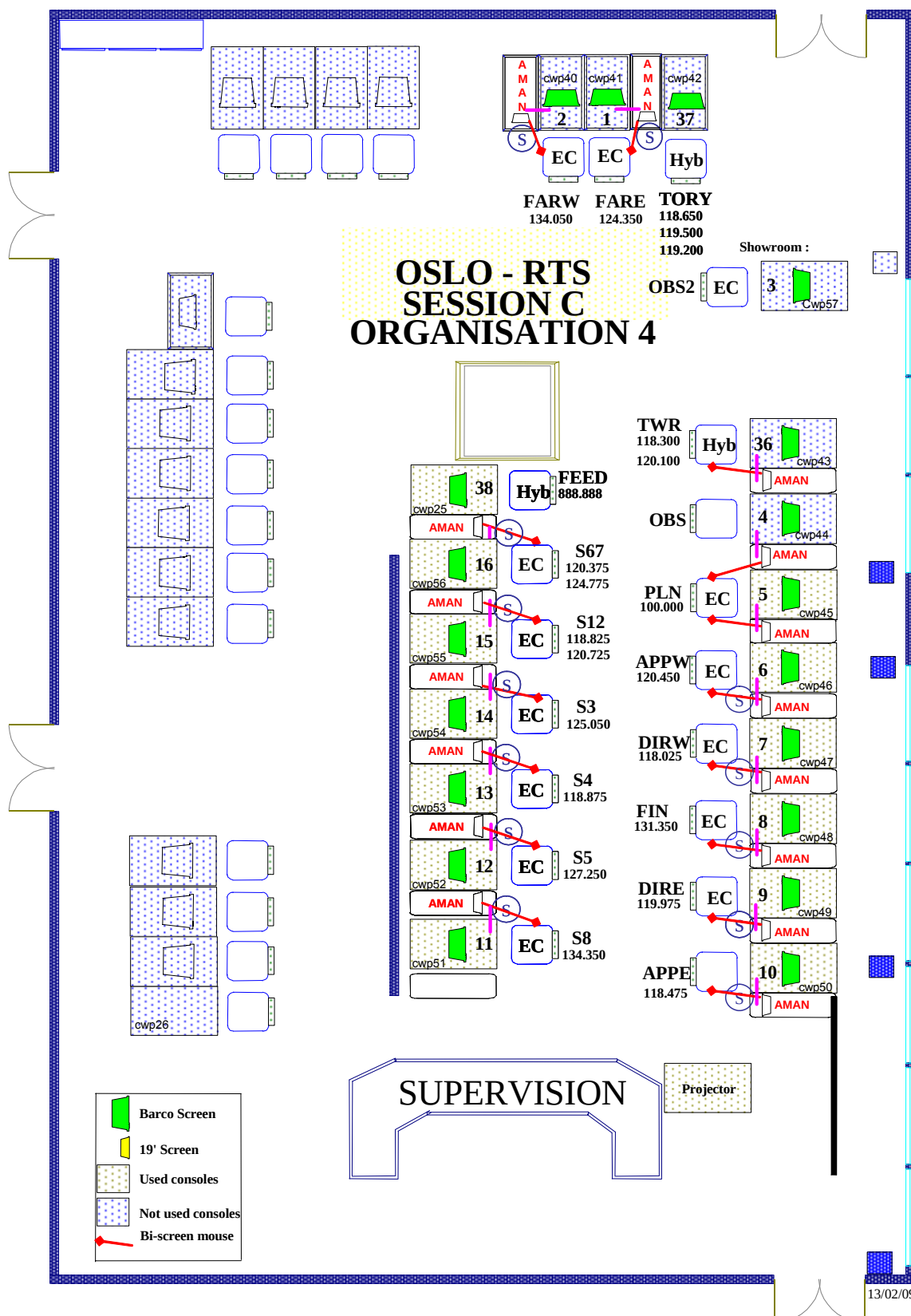
ANNEXES

Page intentionally left blank

ANNEX A: SIMULATION ROOM LAYOUT







13/02/09

Page intentionally left blank

ANNEX B: BLANK QUESTIONNAIRES

Post-Exercise Questionnaire

Oslo ACC

Purpose of this questionnaire

The purpose of this questionnaire is to collect information both about your perception of your workload and about other elements that could affect, in a positive or negative way, your overall performance in the last exercise you performed.

Method to fill it

Please complete this questionnaire by:

- putting a cross in the box ☐ that corresponds to your answer for the exercise run that you have just completed. If you make a mistake, please fill the box in completely and put a cross in the correct box. (This is an example of a crossed box ☒ and this is an example of a filled-in box ☐):
- filling the open-ended questions in.

If you need help, please, ask the analysis team representatives.

Thank you very much for your co-operation and contribution!

Controller Name : _____

Sector : _____

Date : _____

Exercise number : _____

Note

All the individual data collected during this simulation, including the responses to this questionnaire, will be treated in the ***strictest confidentiality***. Although your name is requested on each questionnaire form, for convenience, only ID numbers will be used to report individual results so that nobody can identify the respondent. Once this questionnaire has been filled in, only members of the simulation team will be allowed to see it. They will not pass any personal details to anyone outside the team.

1. Which of the following phrases best defines your perceived overall workload?	☐	☐	☐	☐	☐
	Very little to do	Little to do	Working comfortably	Busy	Lost the picture
2. In your opinion, the traffic load was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
3. In your opinion, the traffic complexity was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
4. How great a part did the radio communications play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
5. How great a part did the co-ordination with the APP PLN play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
6. How great a part did the co-ordination play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
7. How great a part did the radar monitoring play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
8. What was your estimate of your overall situation awareness (clear picture of the situation)?	☐	☐	☐	☐	☐
	Very poor	Poor	Medium	Good	Very good
9. What was your level of job satisfaction applying the proposed working method?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
10. Overall, do you consider that all the tasks you had to carry out were feasible and remained at an acceptable level?	☐				☐
	YES				NO
11. Which event / situation / aspect of the last exercise do you think should be highlighted for being cause of workload (higher than normally required) / frustration / problems to you?					

Please rate your general level of satisfaction with...

12. ... the proposed Airspace structure design.	☐	☐	☐	☐	☐
	Very bad	Bad	Medium	Good	Very good
13. ... the segregation of inbound/outbound routes.	☐	☐	☐	☐	☐
	Very bad	Bad	Medium	Good	Very good

14. The proposed Airspace structure generally reduced coordination if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
15. The proposed Airspace structure contributed to a higher situational awareness on traffic management if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
16. The proposed Airspace structure allowed for good interaction between ACC sectors and Oslo TMA (<i>if applicable</i>).	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
17. The proposed Airspace structure allowed for good interaction between ACC sectors and Farris TMA (<i>if applicable</i>).	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

18. AMAN reduced the coordination with APP sectors for arriving traffic to ENGM if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
19. In case coordination was needed, AMAN made this easier and clearer.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
20. AMAN contributed to a high situational awareness on arriving traffic management.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

In the last exercise, how often...

21. ... was speed control implemented in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always
22. ... was vectoring implemented in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always
23. ... were other actions (hold, orbits, etc) used in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always

24. If you have any comments or suggestions on the Airspace structure or FPB, then please make them here:

25. Do you have any additional comments about the last exercise? If yes, please then make them hereafter.

Post-Exercise Questionnaire

Oslo APP

Purpose of this questionnaire

The purpose of this questionnaire is to collect information both about your perception of your workload and about other elements that could affect, in a positive or negative way, your overall performance in the last exercise you performed.

Method to fill it

Please complete this questionnaire by:

- putting a cross in the box ☐ that corresponds to your answer for the exercise run that you have just completed. If you make a mistake, please fill the box in completely and put a cross in the correct box. (This is an example of a crossed box ☒ and this is an example of a filled-in box ☐):
- filling the open-ended questions in.

If you need help, please, ask the analysis team representatives.

Thank you very much for your co-operation and contribution!

Controller Name : _____

Sector : _____

Date : _____

Exercise number : _____

Note

All the individual data collected during this simulation, including the responses to this questionnaire, will be treated in the **strictest confidentiality**. Although your name is requested on each questionnaire form, for convenience, only ID numbers will be used to report individual results so that nobody can identify the respondent. Once this questionnaire has been filled in, only members of the simulation team will be allowed to see it. They will not pass any personal details to anyone outside the team.

1. Which of the following phrases best defines your perceived overall workload?	☐	☐	☐	☐	☐
	Very little to do	Little to do	Working comfortably	Busy	Lost the picture
2. In your opinion, the traffic load was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
3. In your opinion, the traffic complexity was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
4. How great a part did the radio communications play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
5. How great a part did the co-ordination with the adjacent sectors play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
6. How great a part did the co-ordination with the APP PLN play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
7. How great a part did the radar monitoring play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
8. What was your estimate of your overall situation awareness (clear picture of the situation)?	☐	☐	☐	☐	☐
	Very poor	Poor	Medium	Good	Very good
9. What was your level of job satisfaction applying the proposed working method?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
10. Overall, do you consider that all the tasks you had to carry out were feasible and remained at an acceptable level?	☐		☐		
	YES		NO		
11. Which event / situation / aspect of the last exercise do you think should be highlighted for being cause of workload (higher than normally required) / frustration / problems to you?					

12. The proposed Airspace structure generally reduced coordination if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
13. The proposed Airspace structure allowed for good interaction with ACC sectors.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

14. AMAN reduced the coordination with ACC sectors for arriving traffic to ENGM if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
15. In case coordination was needed, AMAN made this easier and clearer.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
16. AMAN contributed to a high situational awareness on ENGM arriving traffic management.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

In the last exercise, how often...

17. ... was speed control implemented in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always
18. ... was vectoring implemented in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always
19. ... were other actions (hold, orbits, etc) used in order to meet the AMAN requirements?	☐	☐	☐	☐	☐	☐	☐
	Never	Seldom	Sometimes	Often	More often	Very often	Always

20. How easy/difficult was it for you to manage departing traffic conflicting with arriving traffic?	☐	☐	☐	☐	☐
	Very easy	Easy	Medium	Difficult	Very difficult

21. How easy/difficult was it for you to manage transit flights?	☐	☐	☐	☐	☐
	Very easy	Easy	Medium	Difficult	Very difficult

22. Do you have any additional comments about the last exercise? If yes, please then make them hereafter.	
---	--

Post-Exercise Questionnaire

Oslo DIR / FIN

Purpose of this questionnaire

The purpose of this questionnaire is to collect information both about your perception of your workload and about other elements that could affect, in a positive or negative way, your overall performance in the last exercise you performed.

Method to fill it

Please complete this questionnaire by:

- putting a cross in the box ☐ that corresponds to your answer for the exercise run that you have just completed. If you make a mistake, please fill the box in completely and put a cross in the correct box. (This is an example of a crossed box ☒ and this is an example of a filled-in box ☐):
- filling the open-ended questions in.

If you need help, please, ask the analysis team representatives.

Thank you very much for your co-operation and contribution!

Controller Name : _____

Sector : _____

Date : _____

Exercise number : _____

Note

All the individual data collected during this simulation, including the responses to this questionnaire, will be treated in the **strictest confidentiality**. Although your name is requested on each questionnaire form, for convenience, only ID numbers will be used to report individual results so that nobody can identify the respondent. Once this questionnaire has been filled in, only members of the simulation team will be allowed to see it. They will not pass any personal details to anyone outside the team.

1. Which of the following phrases best defines your perceived overall workload?	☐	☐	☐	☐	☐
	Very little to do	Little to do	Working comfortably	Busy	Lost the picture
2. In your opinion, the traffic load was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
3. In your opinion, the traffic complexity was:	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
4. How great a part did the radio communications play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
5. How great a part did the co-ordination with the adjacent sectors play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
6. How great a part did the co-ordination with the APP PLN play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
7. How great a part did the radar monitoring play in your overall workload?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
8. What was your estimate of your overall situation awareness (clear picture of the situation)?	☐	☐	☐	☐	☐
	Very poor	Poor	Medium	Good	Very good
9. What was your level of job satisfaction applying the proposed working method?	☐	☐	☐	☐	☐
	Very low	Low	Medium	High	Very high
10. Overall, do you consider that all the tasks you had to carry out were feasible and remained at an acceptable level?	☐		☐		
	YES		NO		
11. Which event / situation / aspect of the last exercise do you think should be highlighted for being cause of workload (higher than normally required) / frustration / problems to you?					

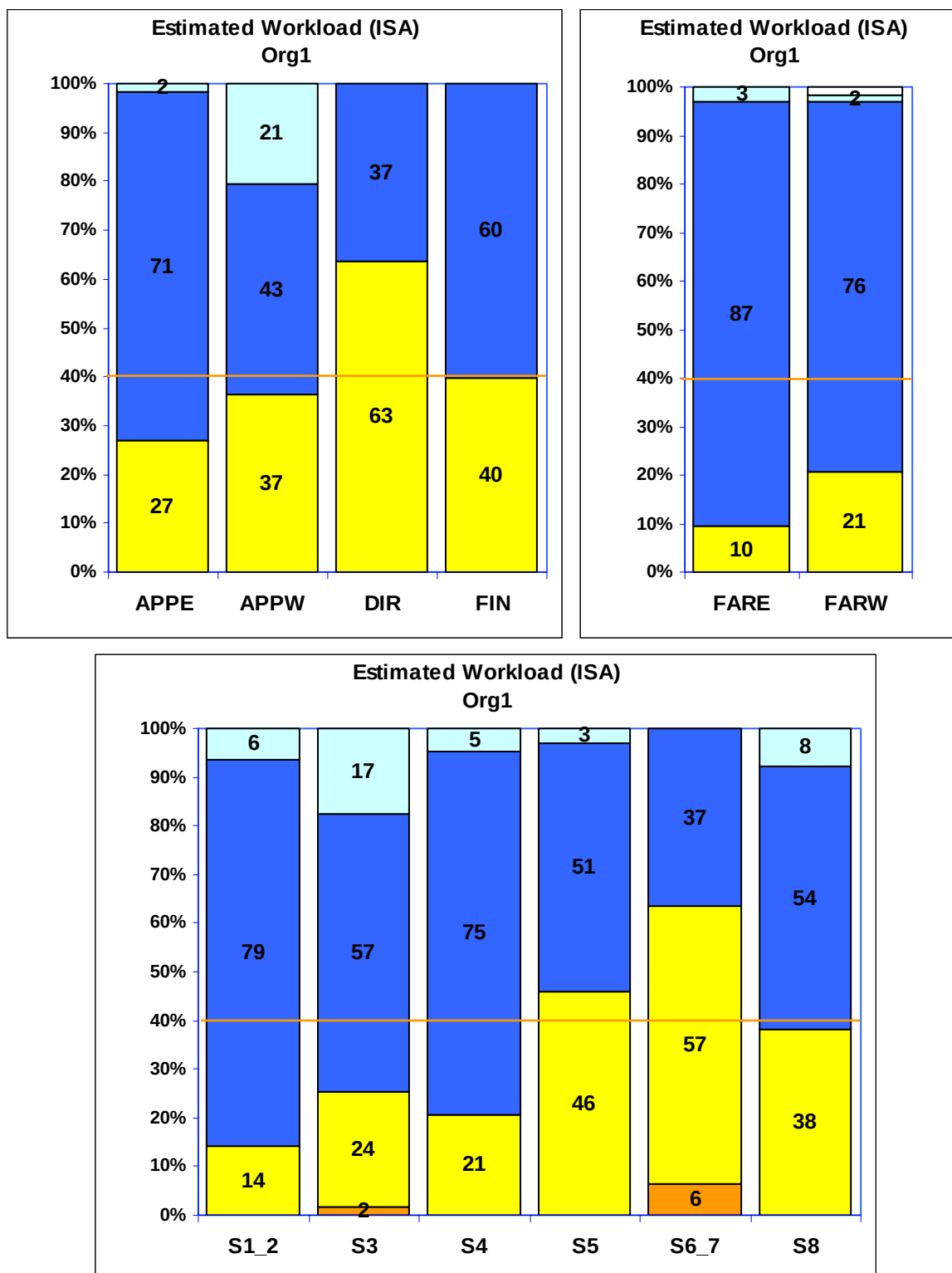
12. The proposed Airspace structure generally reduced coordination if compared to today's situation.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
13. The proposed Airspace structure allowed for good interaction with ACC sectors as for arriving traffic (if applicable).	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
14. How easy/difficult was it for you to sequence the traffic and/or monitor the sequence?	☐	☐	☐	☐	☐
	Very easy	Easy	Medium	Difficult	Very difficult
15. How easy/difficult was it for you to obtain/maintain required spacing between a/c?	☐	☐	☐	☐	☐
	Very easy	Easy	Medium	Difficult	Very difficult
16. Building the a/c sequence with PMS required less effort if compared with the current working method based on radar vectoring.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
17. Maintaining the a/c sequence with PMS required less effort if compared with the current working method based on radar vectoring.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
18. How easy/difficult was it for you to manage the differences in a/c performances?	☐	☐	☐	☐	☐
	Very easy	Easy	Medium	Difficult	Very difficult
19. PMS was flexible enough to cope with heterogeneous traffic (e.g. non P-RNAV equipped, slow moving).	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
20. Reverting to radar vectoring when necessary was not a problem.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
21. AMAN contributed to a high situational awareness on arriving traffic management.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
22. PMS increases trajectory predictability and in turn traffic situation awareness.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

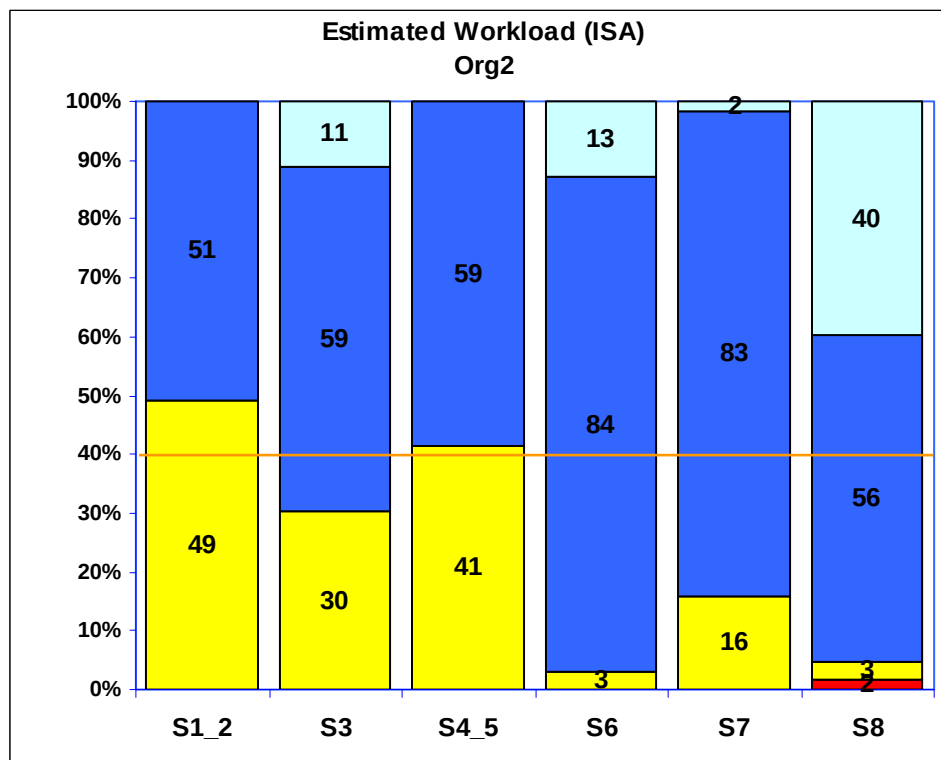
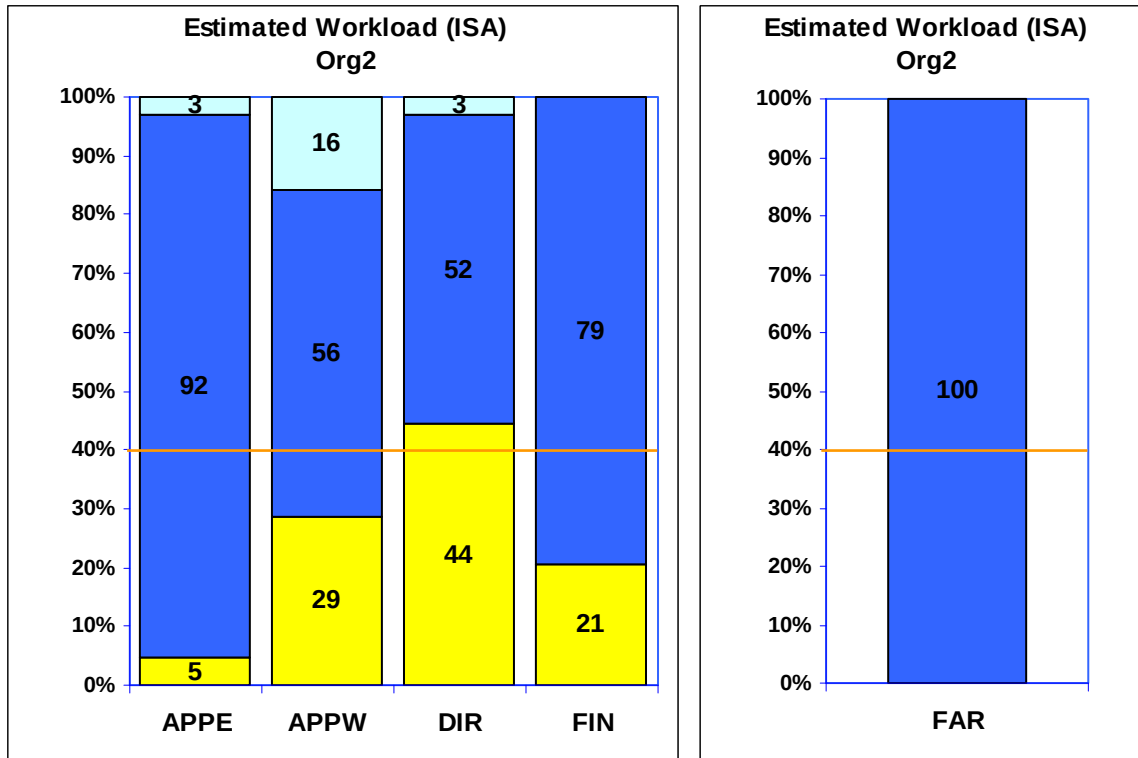
23. PMS did not decrease the level of job satisfaction.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
24. PMS did not decrease safety.	☐	☐	☐	☐	☐
	Strongly disagree	Disagree	Neutral	Agree	Strongly agree

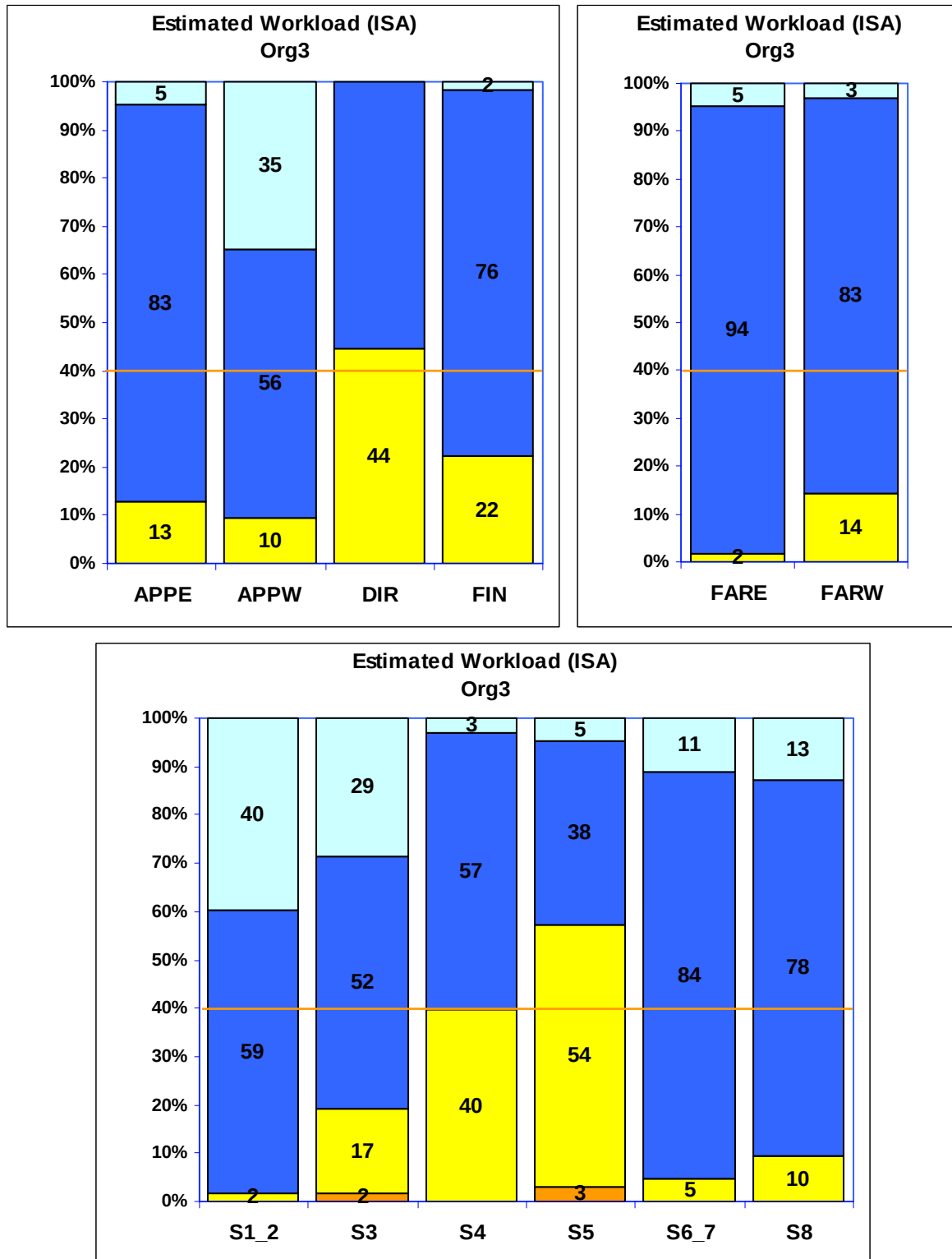
25. Do you have any additional comments about the last exercise? If yes, please then make them hereafter.

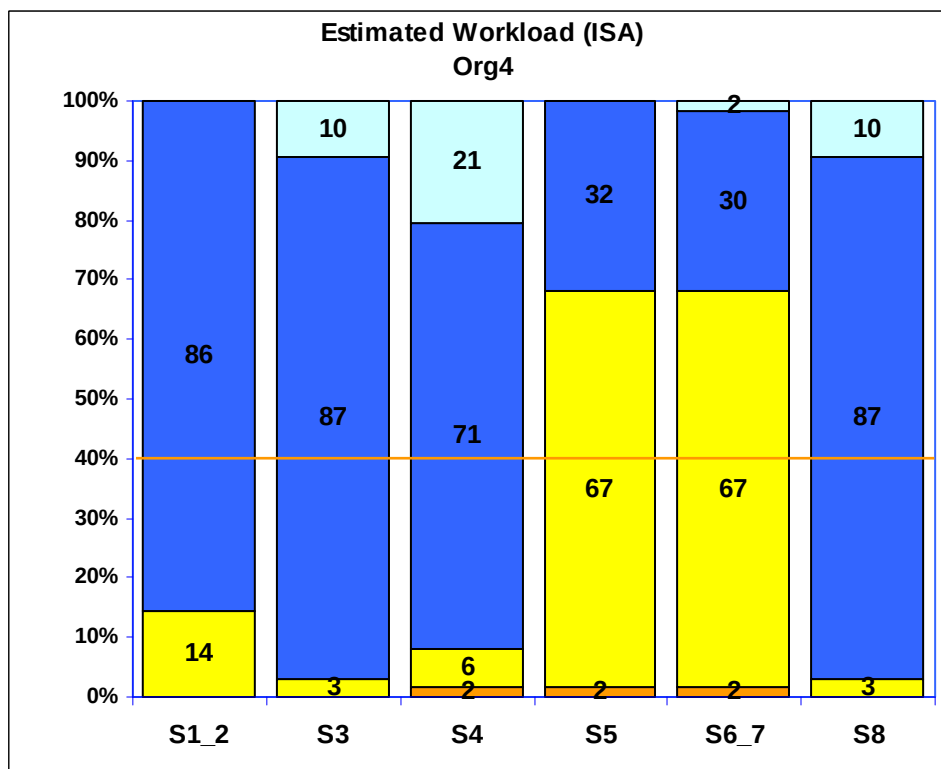
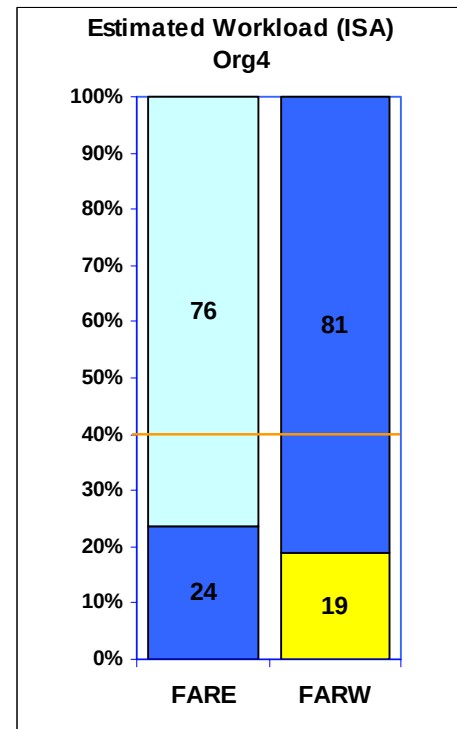
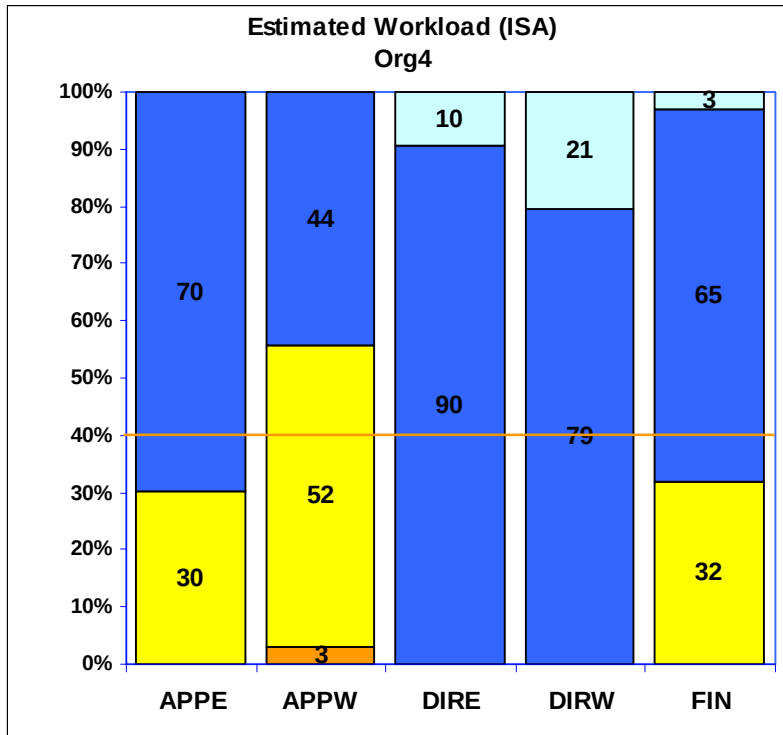
Page intentionally left blank

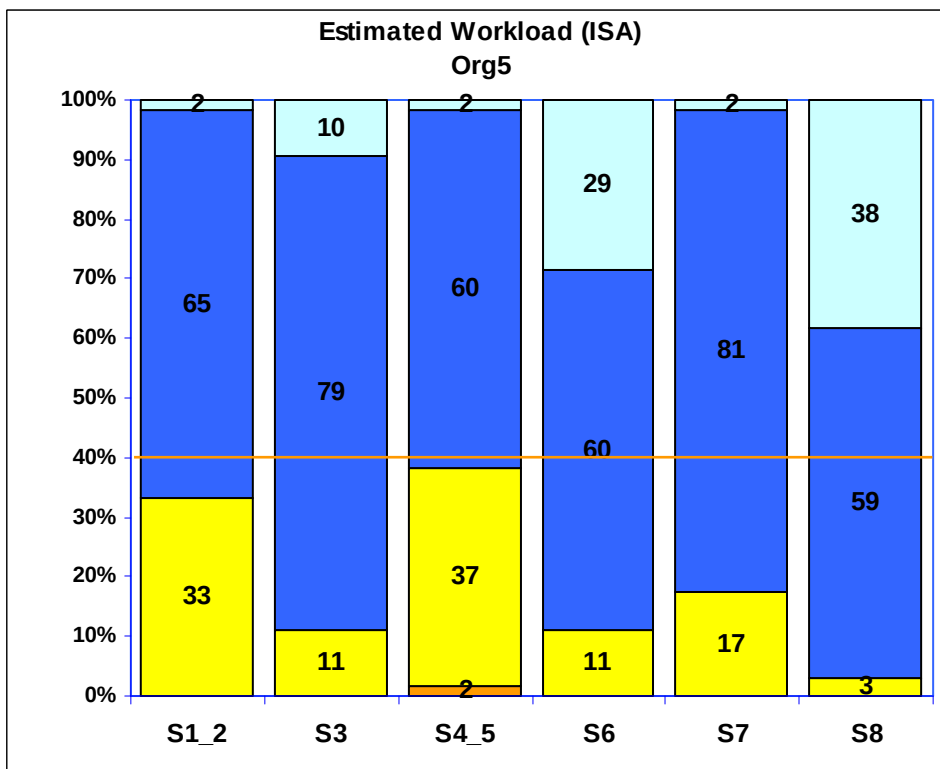
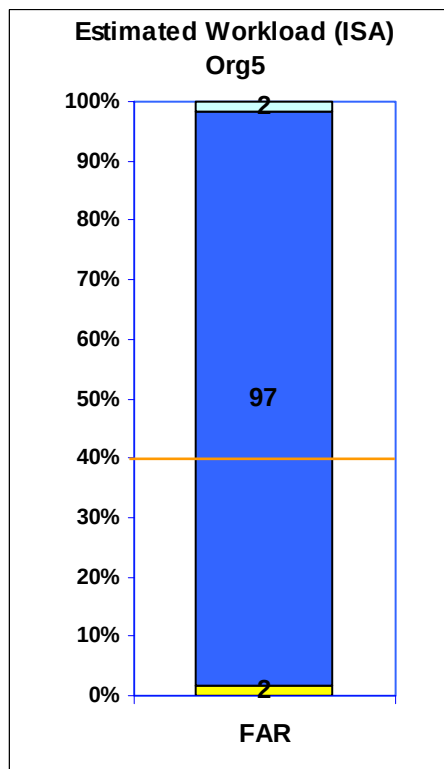
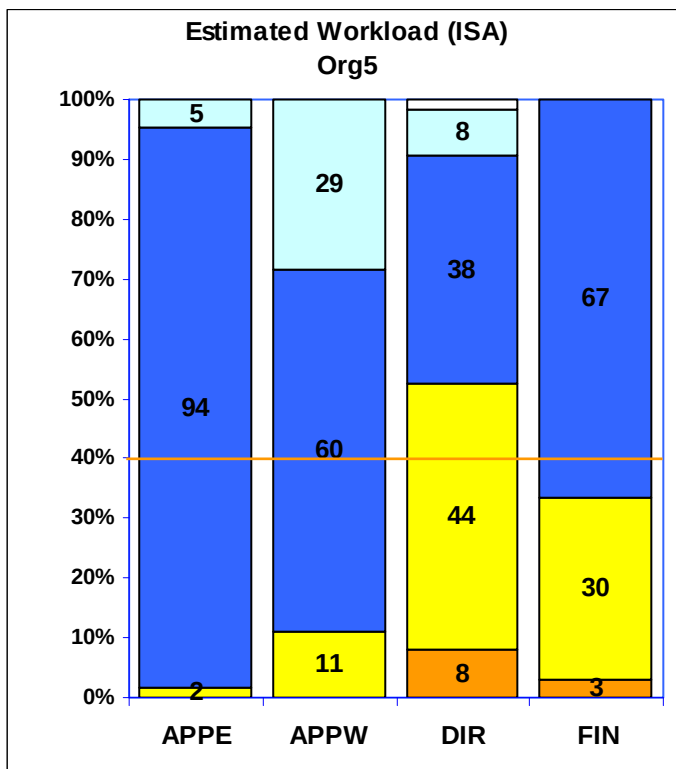
ANNEX C: INSTANTANEOUS SELF-ASSESSMENT (ISA)

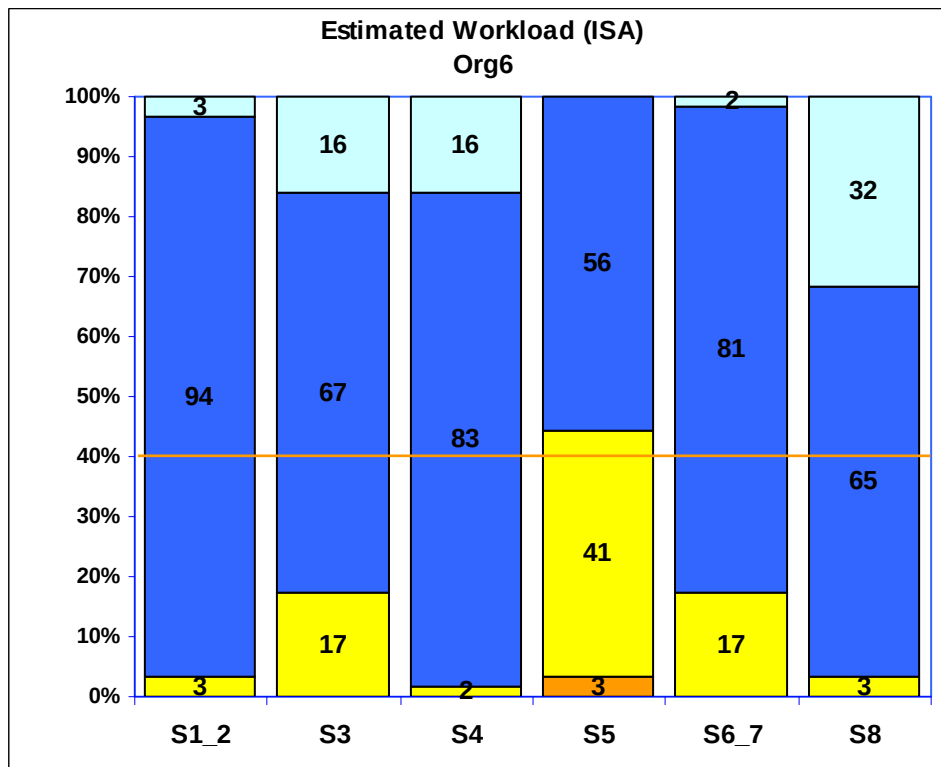
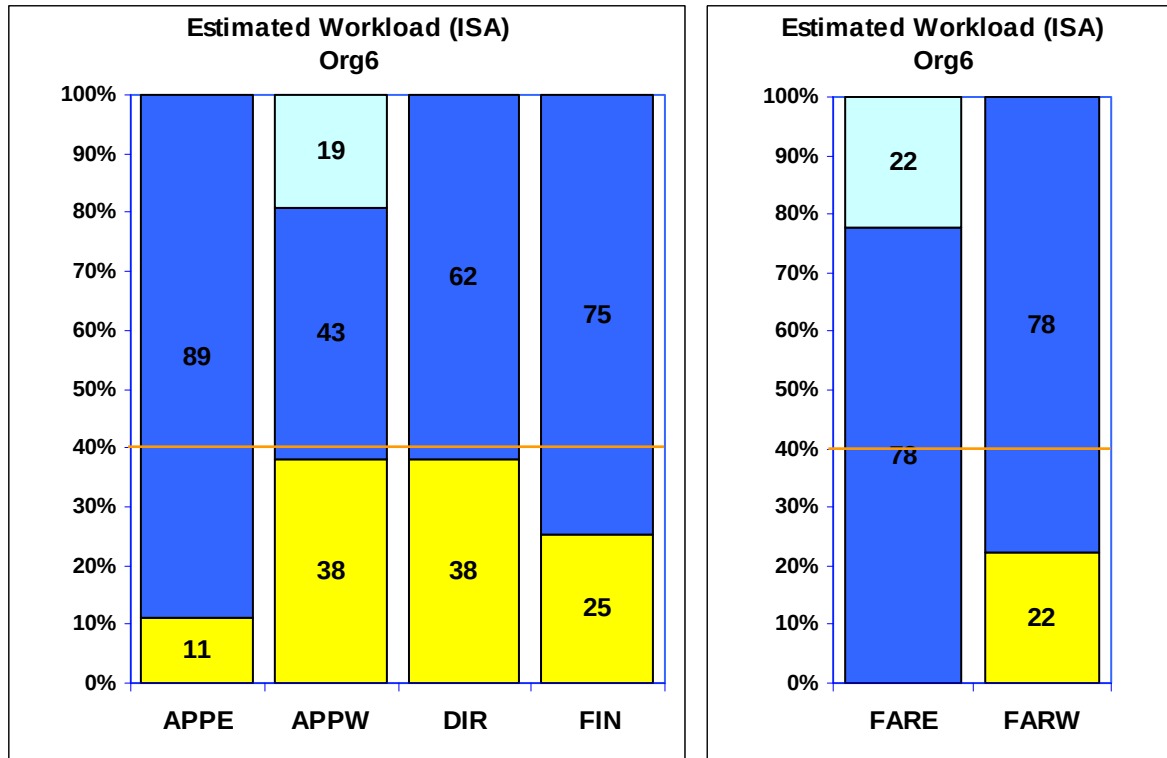


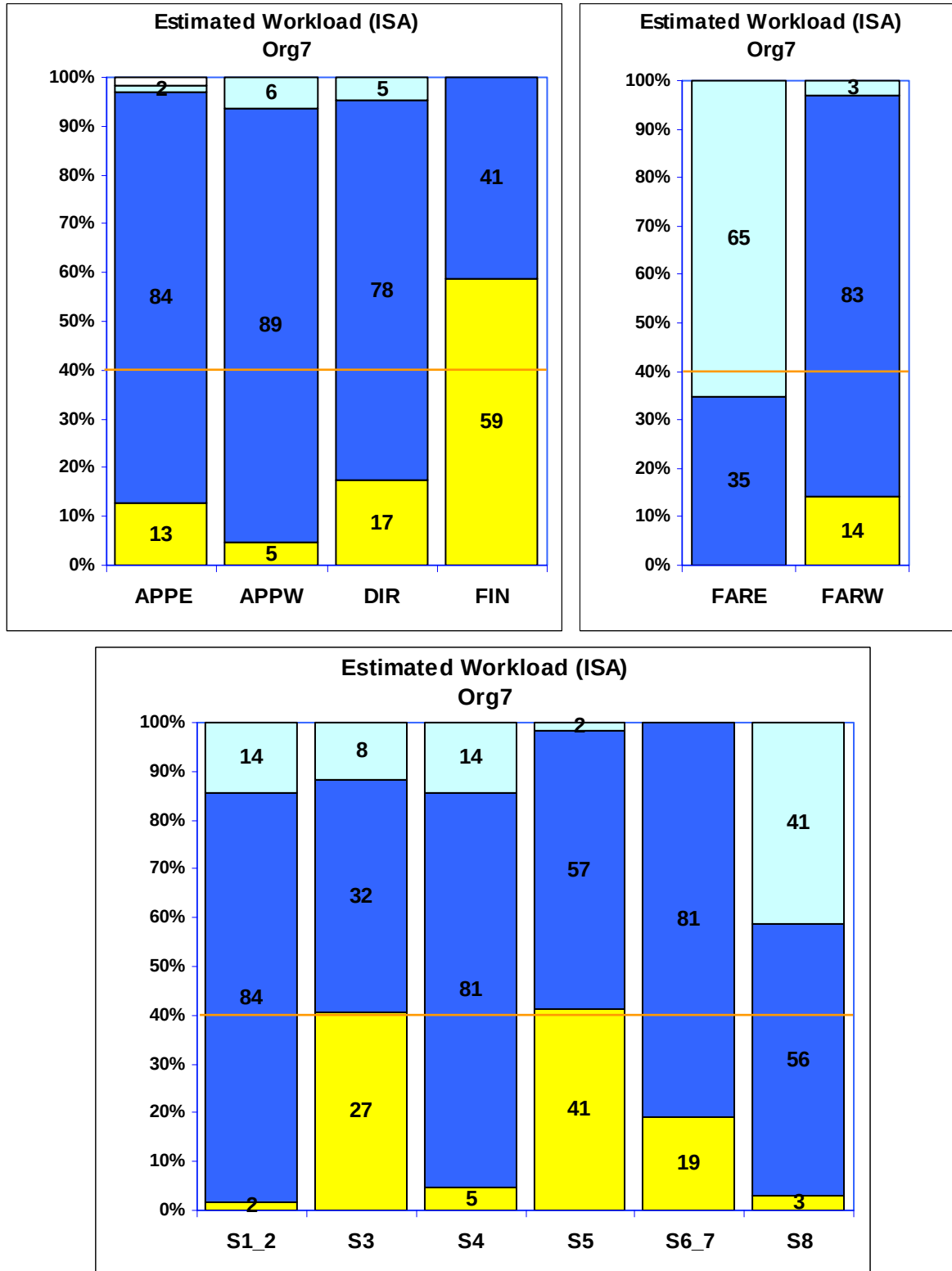






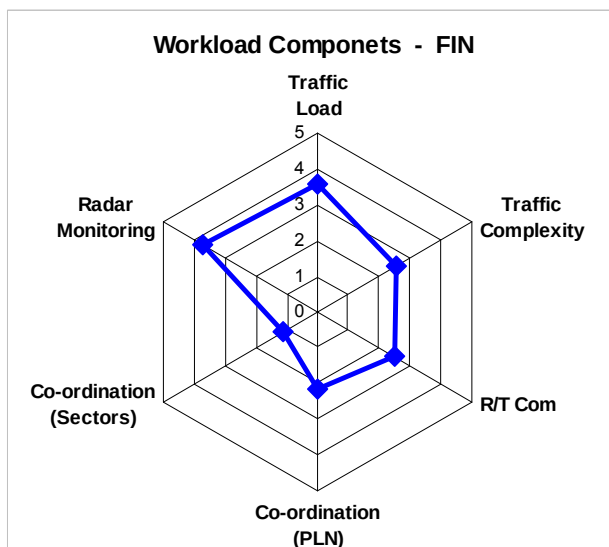
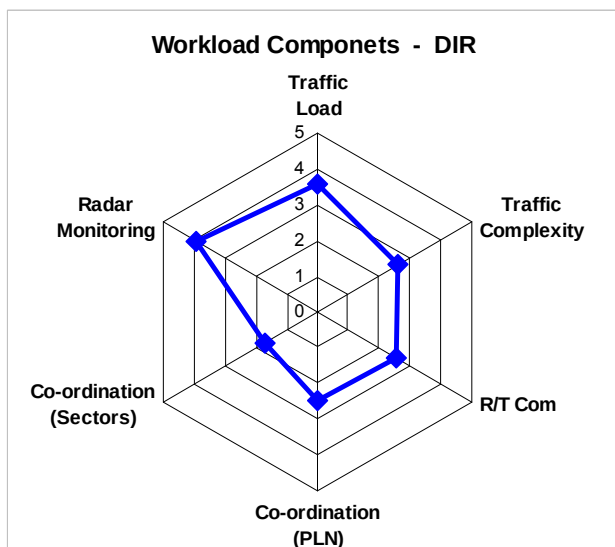
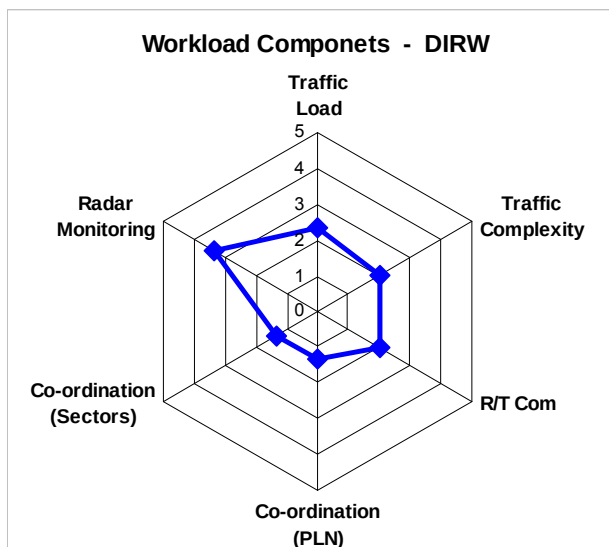
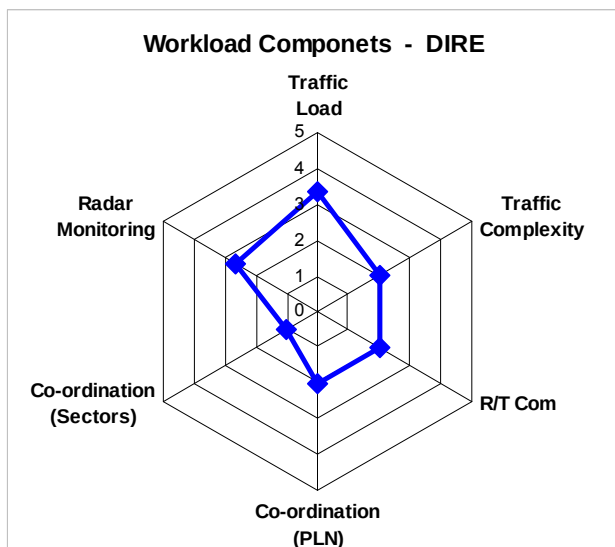
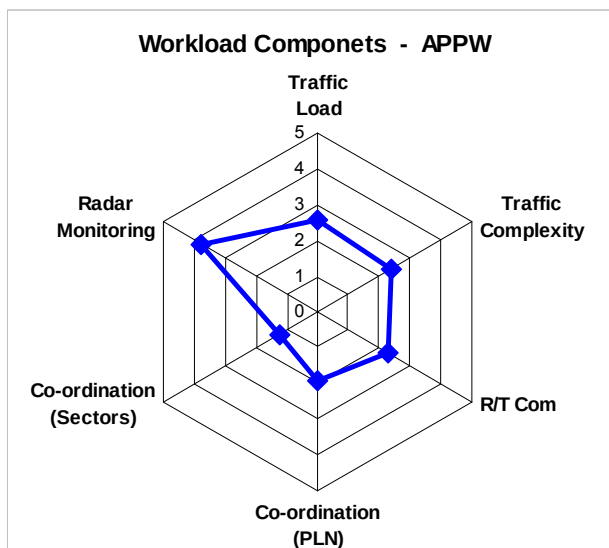
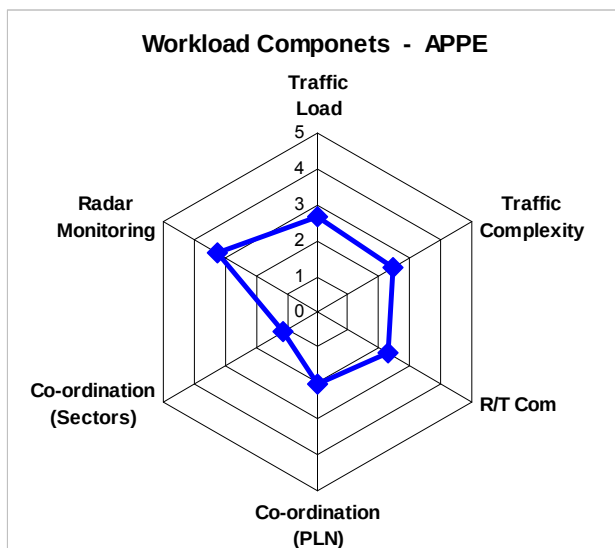


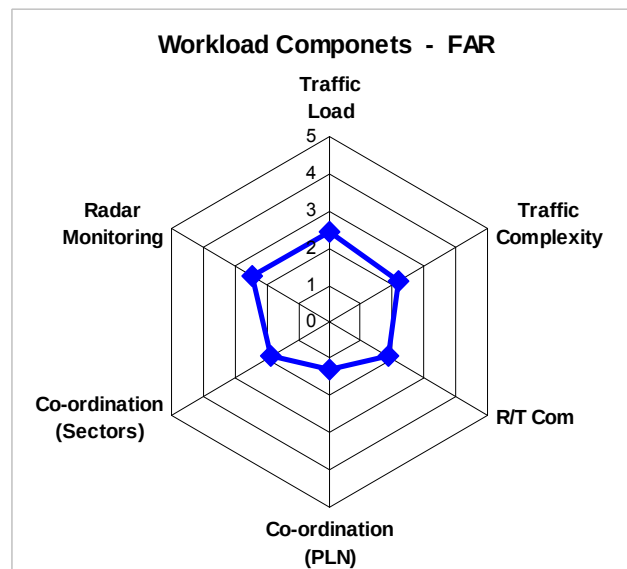
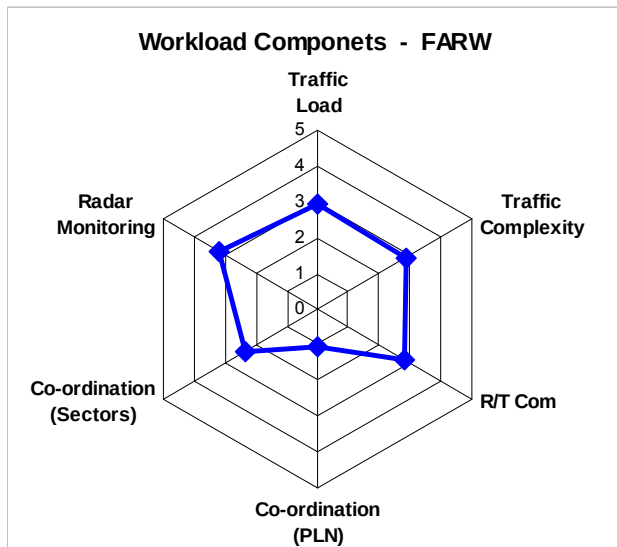
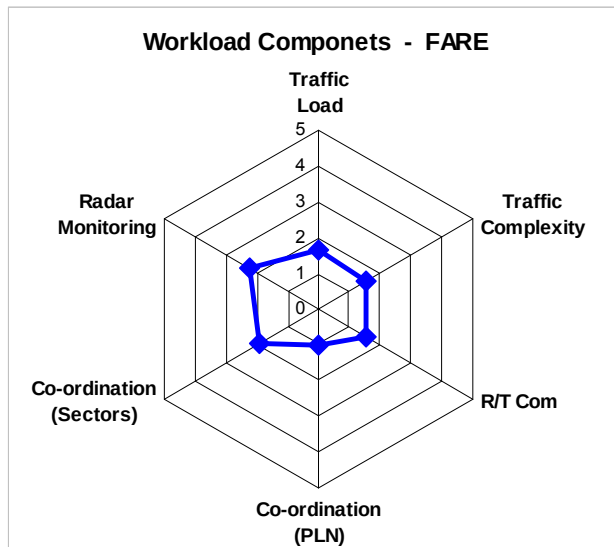


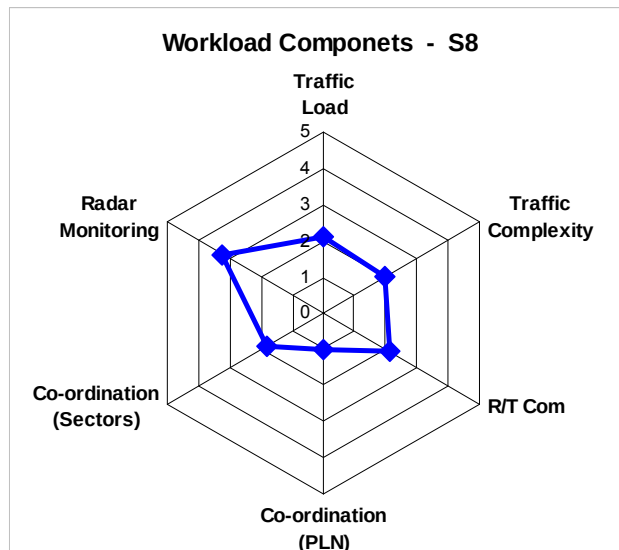
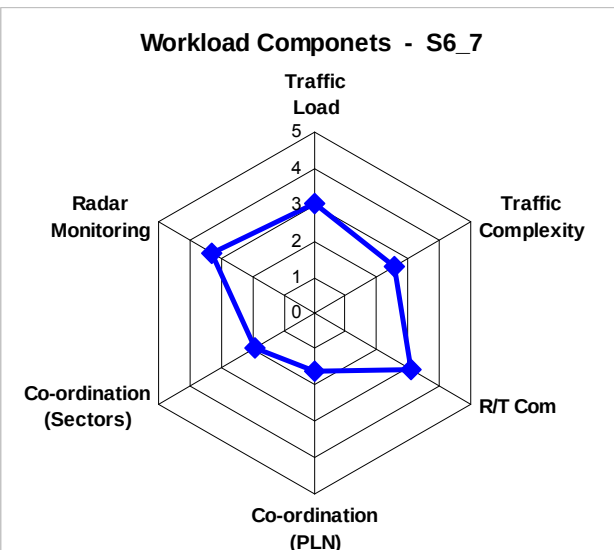
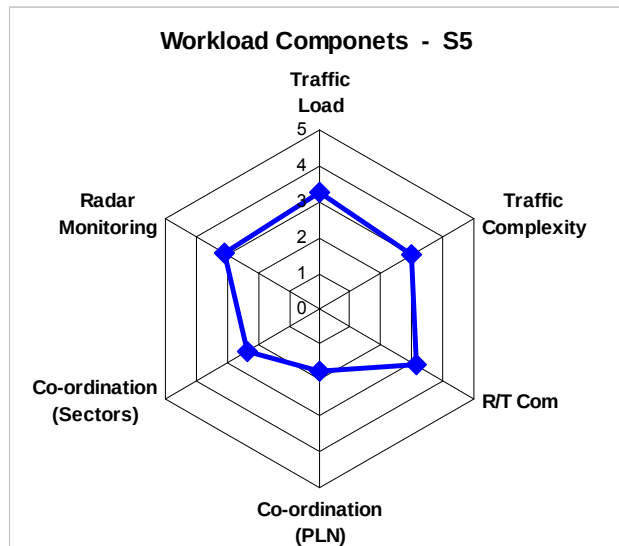
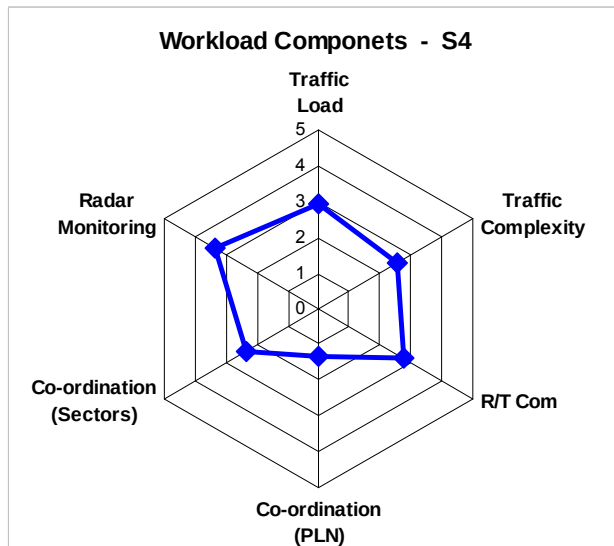
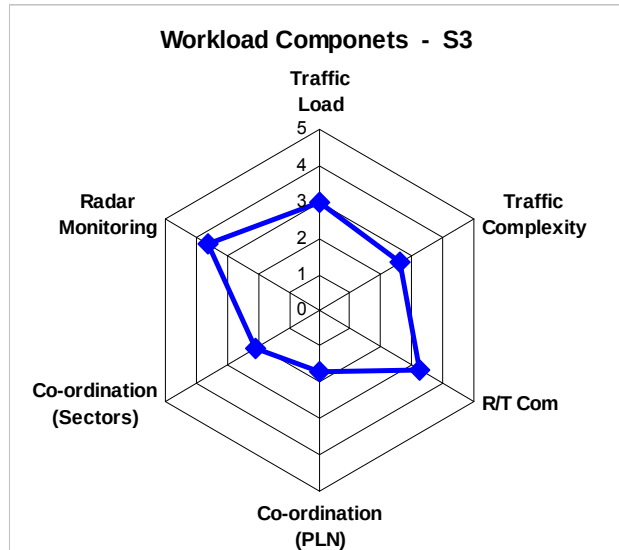
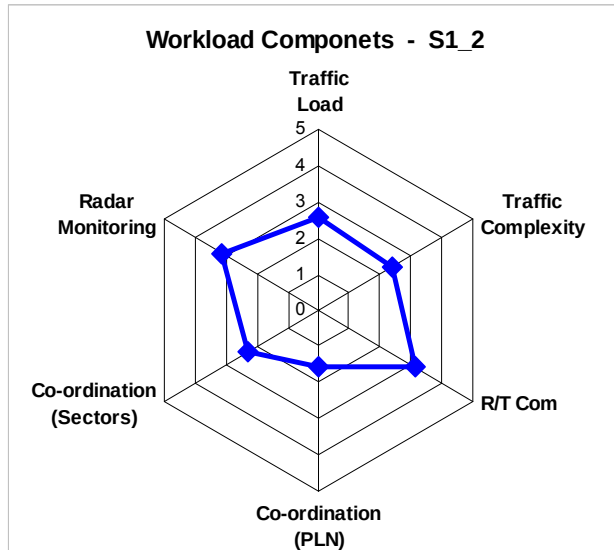


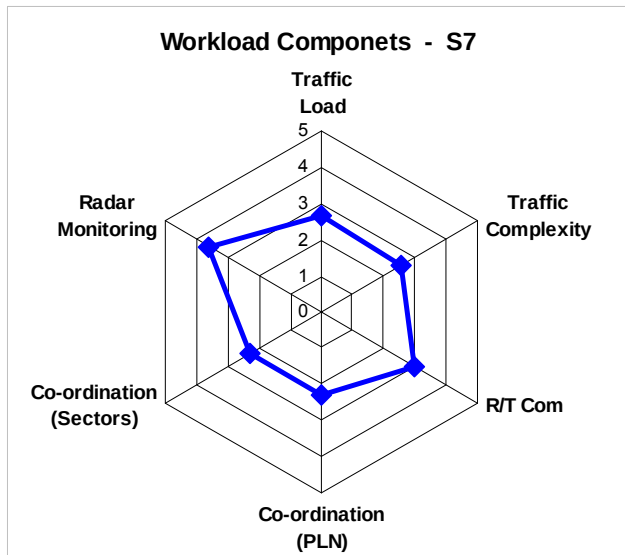
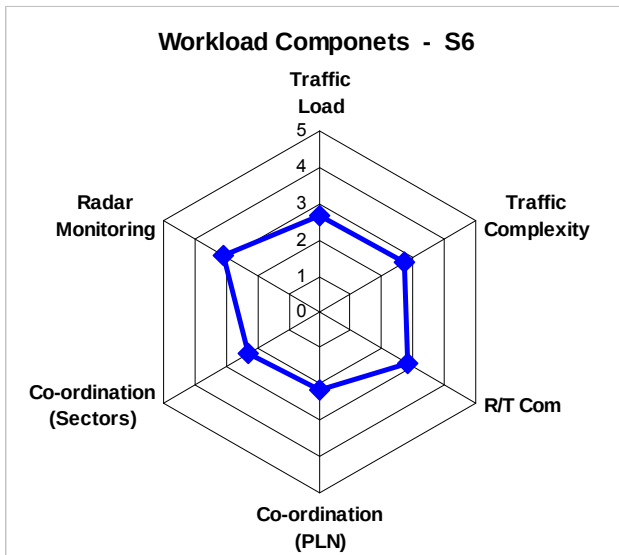
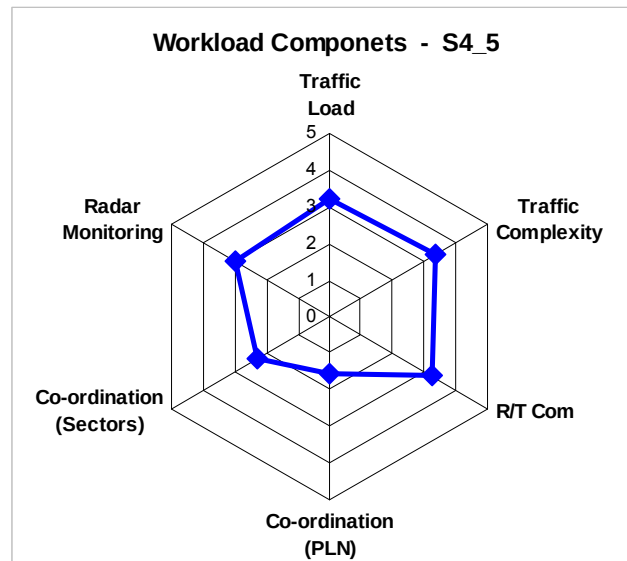
Page intentionally left blank

ANNEX D: WORKLOAD COMPONENTS

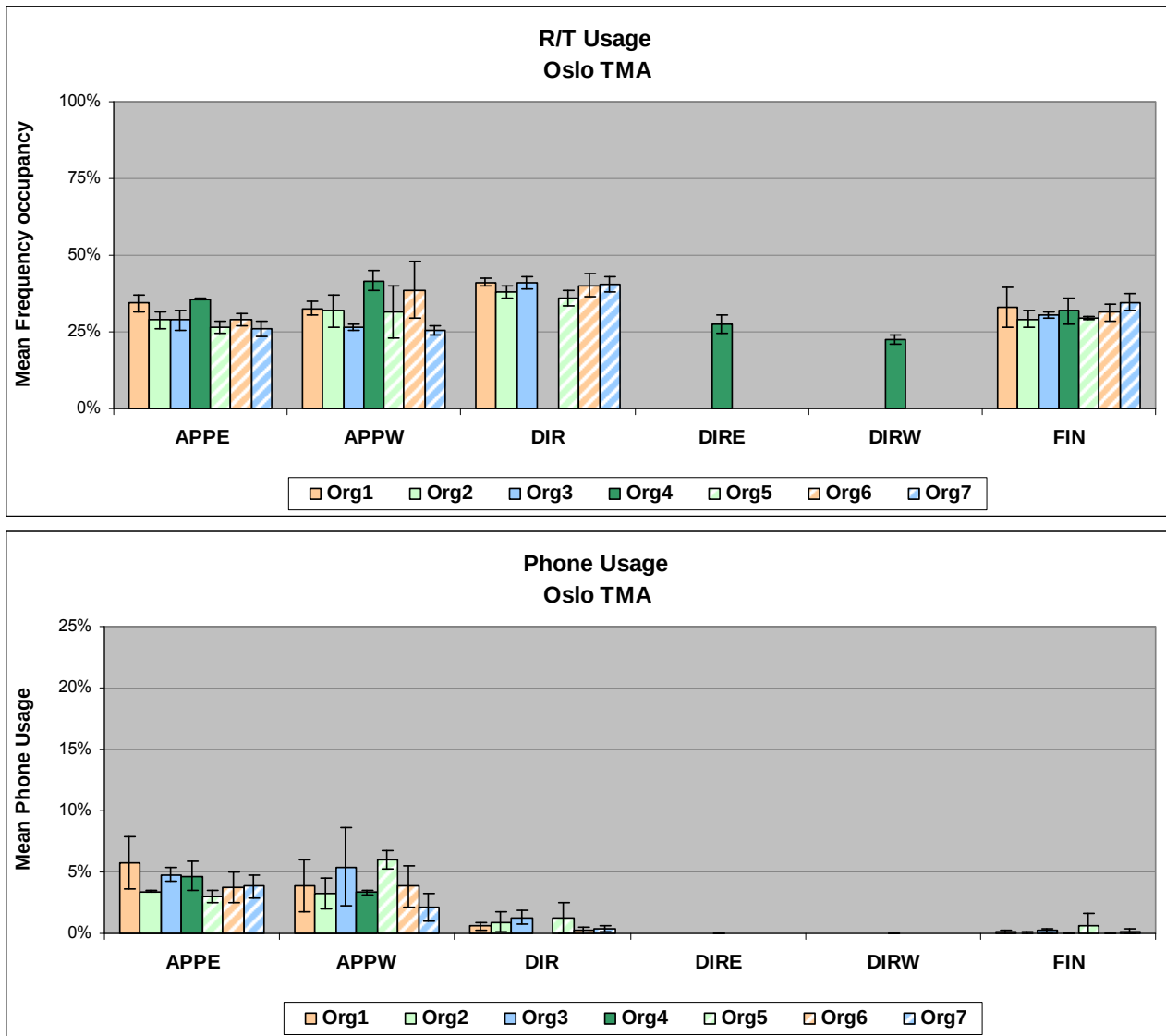


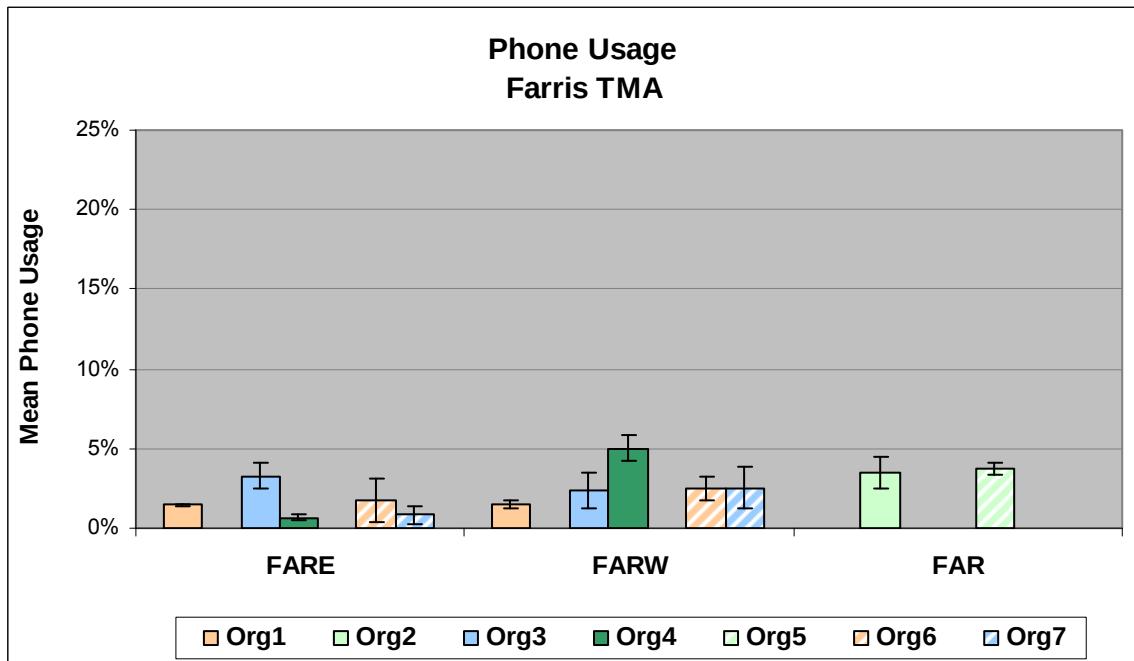
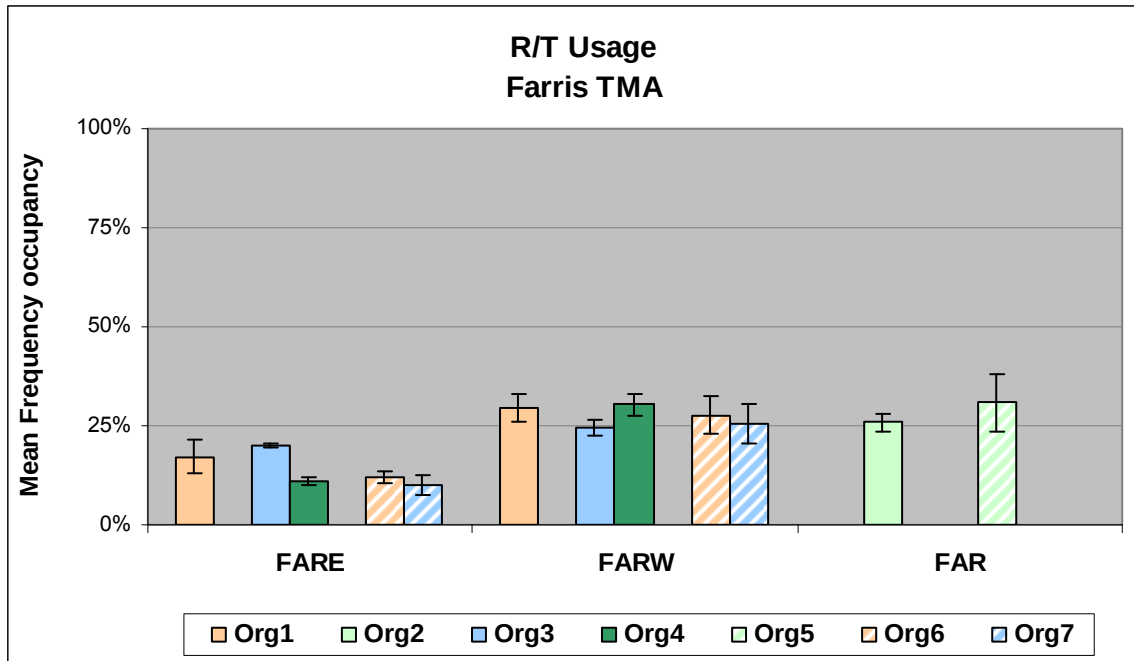




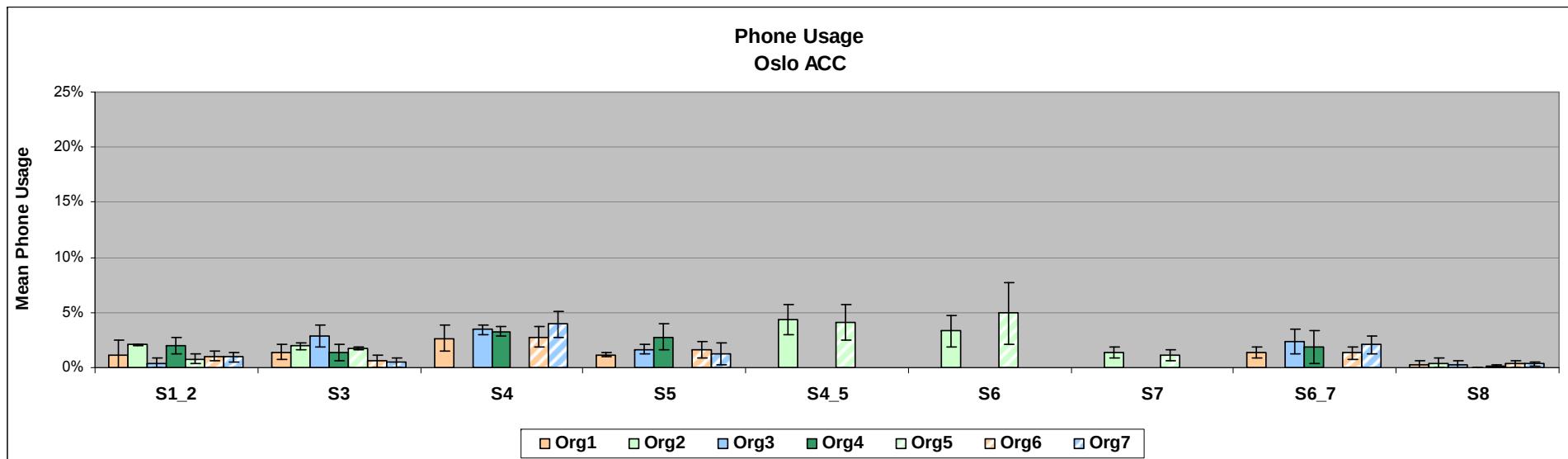
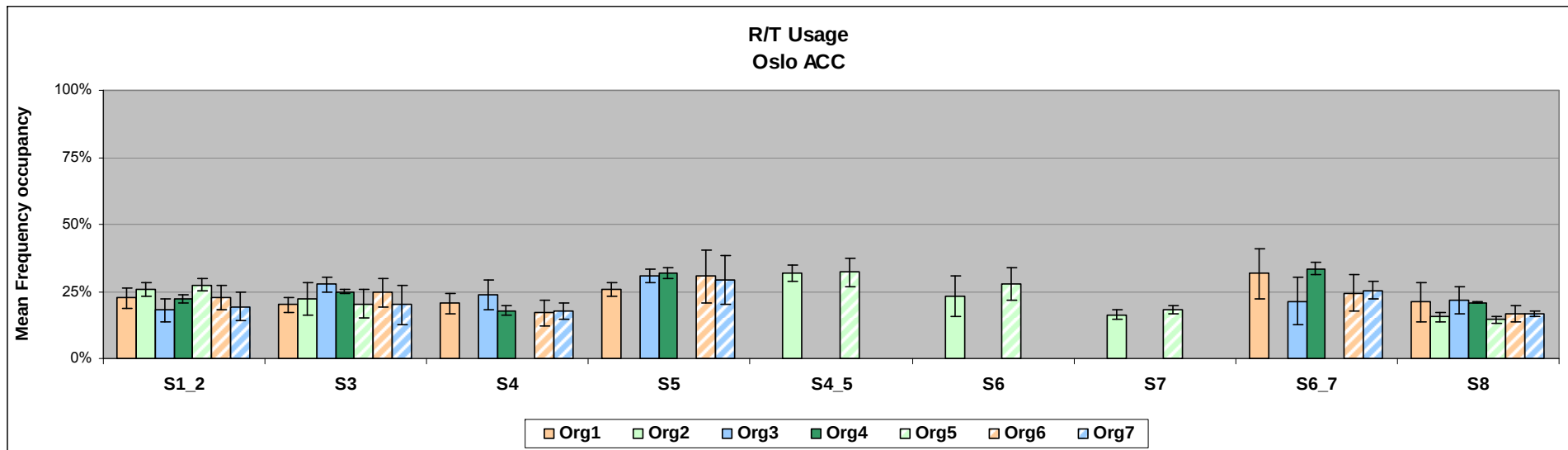


ANNEX E: R/T AND PHONE USAGE



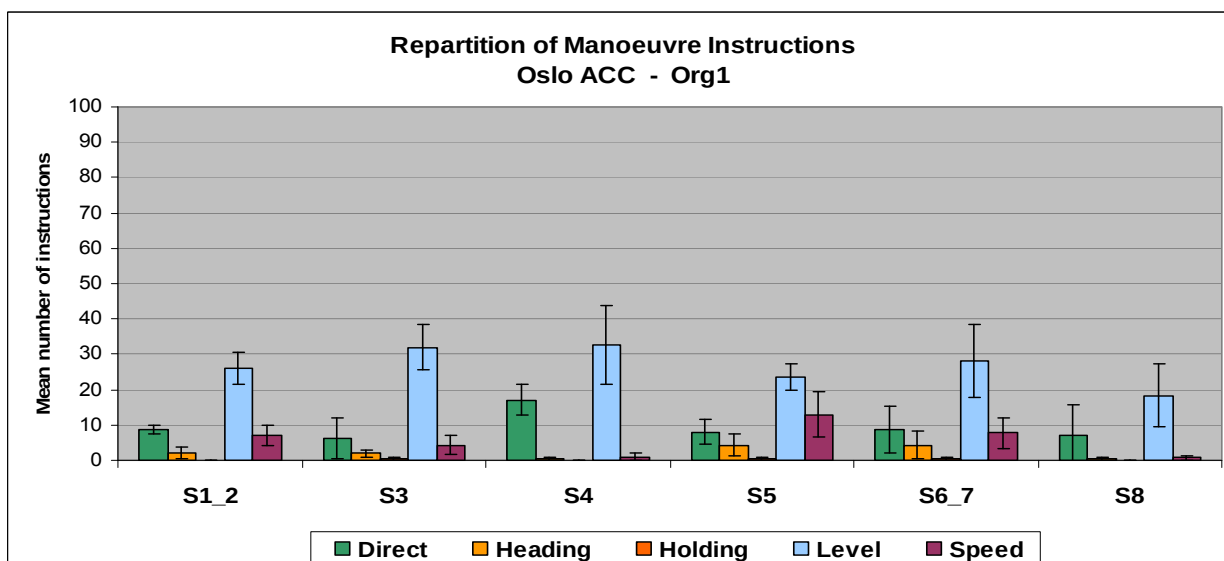
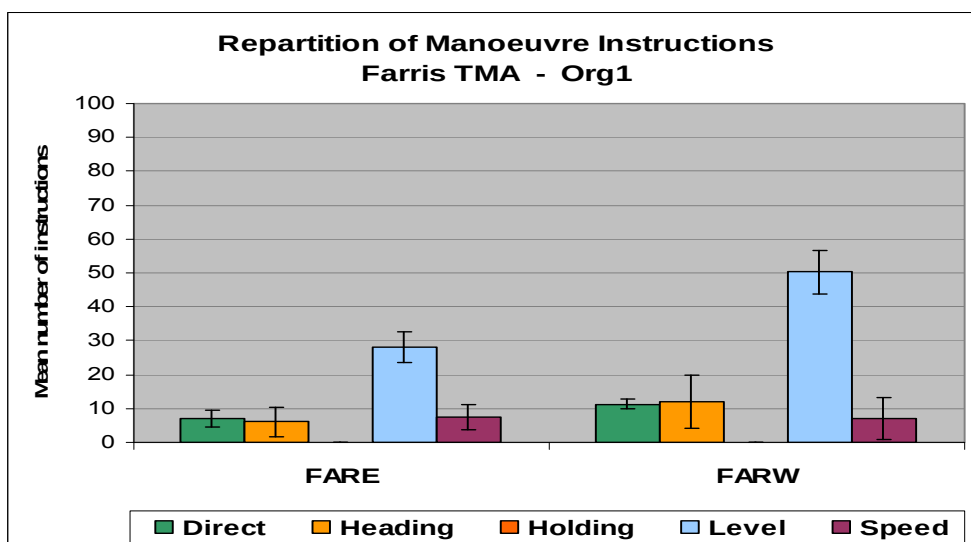
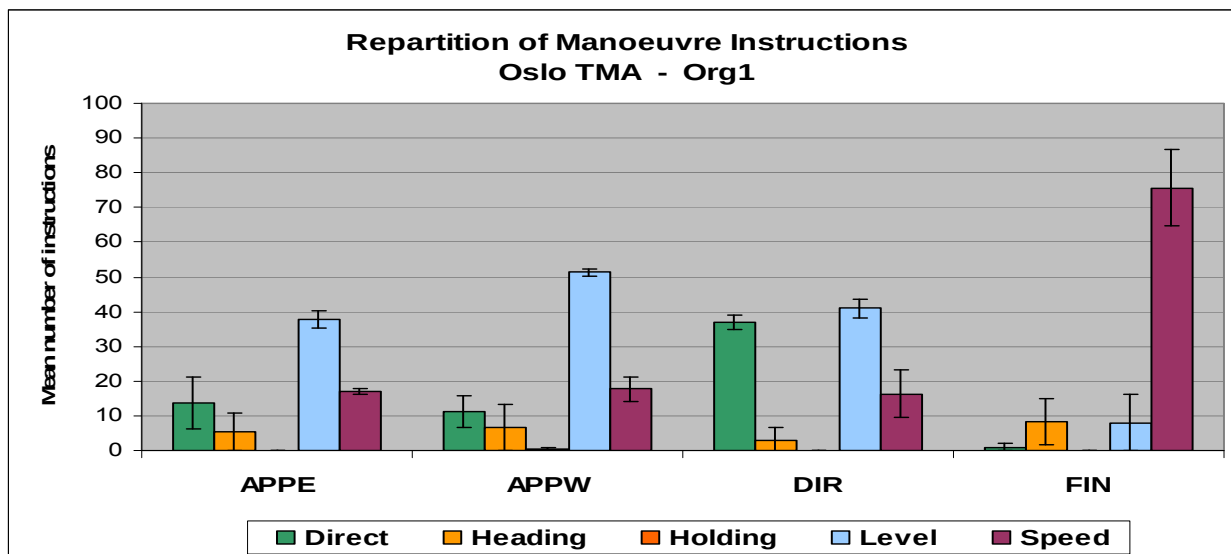


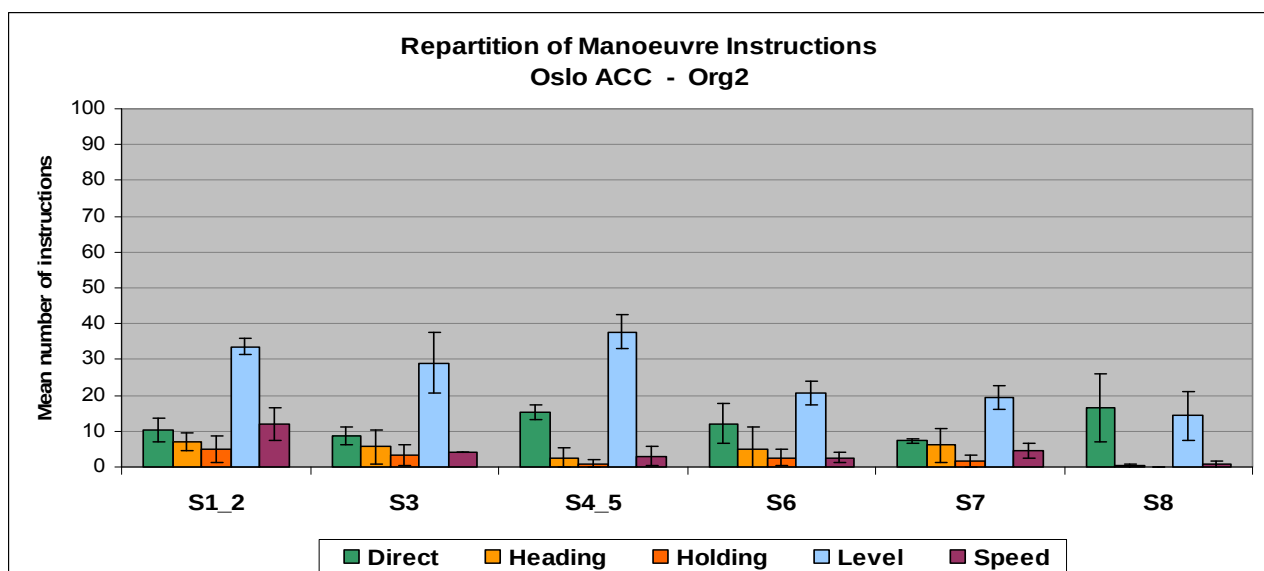
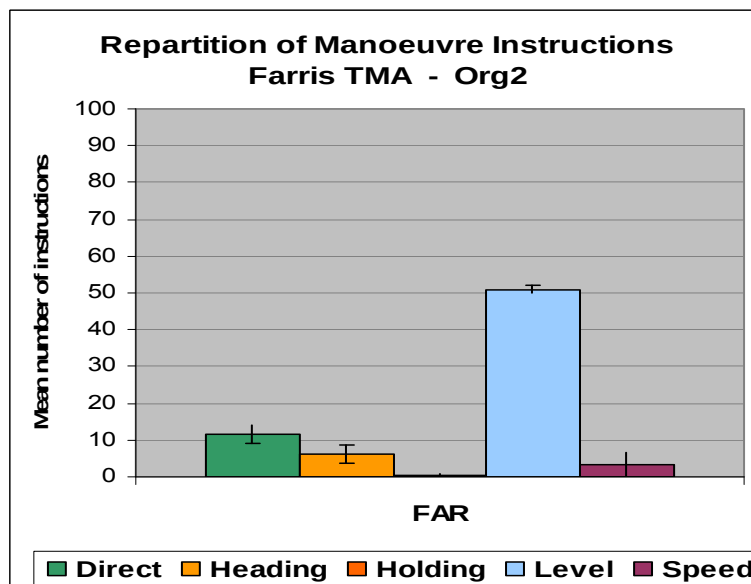
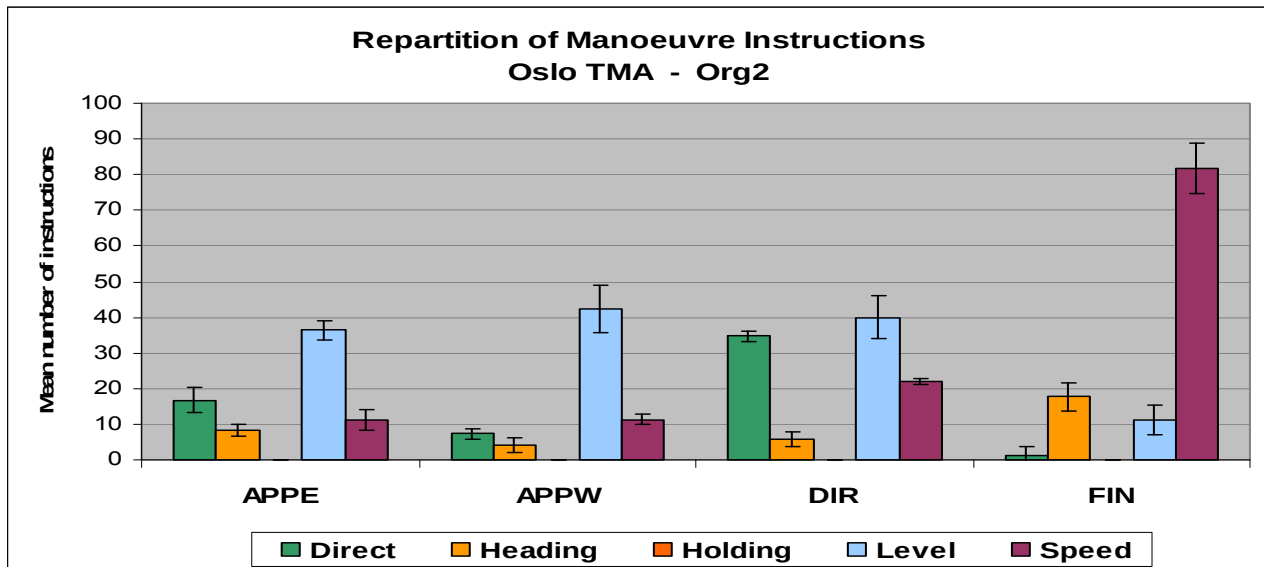
Real Time Simulation Oslo ASAP

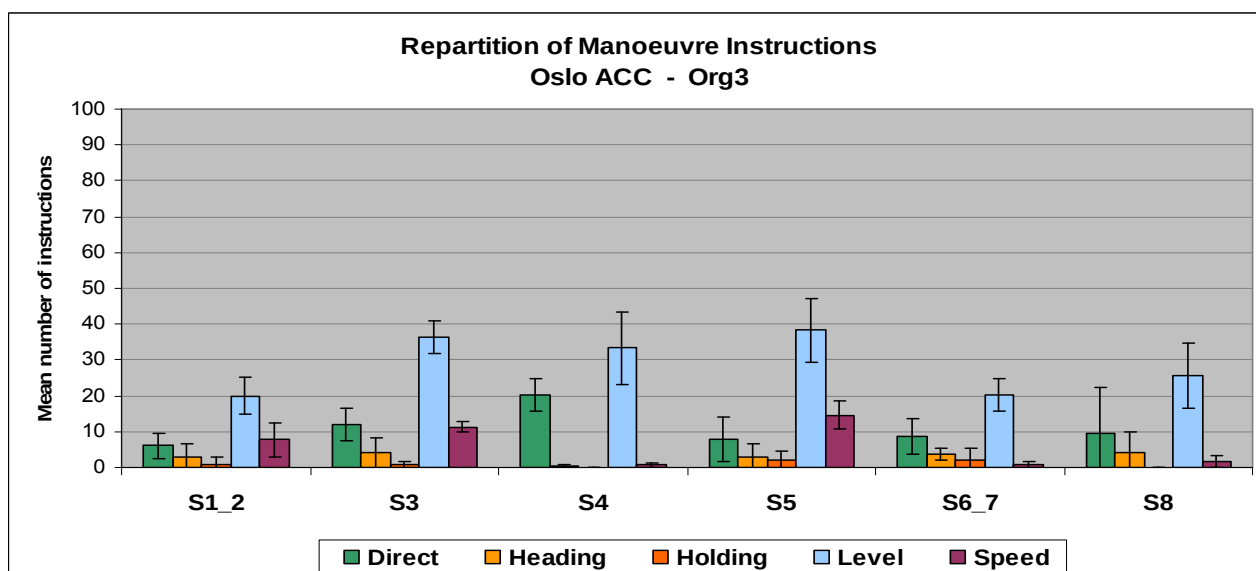
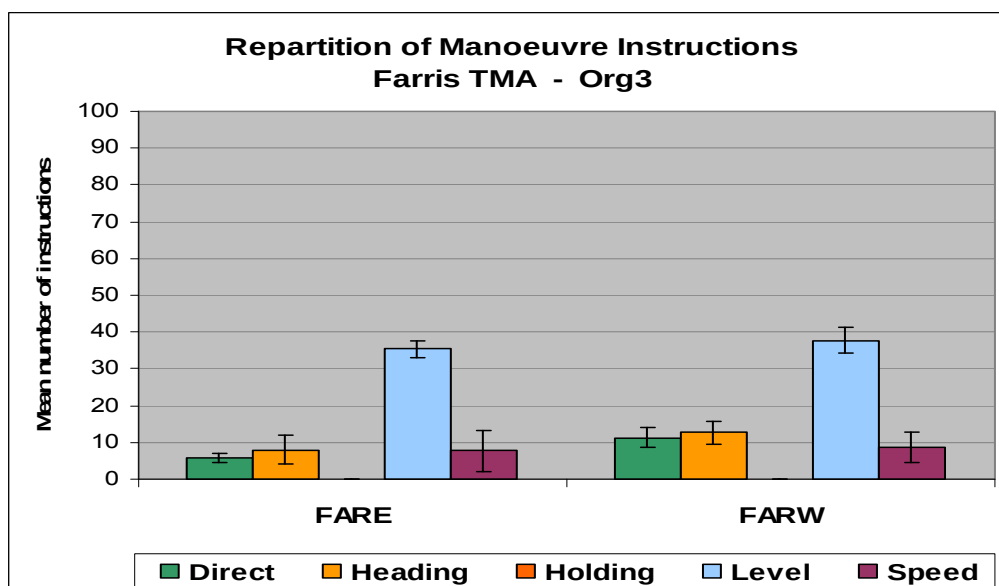
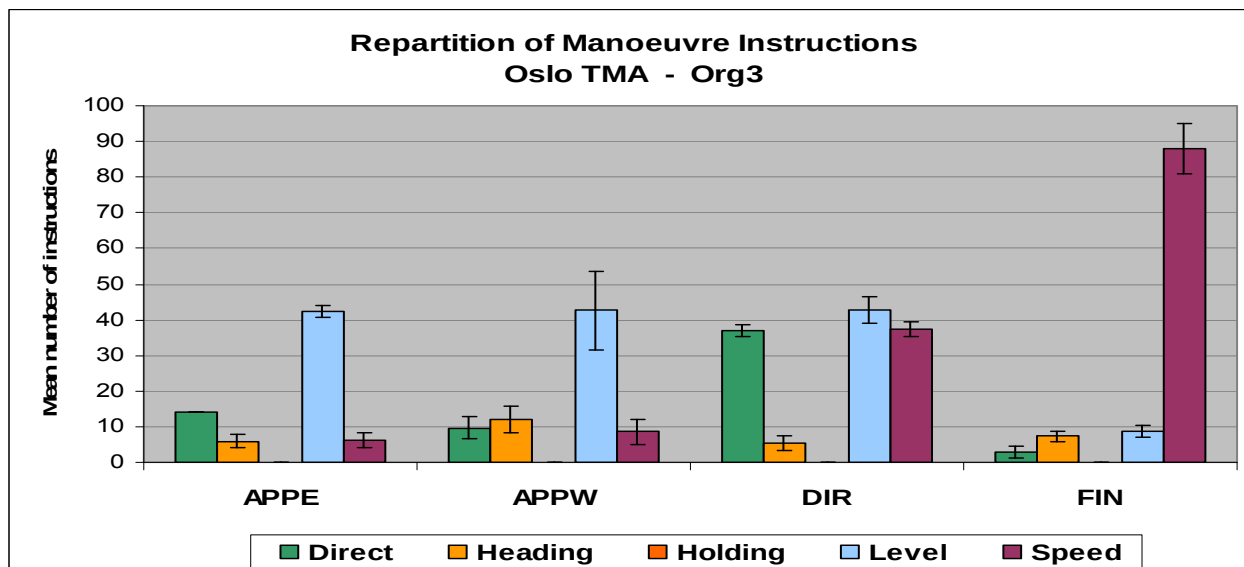


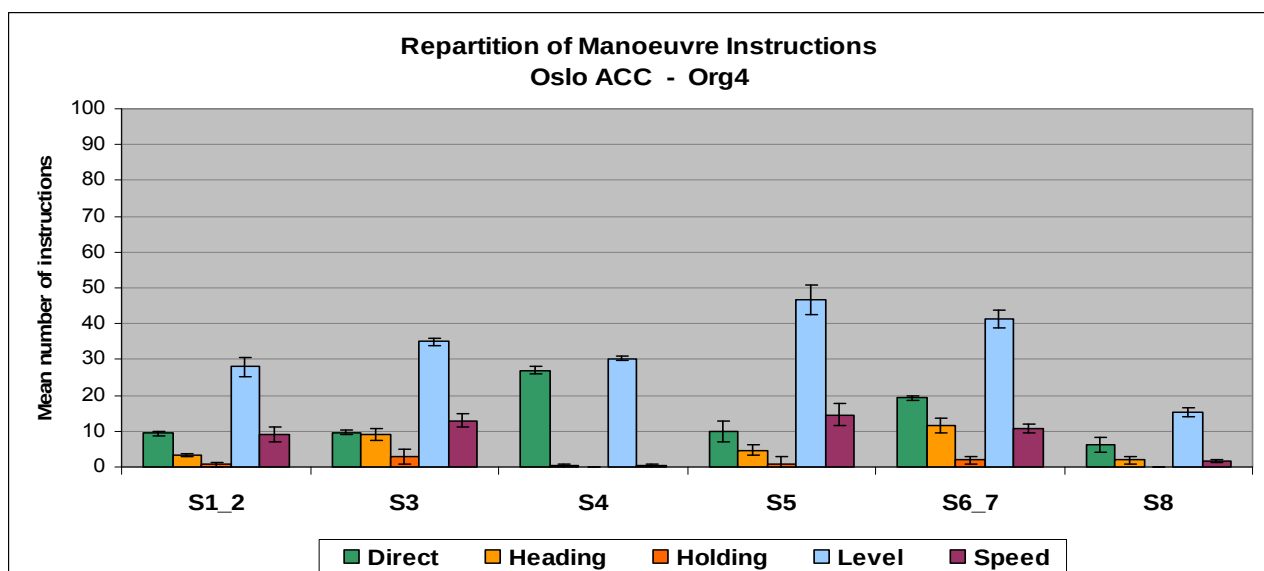
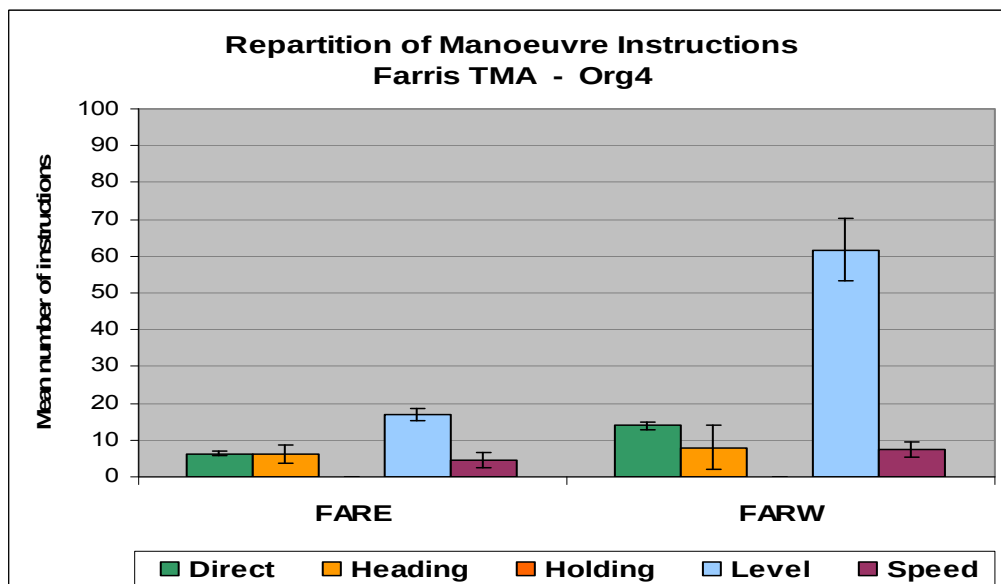
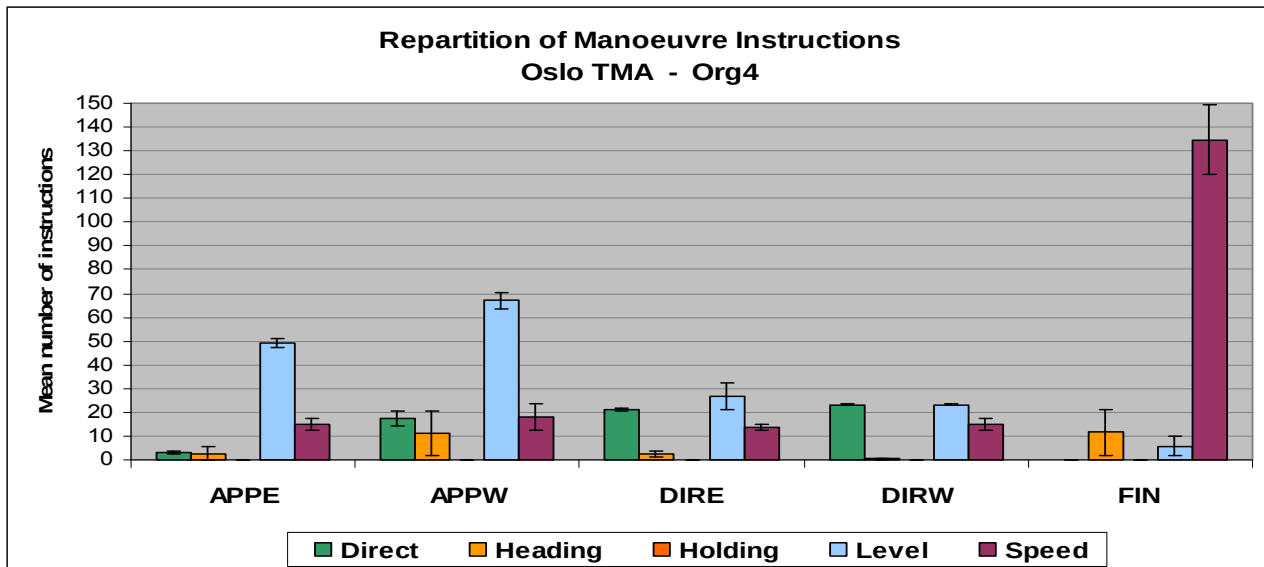
Page intentionally left blank

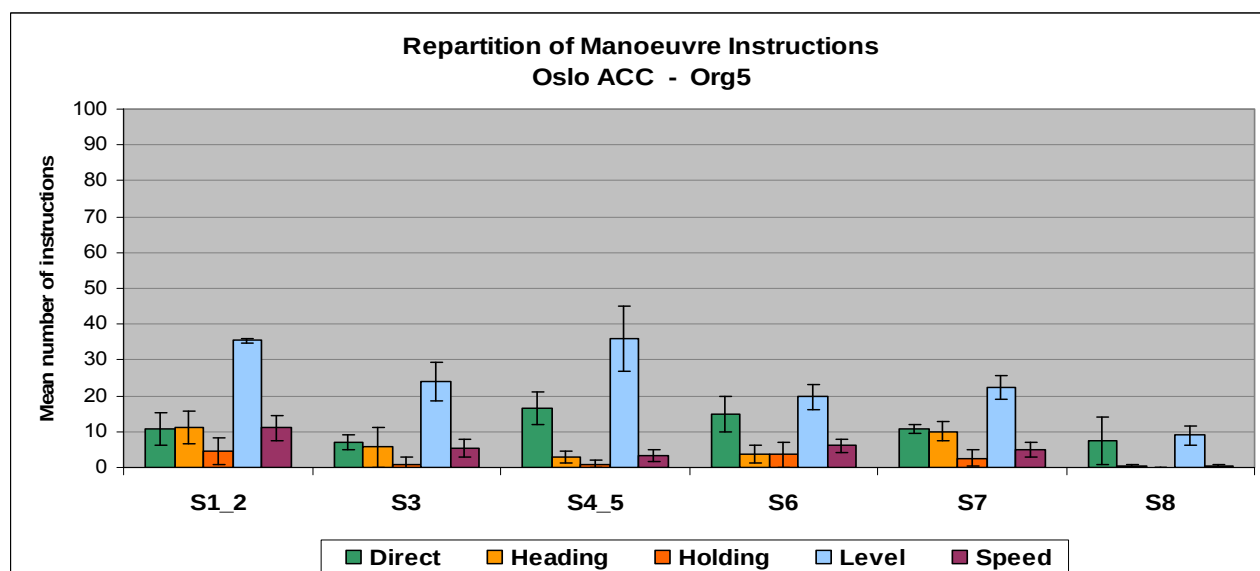
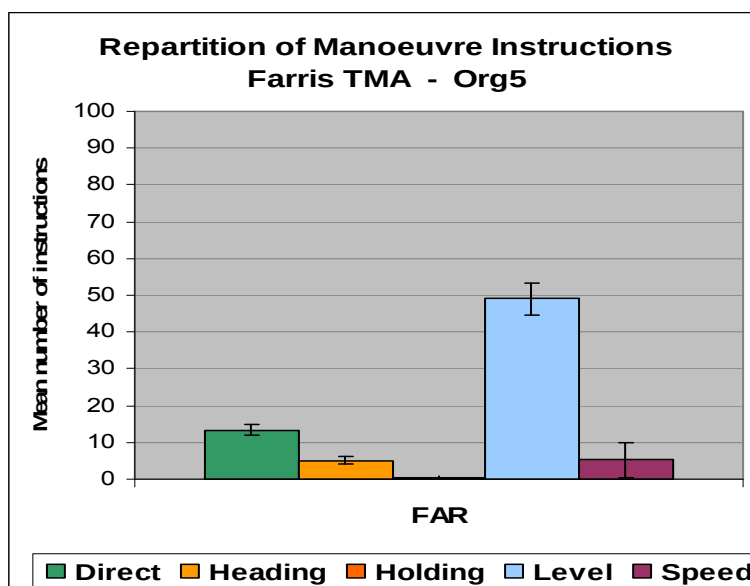
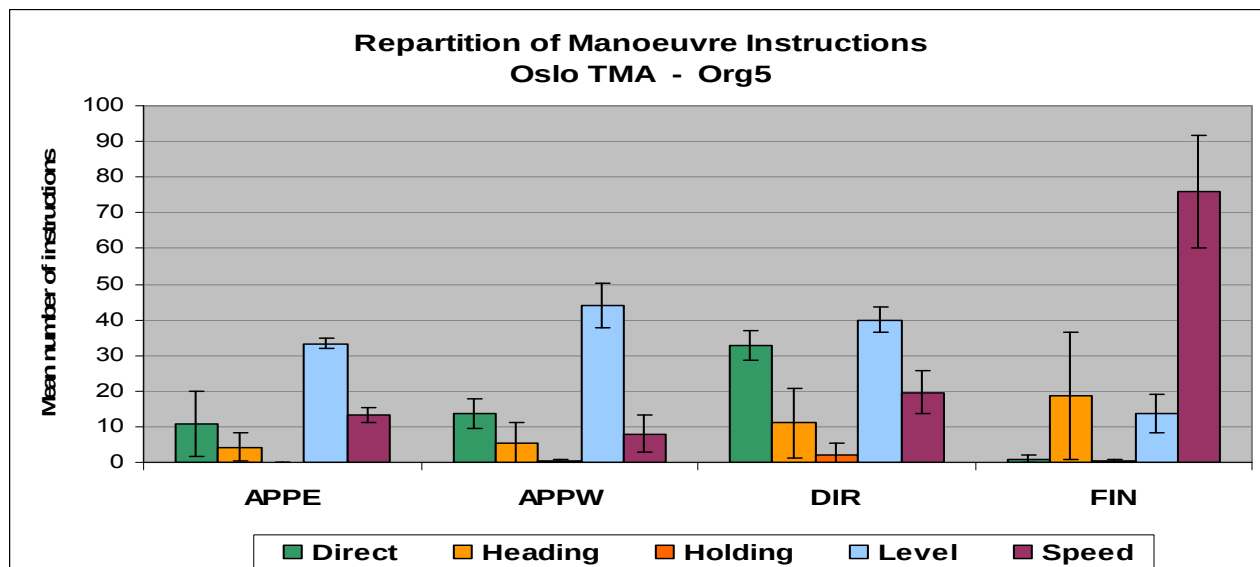
ANNEX F: REPARTITION OF INSTRUCTIONS

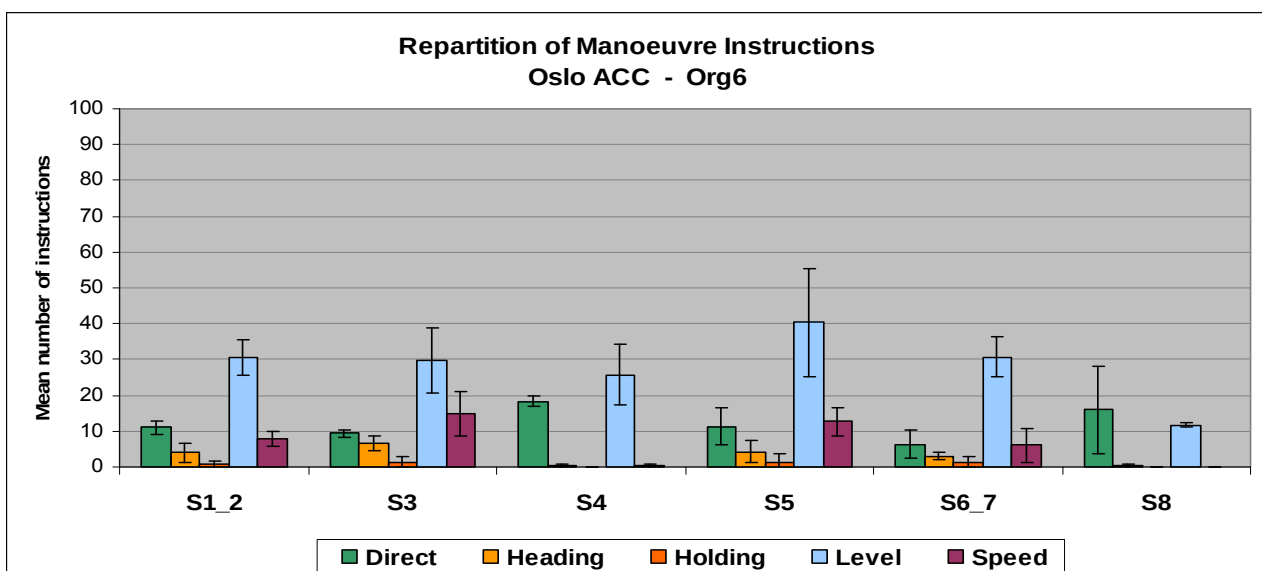
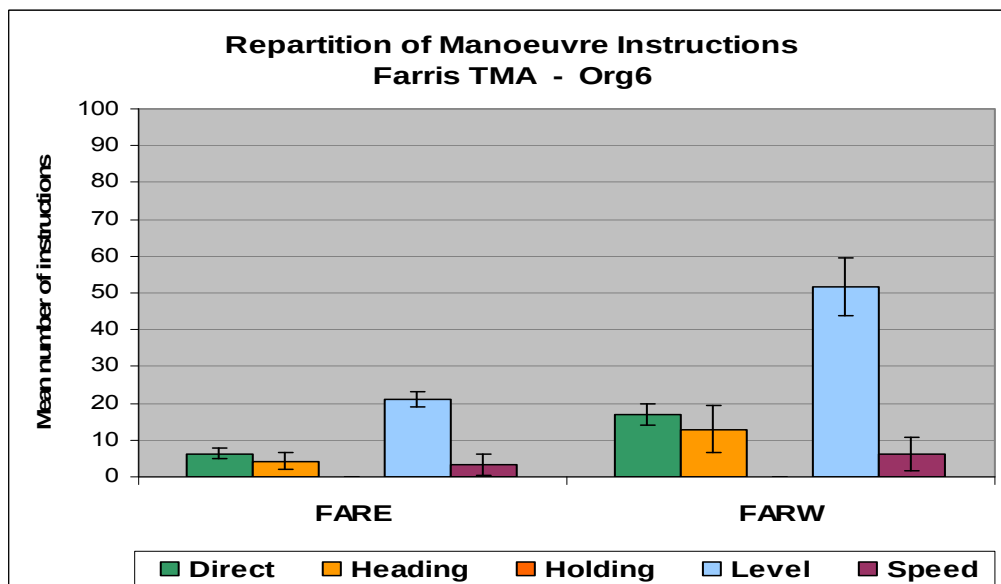
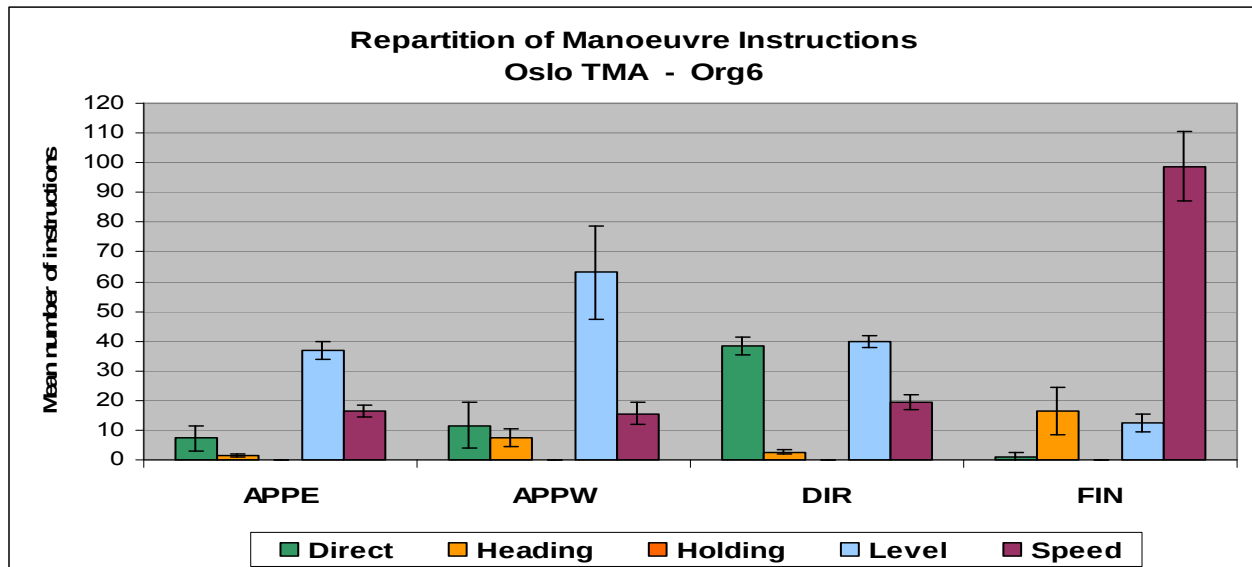


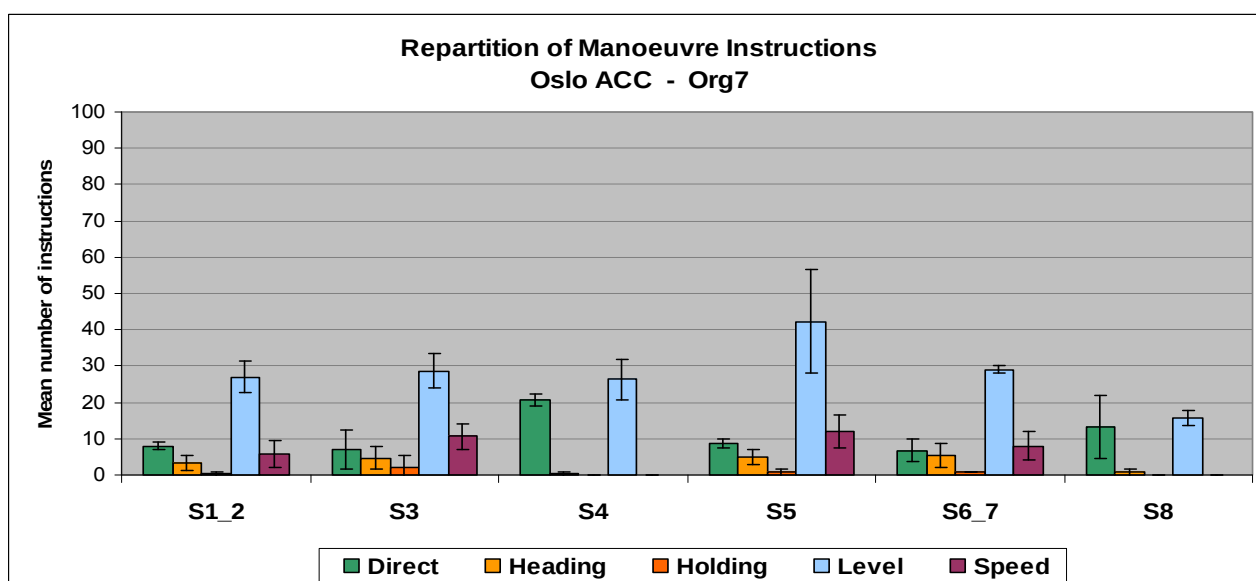
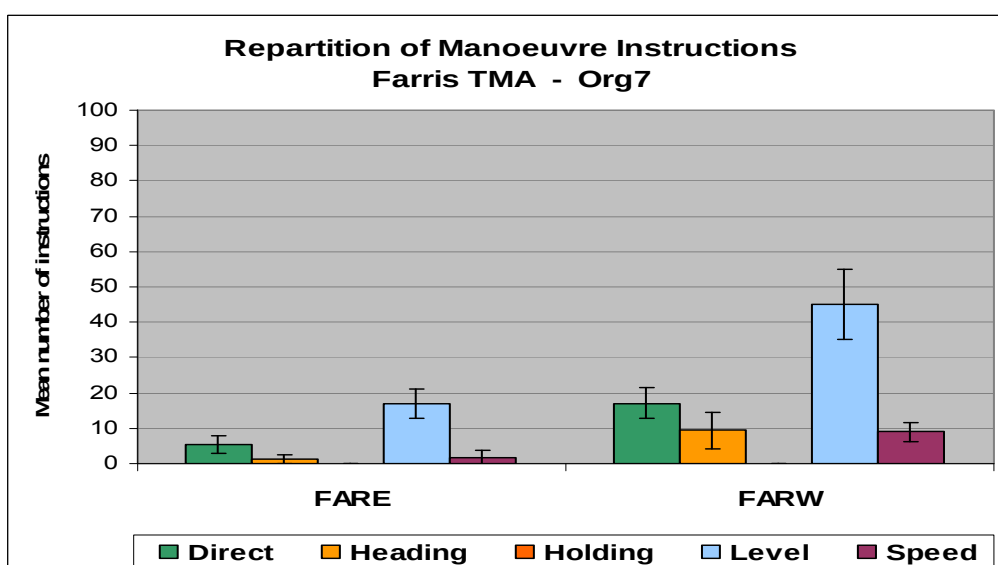
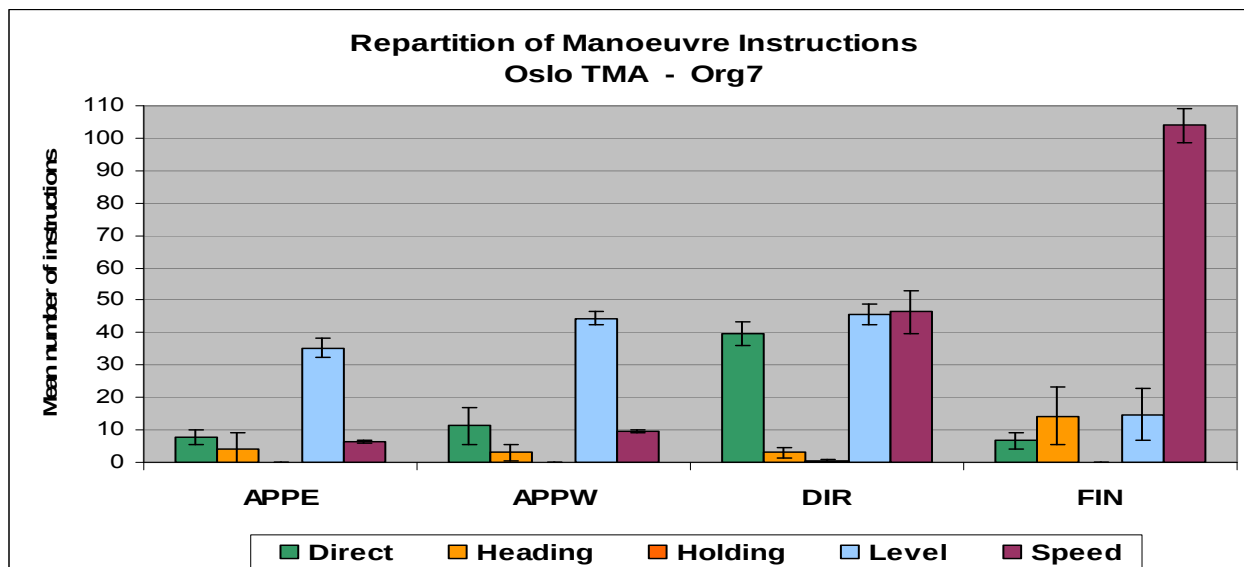






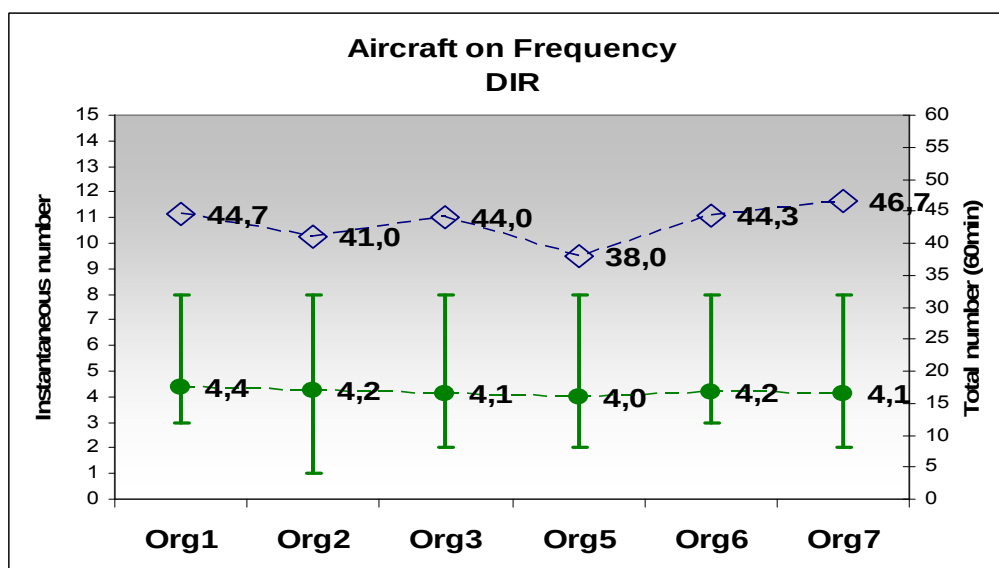
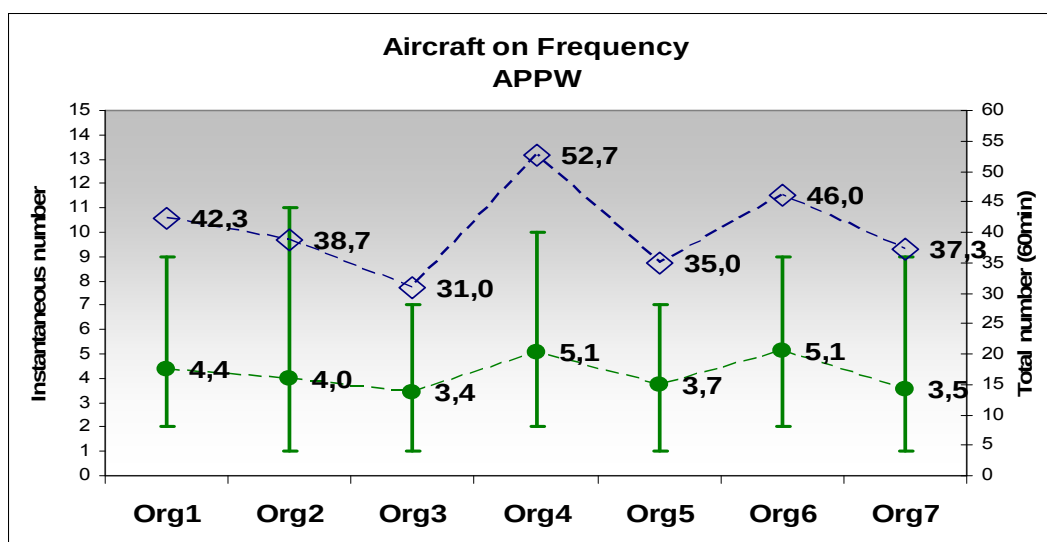
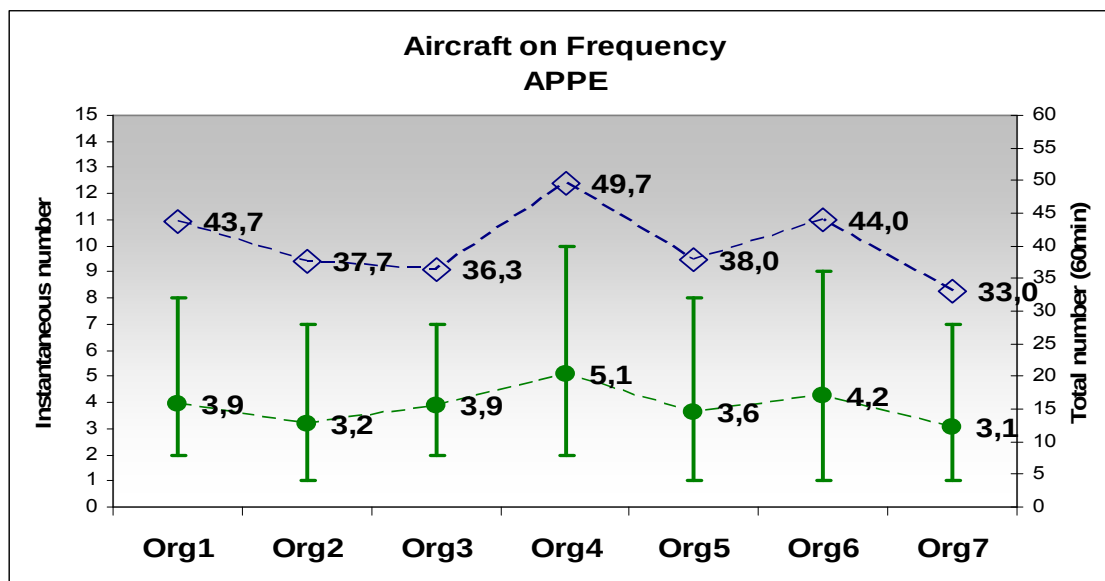


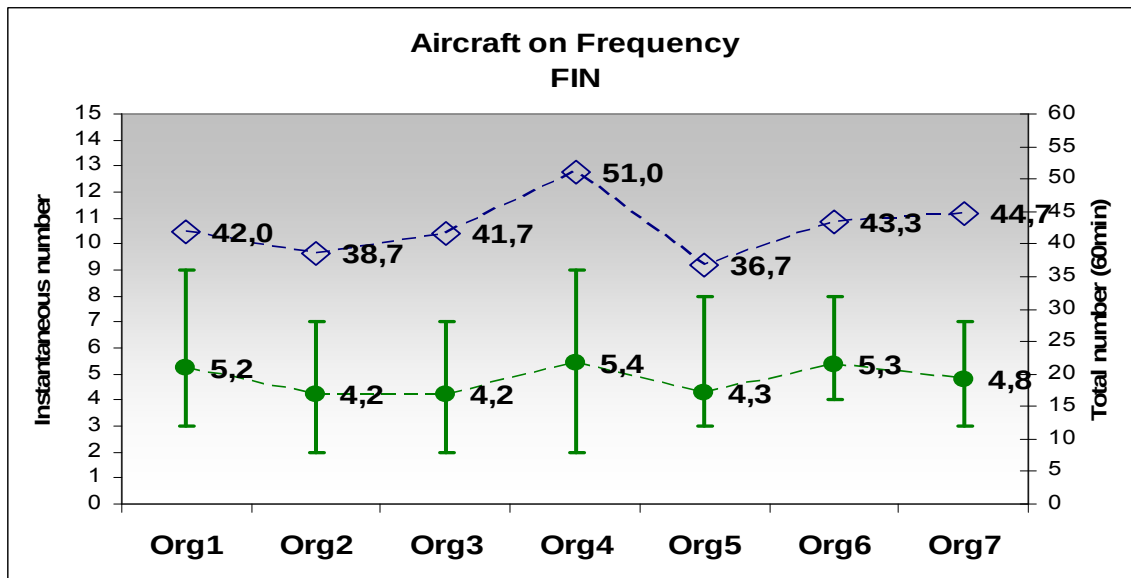
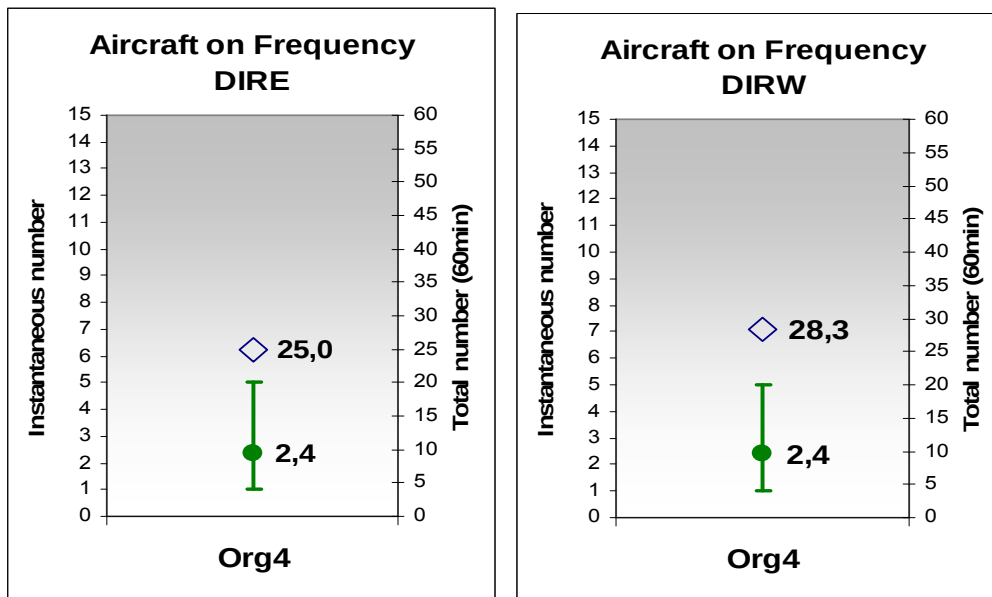


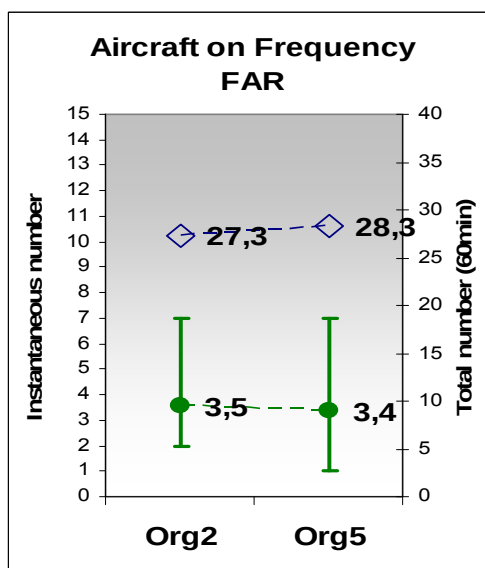
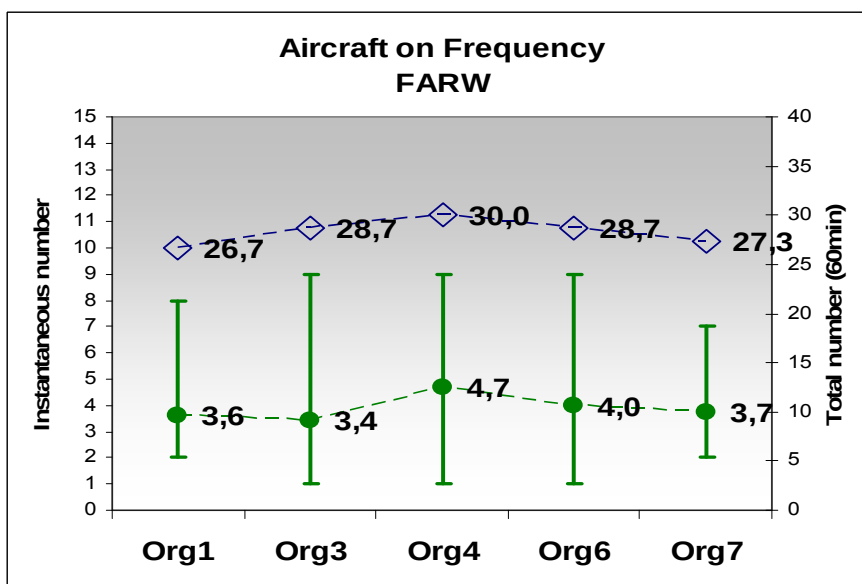
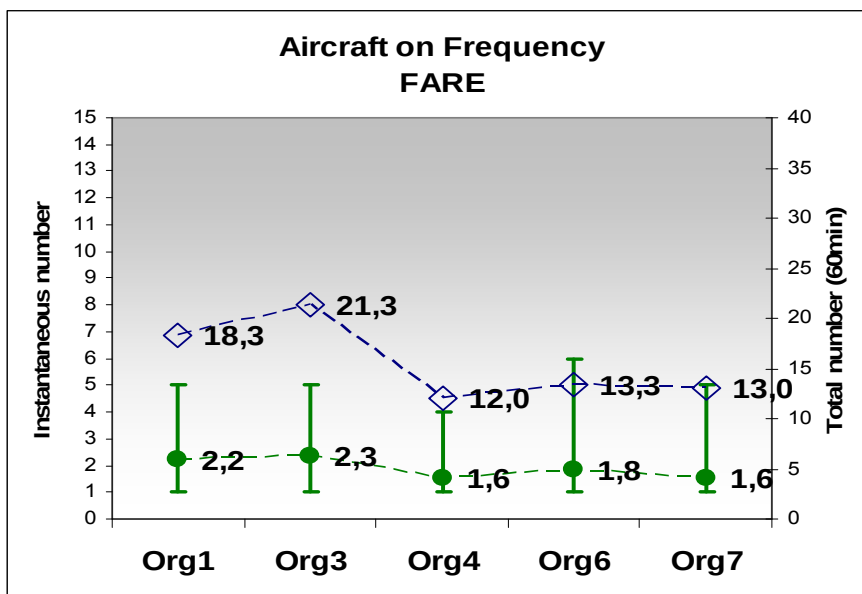


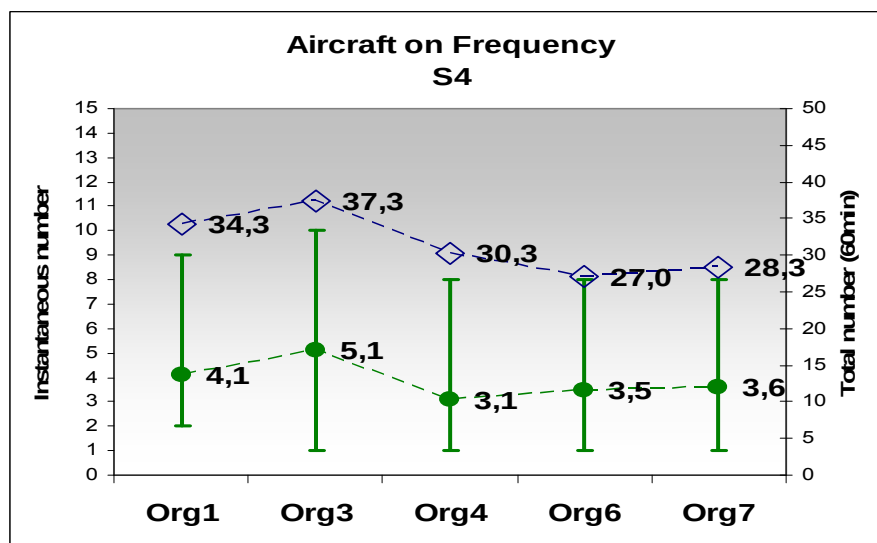
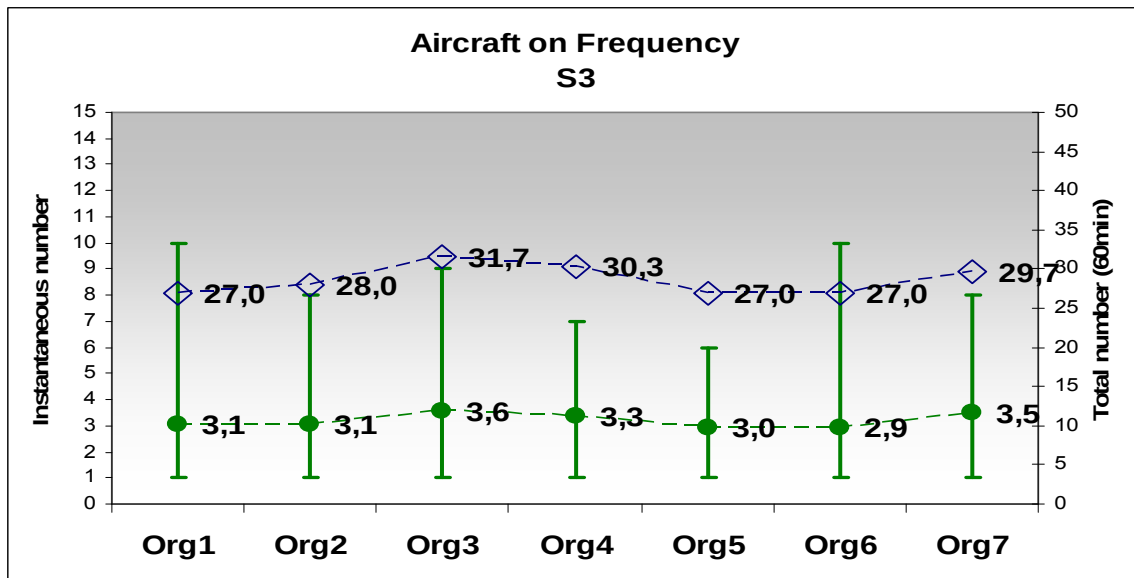
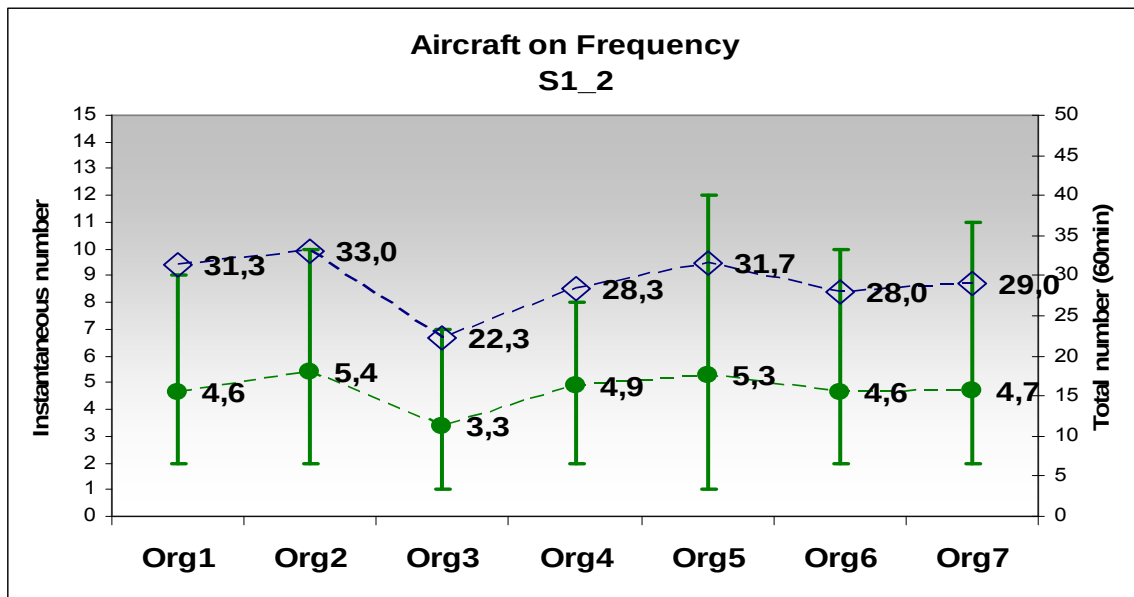
Page intentionally left blank

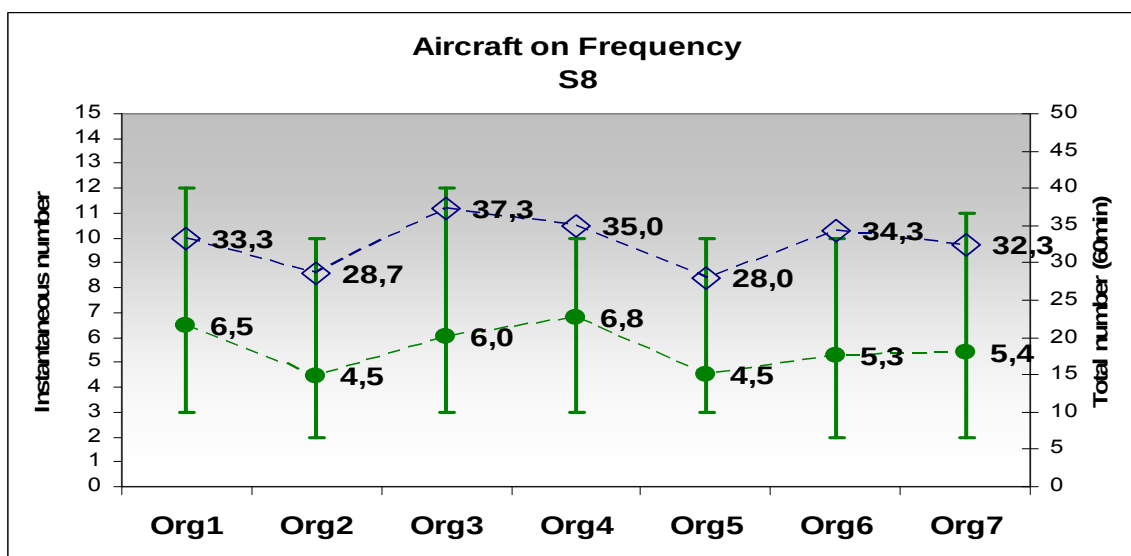
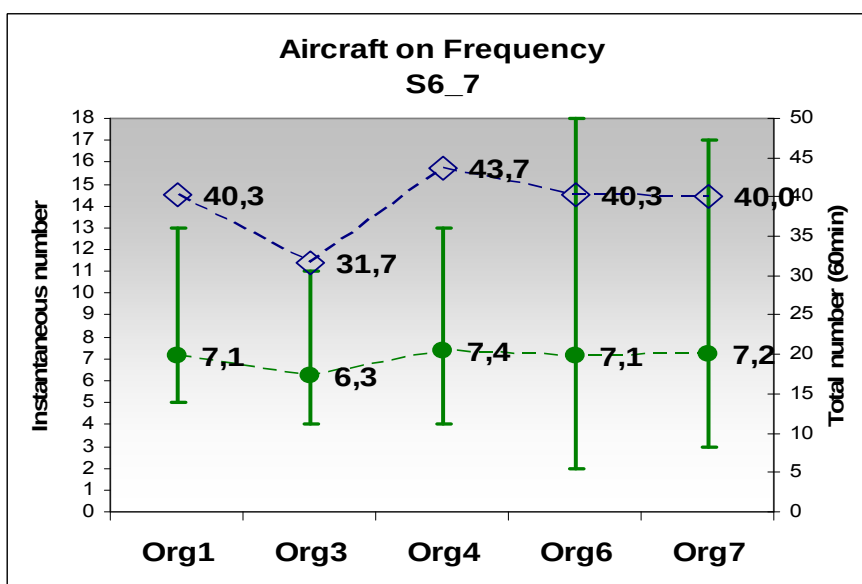
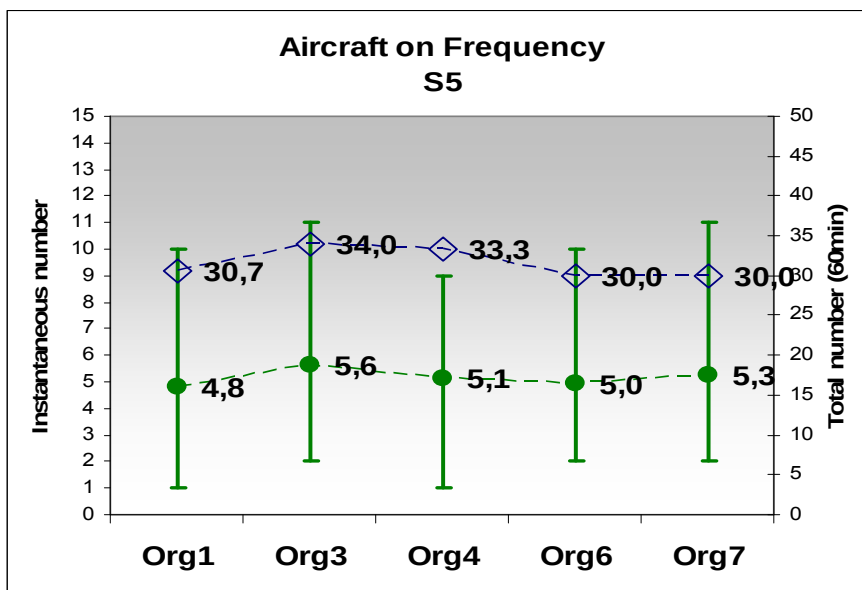
ANNEX G: AIRCRAFT ON FREQUENCY

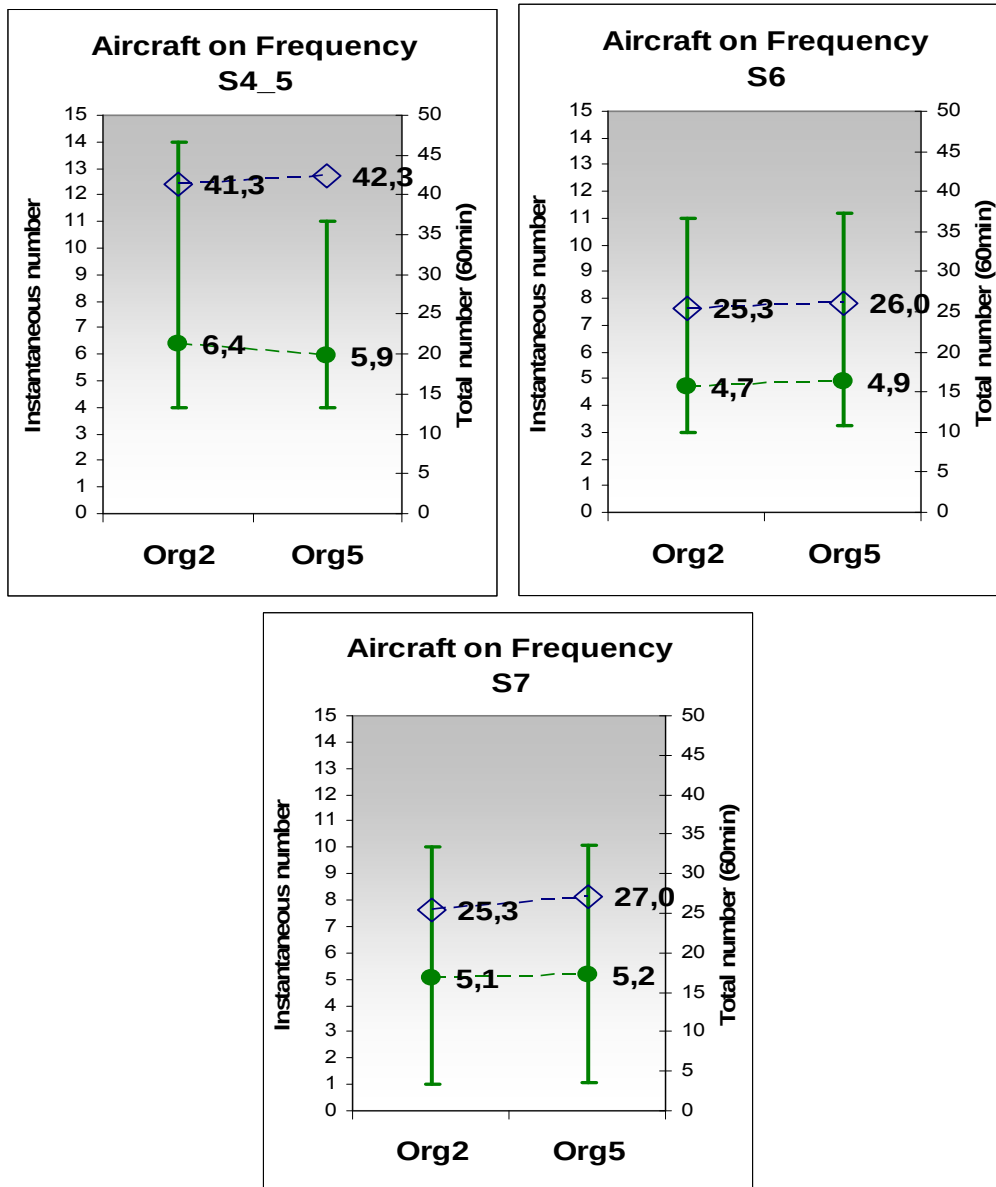




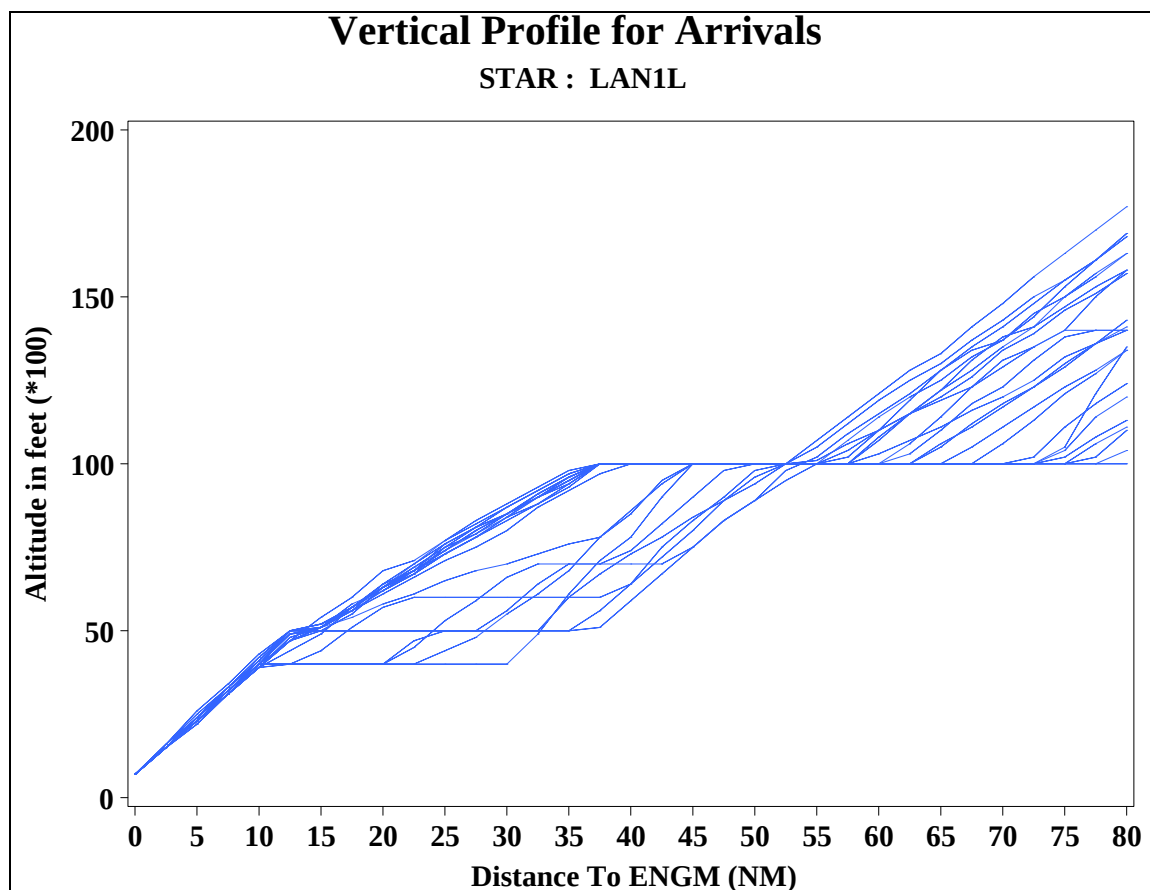
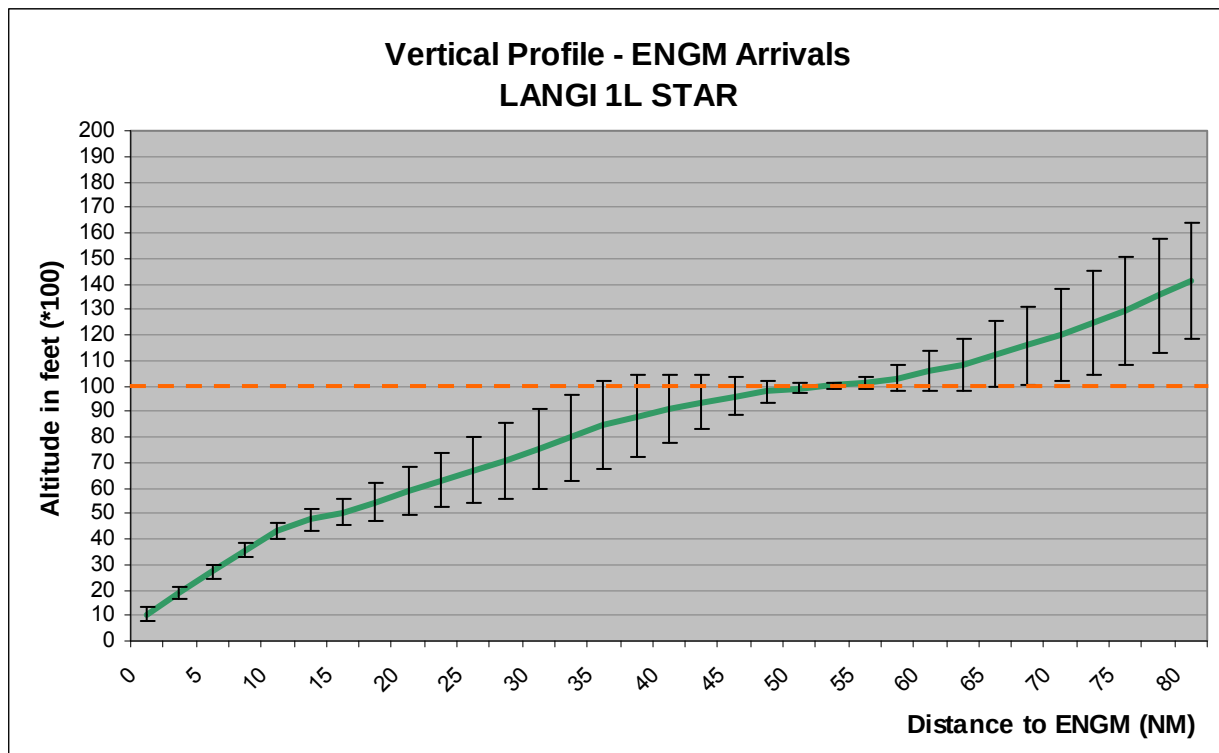


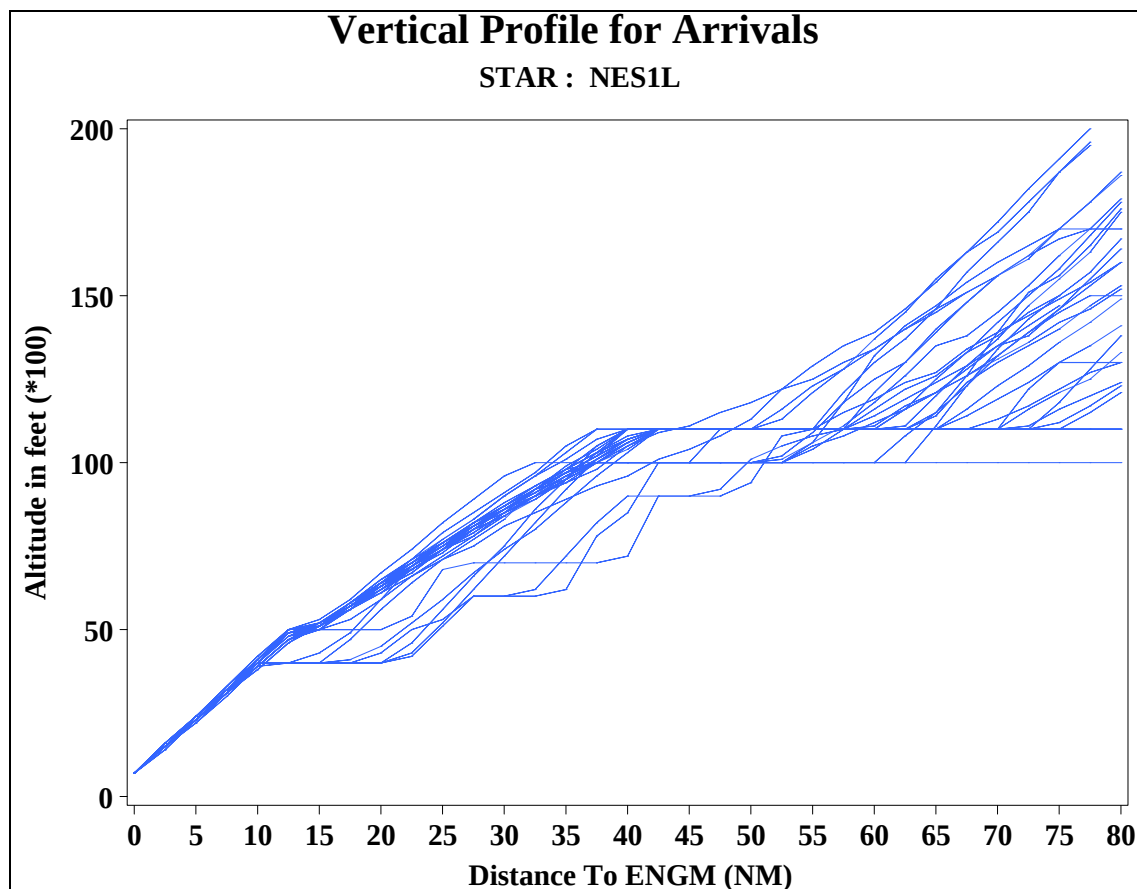
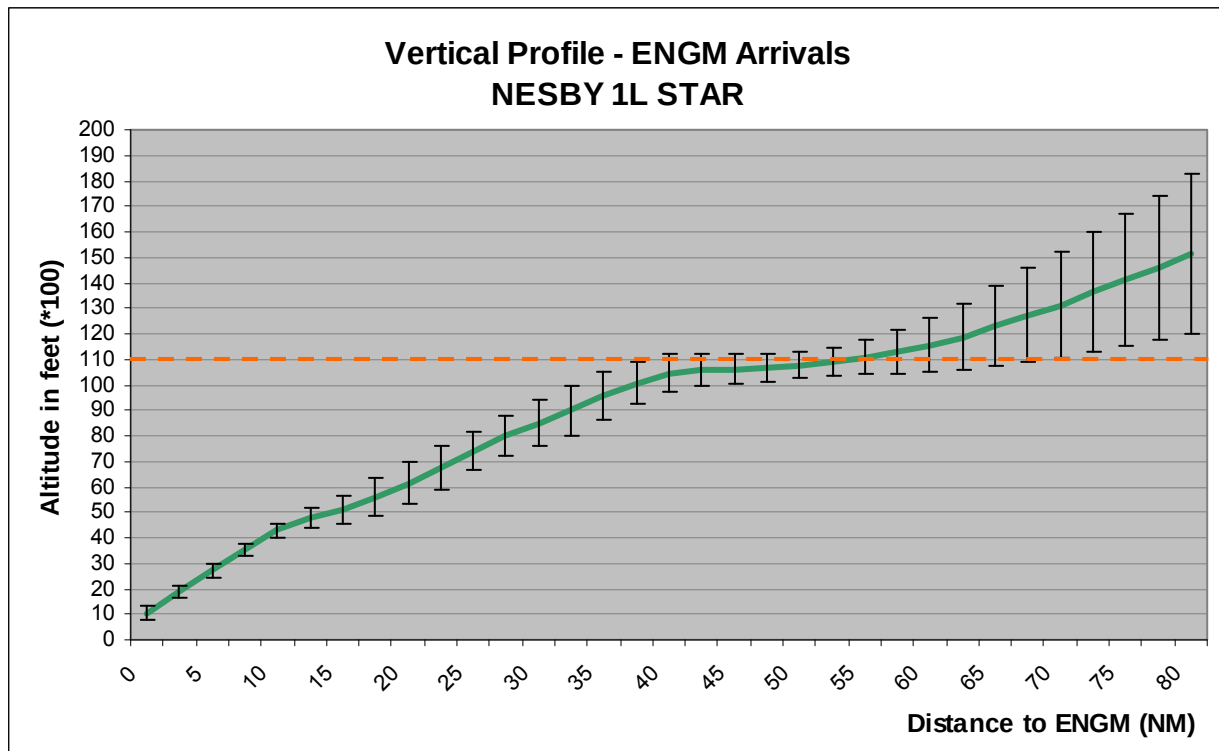


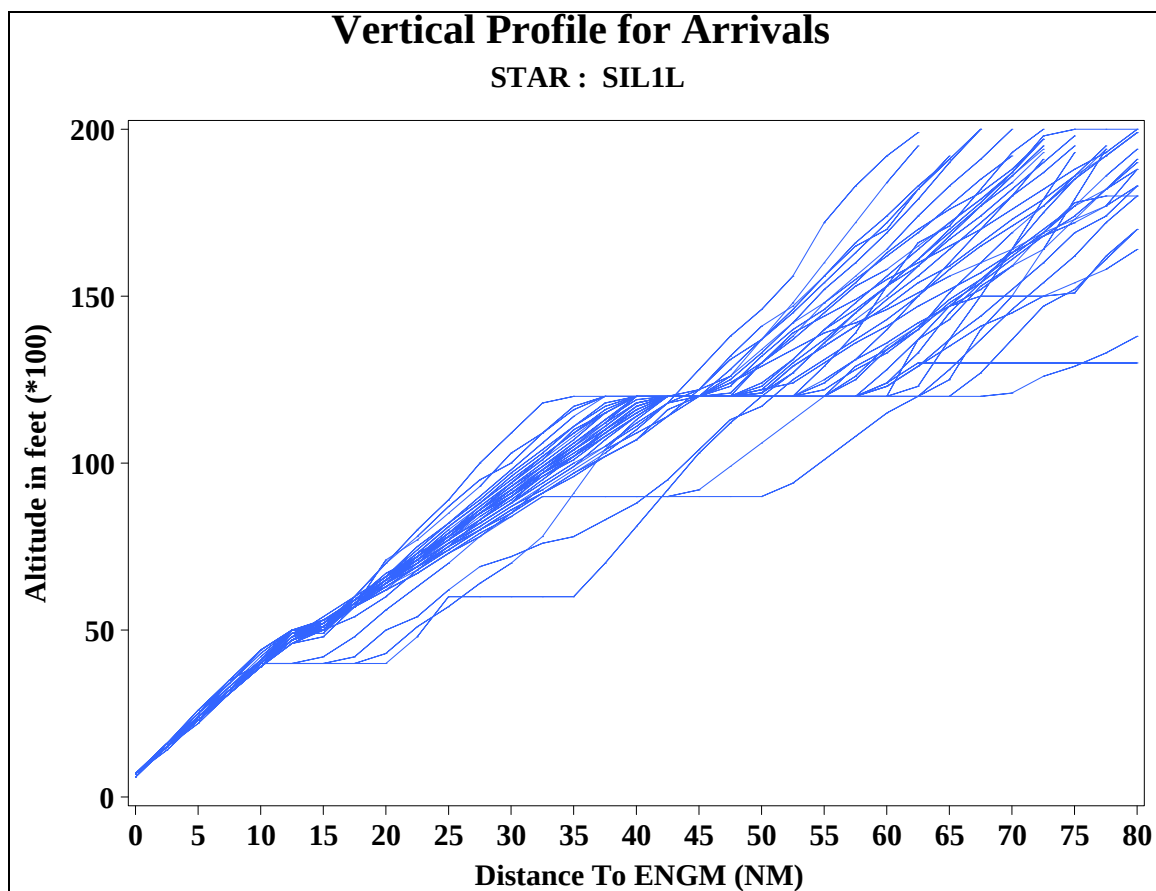
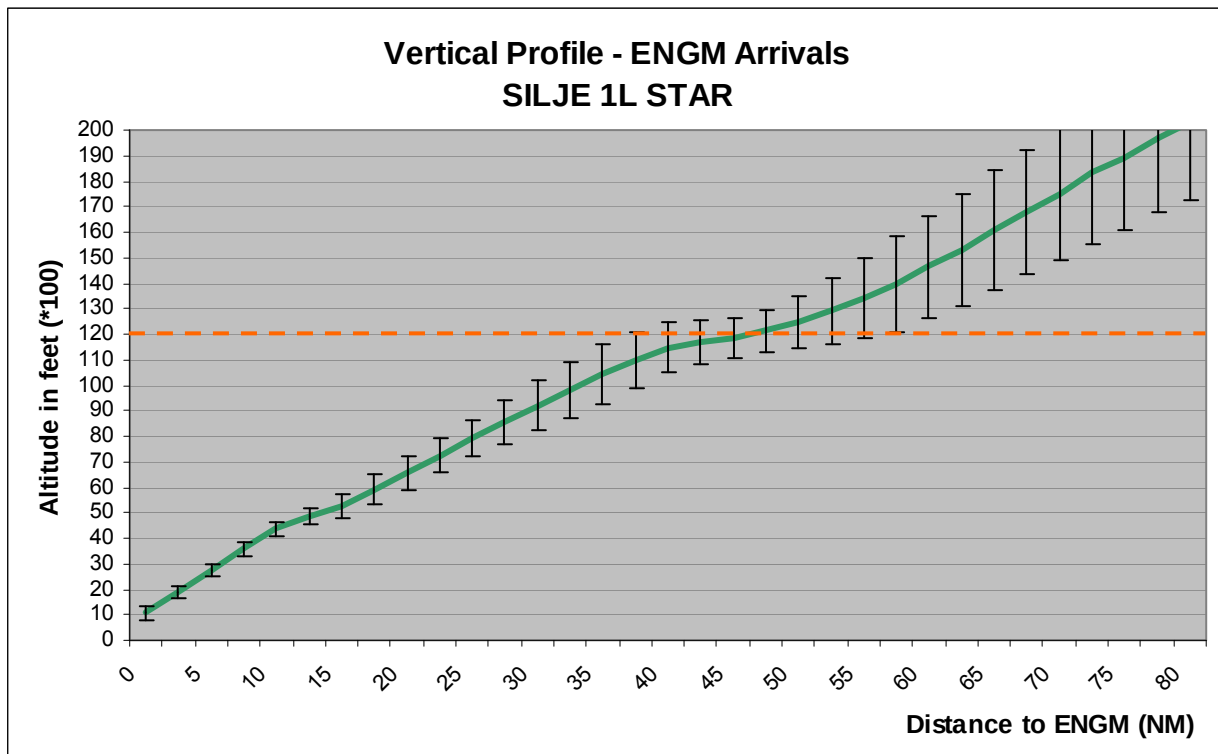


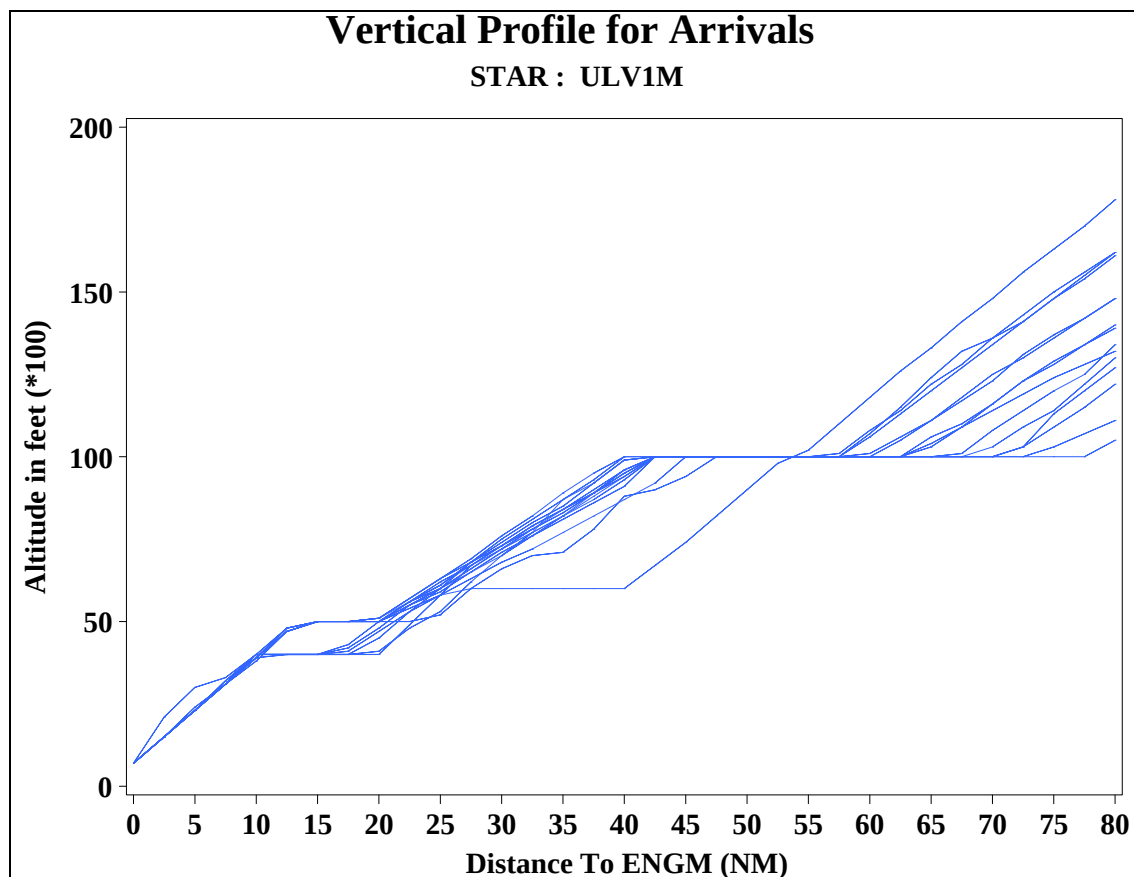
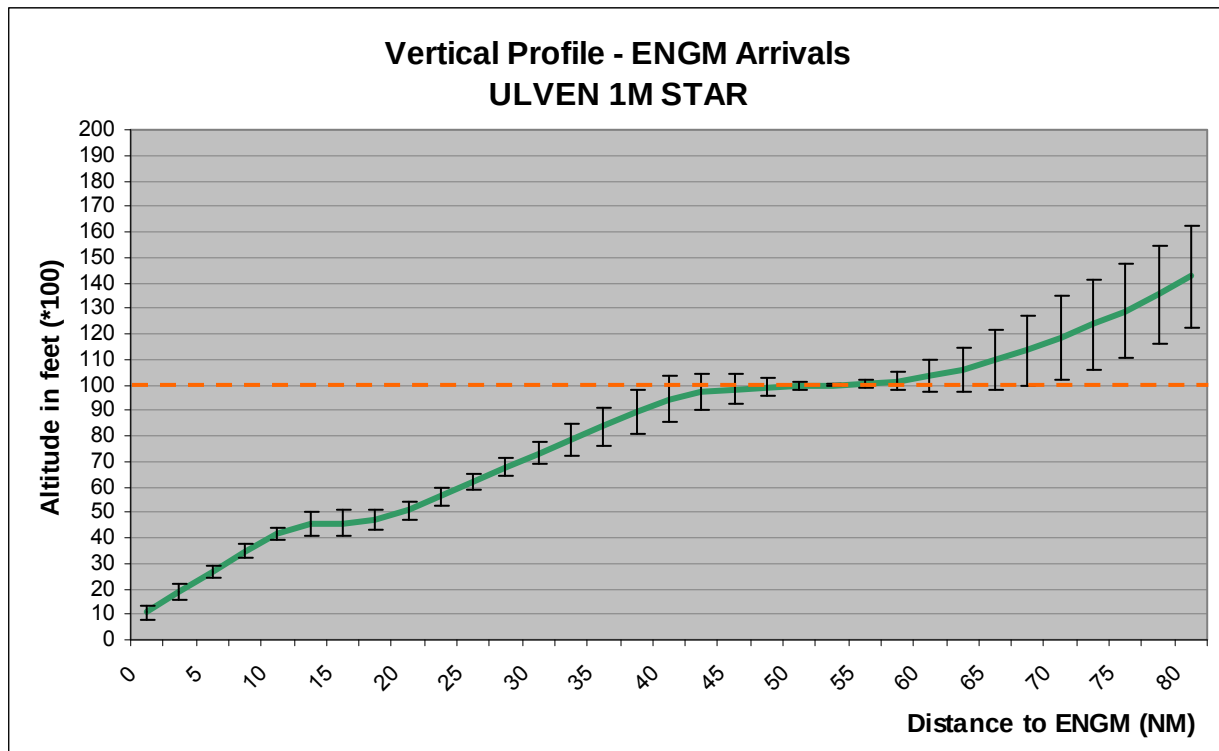


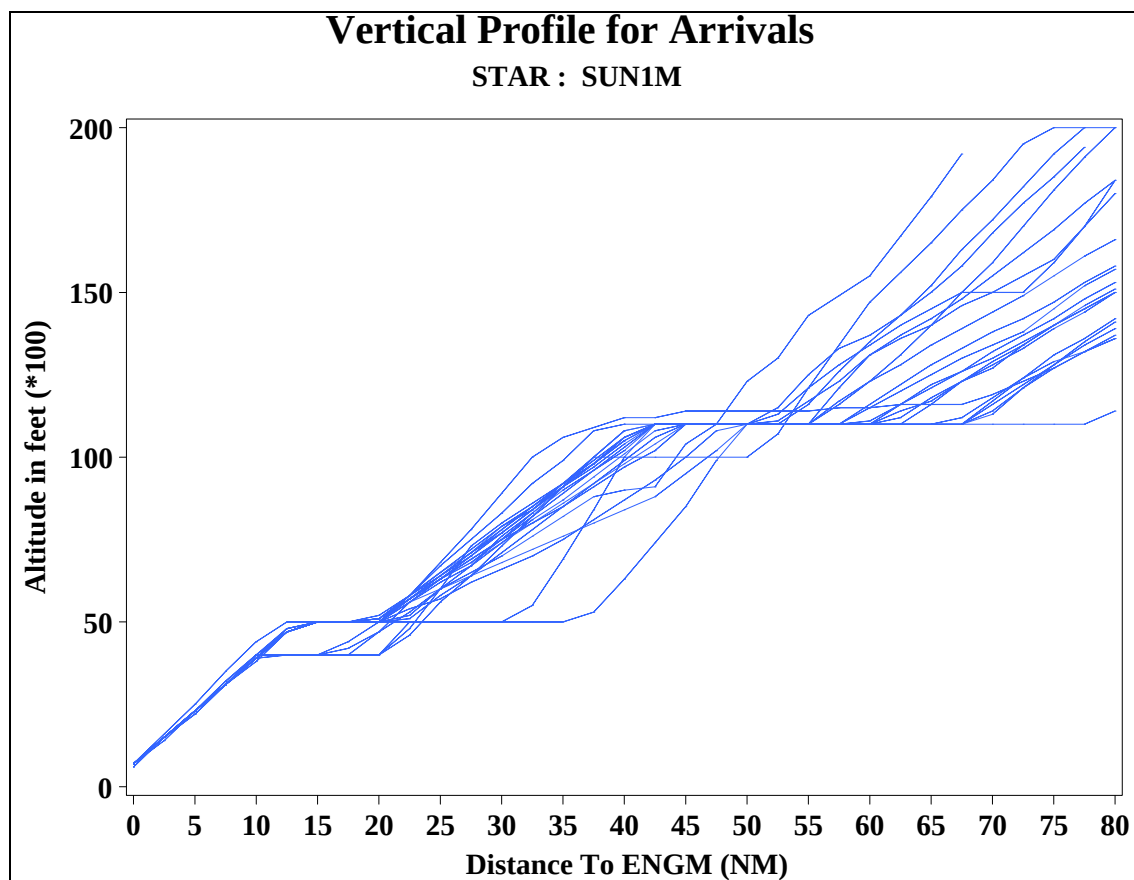
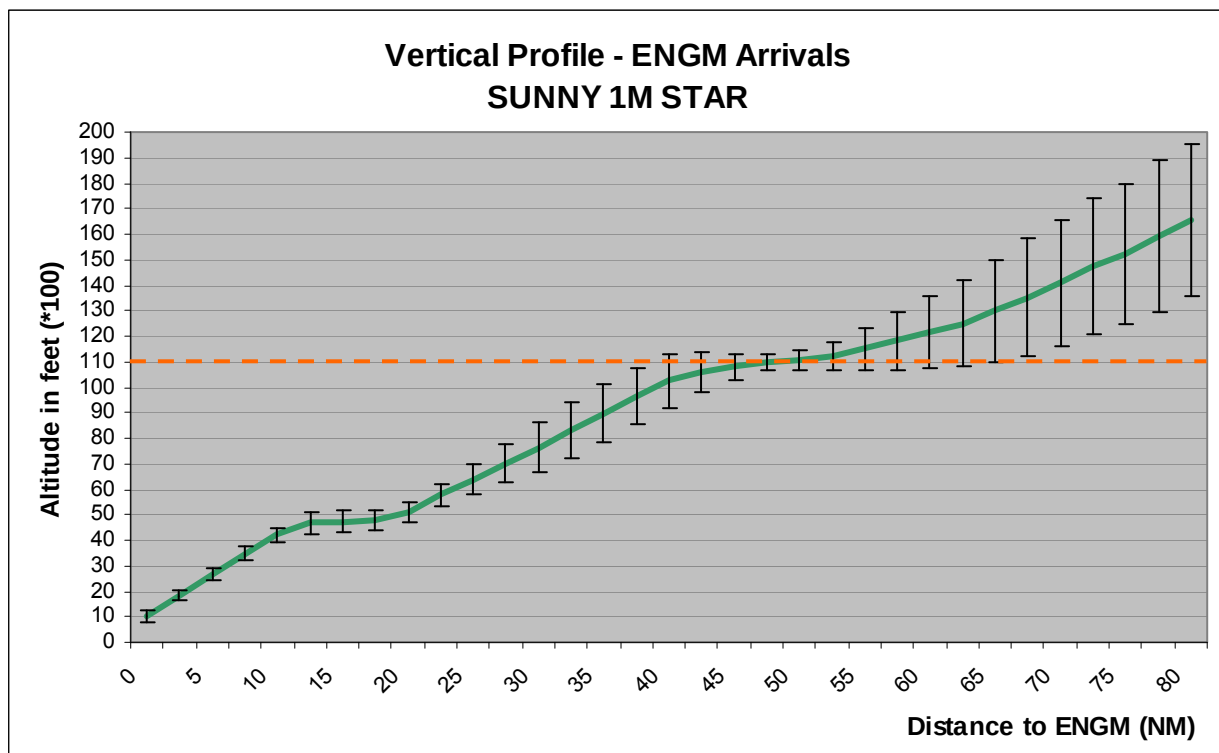
ANNEX H: ENGM STARS - ARRIVALS VERTICAL PROFILES

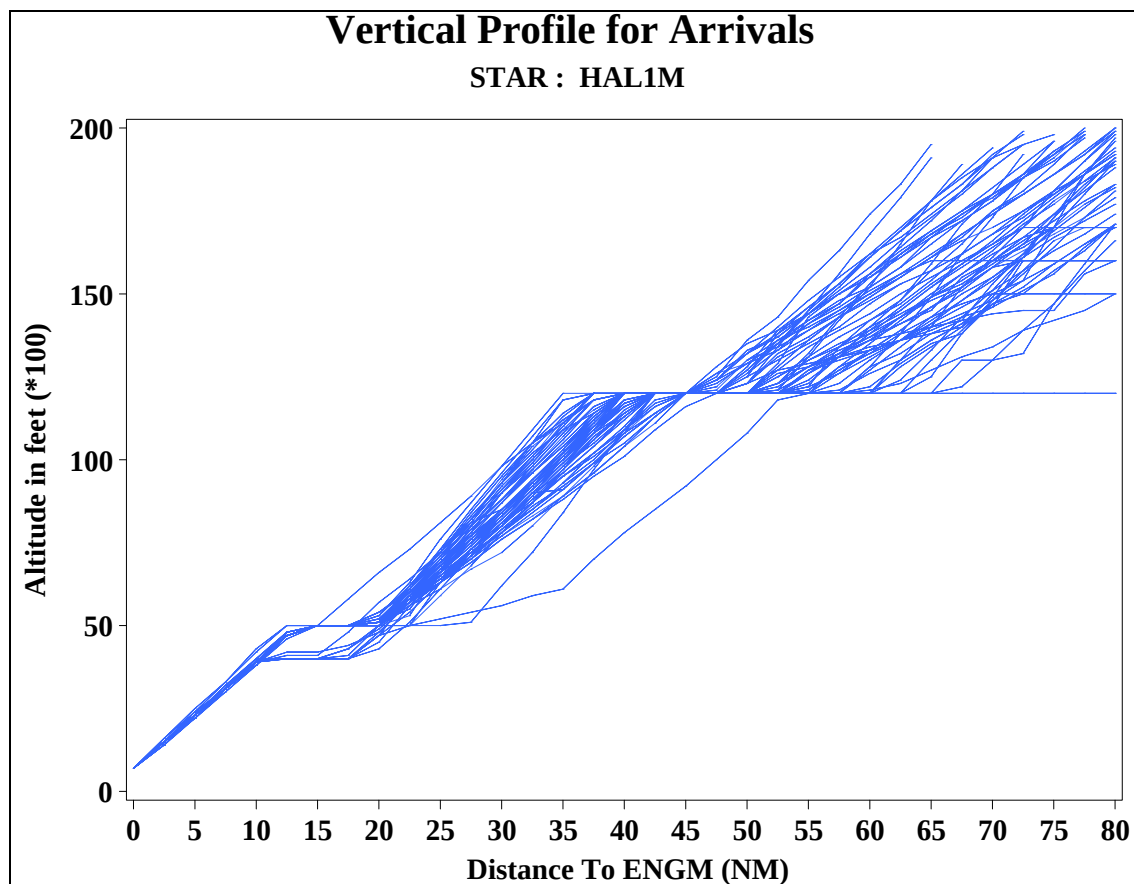
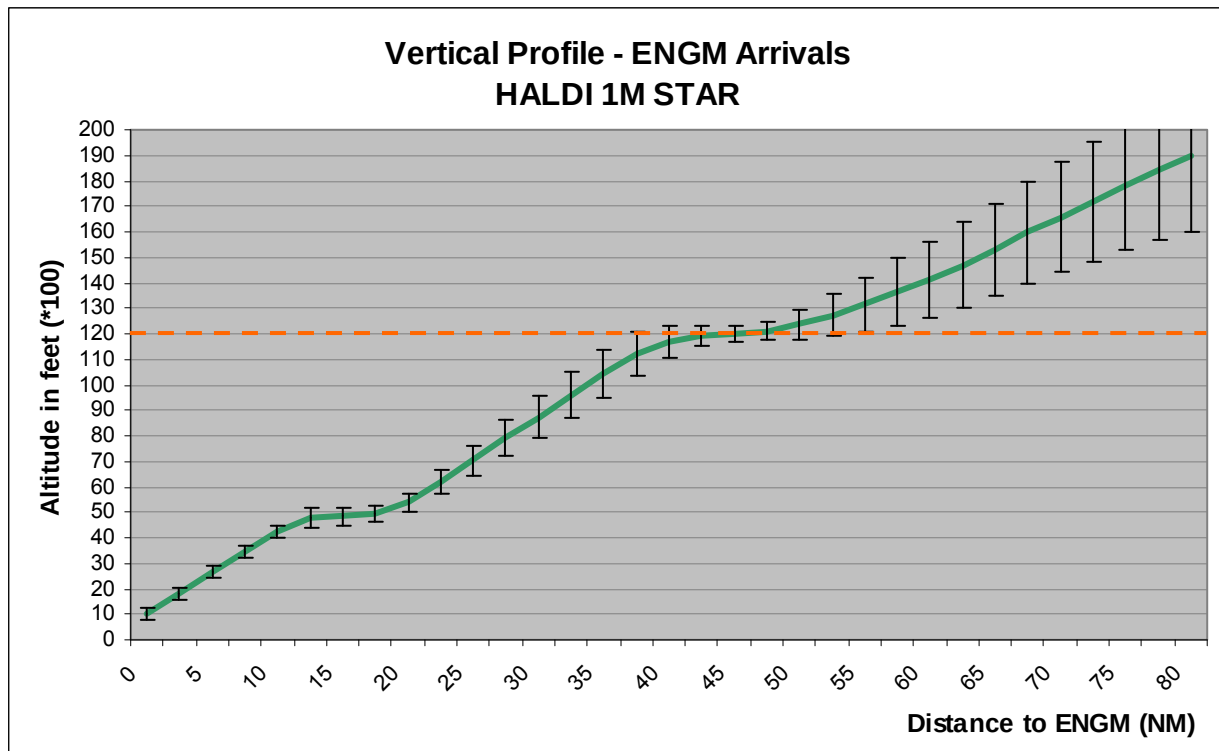


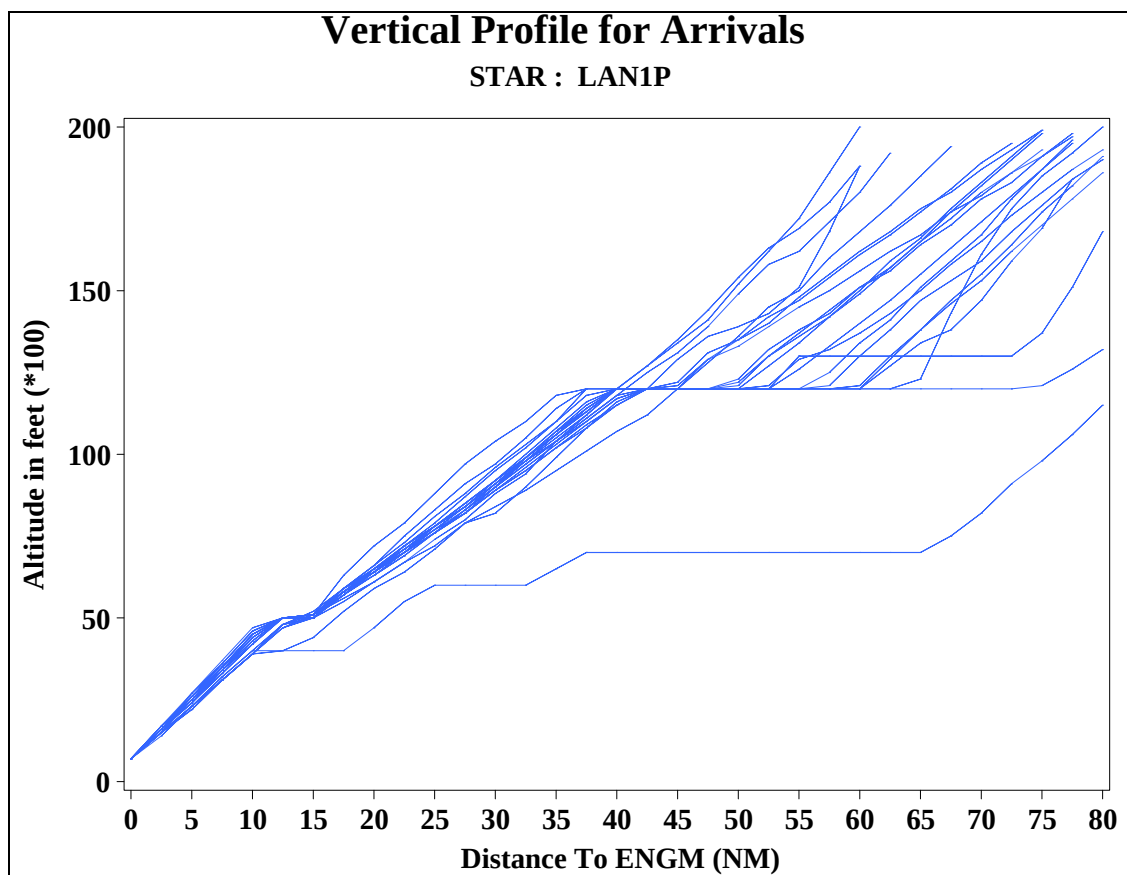
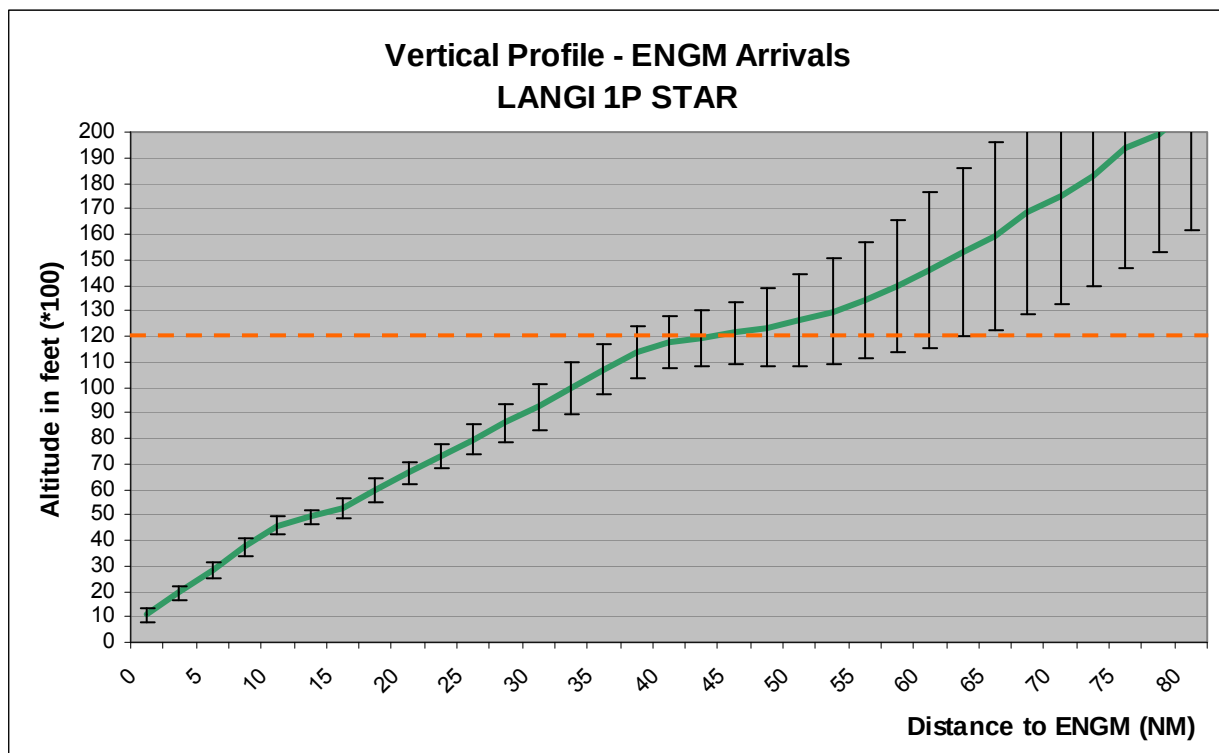


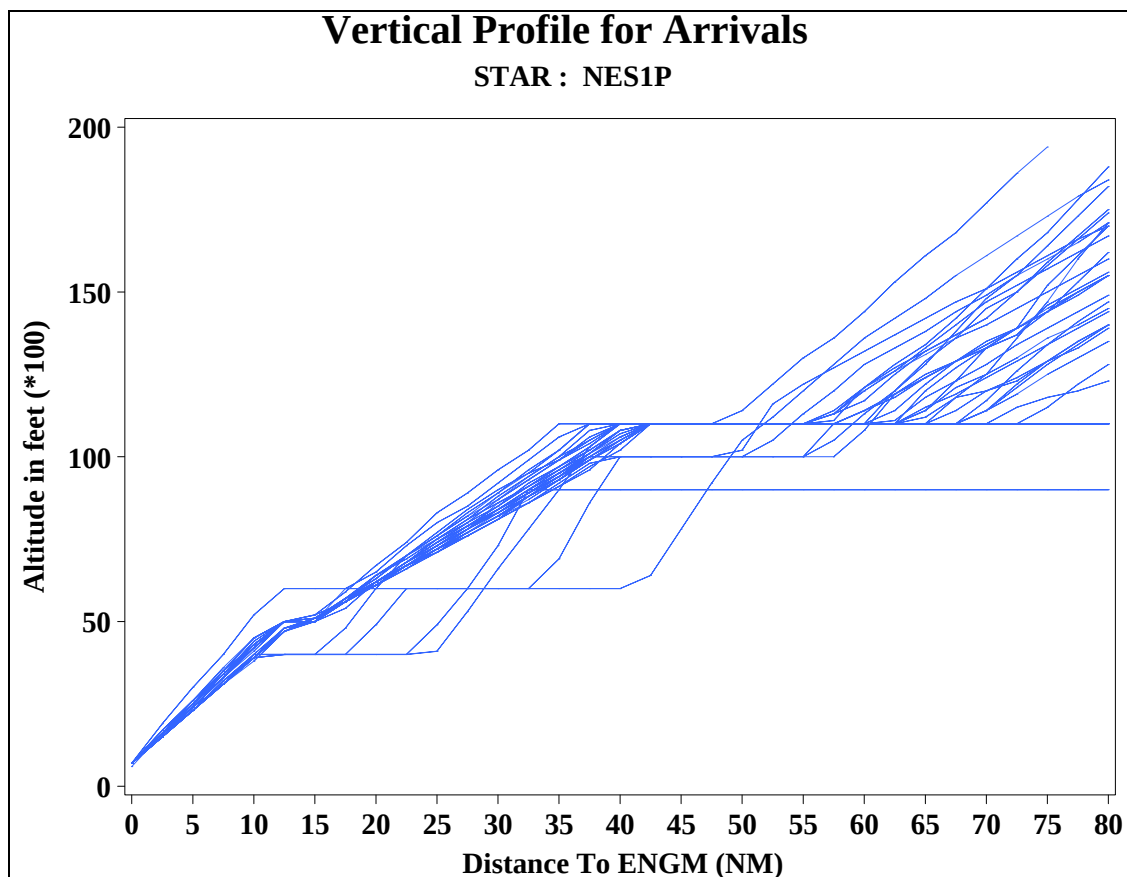
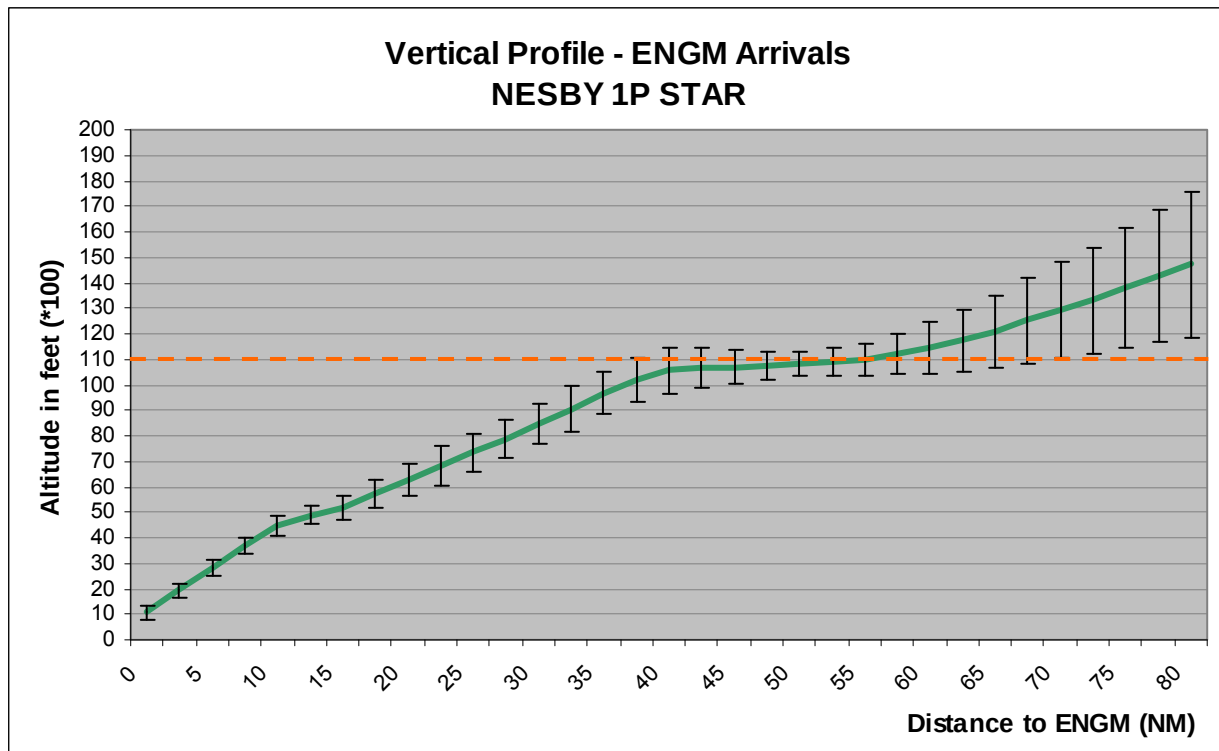


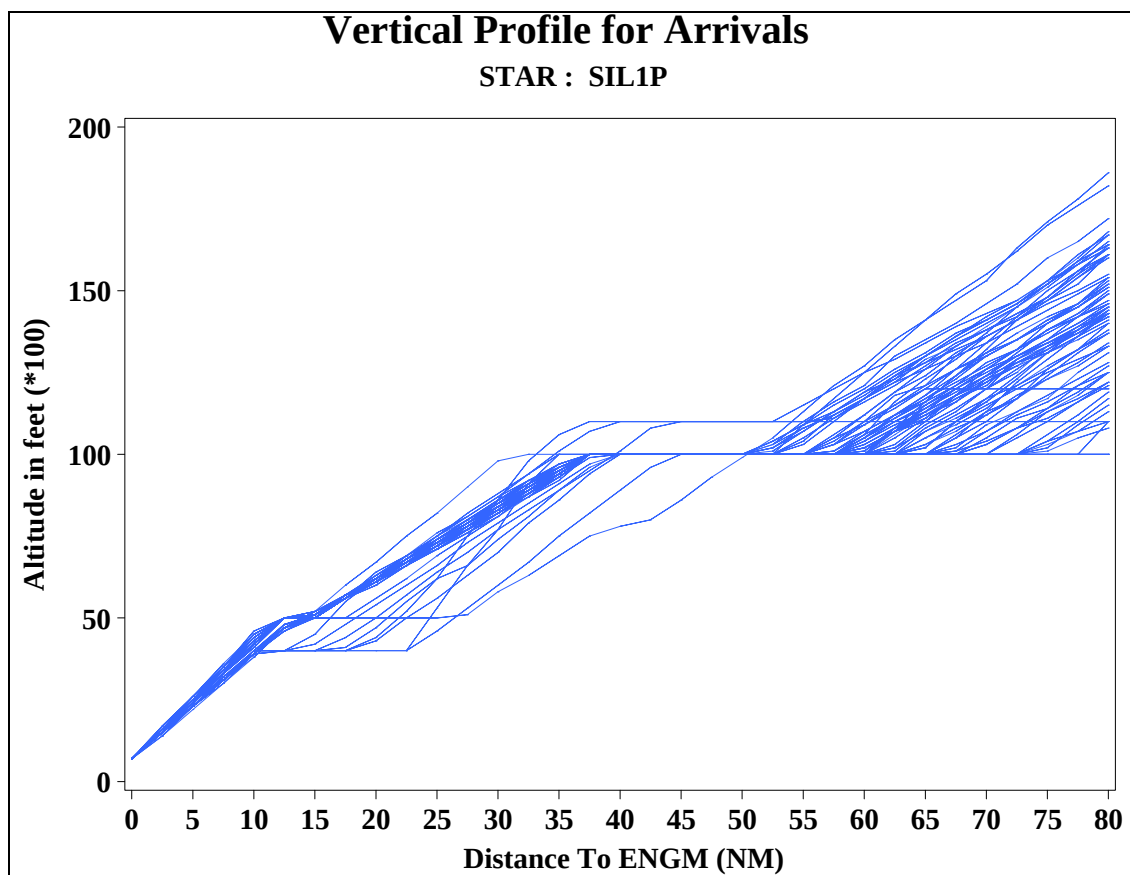
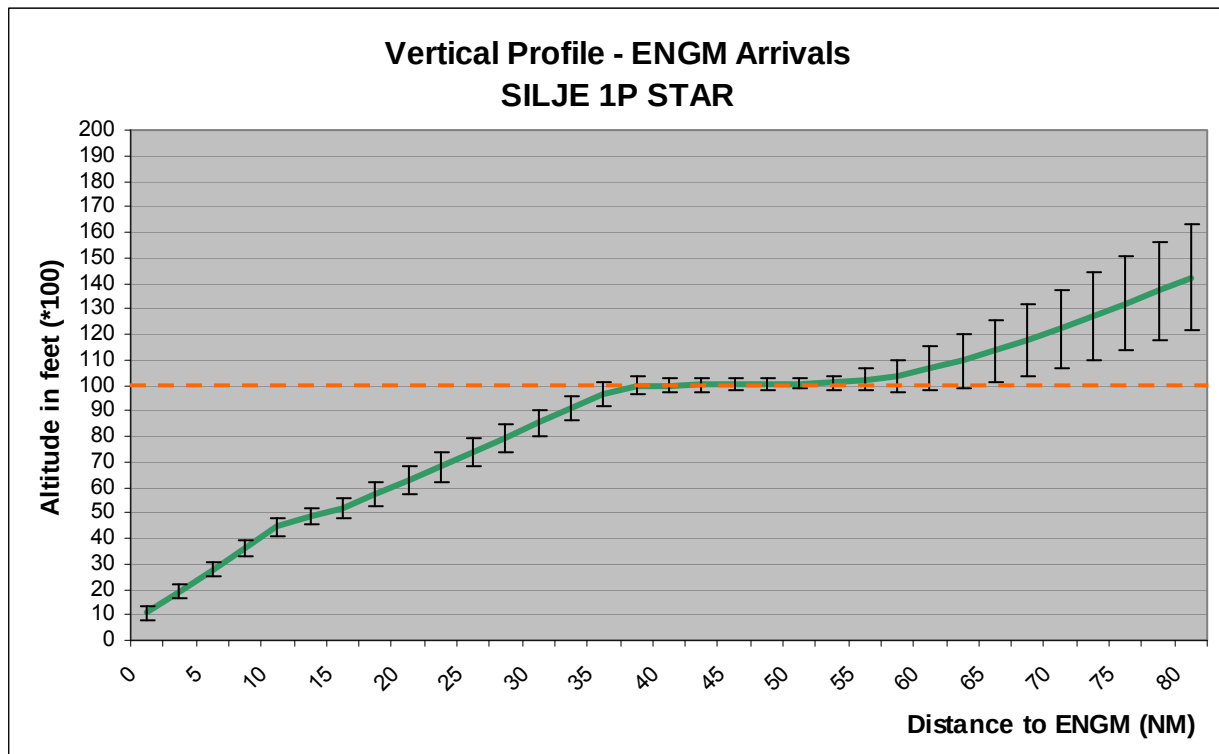


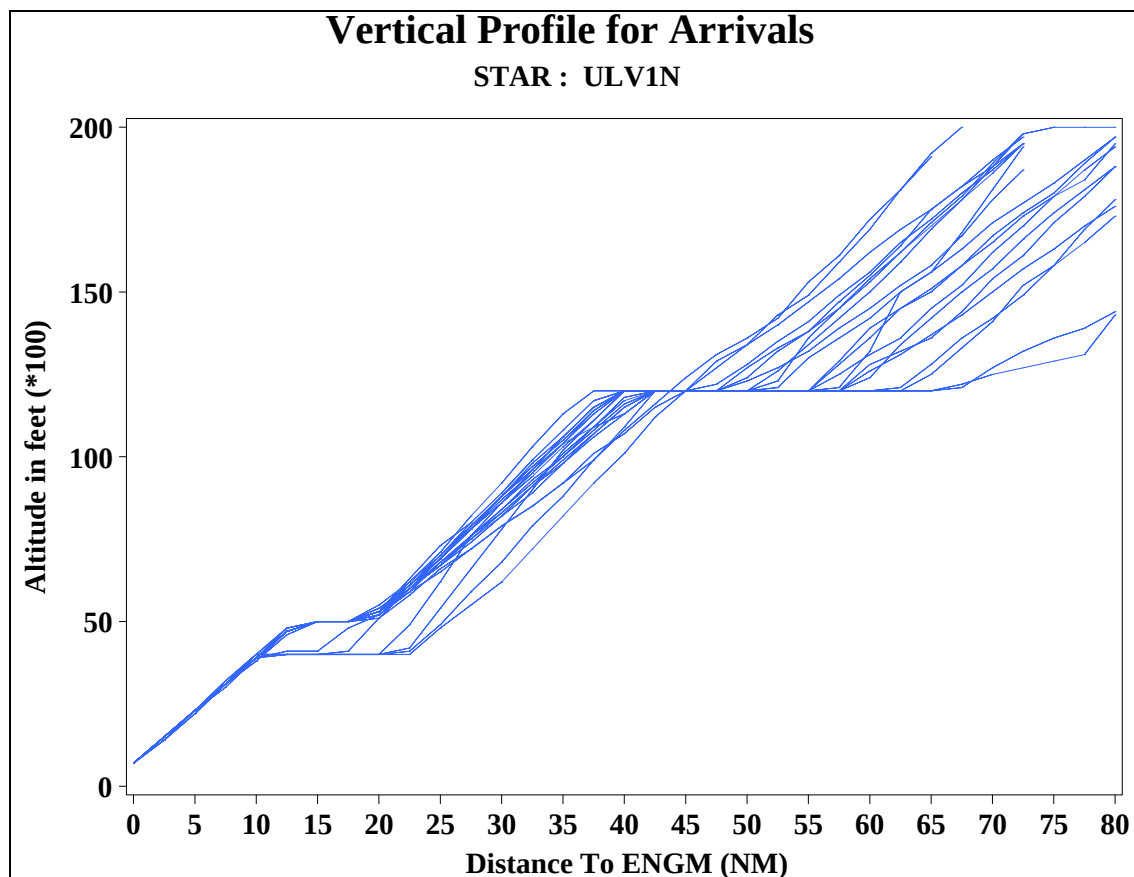
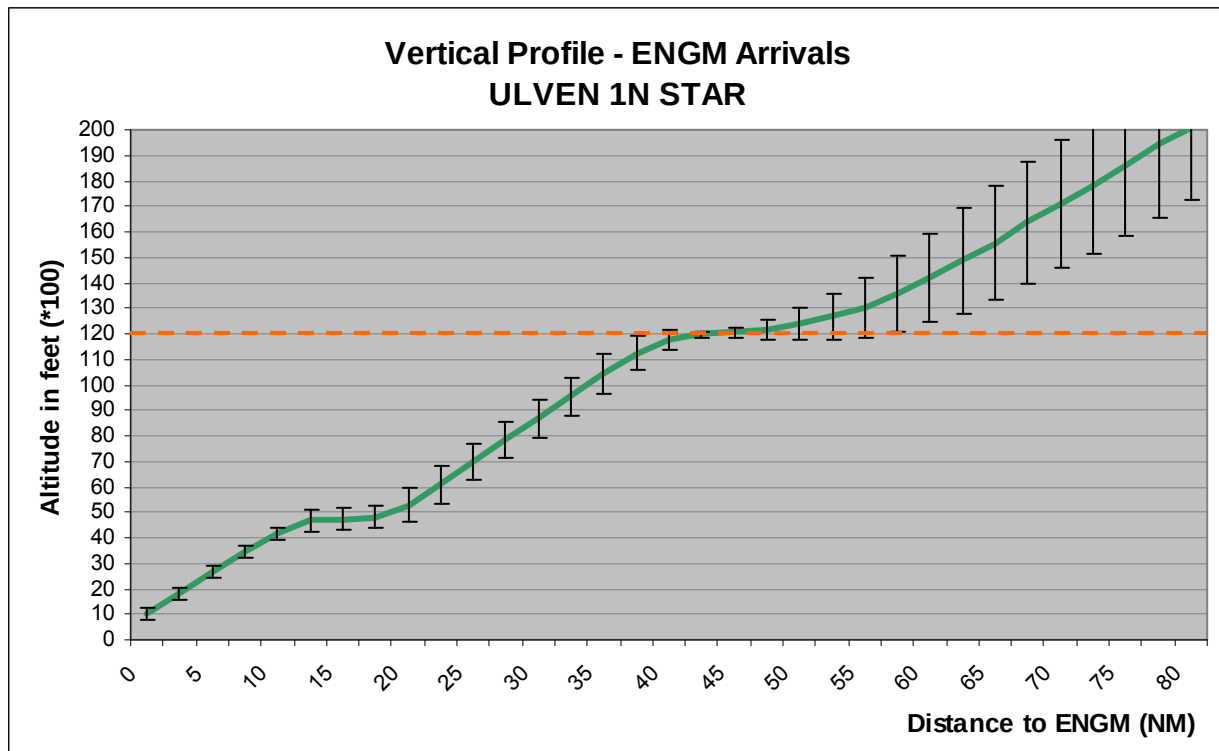


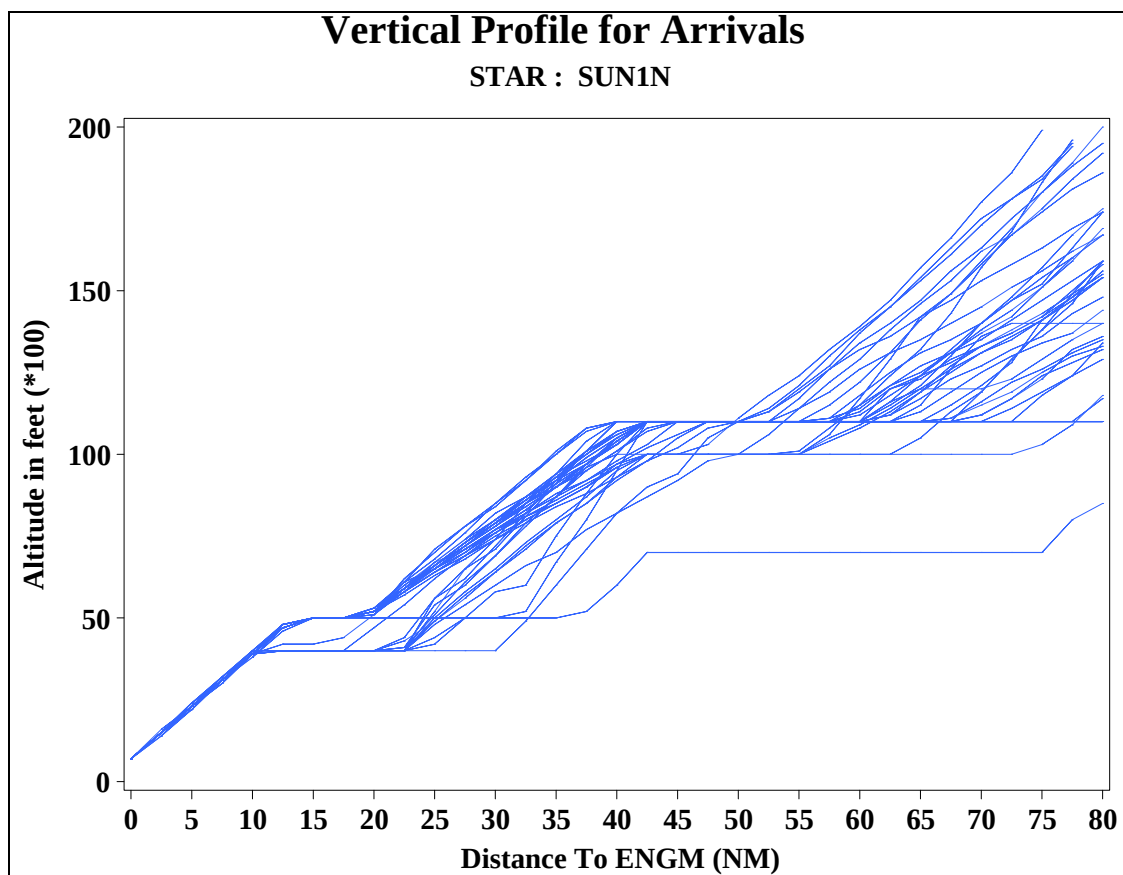
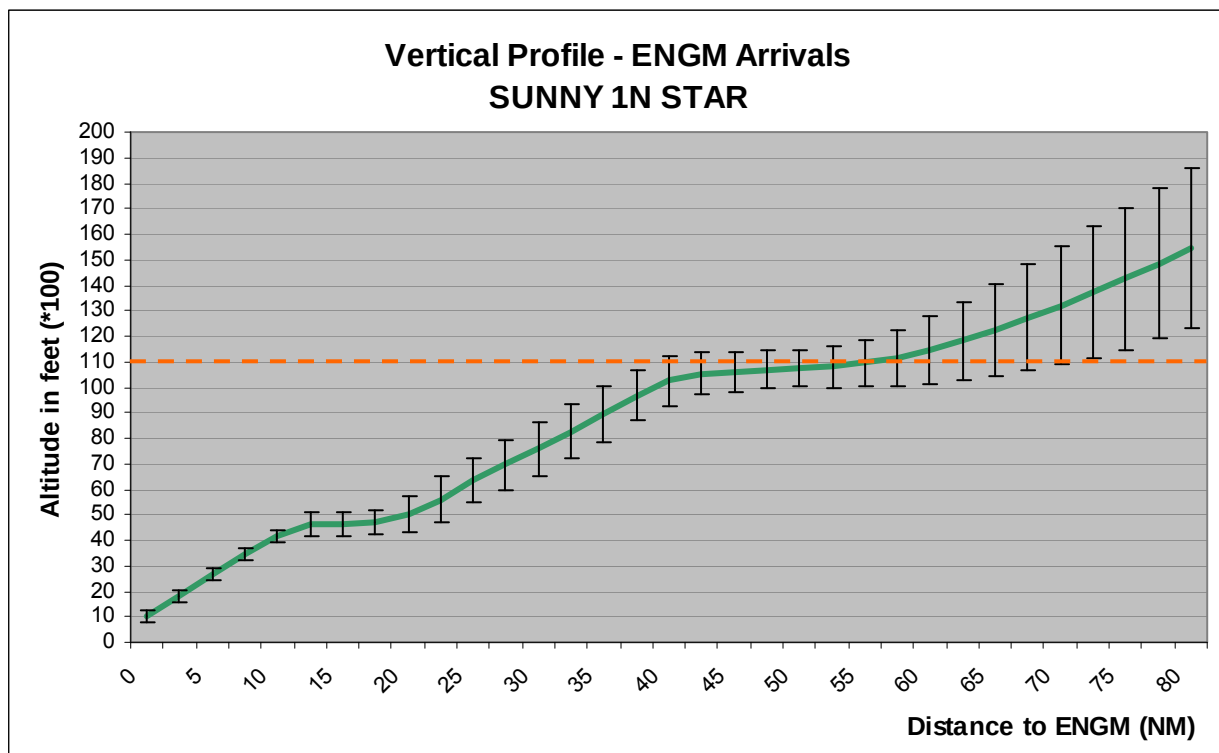


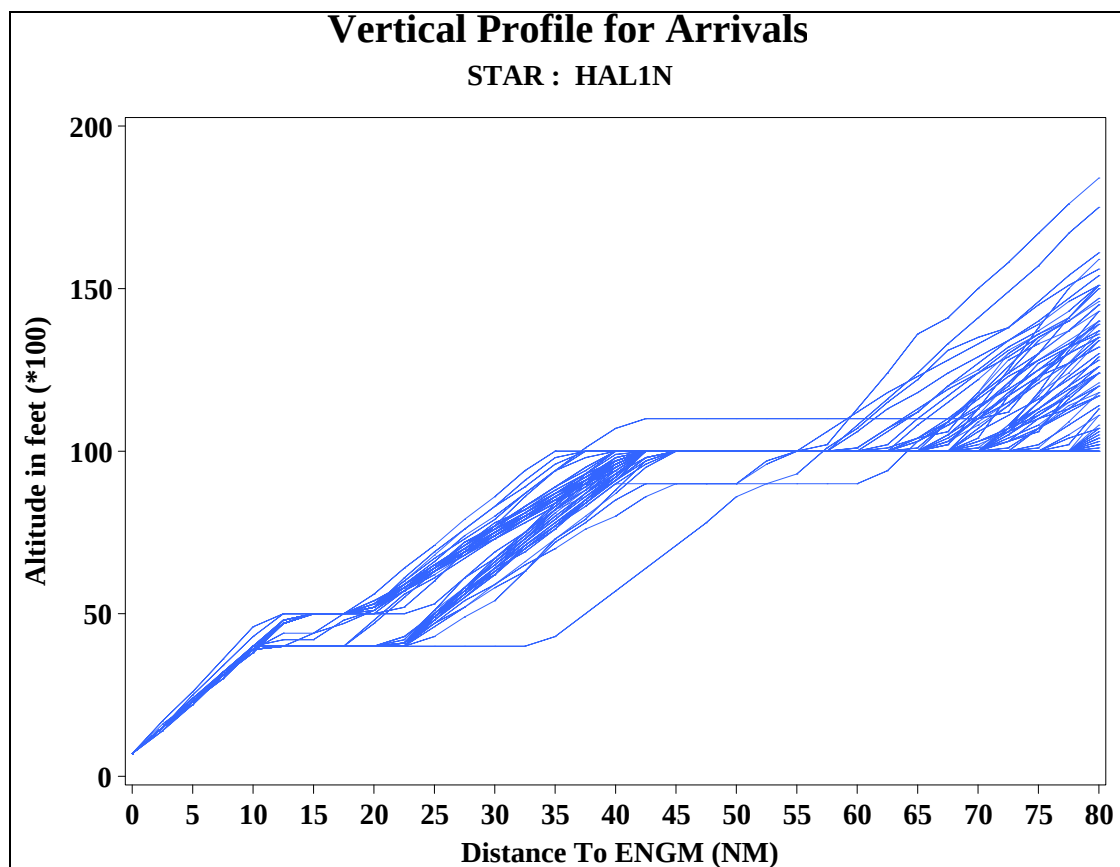
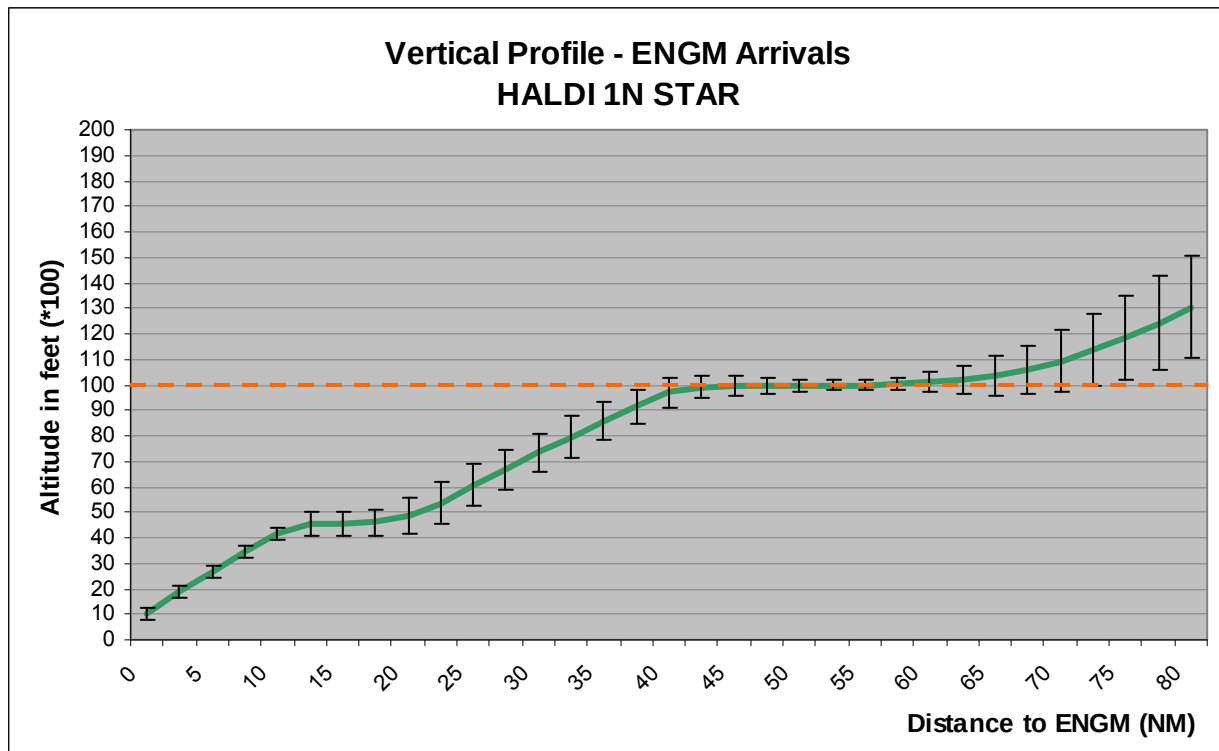


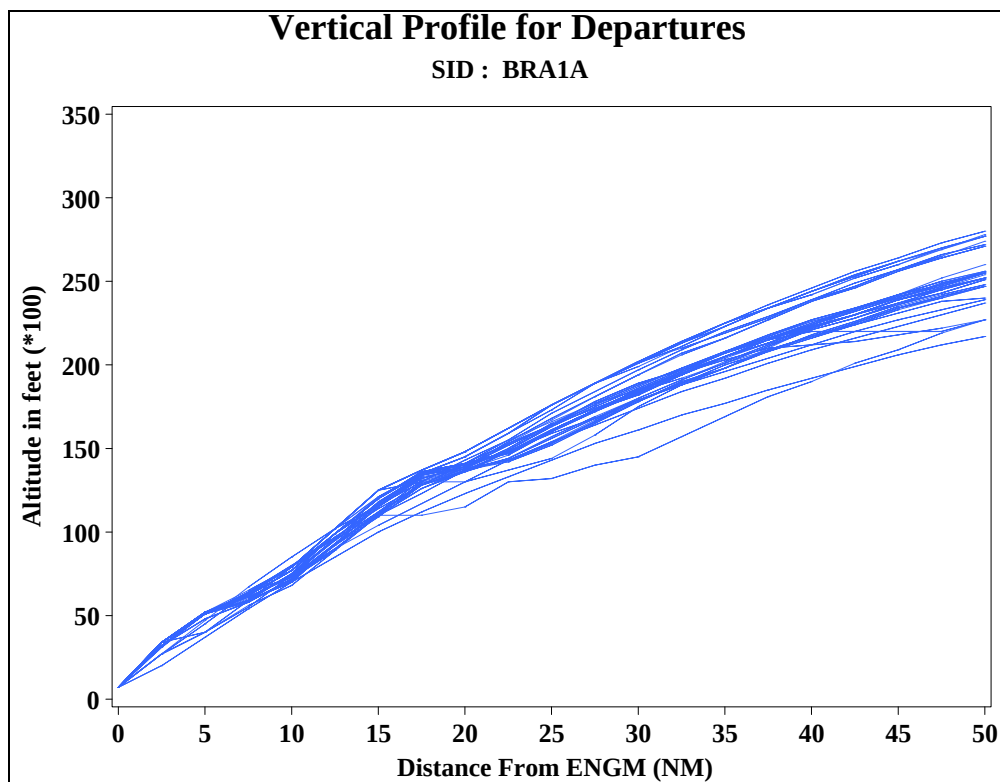
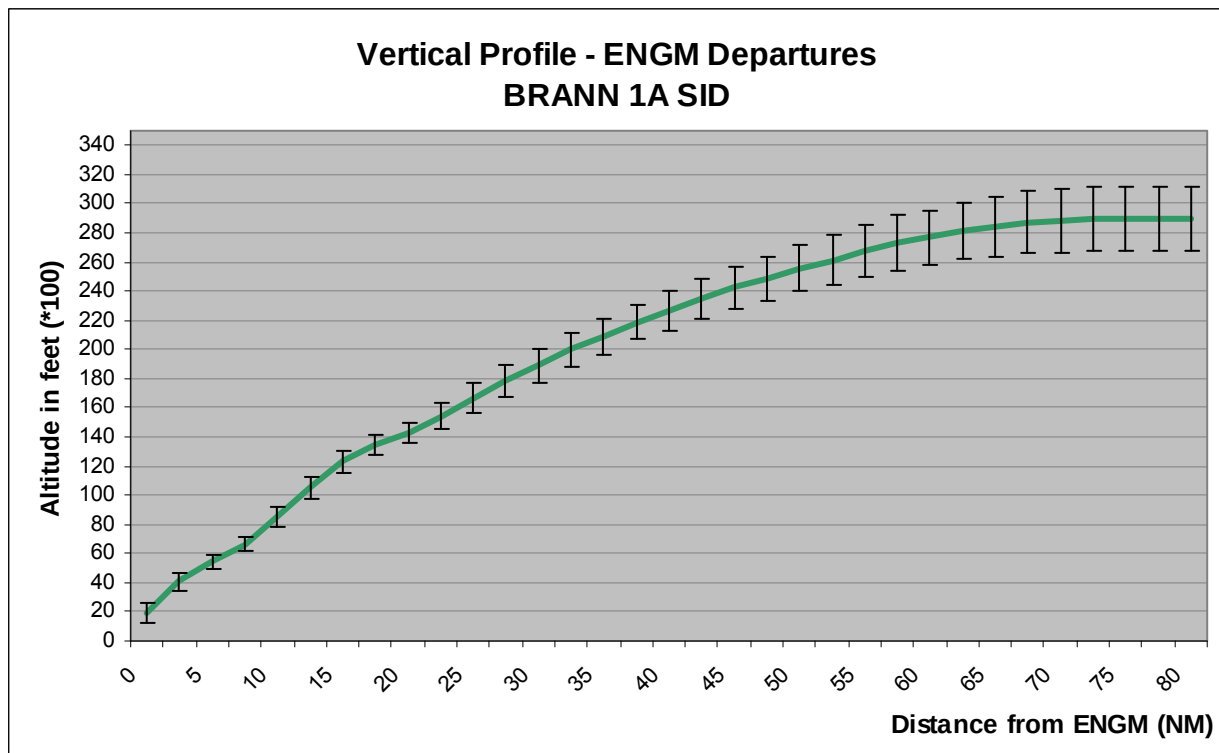


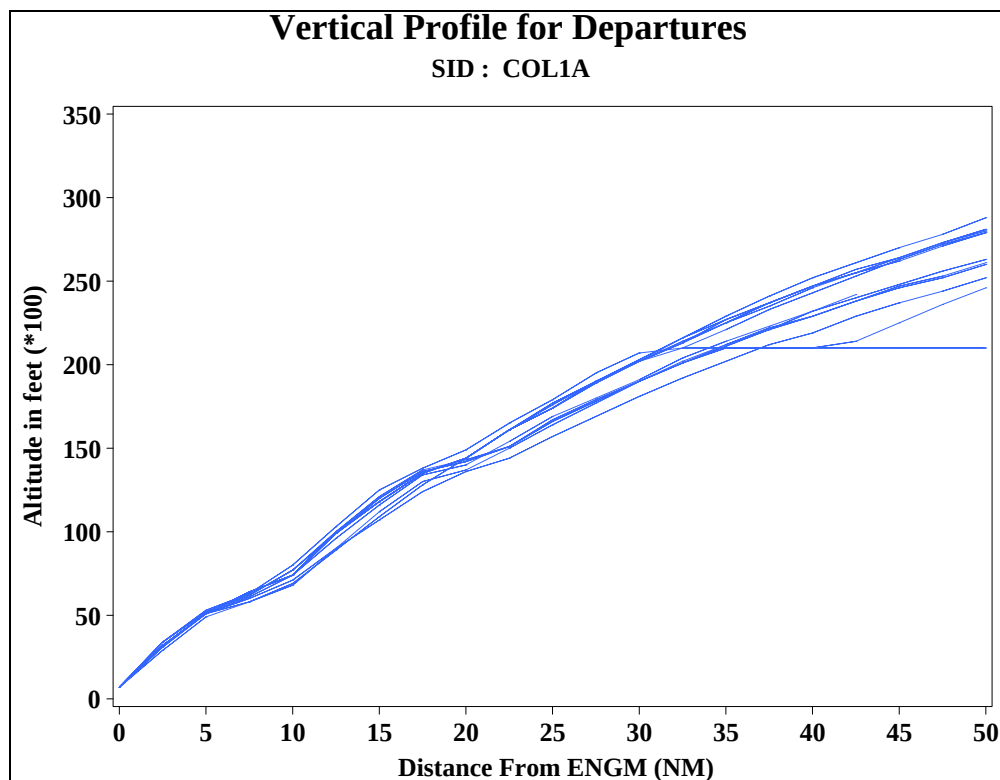
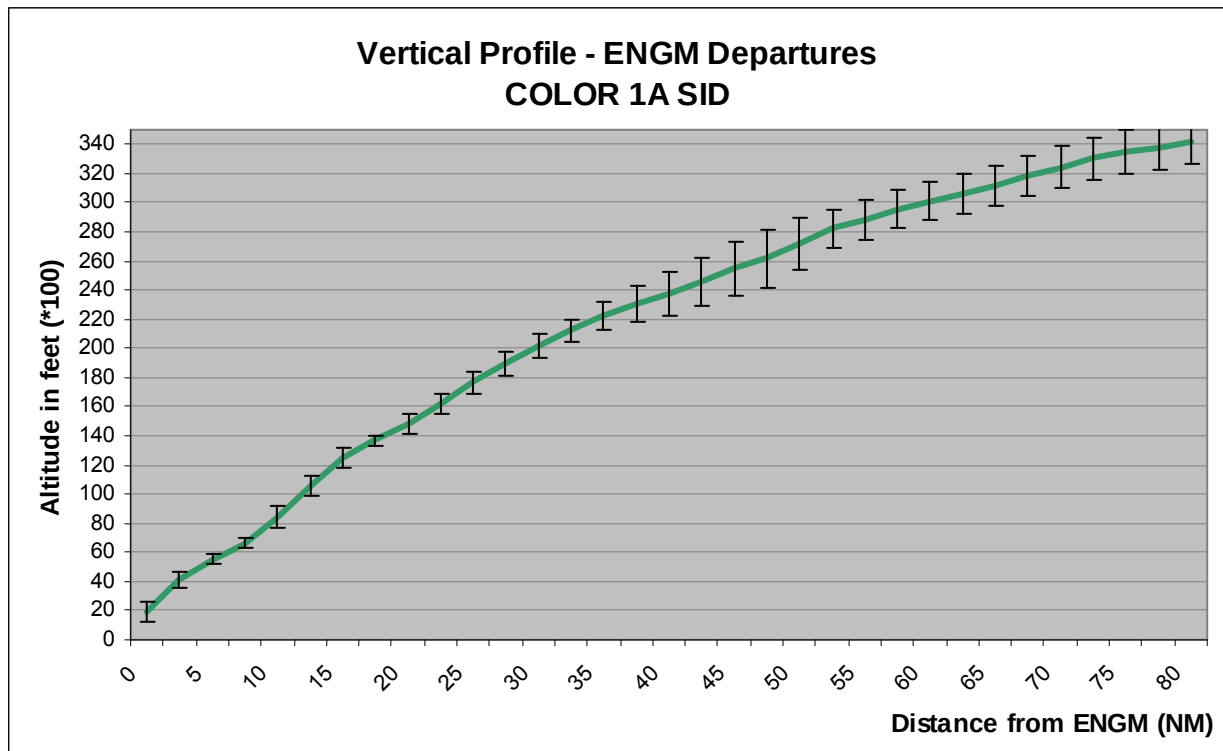


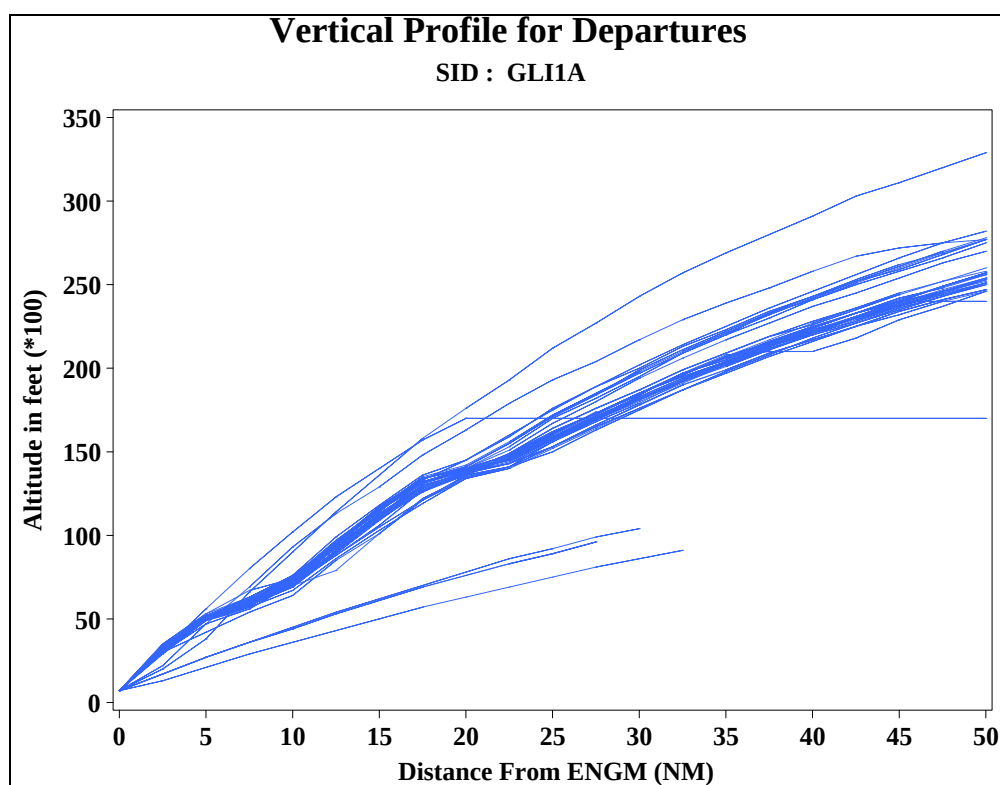
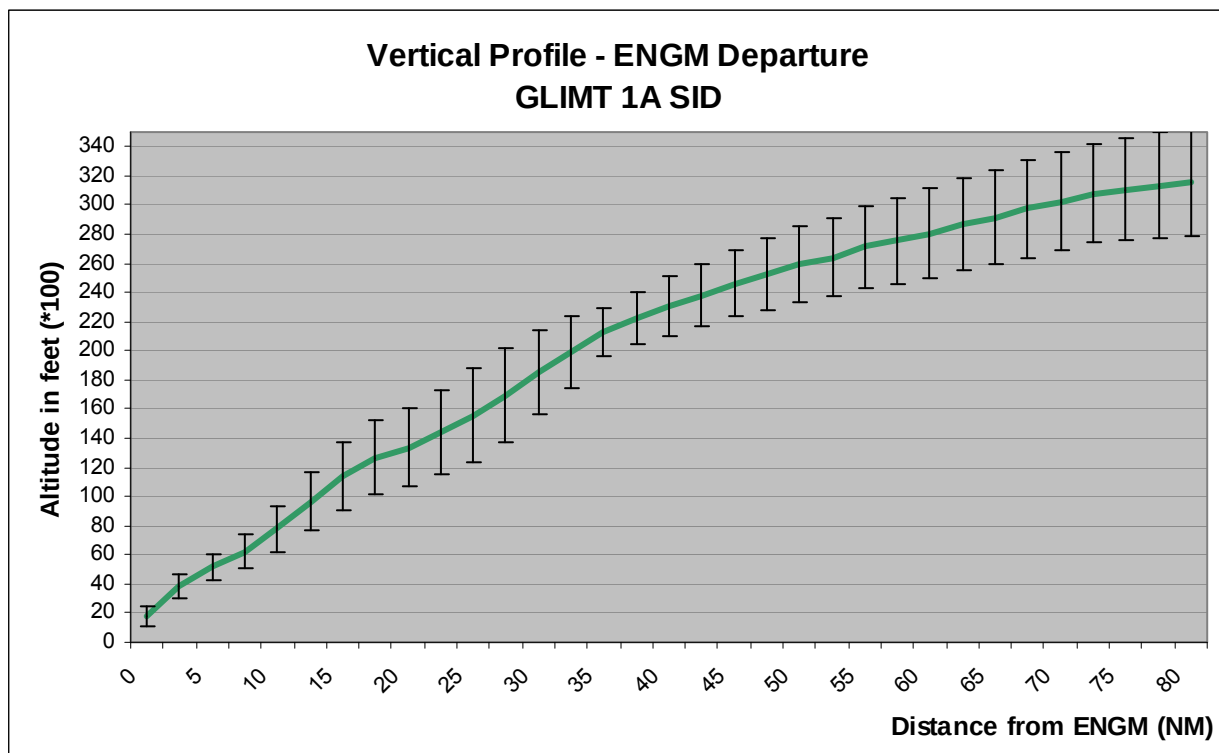


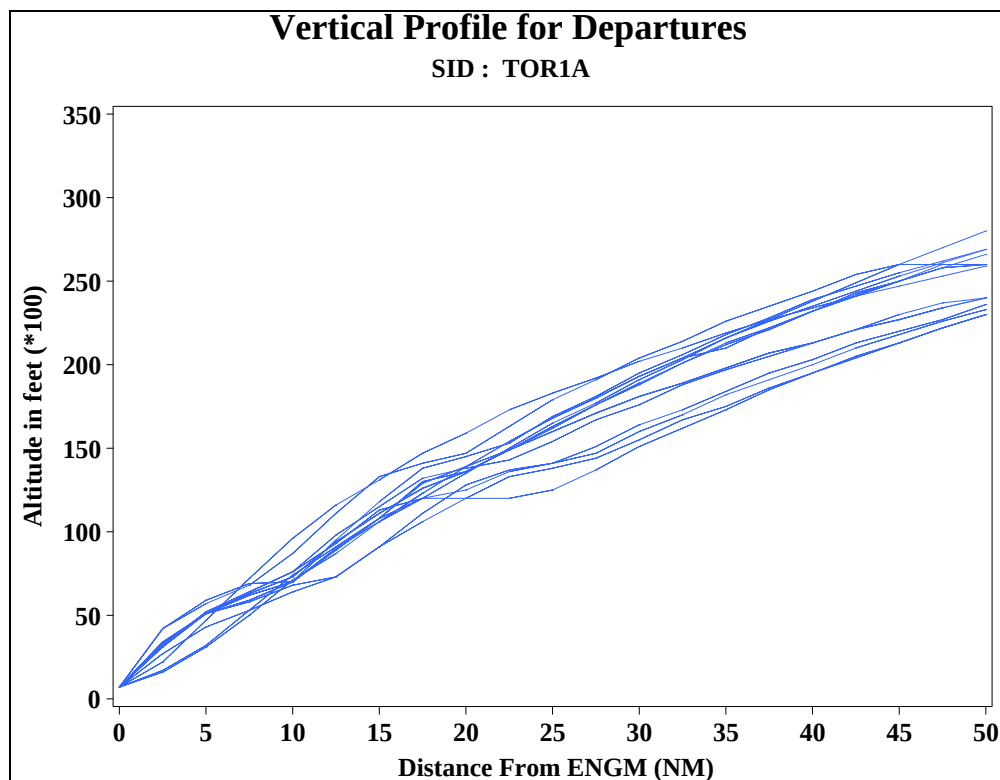
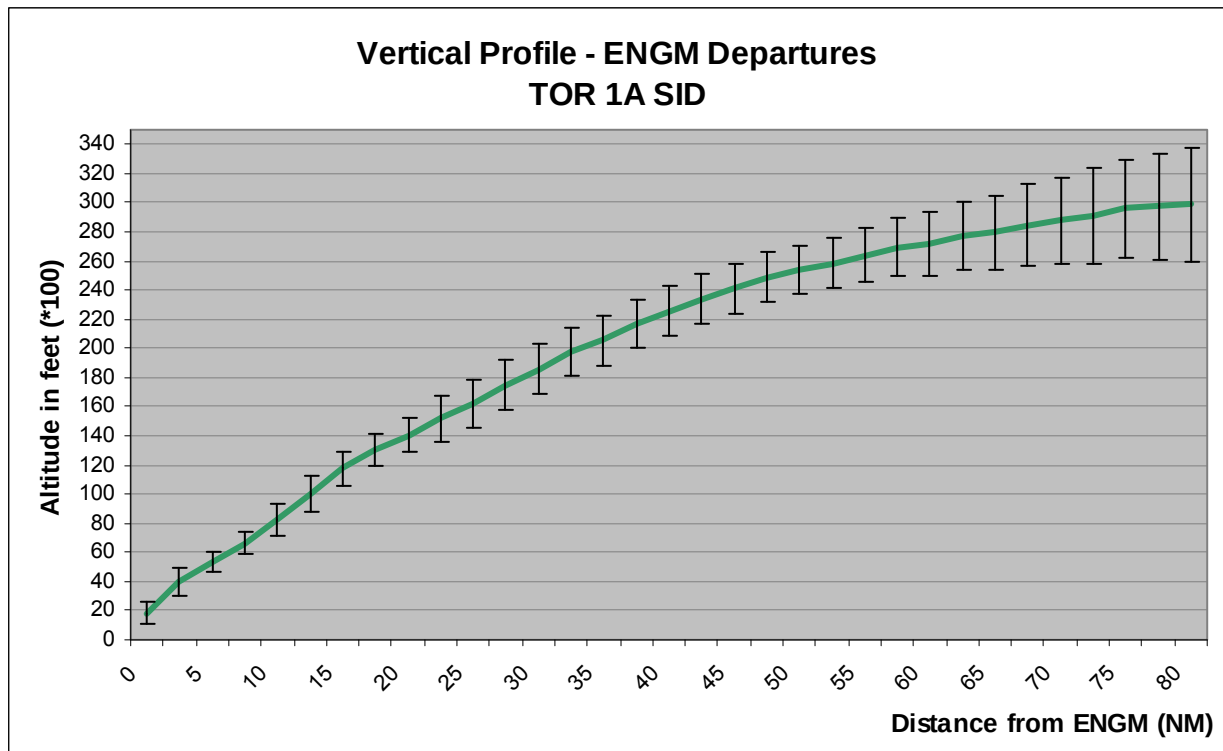


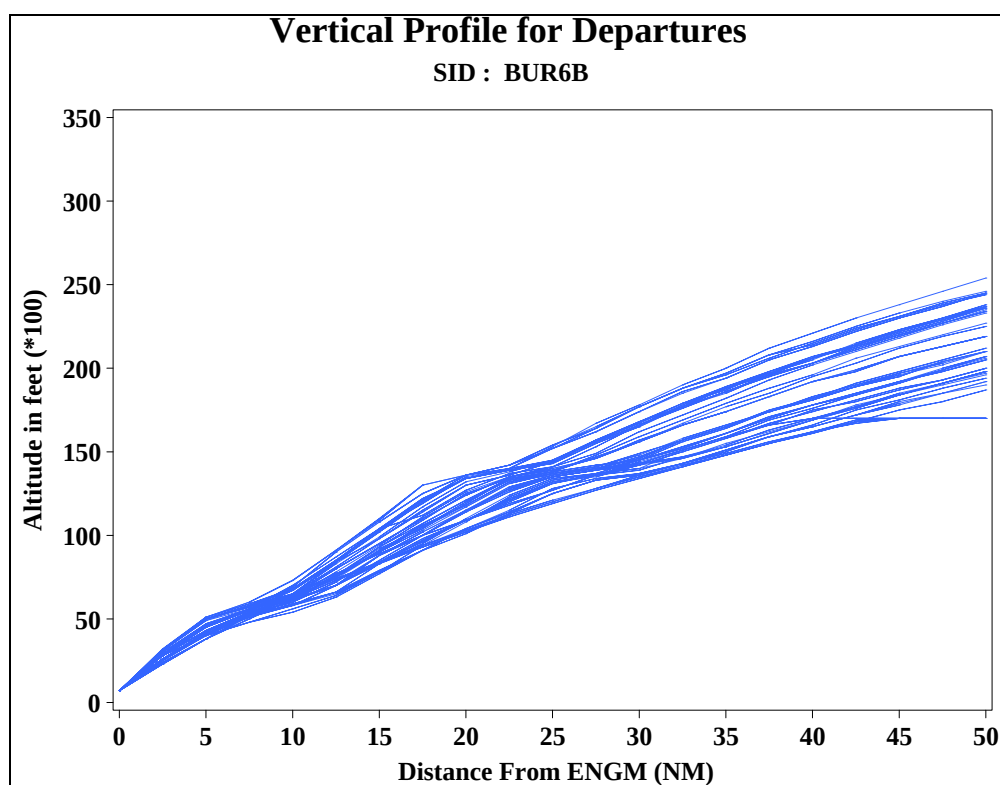
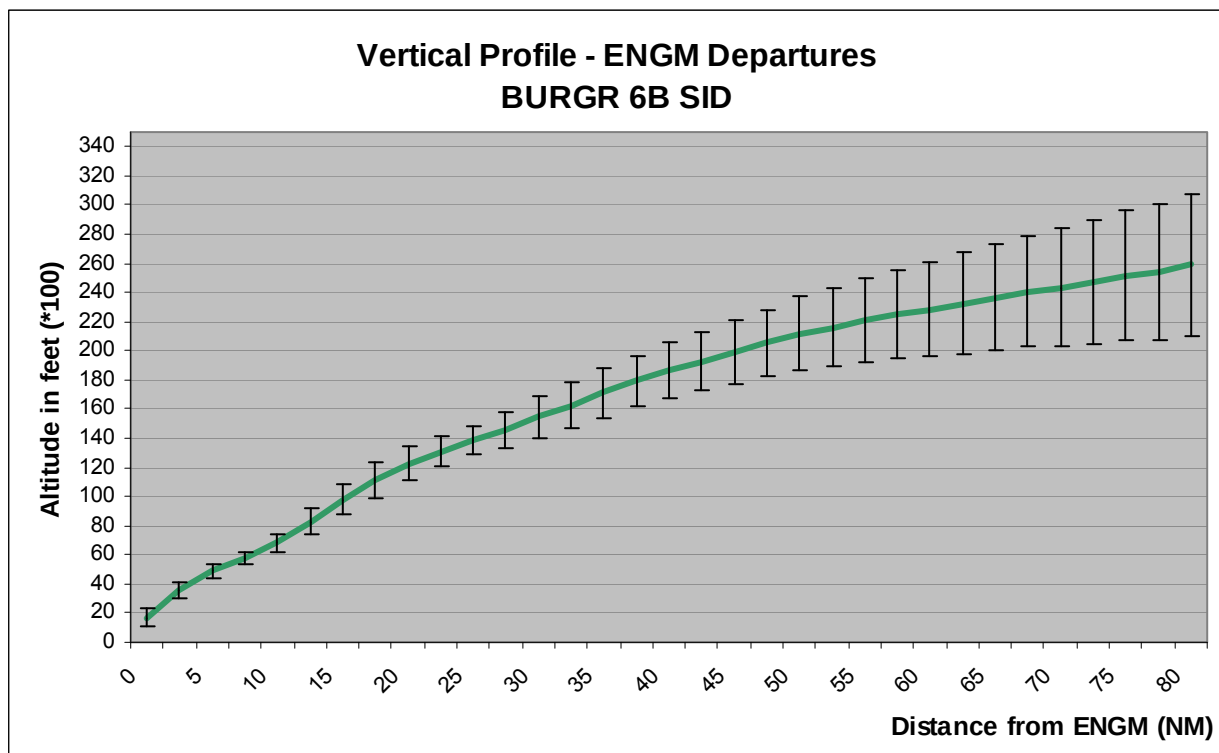


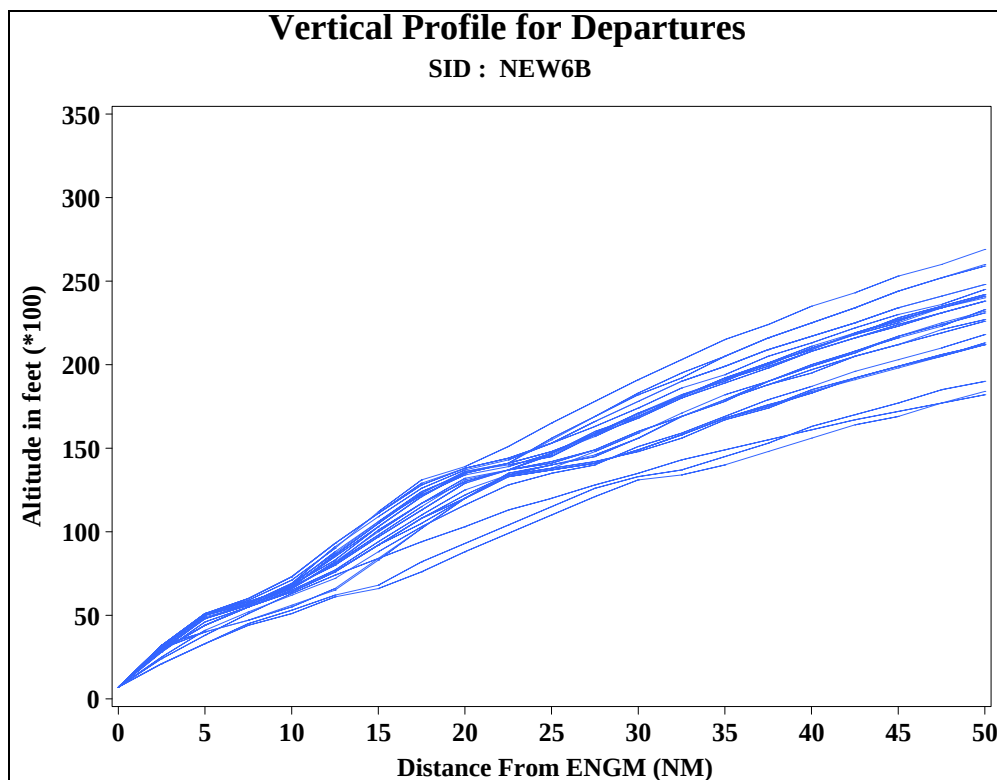
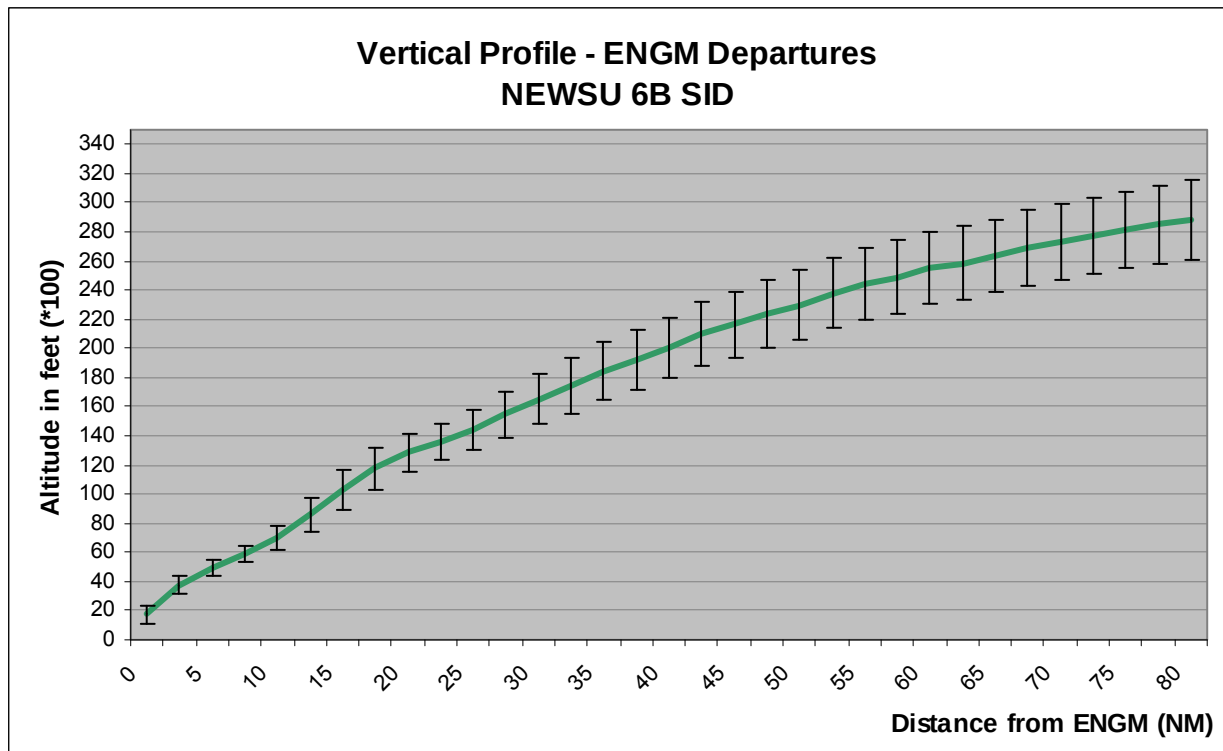
ANNEX I: ENGM SIDS - DEPARTURES VERTICAL PROFILES

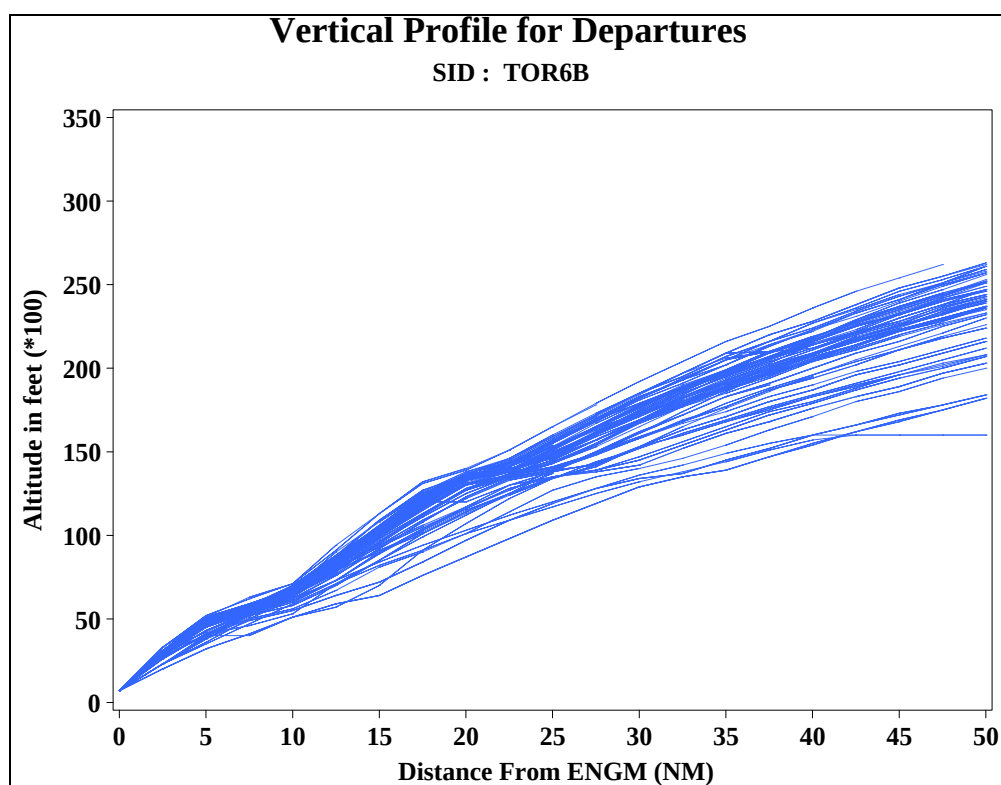
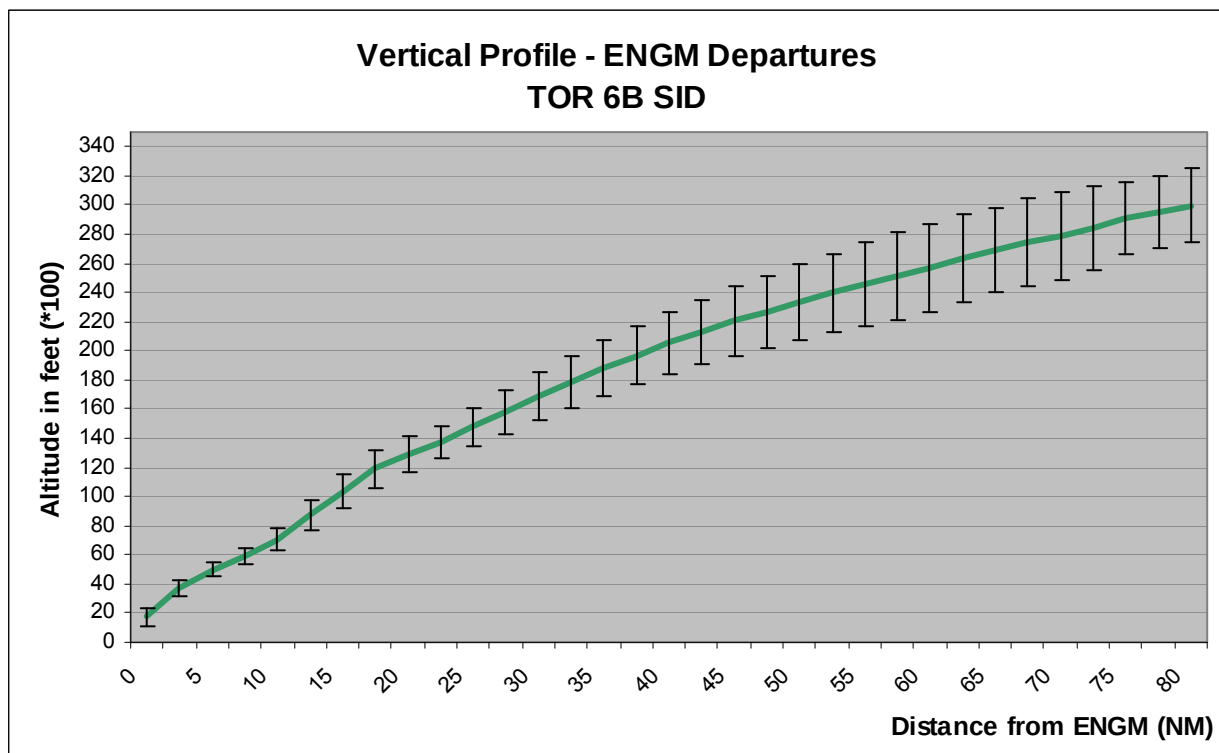


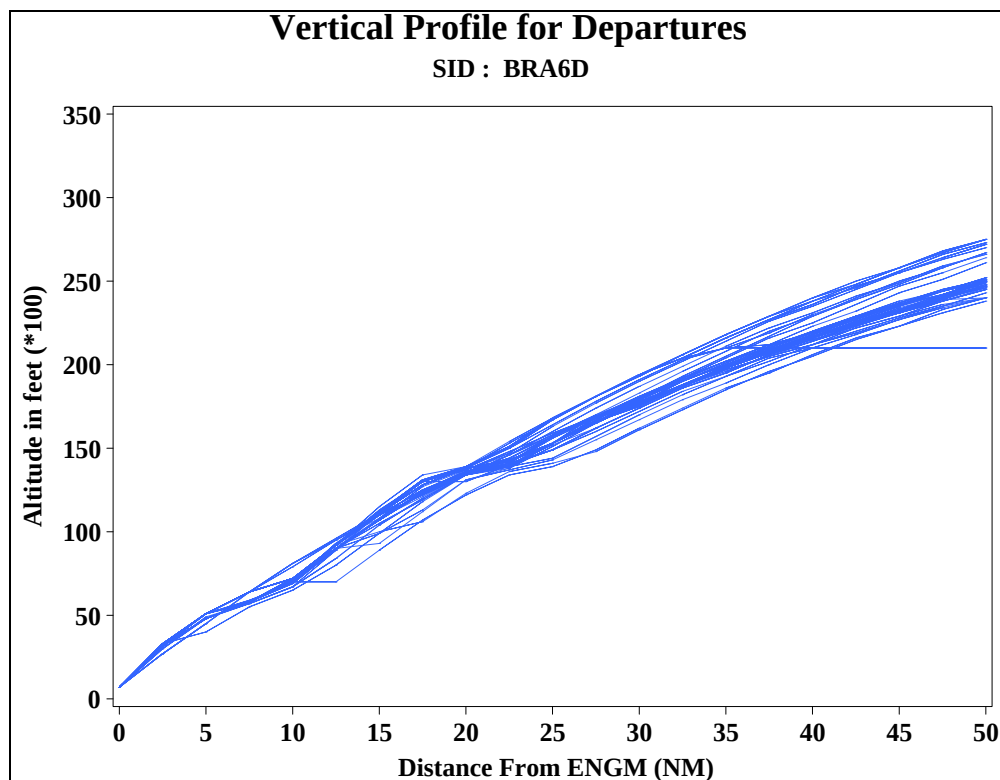
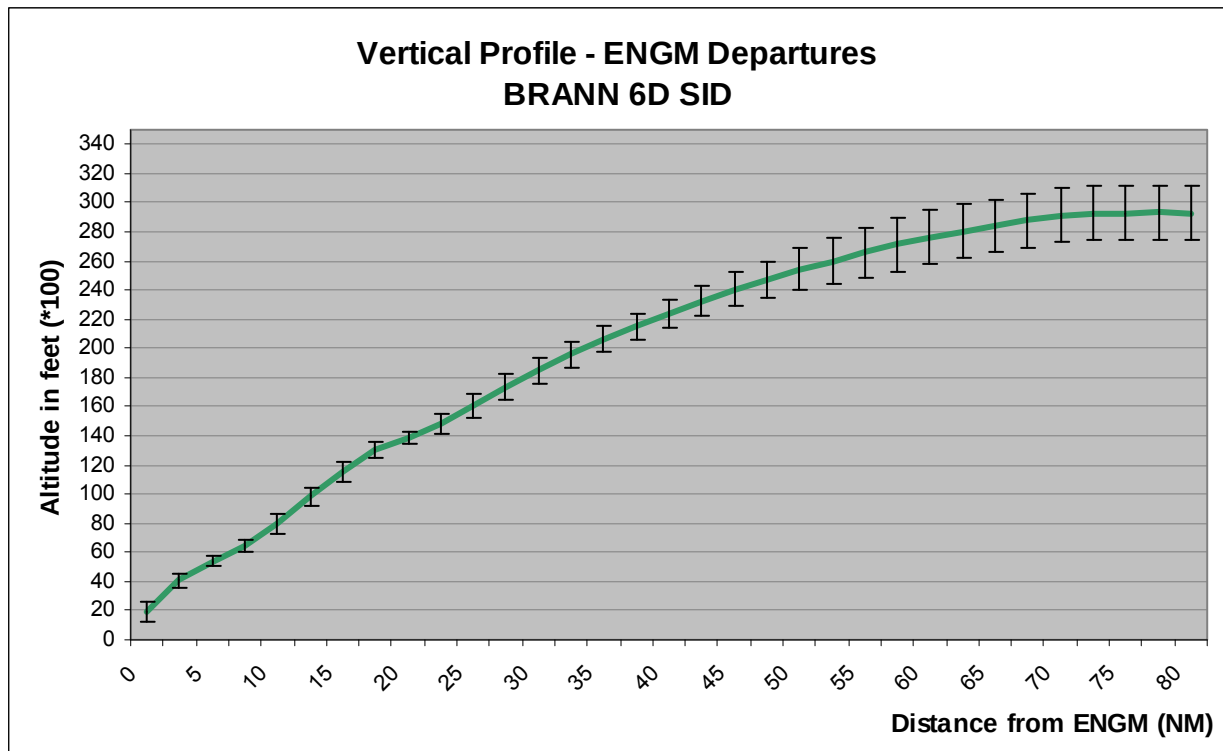


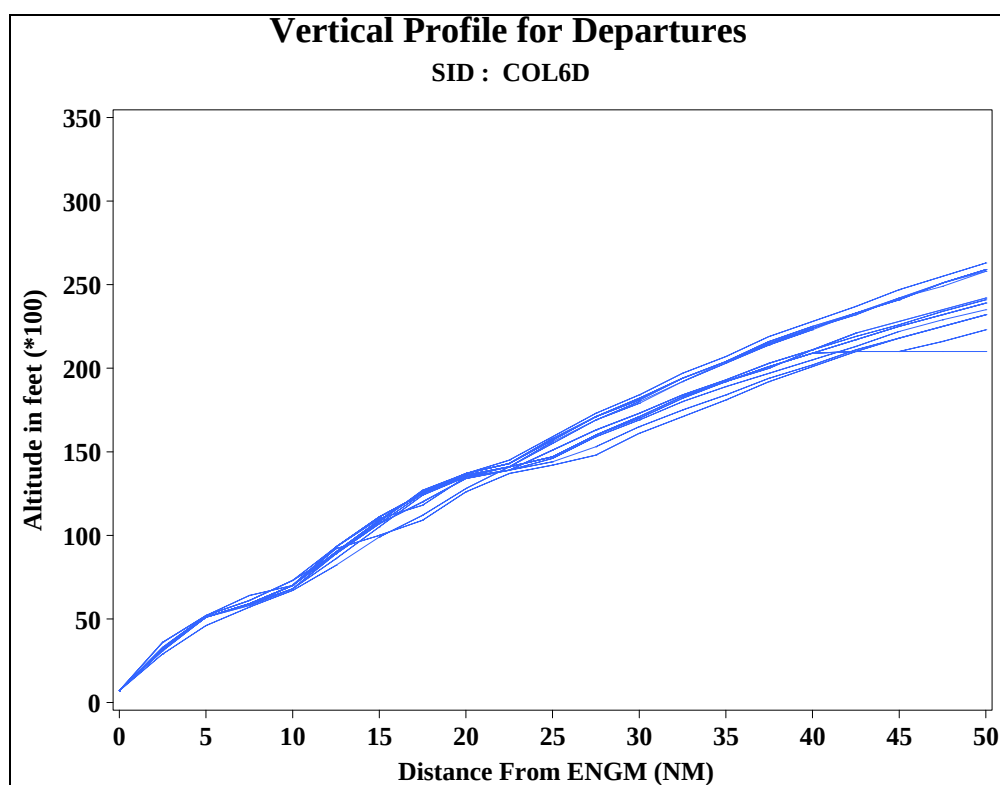
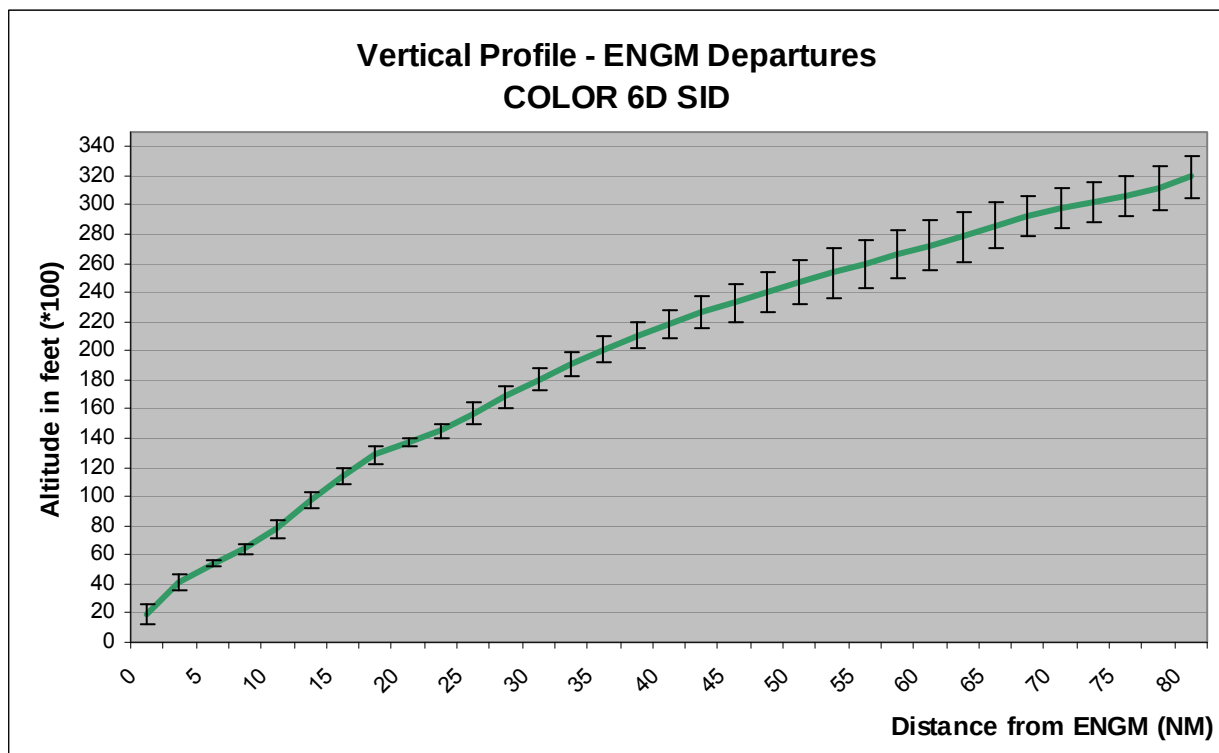


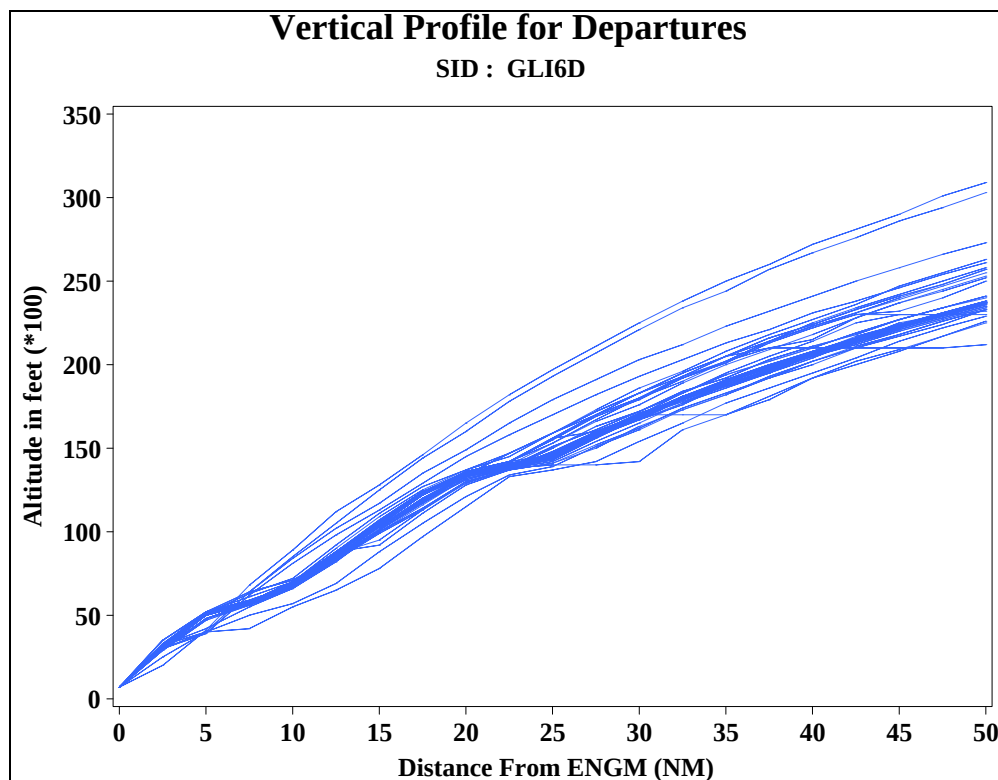
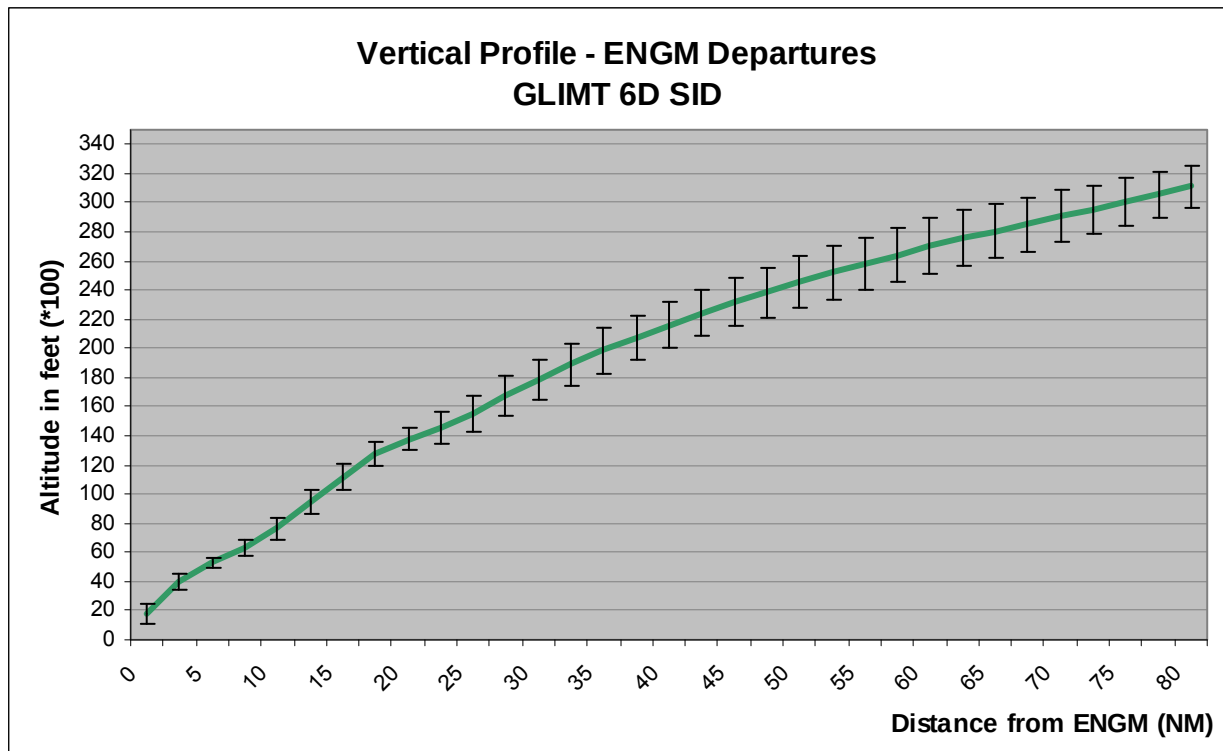


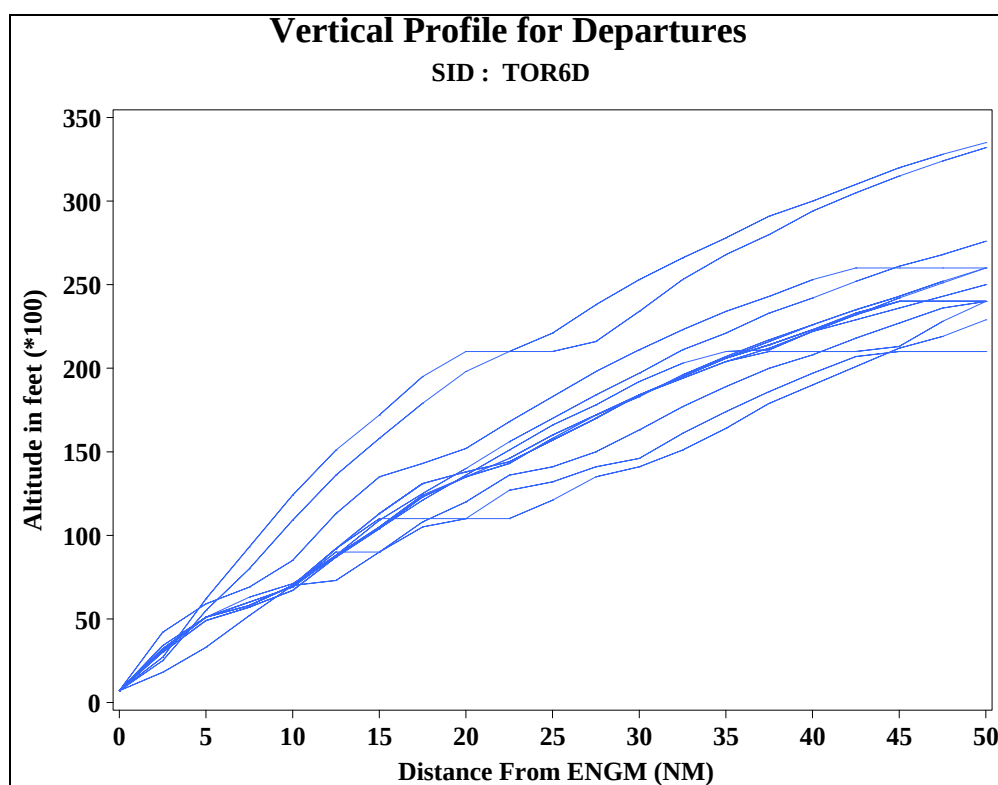
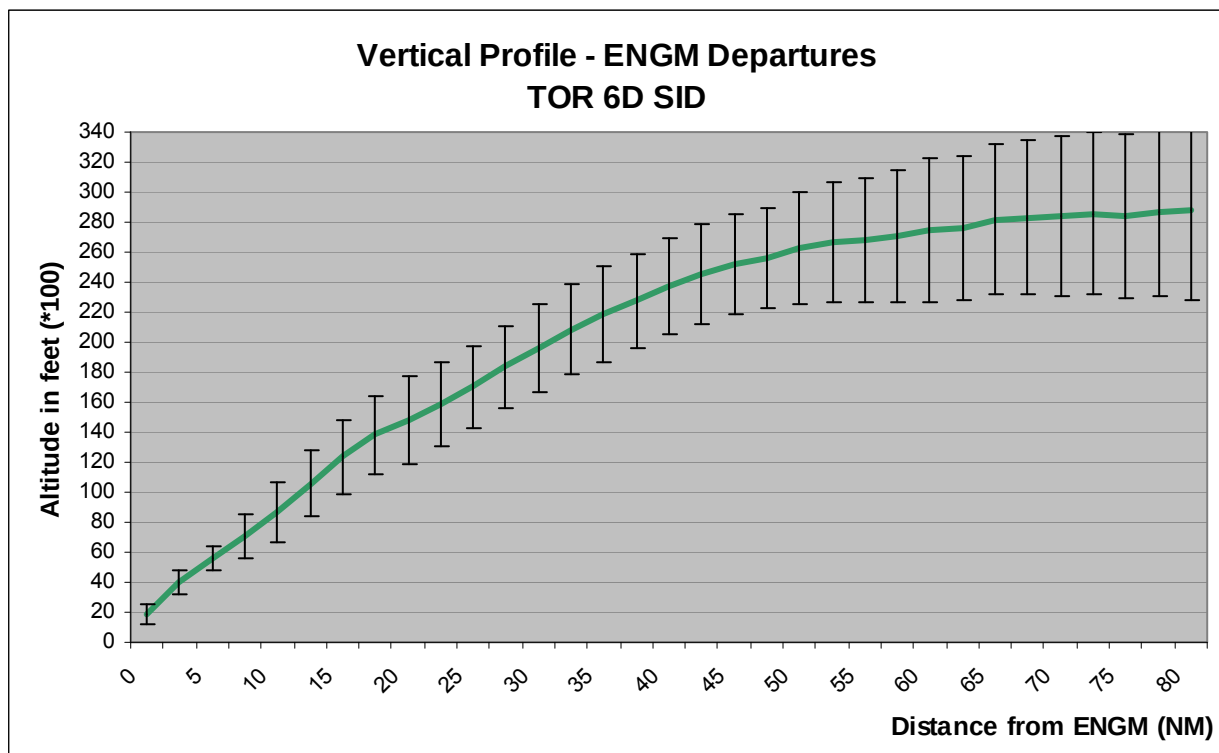


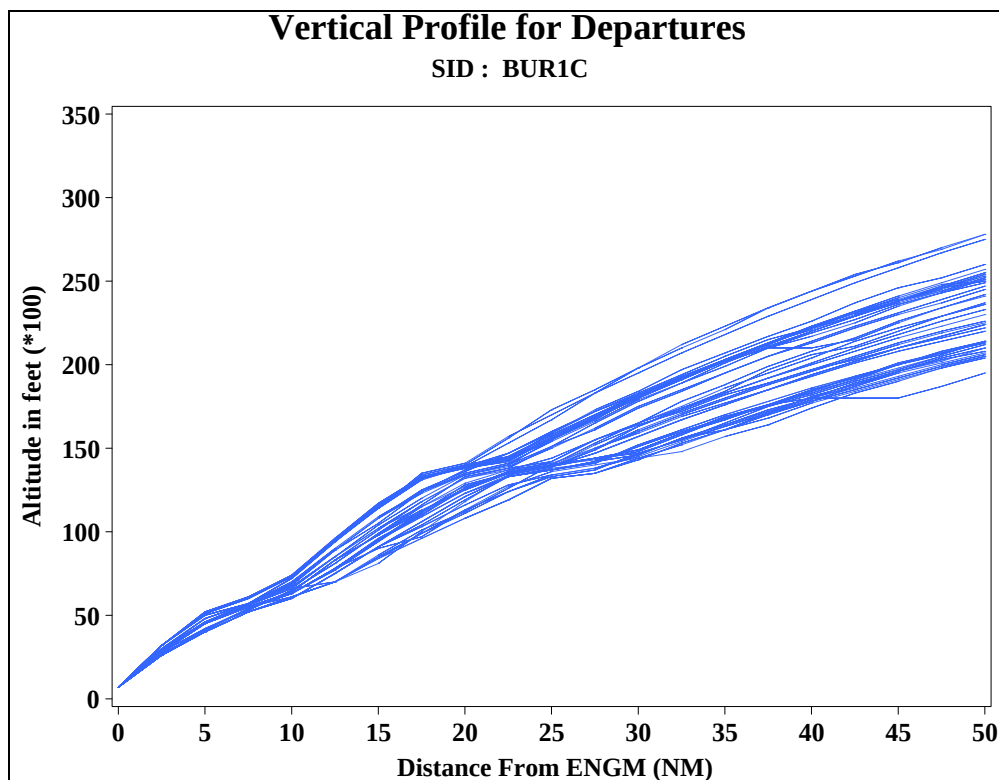
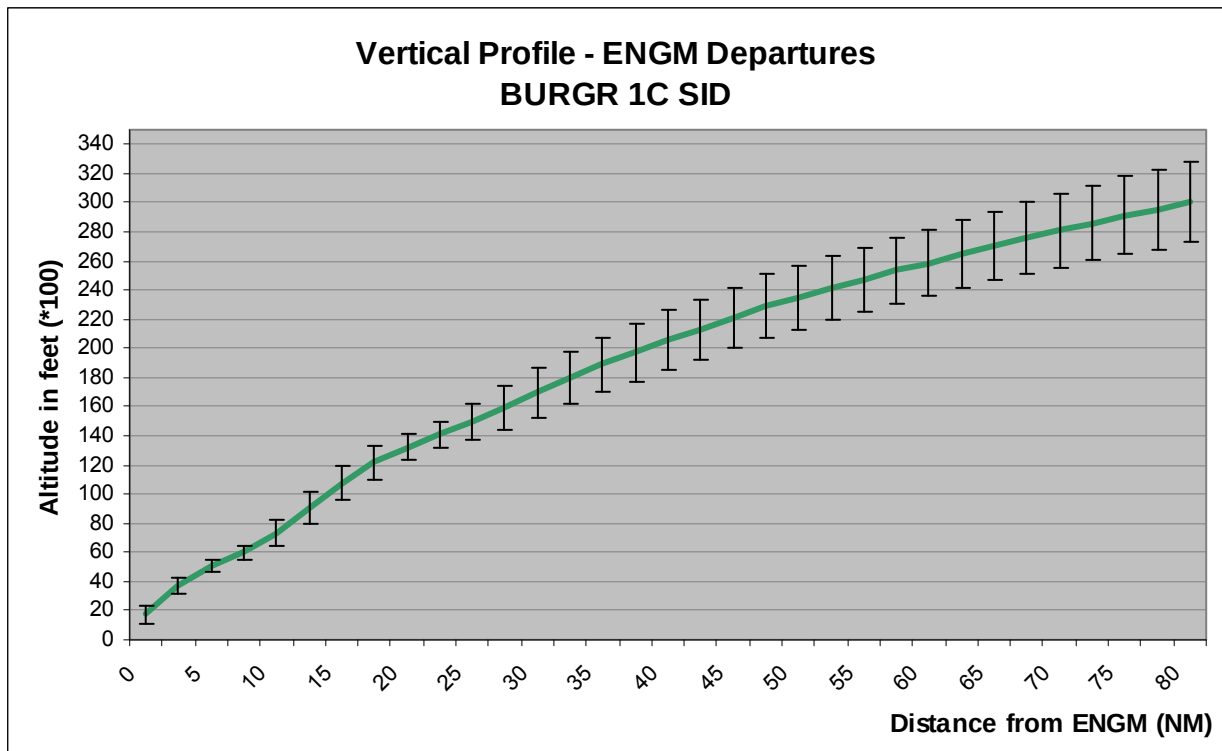


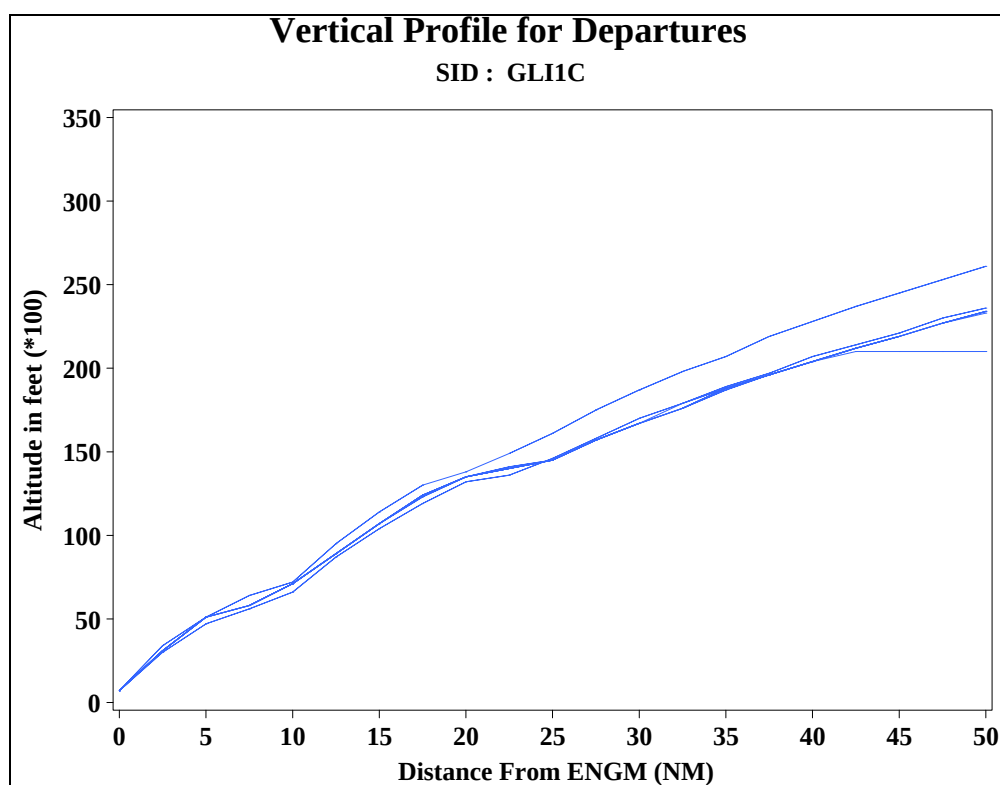
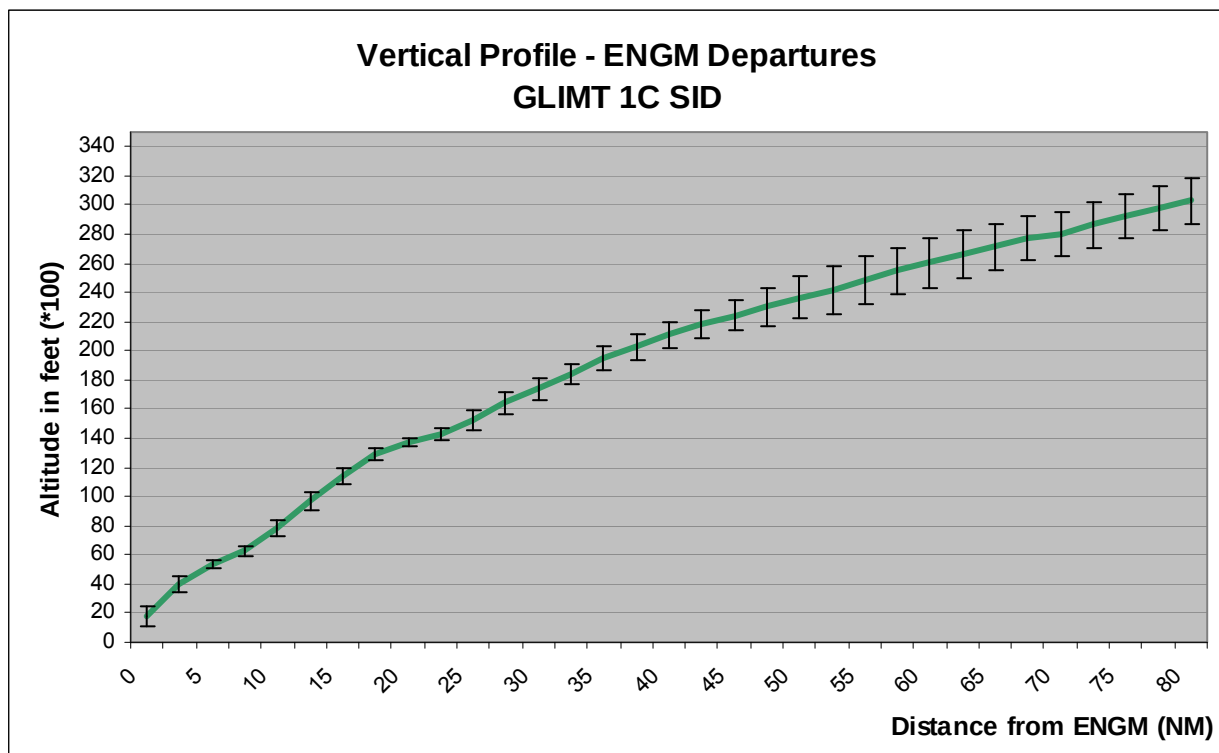


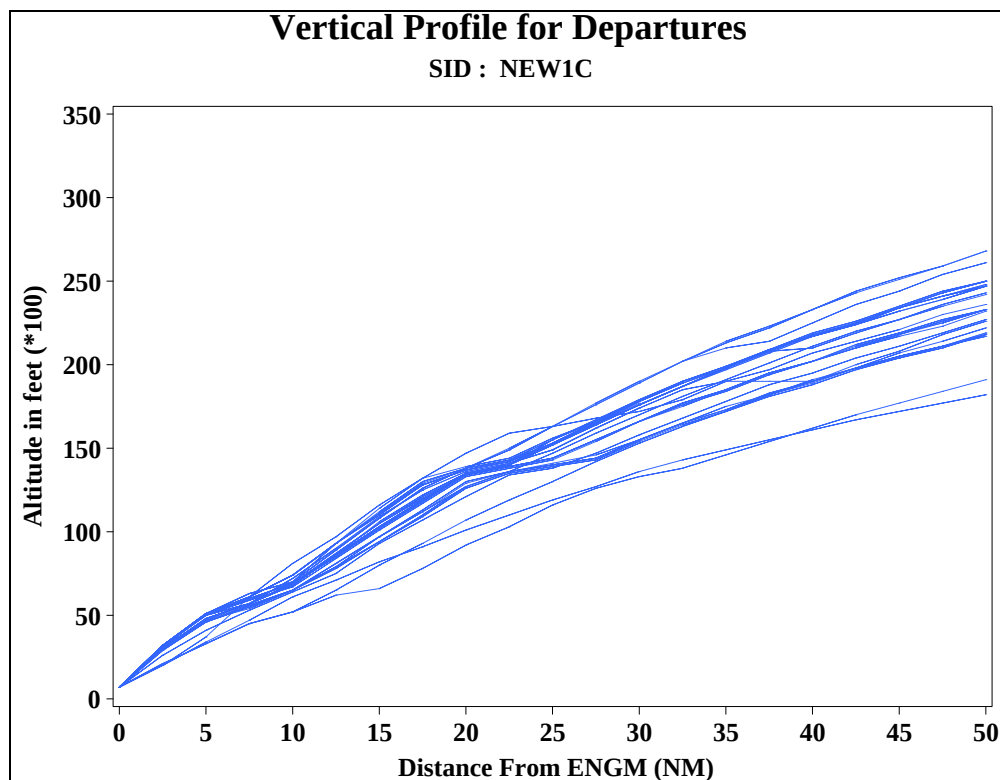
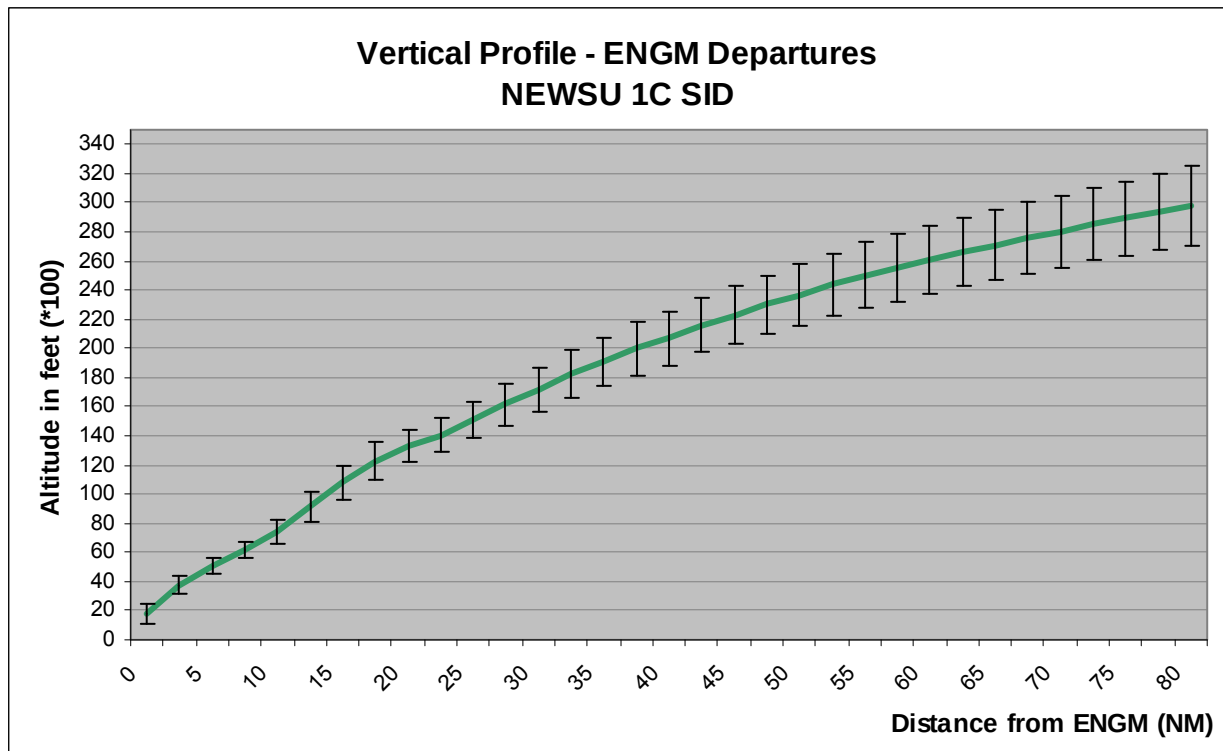


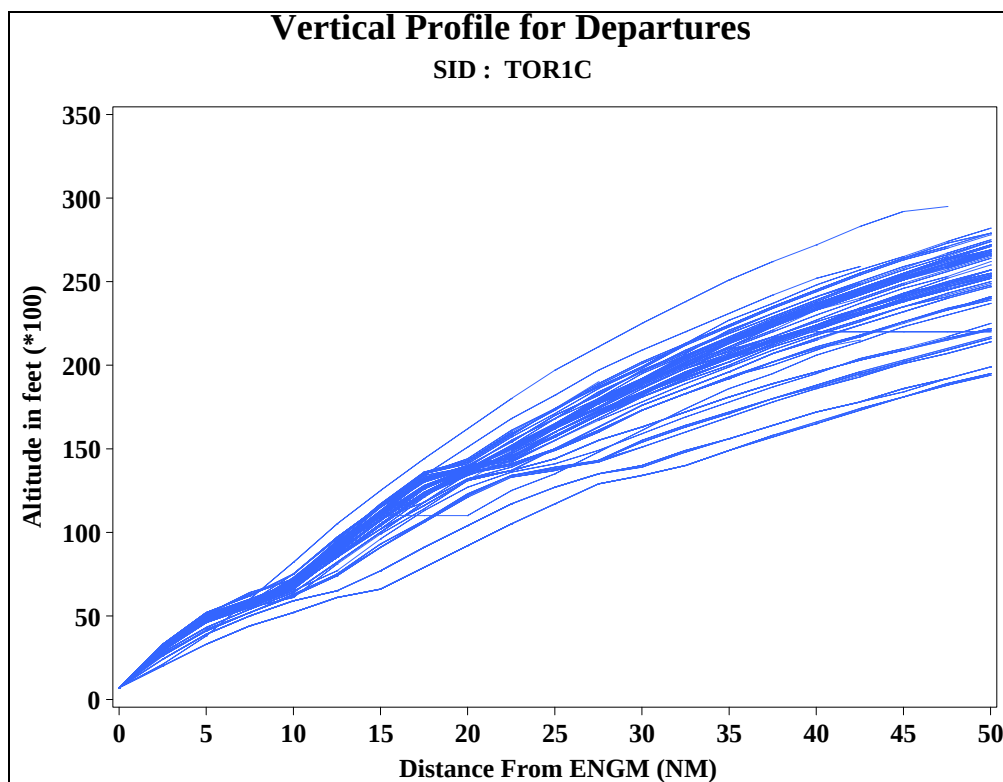
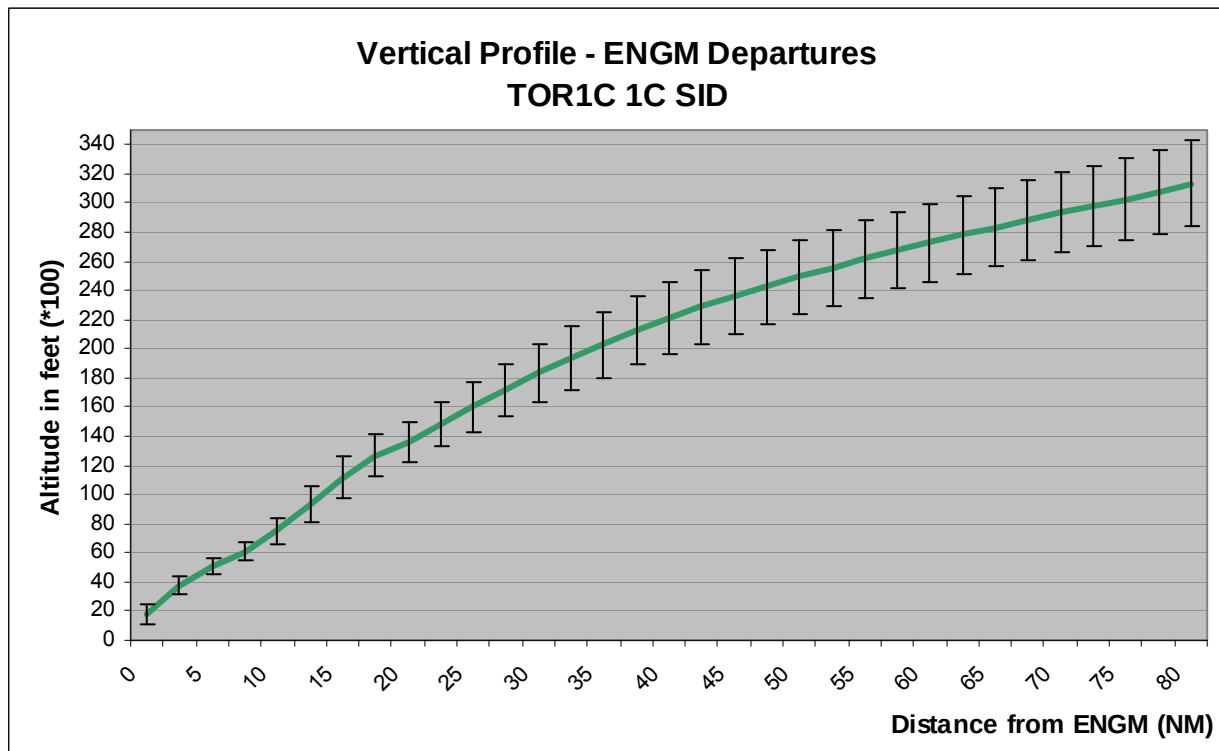








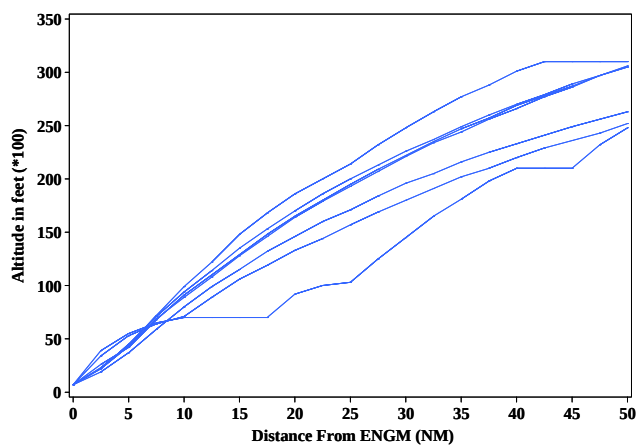




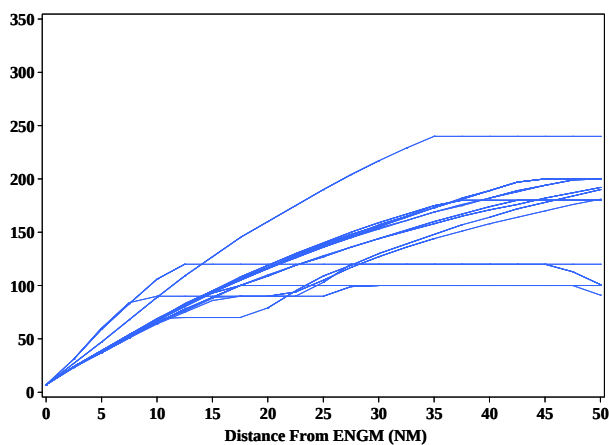
Page intentionally left blank

ANNEX J: ENGM SPECIAL SIDS - DEPARTURES VERTICAL PROFILES

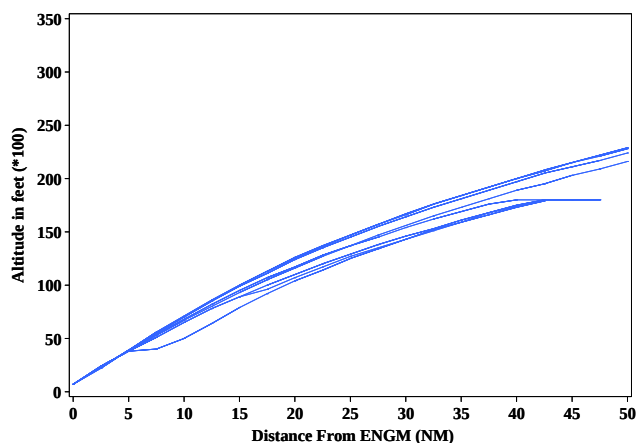
Vertical Profile for Departures
SID : BUR1A



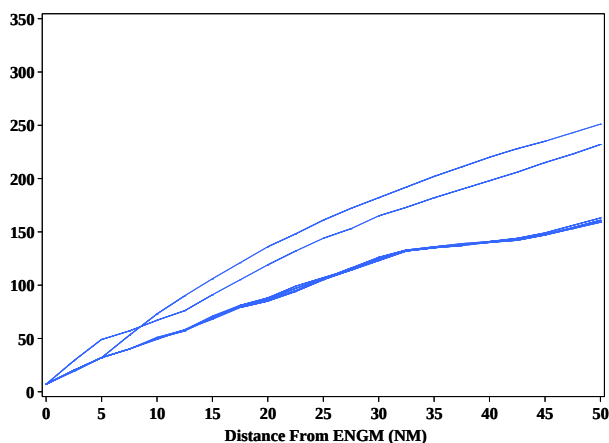
Vertical Profile for Departures
SID : CAC1A



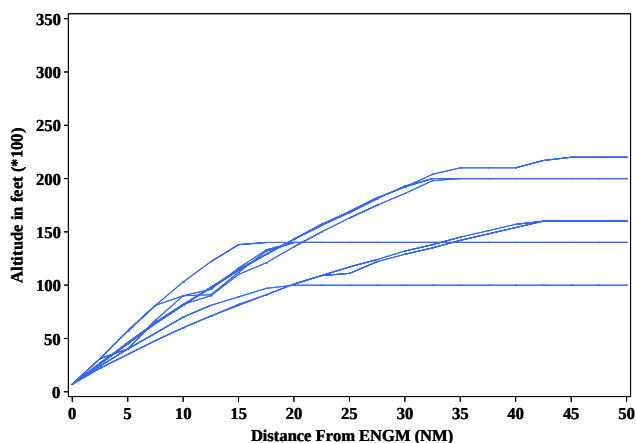
Vertical Profile for Departures
SID : CAB1A



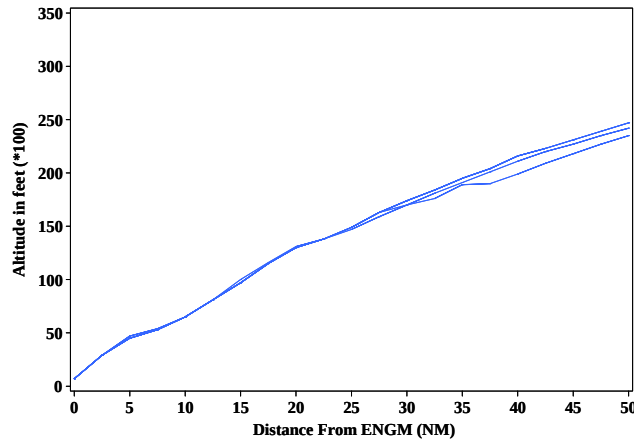
Vertical Profile for Departures
SID : NEW1A



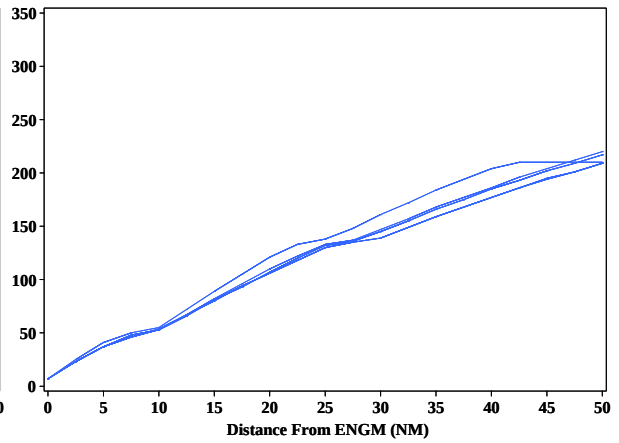
Vertical Profile for Departures
SID : CAT1A



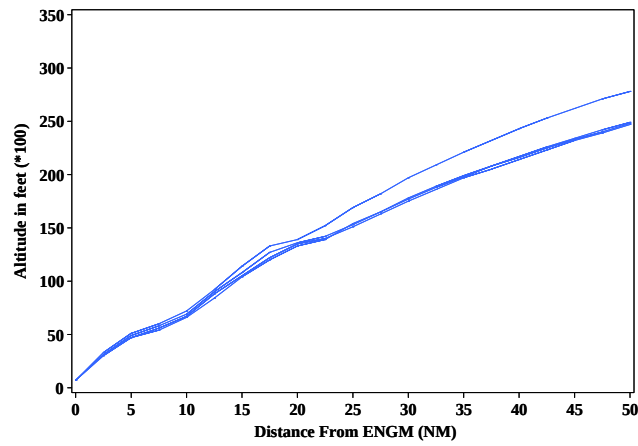
Vertical Profile for Departures
SID : BRA6B



Vertical Profile for Departures
SID : COL6B

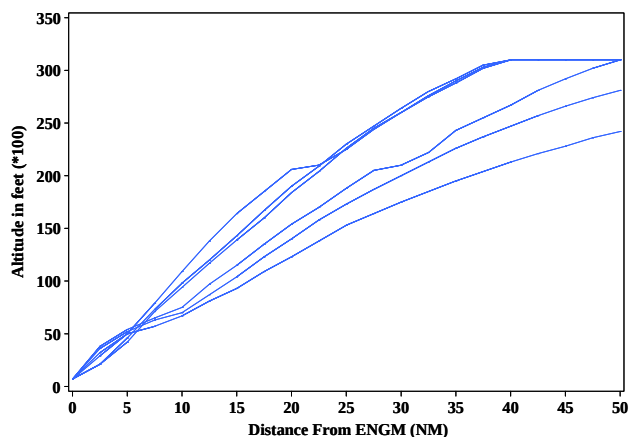


Vertical Profile for Departures
SID : GLI6B



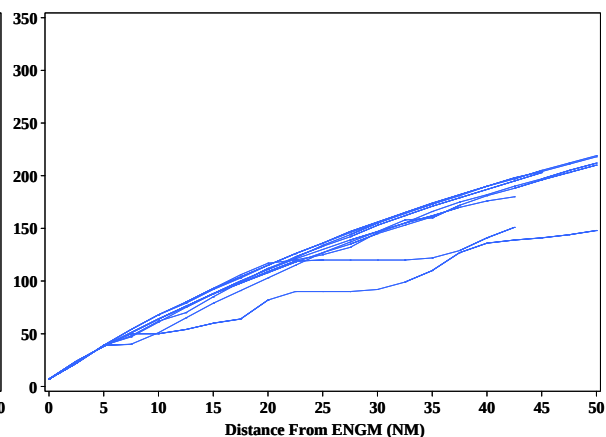
Vertical Profile for Departures

SID : BUR6D



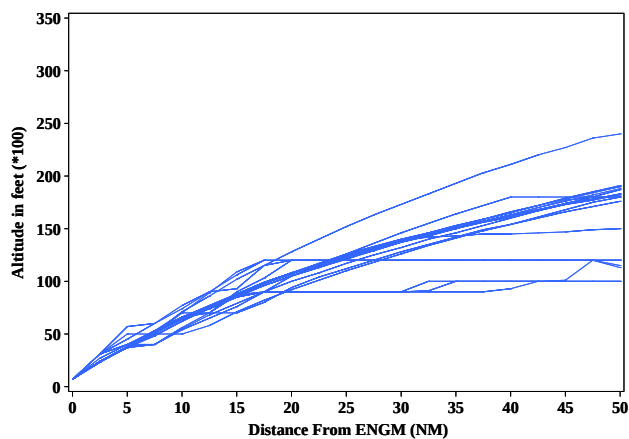
Vertical Profile for Departures

SID : CAB6D



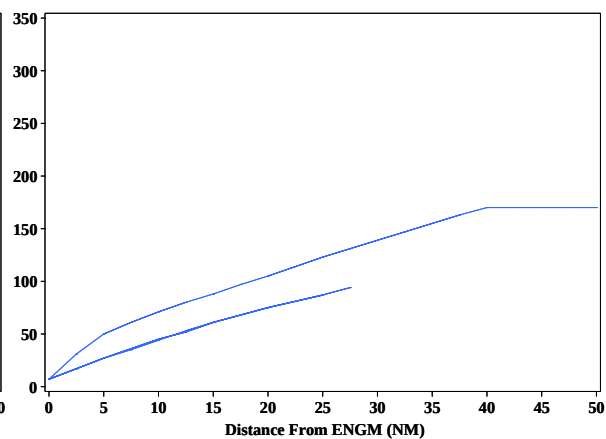
Vertical Profile for Departures

SID : CAC6D



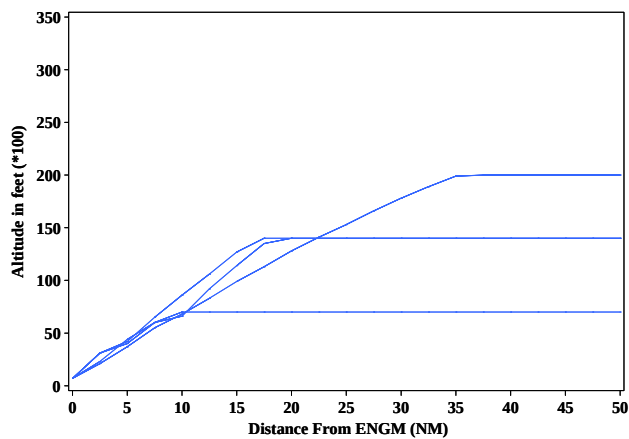
Vertical Profile for Departures

SID : CAG6D



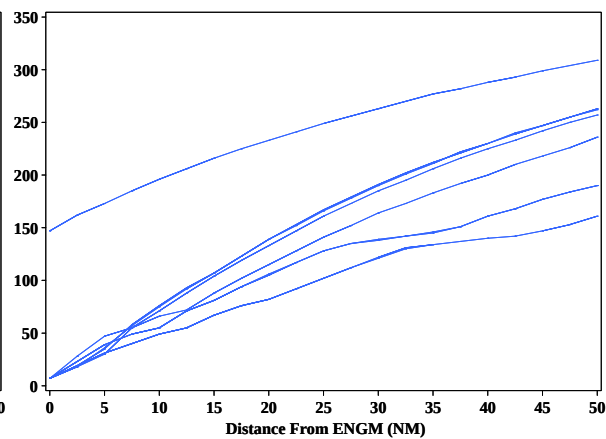
Vertical Profile for Departures

SID : CAT6D

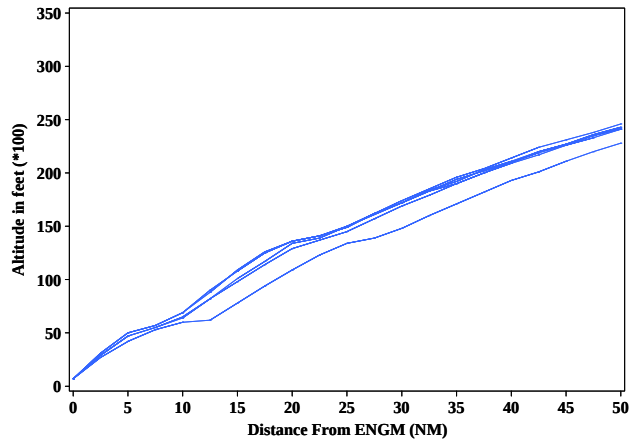


Vertical Profile for Departures

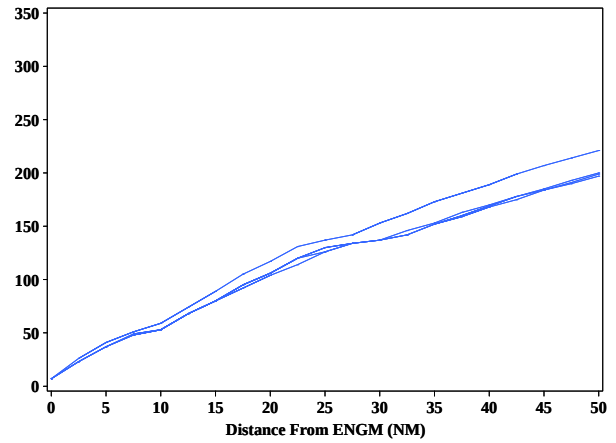
SID : NEW6D



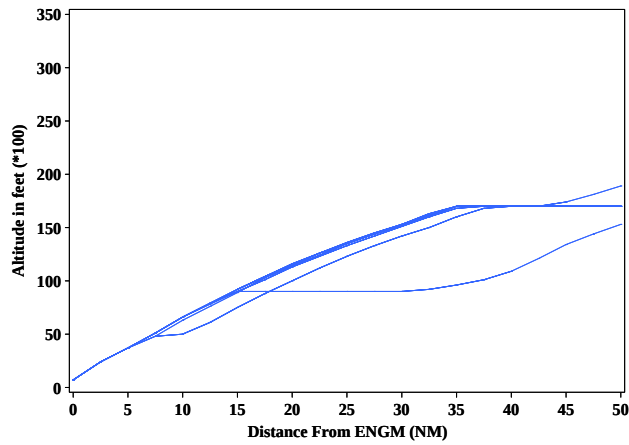
Vertical Profile for Departures
SID : BRA1C



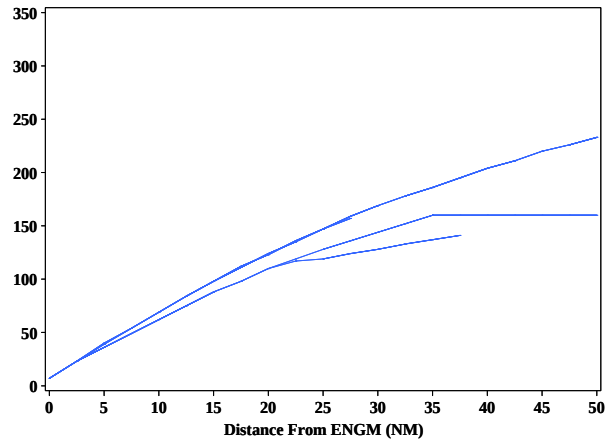
Vertical Profile for Departures
SID : COL1C



Vertical Profile for Departures
SID : VOB1C

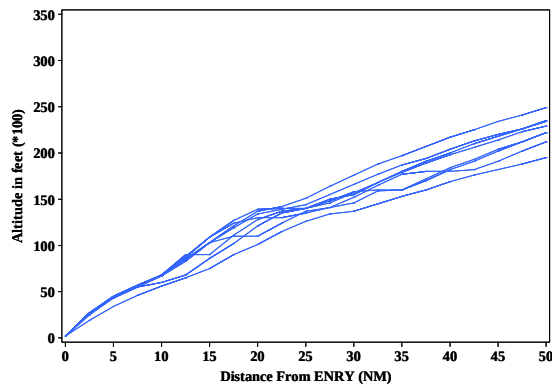


Vertical Profile for Departures
SID : RYG1C

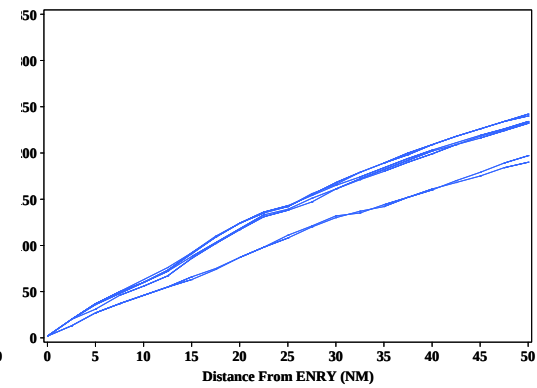


ANNEX K: ENRY SIDS - DEPARTURES VERTICAL PROFILES

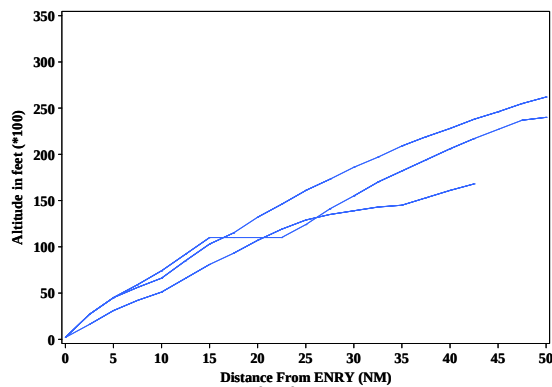
Vertical Profile for Departures
SID : BAM1E



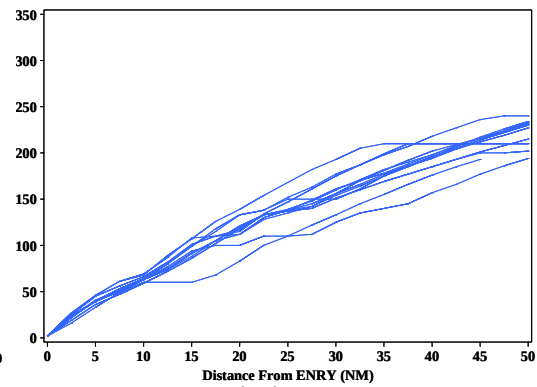
Vertical Profile for Departures
SID : COS1E



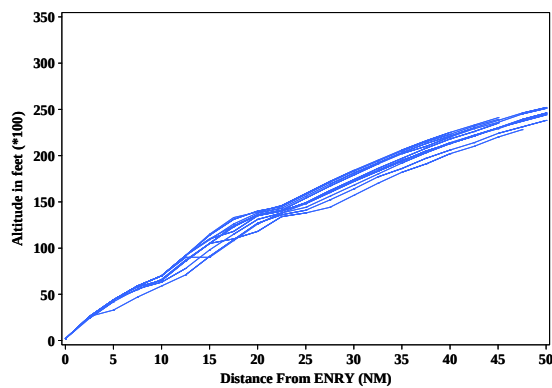
Vertical Profile for Departures
SID : GOK1E



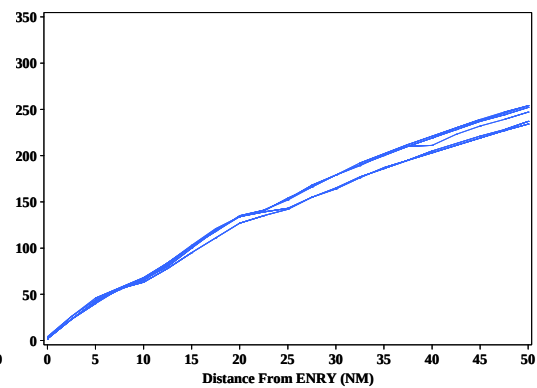
Vertical Profile for Departures
SID : OSL1E



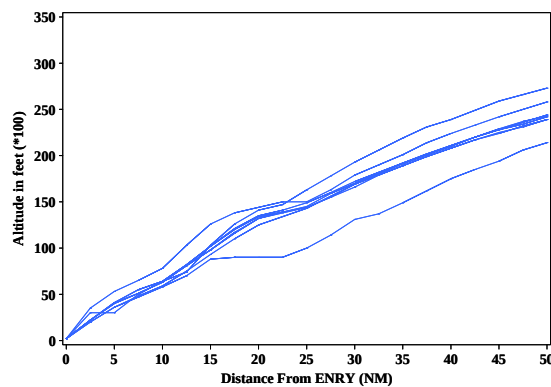
Vertical Profile for Departures
SID : MAR1E



Vertical Profile for Departures
SID : ING1E

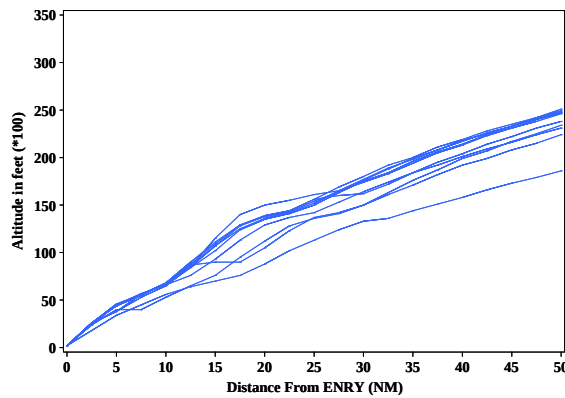


Vertical Profile for Departures
SID : WAC1E



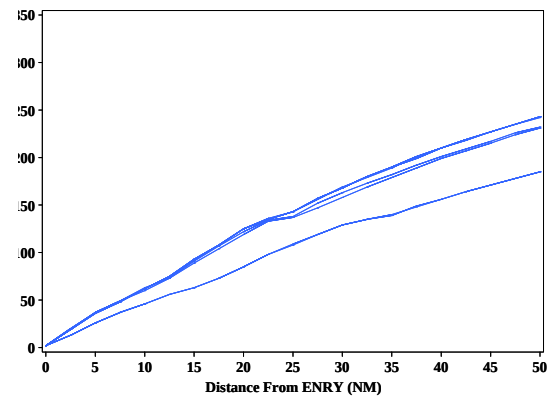
Vertical Profile for Departures

SID : BAM1F



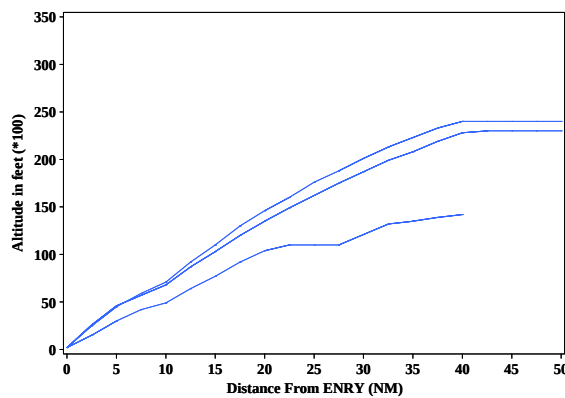
Vertical Profile for Departures

SID : COS1F



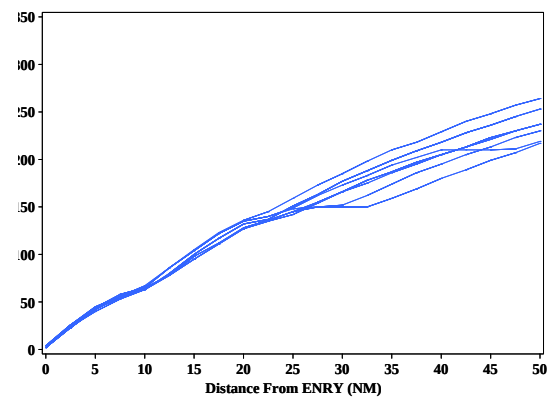
Vertical Profile for Departures

SID : GOK1F



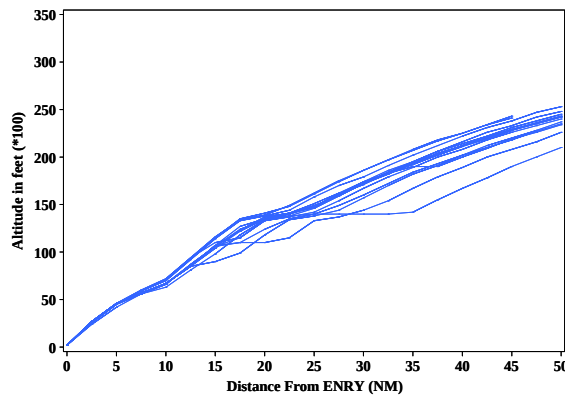
Vertical Profile for Departures

SID : ING1F



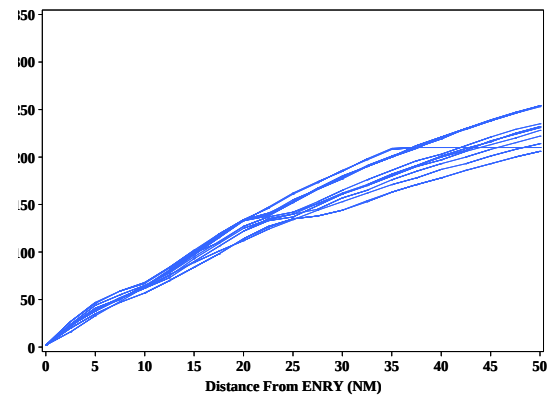
Vertical Profile for Departures

SID : MAR1F



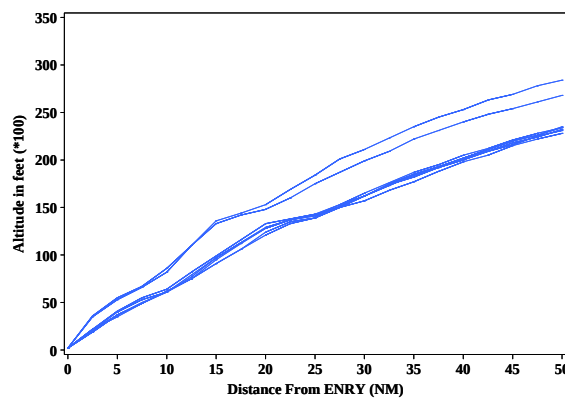
Vertical Profile for Departures

SID : OSL1F



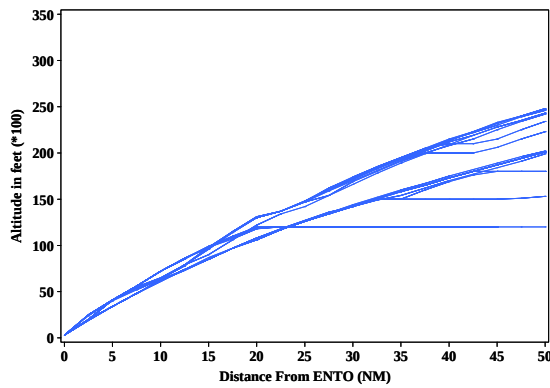
Vertical Profile for Departures

SID : WAC1F

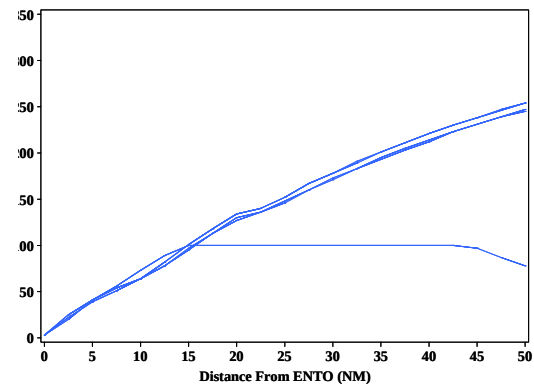


ANNEX L: ENTO SIDS - DEPARTURES VERTICAL PROFILES

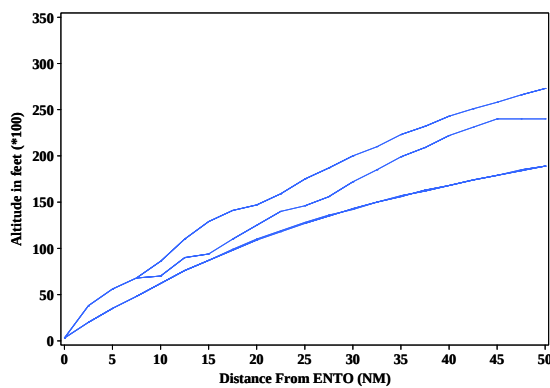
Vertical Profile for Departures
SID : BAM1G



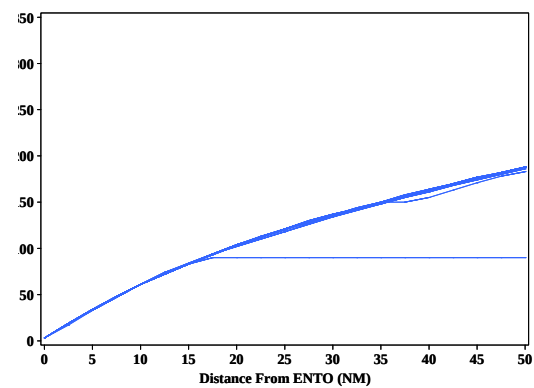
Vertical Profile for Departures
SID : COS1G



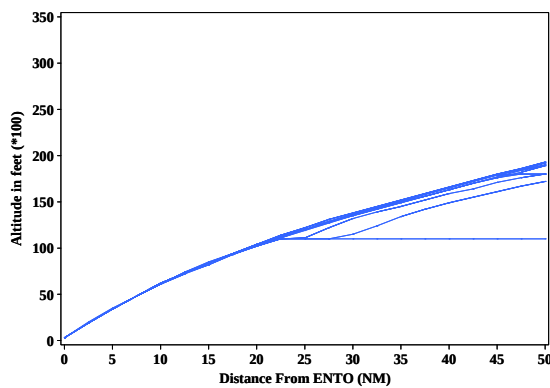
Vertical Profile for Departures
SID : GOK1G



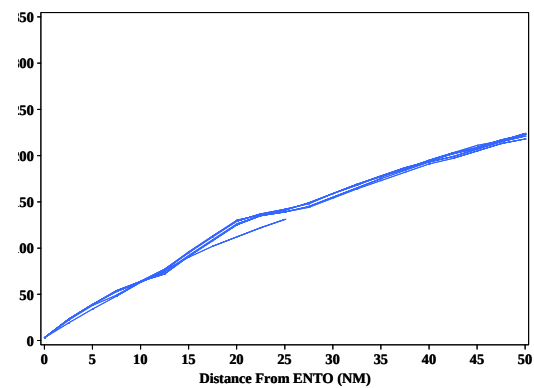
Vertical Profile for Departures
SID : ING1G



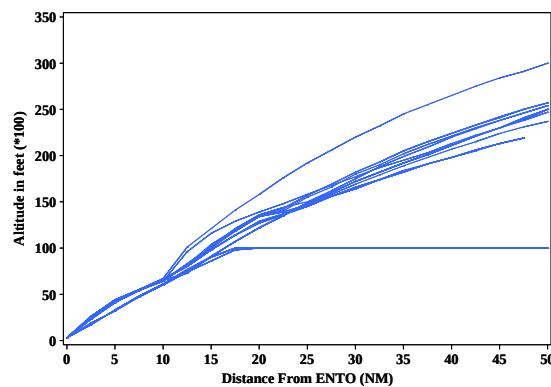
Vertical Profile for Departures
SID : MAR1G



Vertical Profile for Departures
SID : SV11G

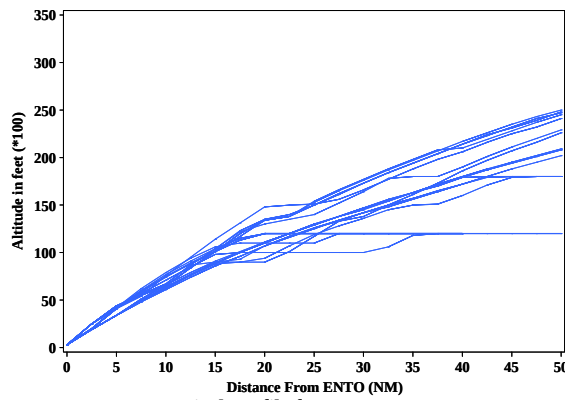


Vertical Profile for Departures
SID : WAC1G



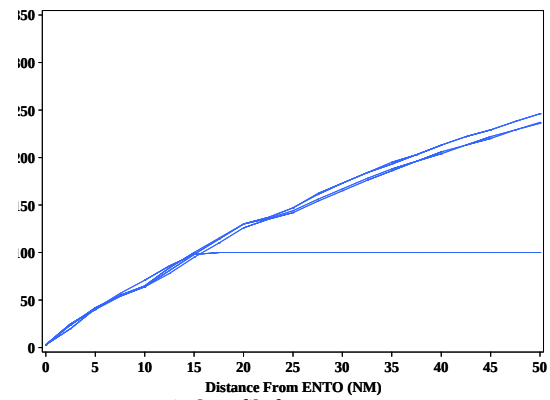
Vertical Profile for Departures

SID : BAM1H



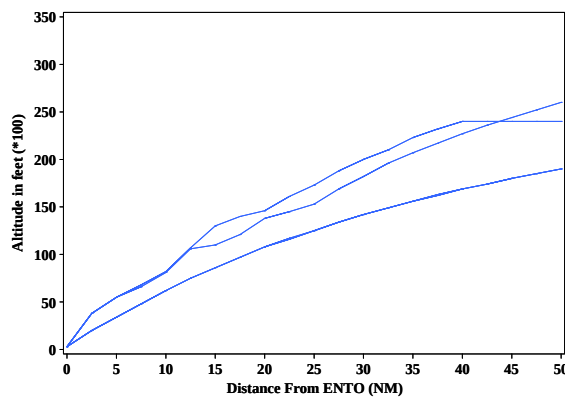
Vertical Profile for Departures

SID : COS1H



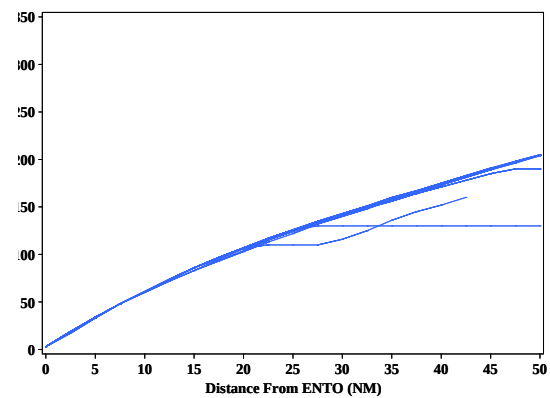
Vertical Profile for Departures

SID : GOK1H



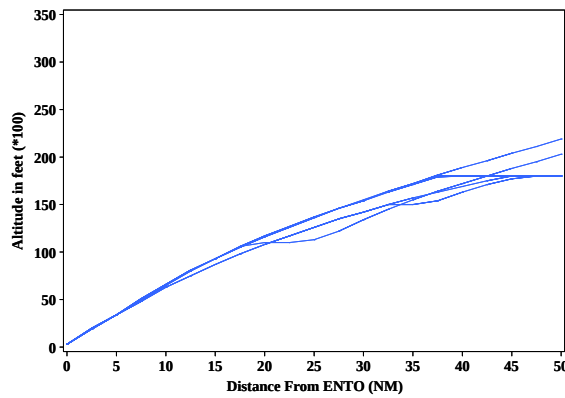
Vertical Profile for Departures

SID : ING1H



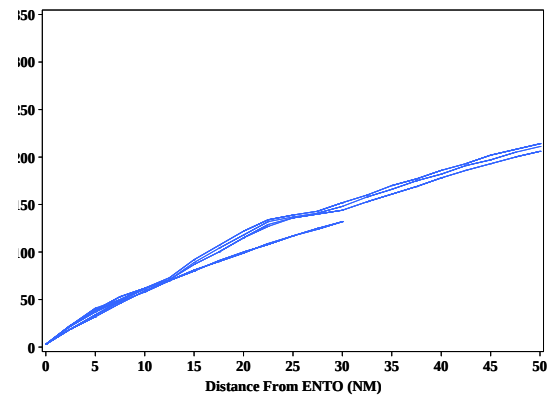
Vertical Profile for Departures

SID : MAR1H



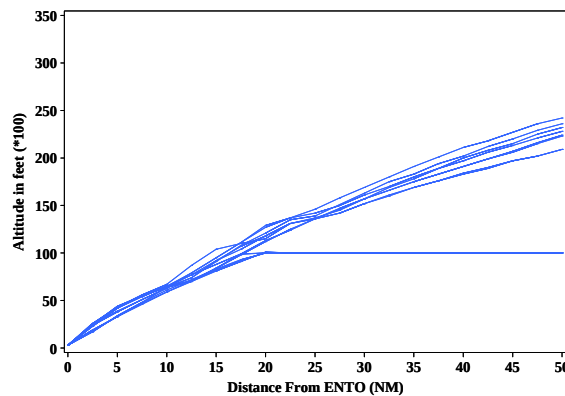
Vertical Profile for Departures

SID : SV11H



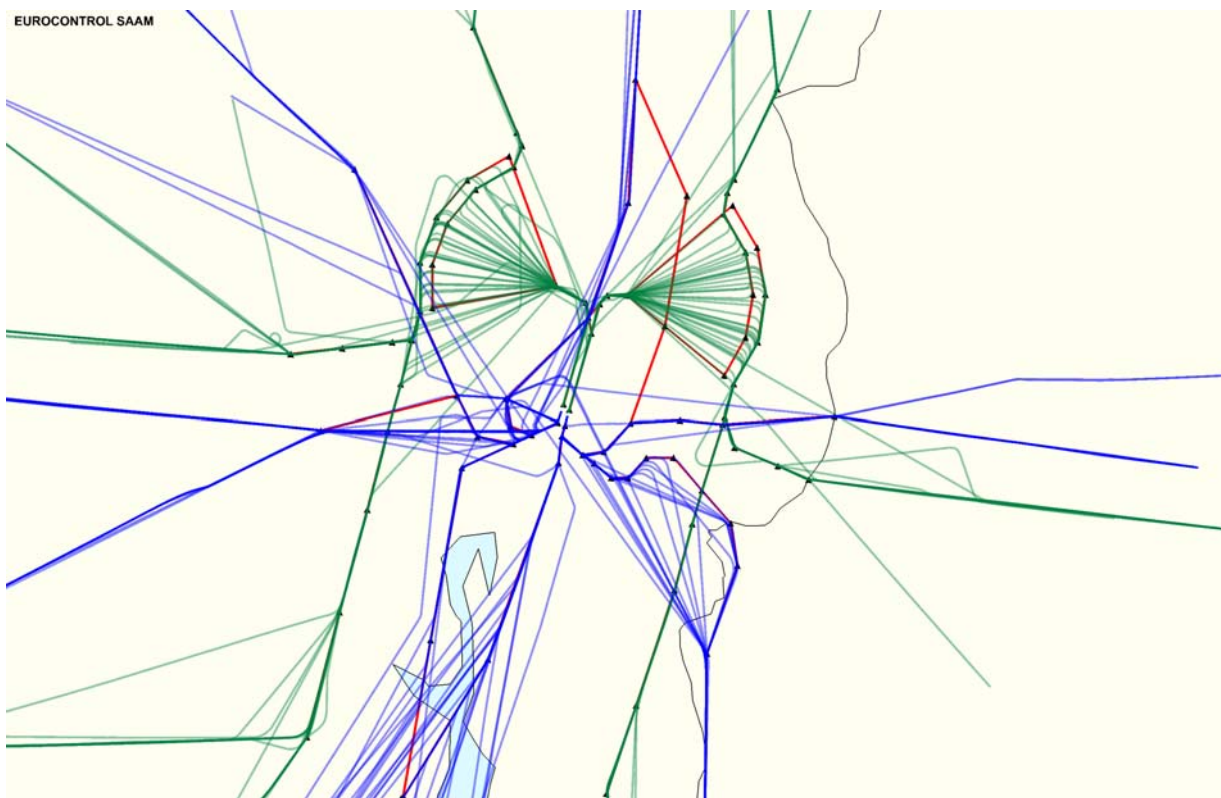
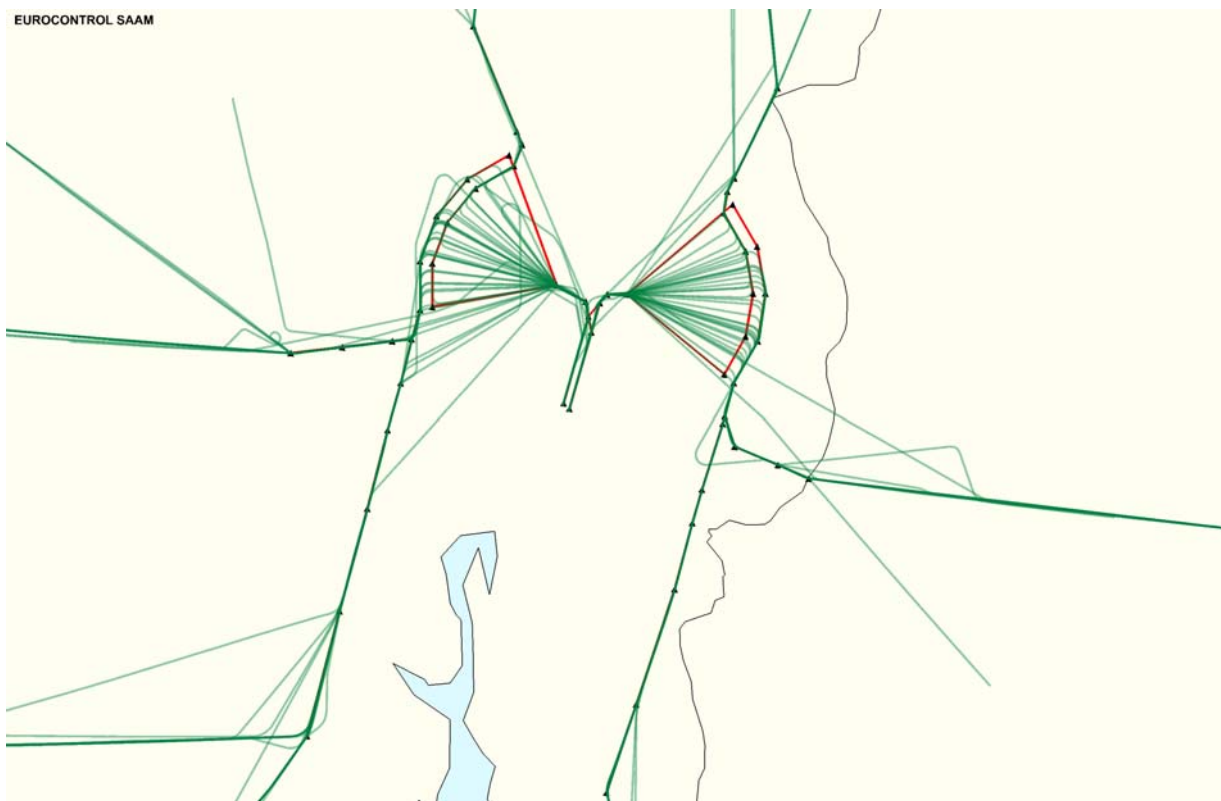
Vertical Profile for Departures

SID : WAC1H

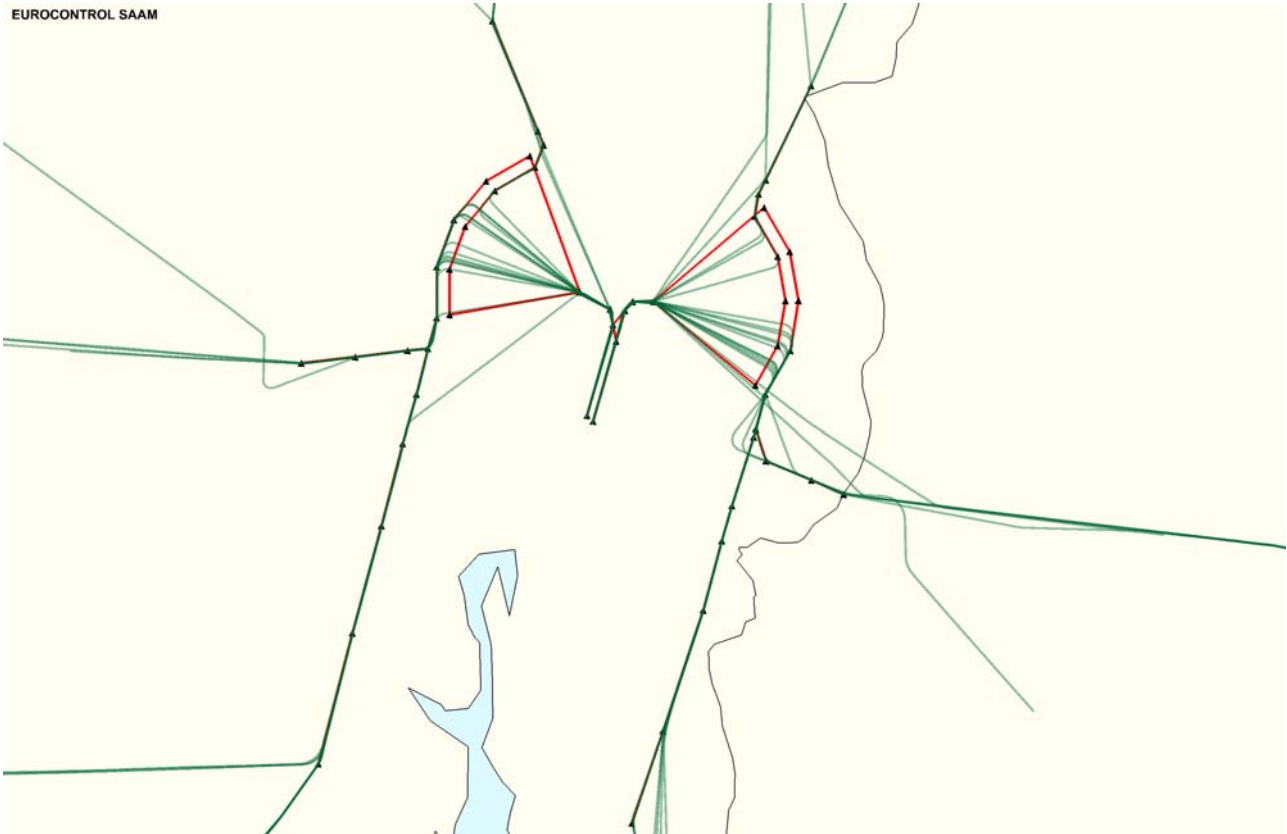


ANNEX M: OSLO TMA - 2D FLOWN TRAJECTORIES

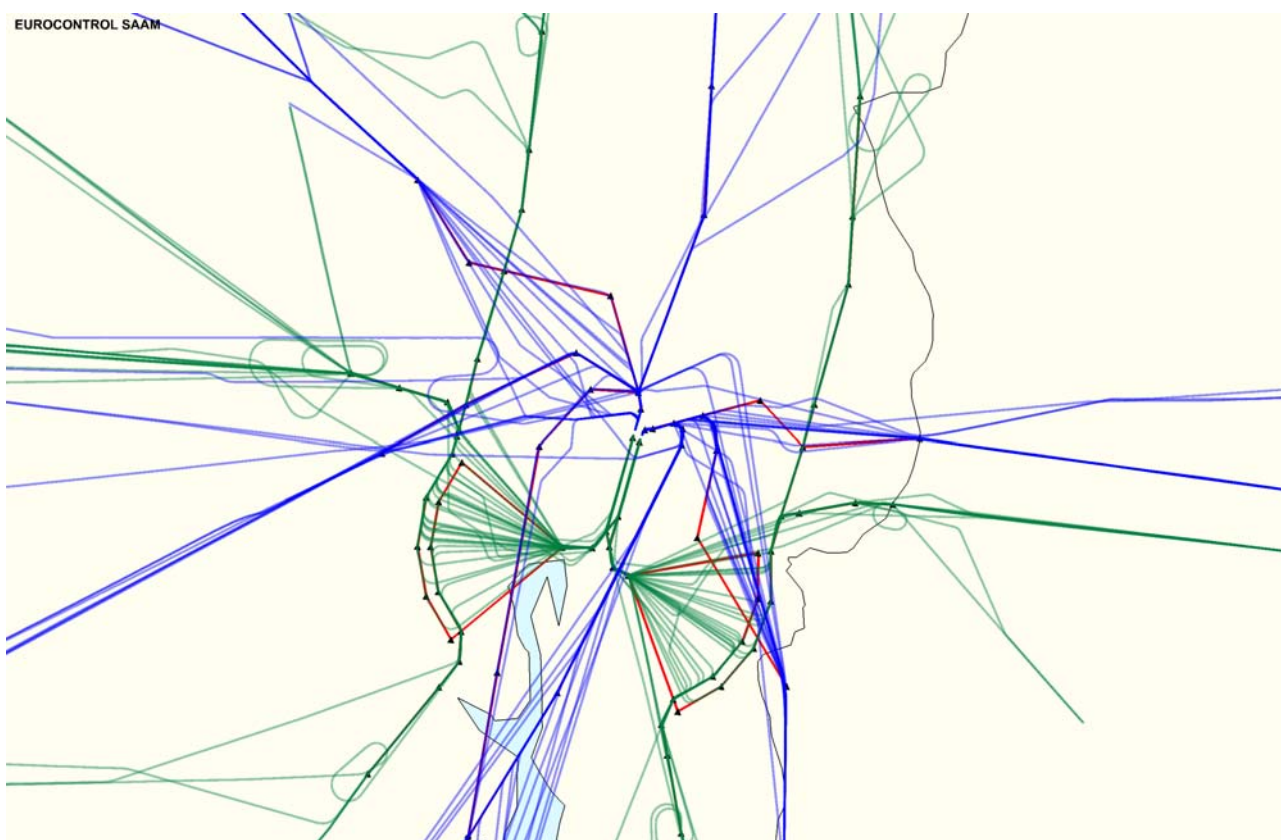
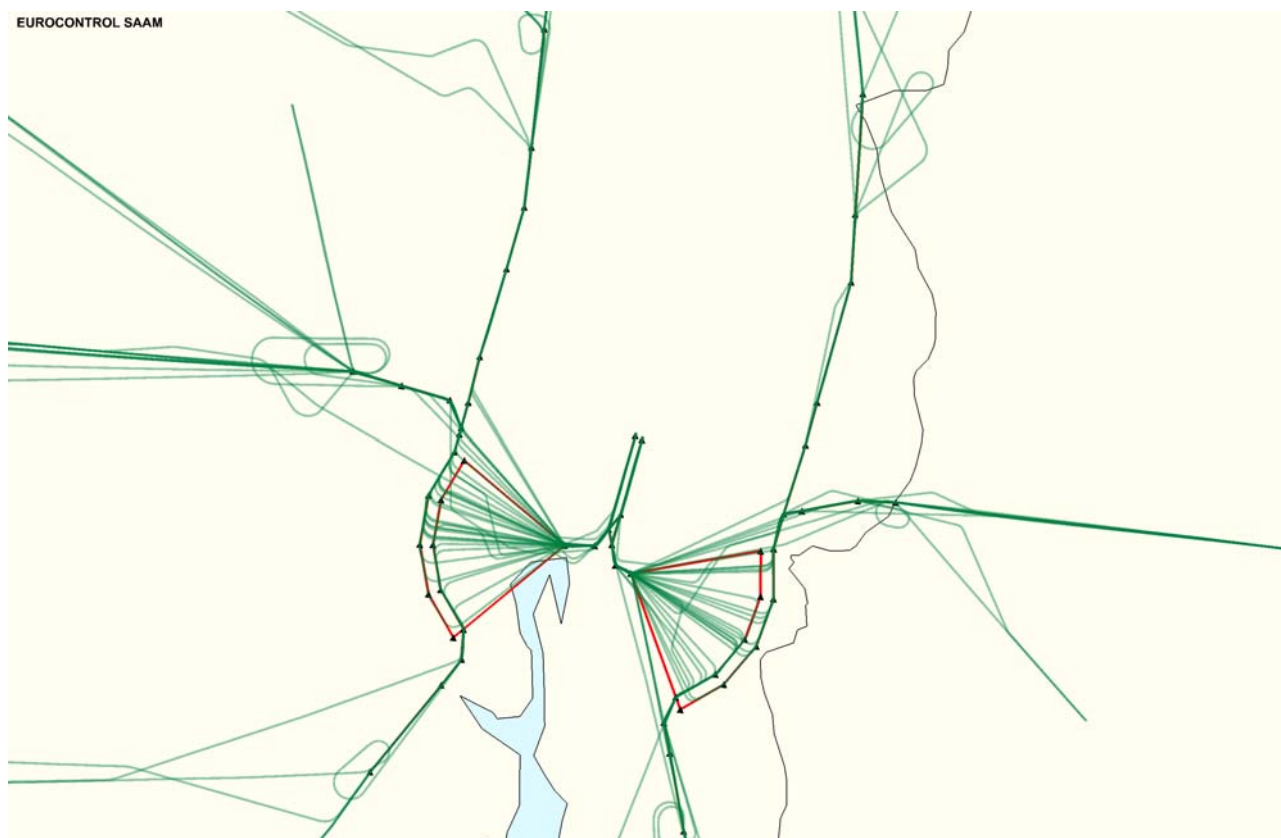
EXE 1 – ENGM RWY 19 (MPO)



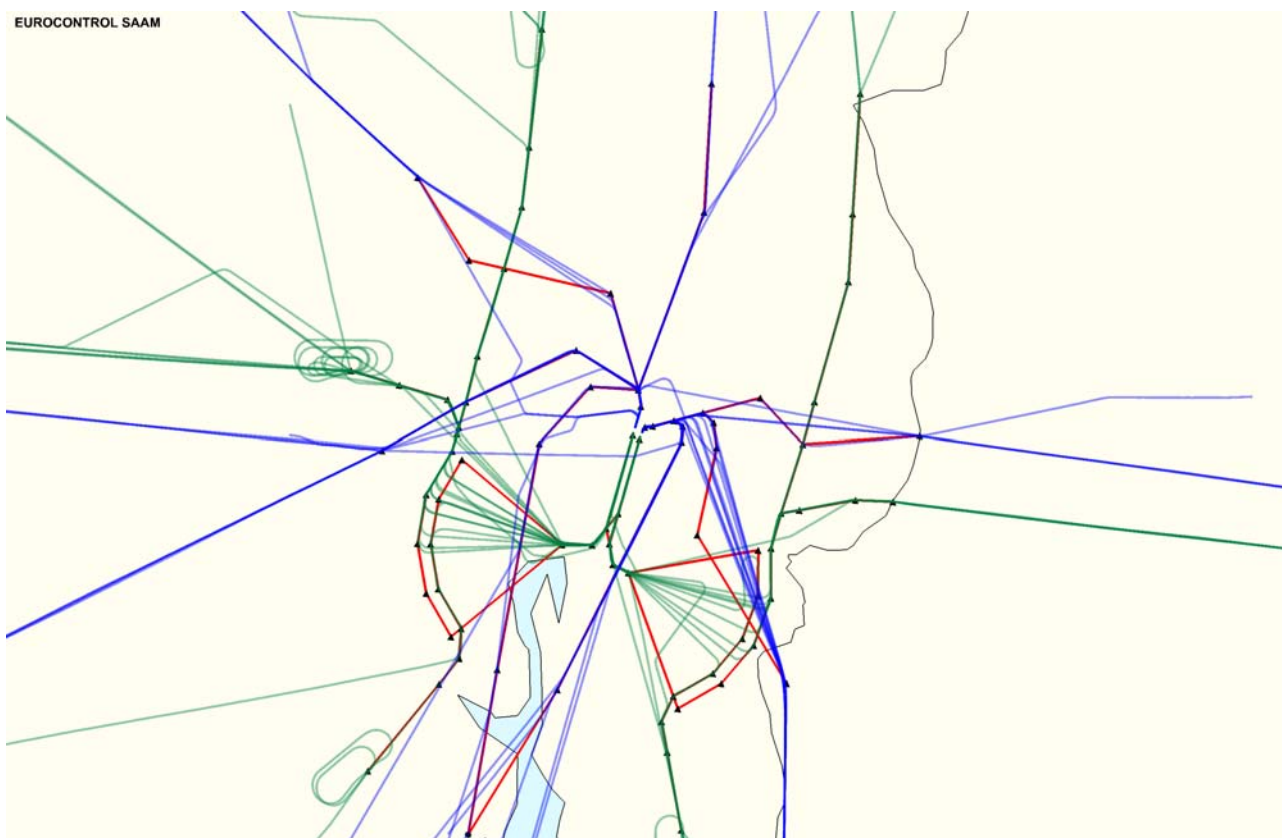
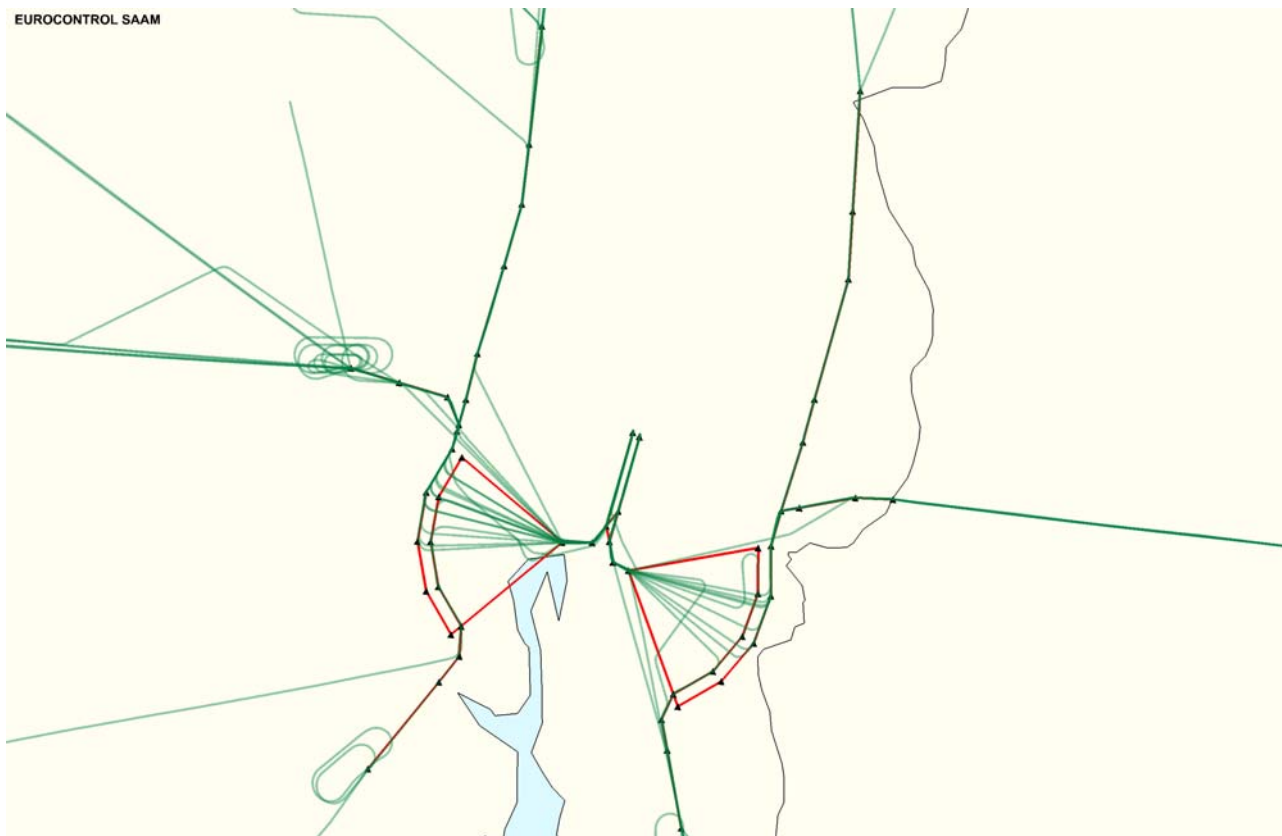
EXE 1-1 – ENGM RWY 19 (MPO)



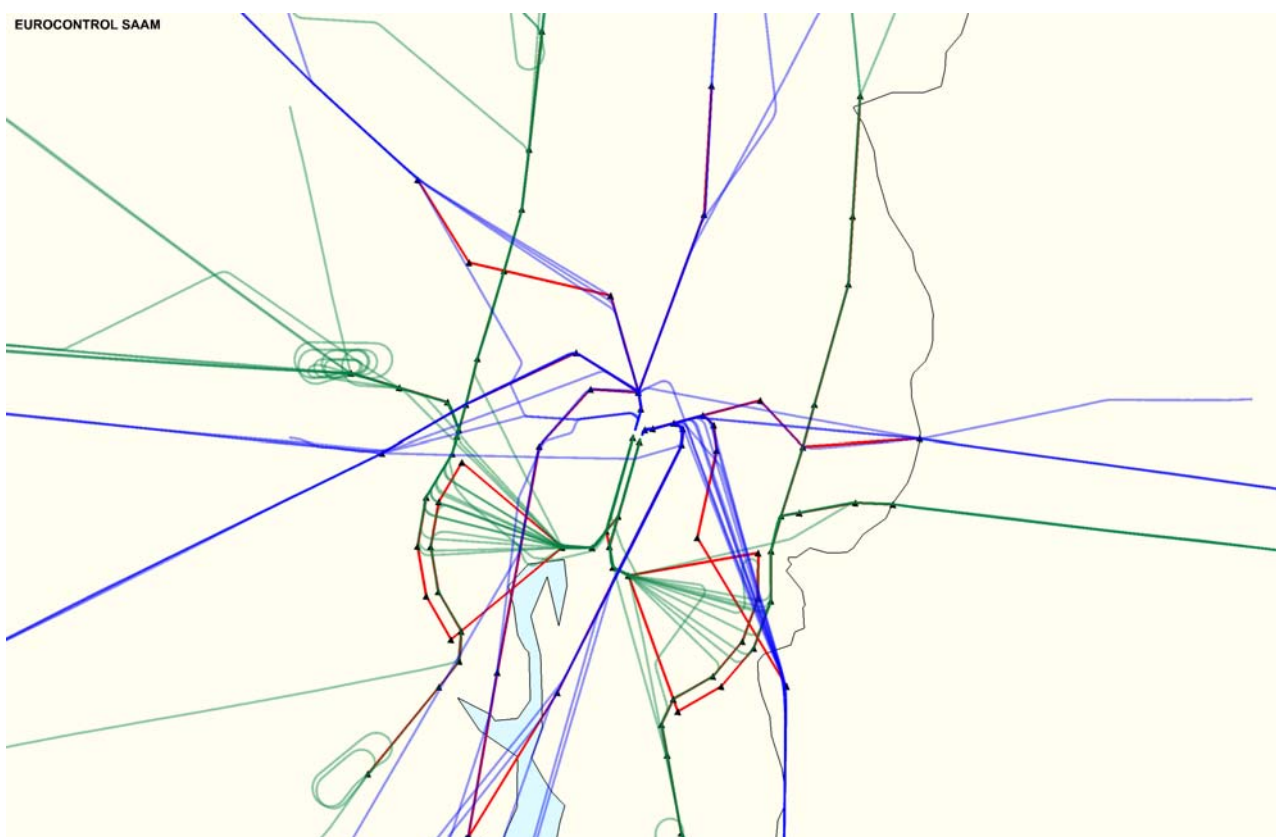
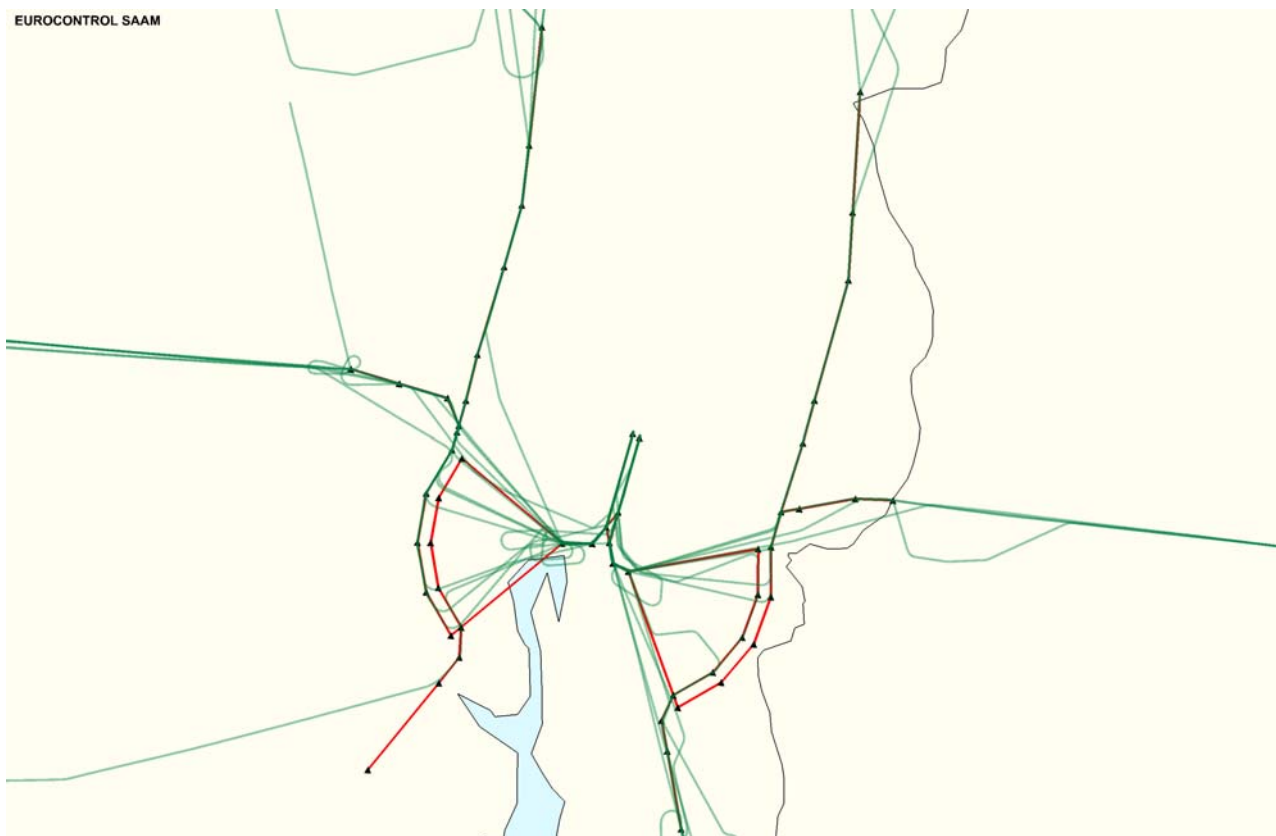
EXE 2 – ENGM RWY 01 (MPO)



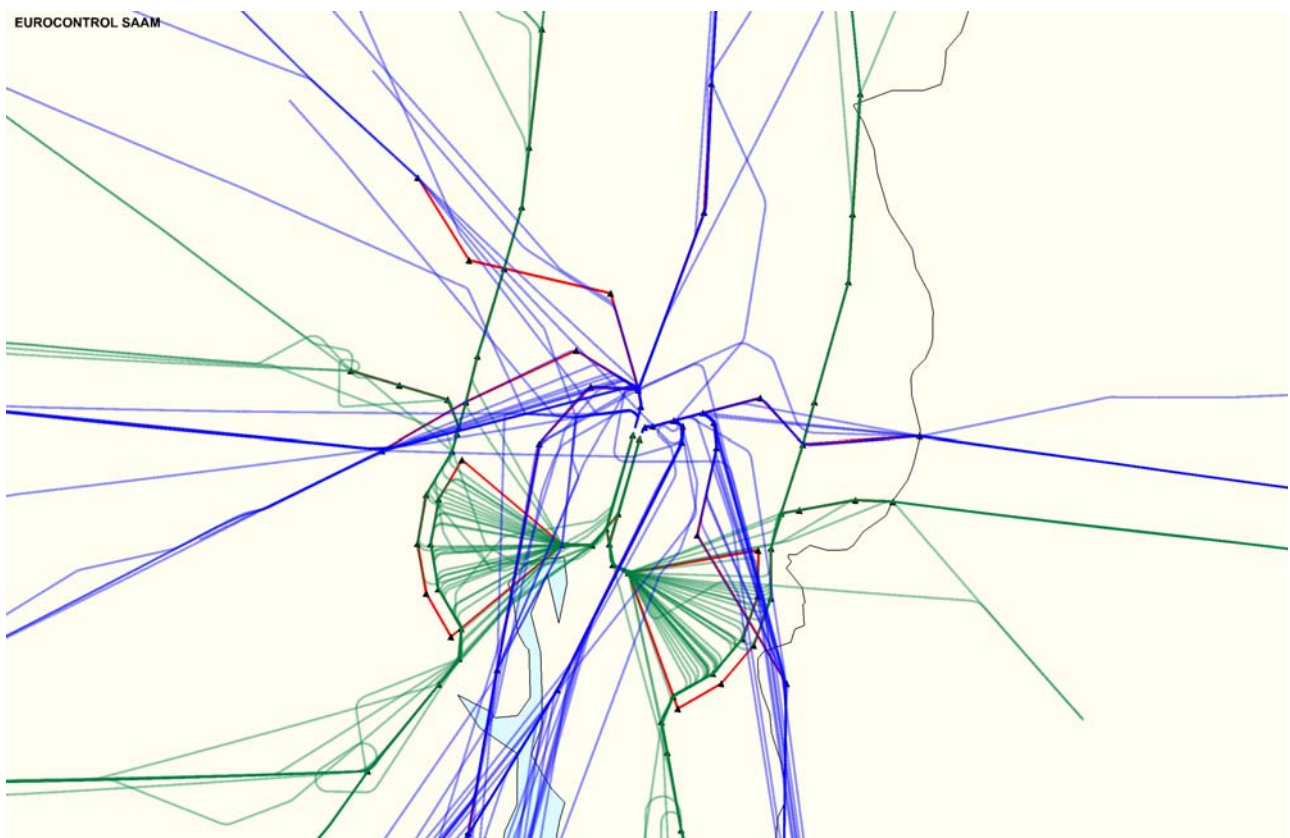
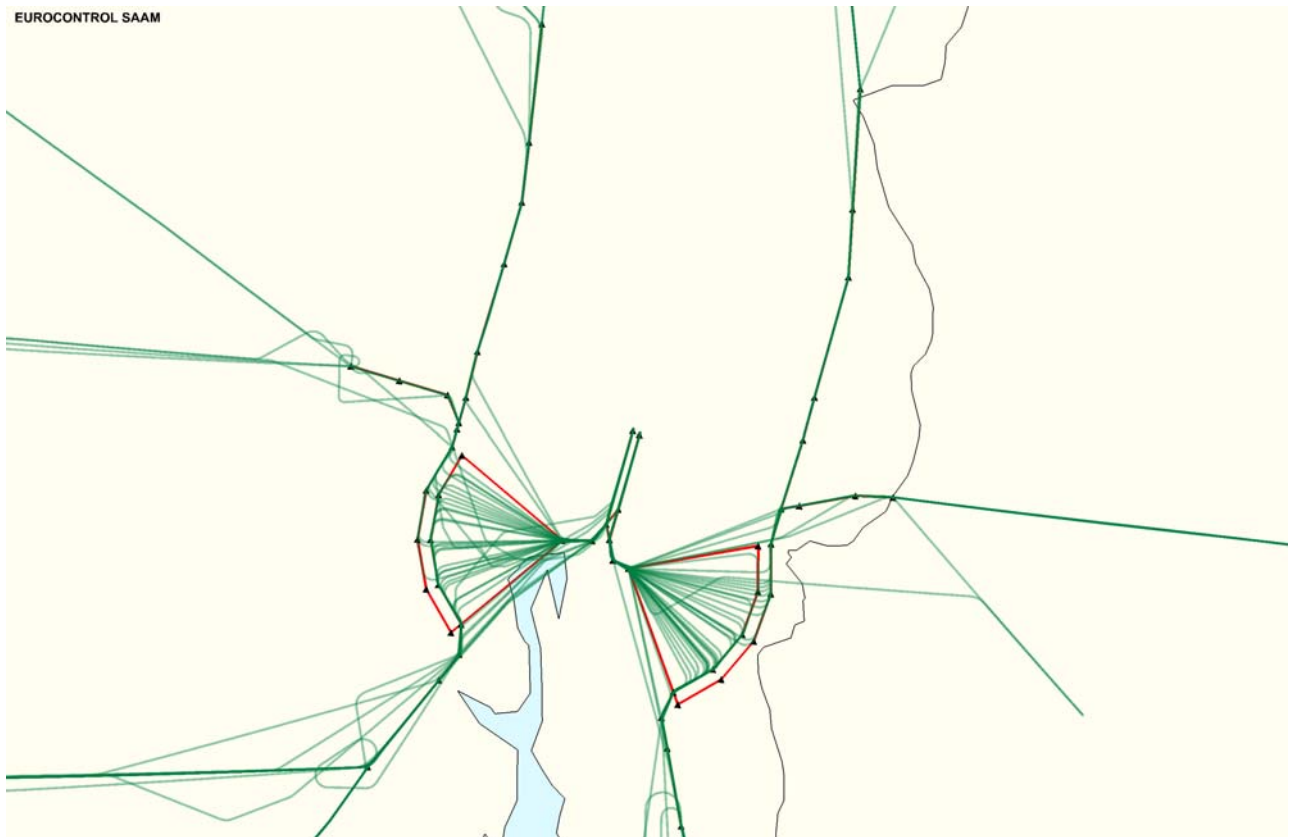
EXE 2-1 (13th March) – ENGM RWY 01 (MPO)



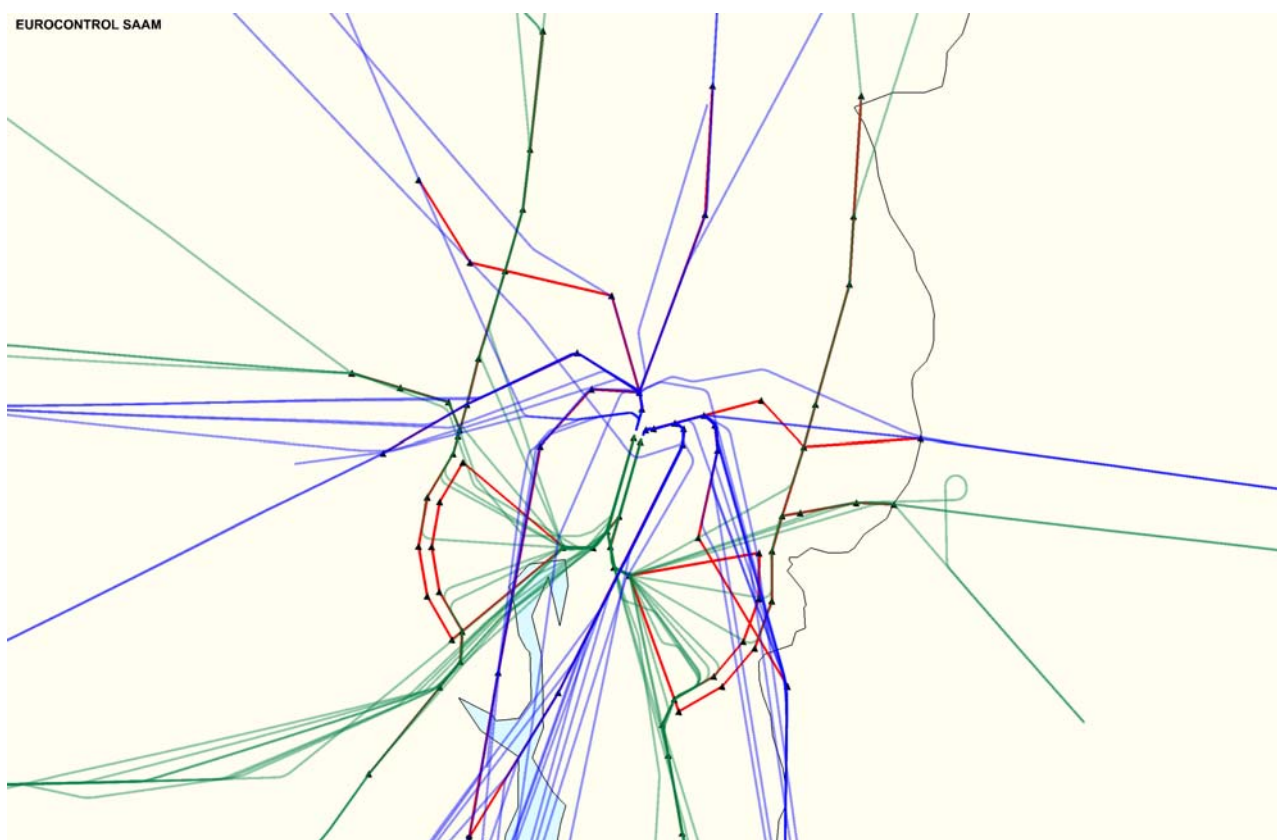
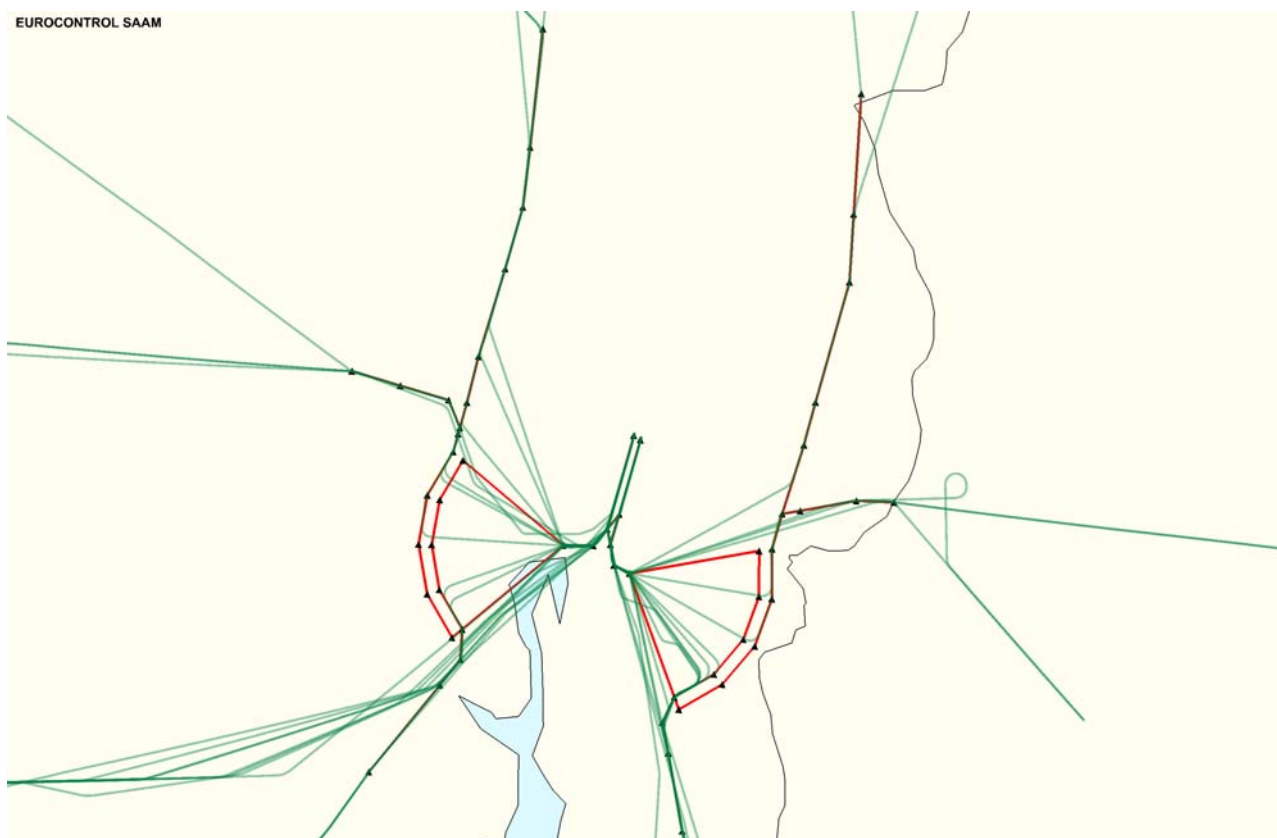
EXE 2-1 (18th March) – ENGM RWY 01 (MPO)



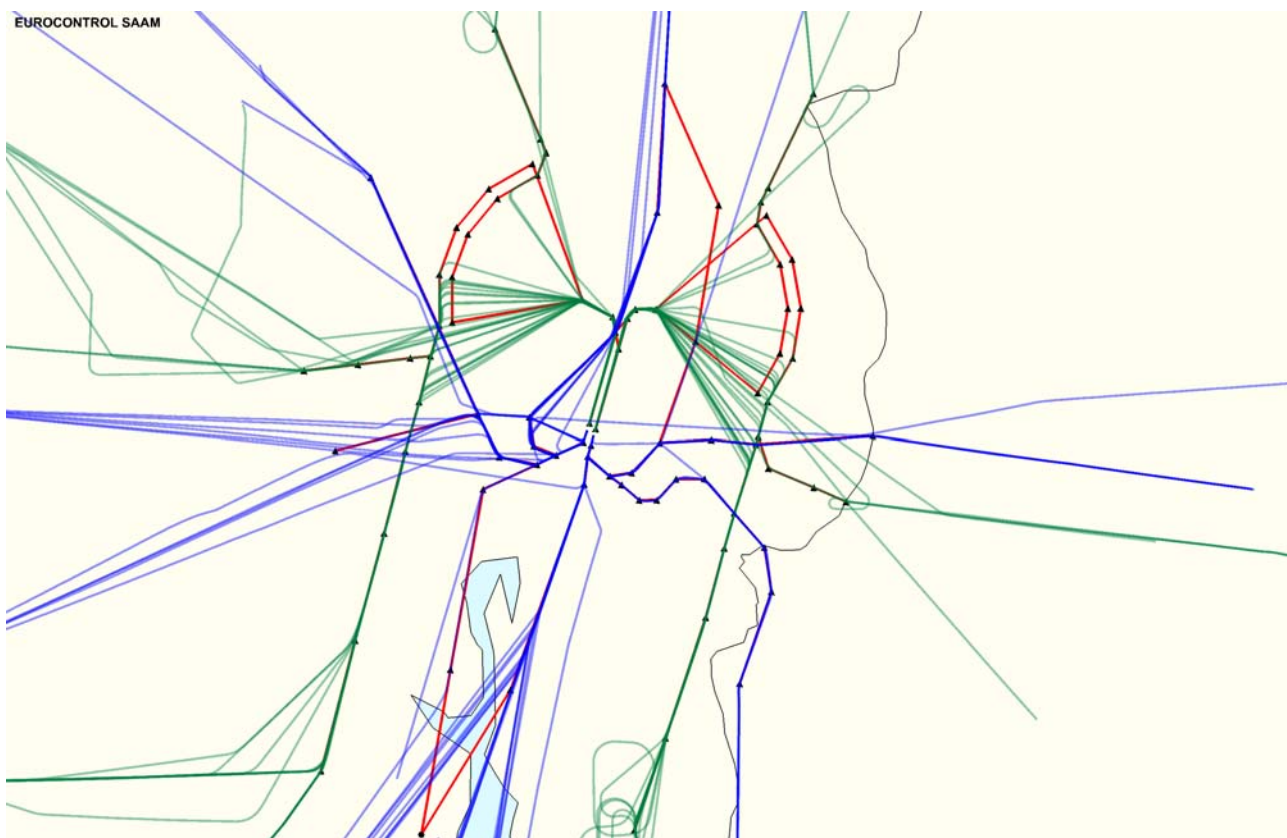
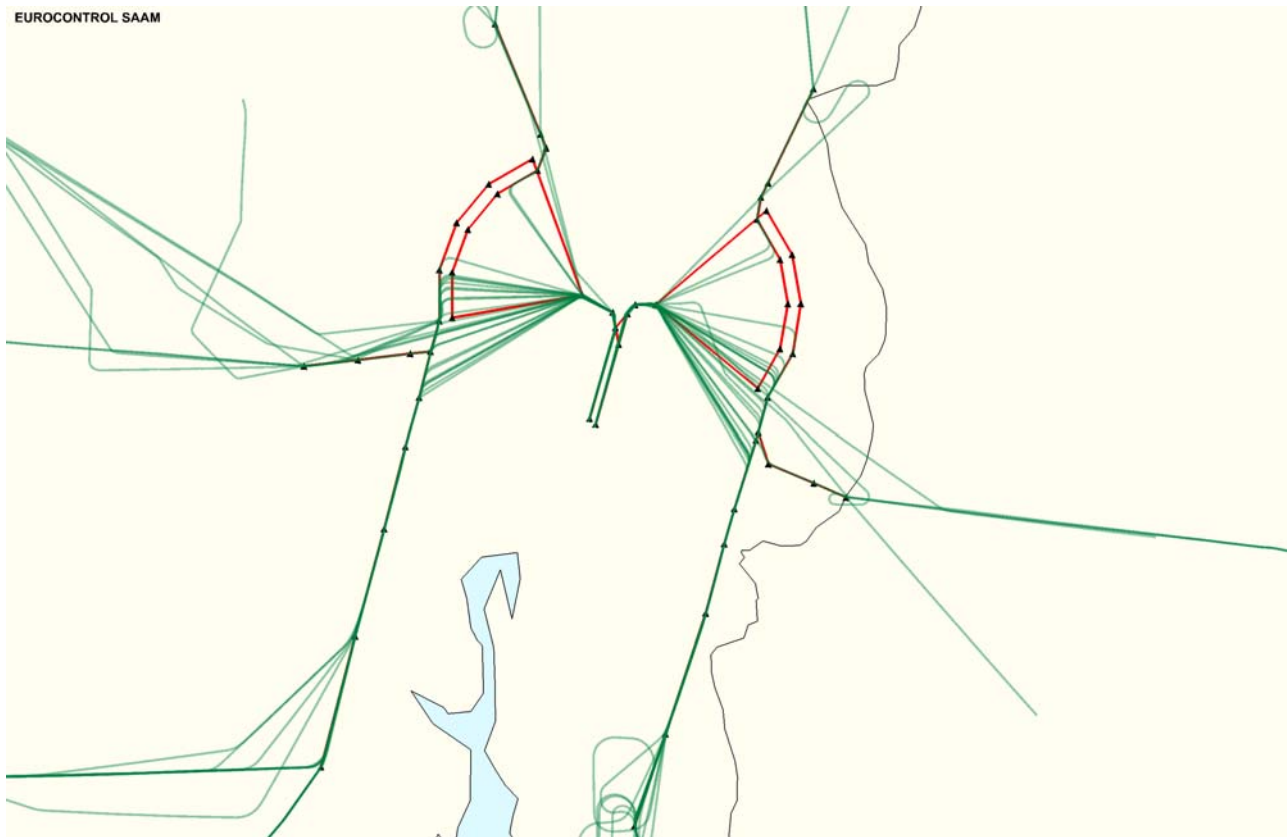
EXE 3 – ENGM RWY 01 (MPO)



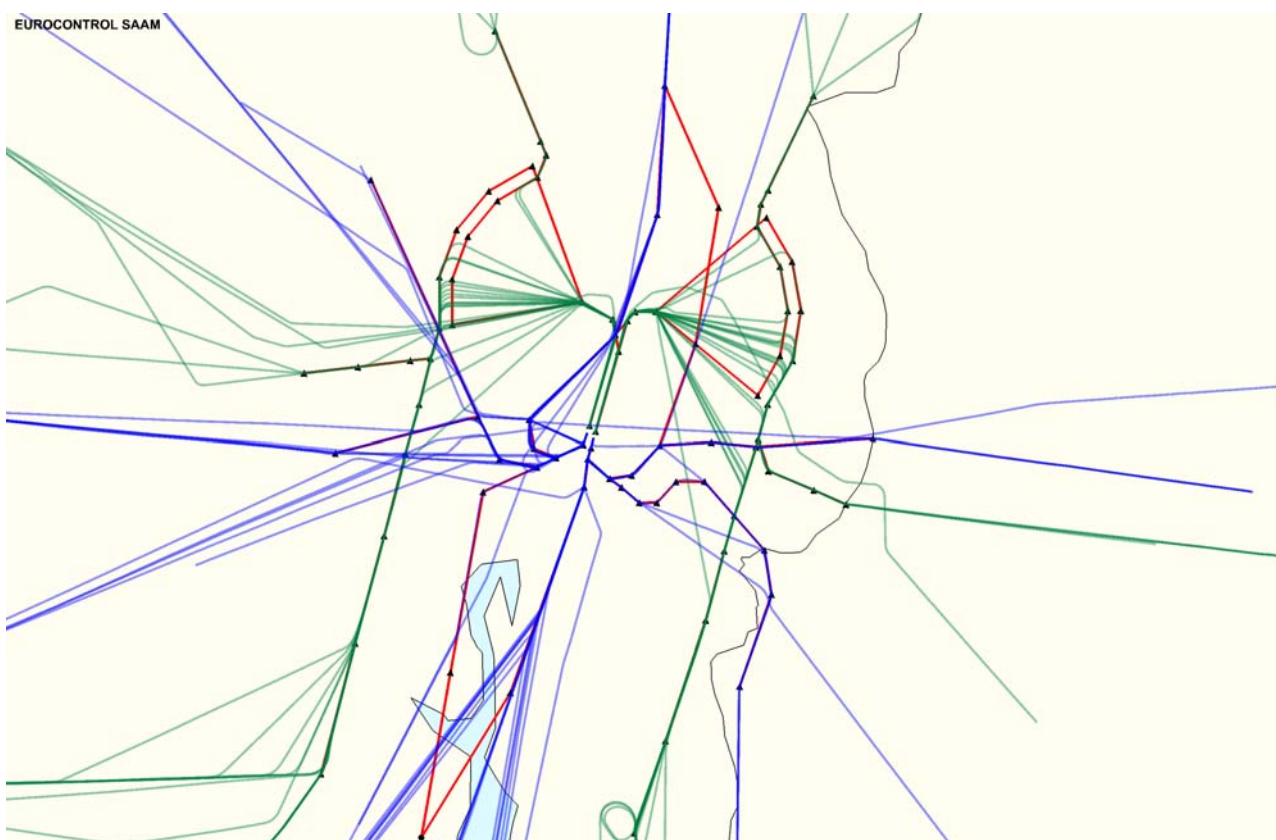
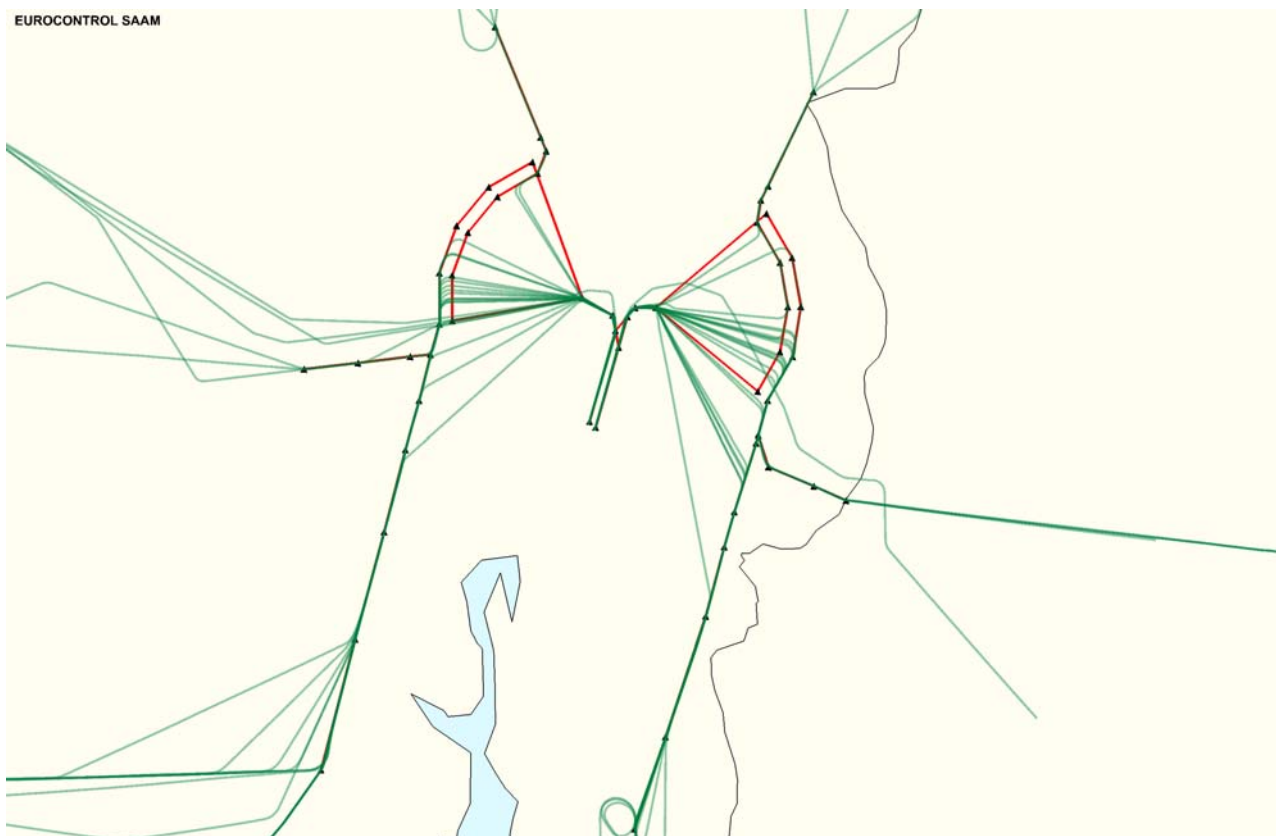
EXE 3-1 – ENGM RWY 01 (MPO)



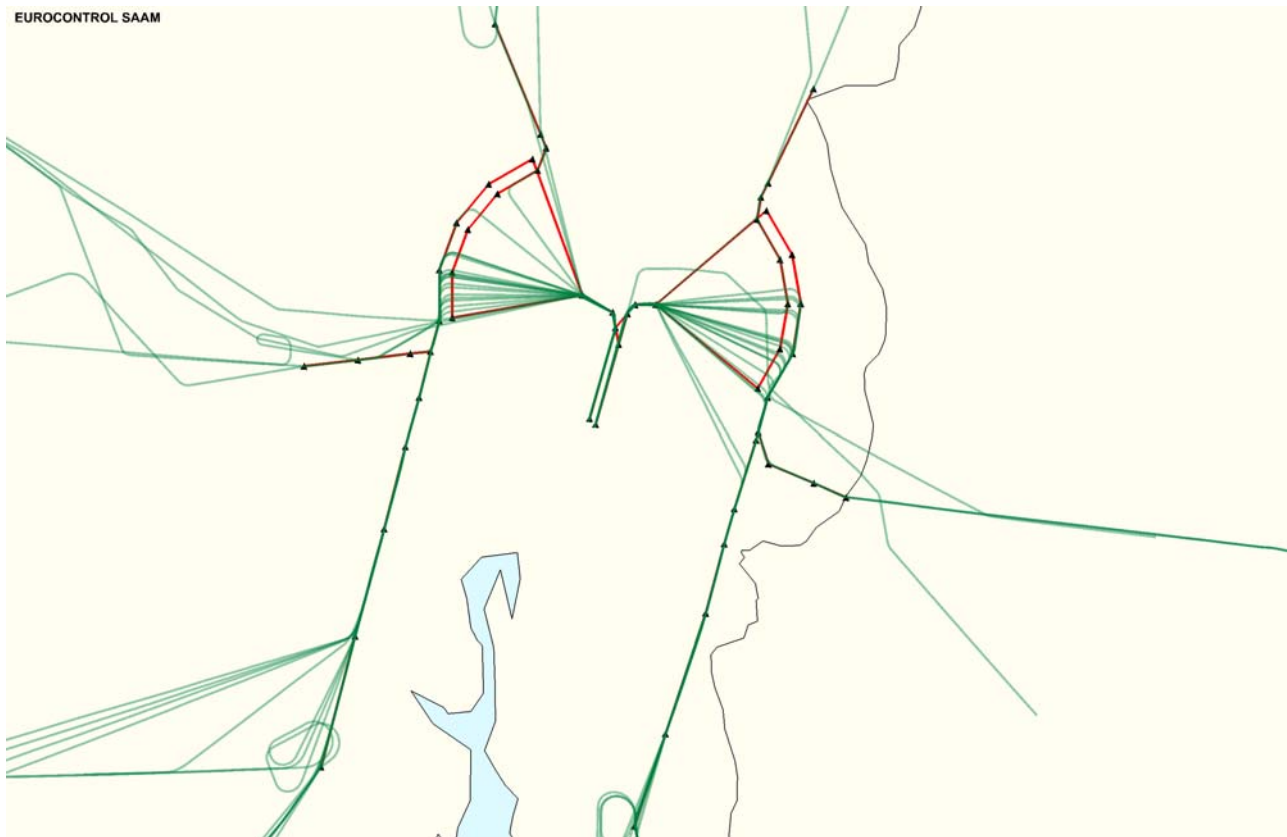
EXE 4 (12th March) – ENGM RWY 19 (IPA)



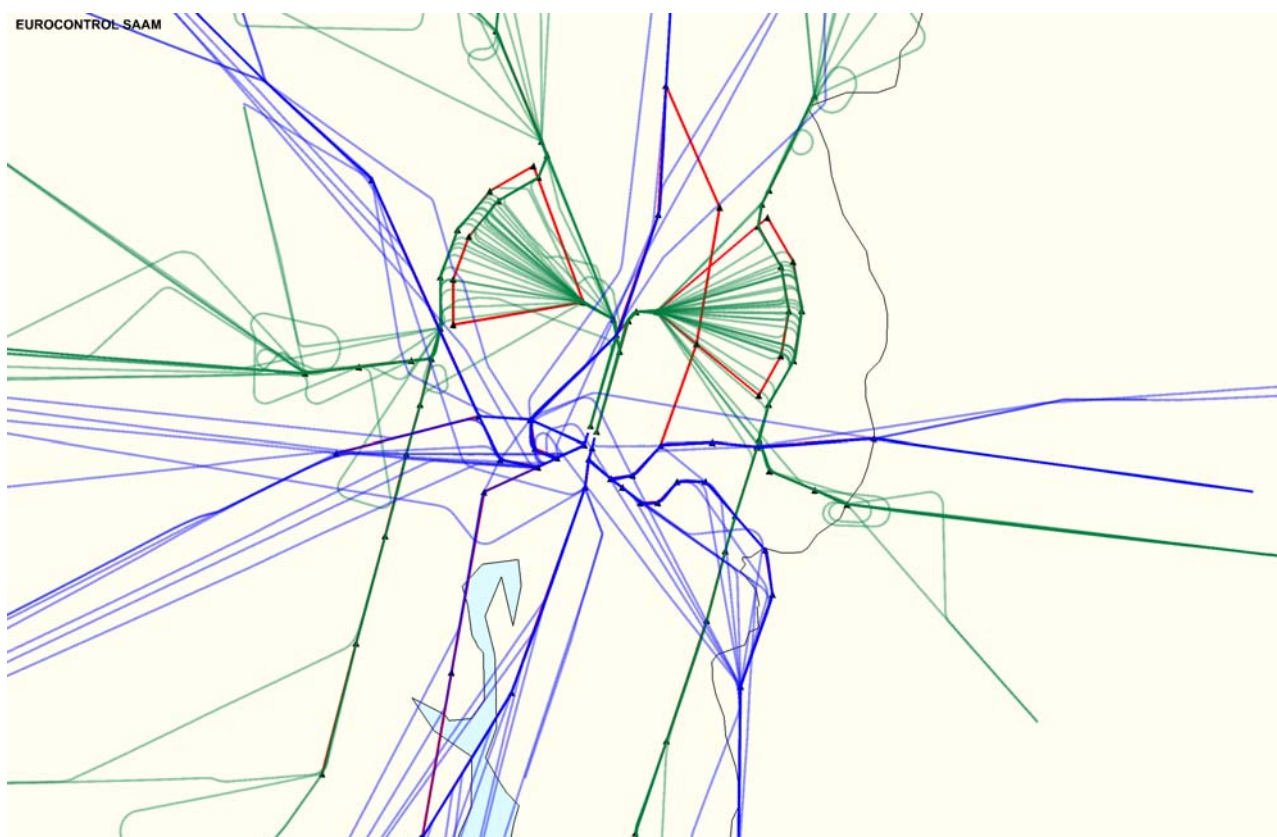
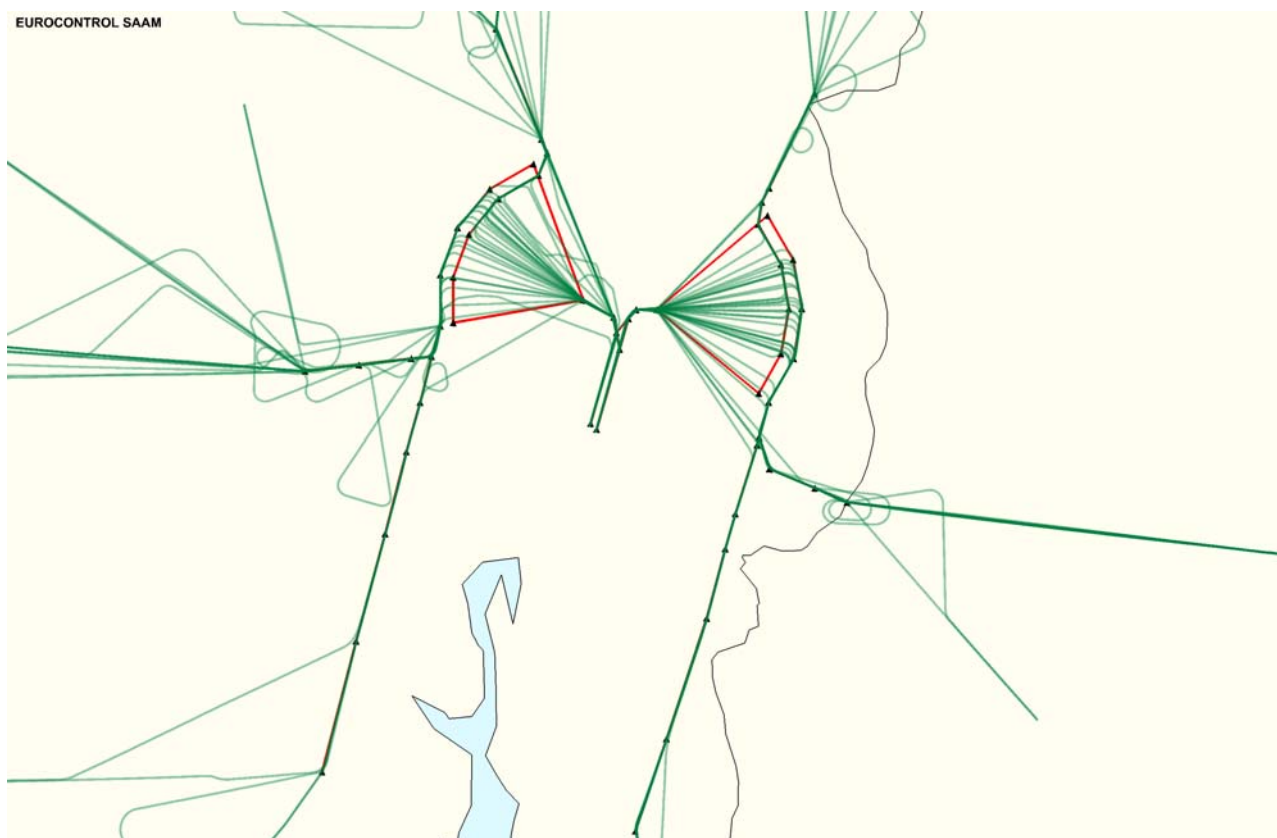
EXE 4 (13th March) – ENGM RWY 19 (IPA)



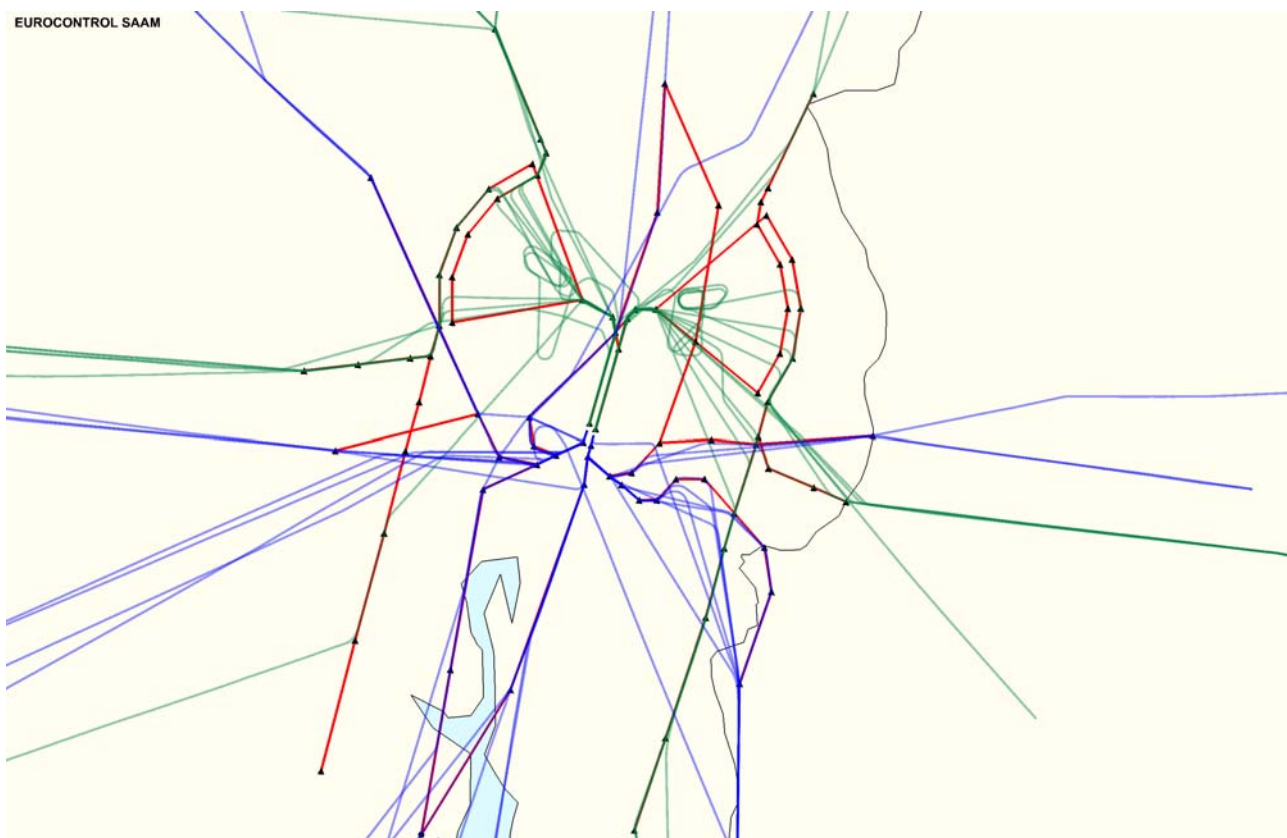
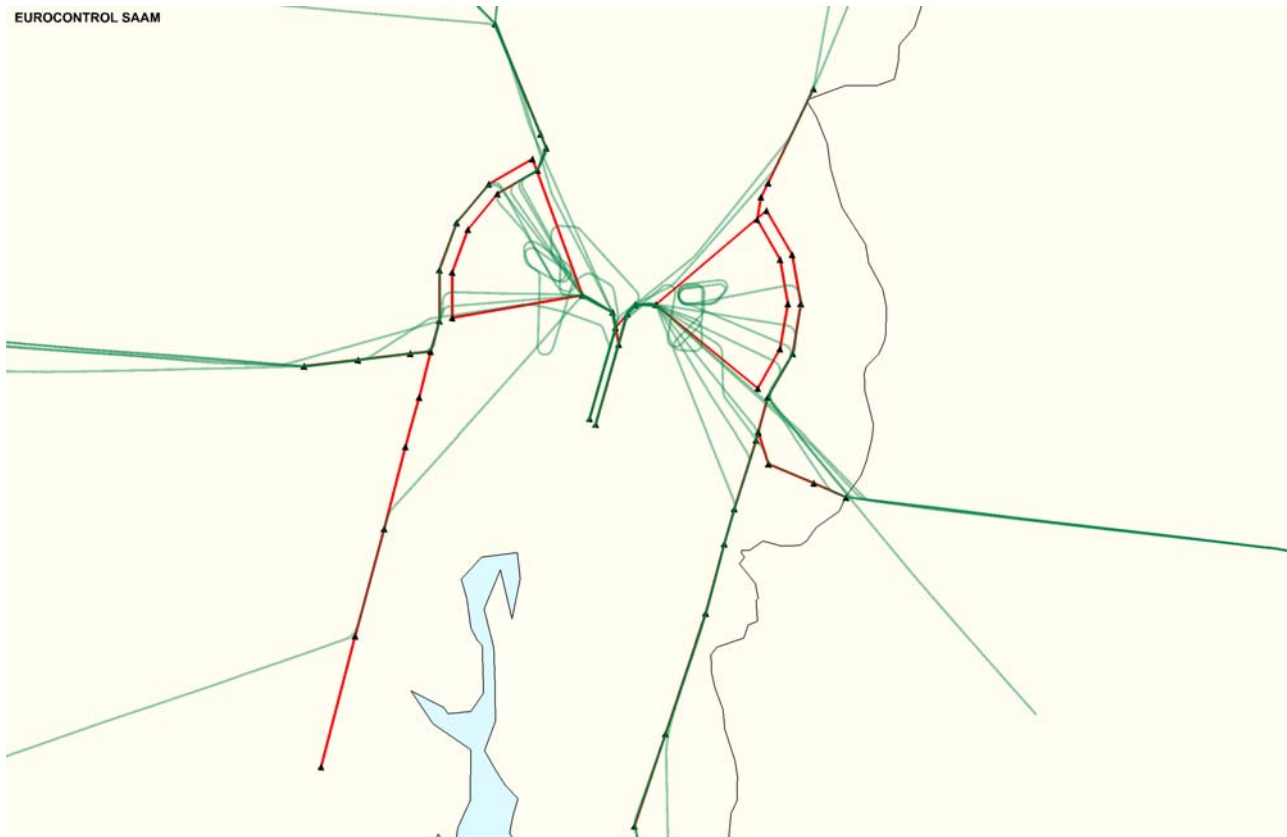
EXE 4 (16th March) – ENGM RWY 19 (IPA)



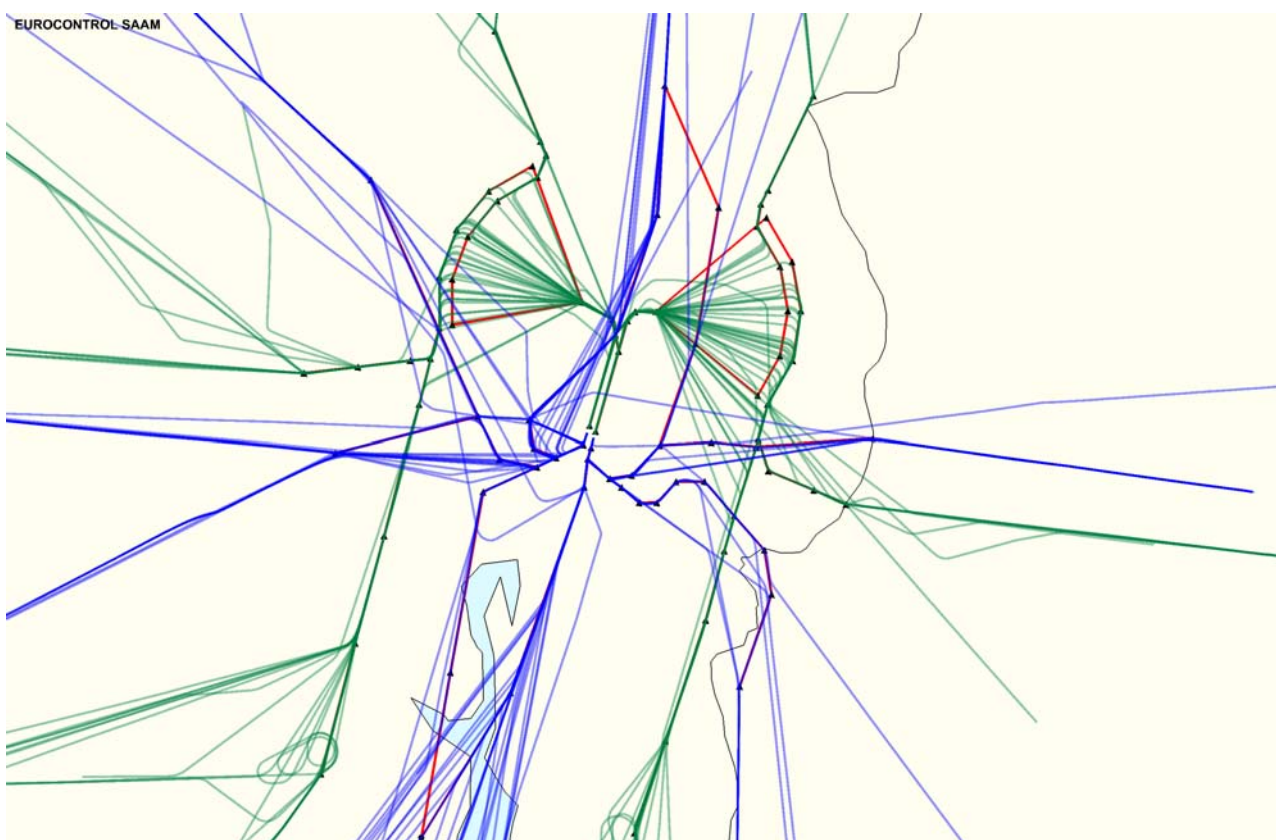
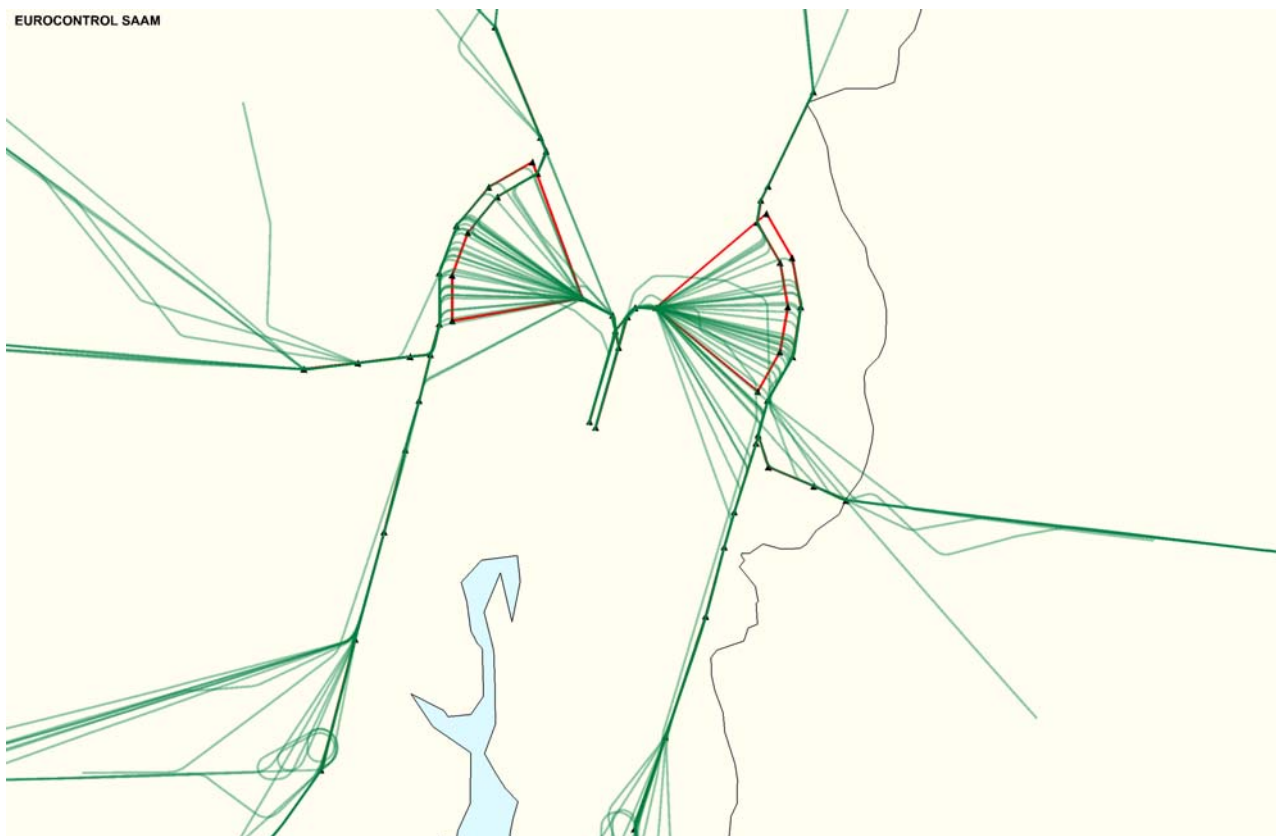
EXE 5 – ENGM RWY 19 (MPO)



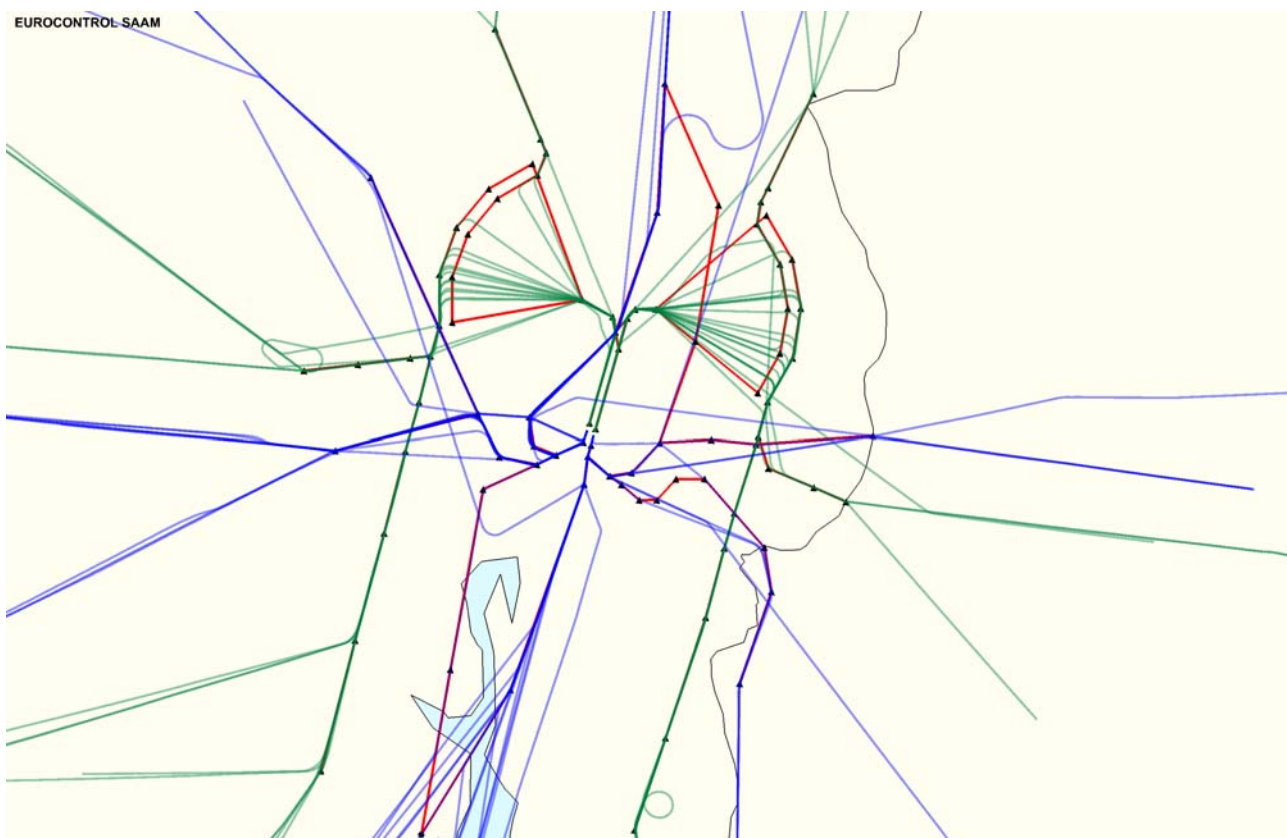
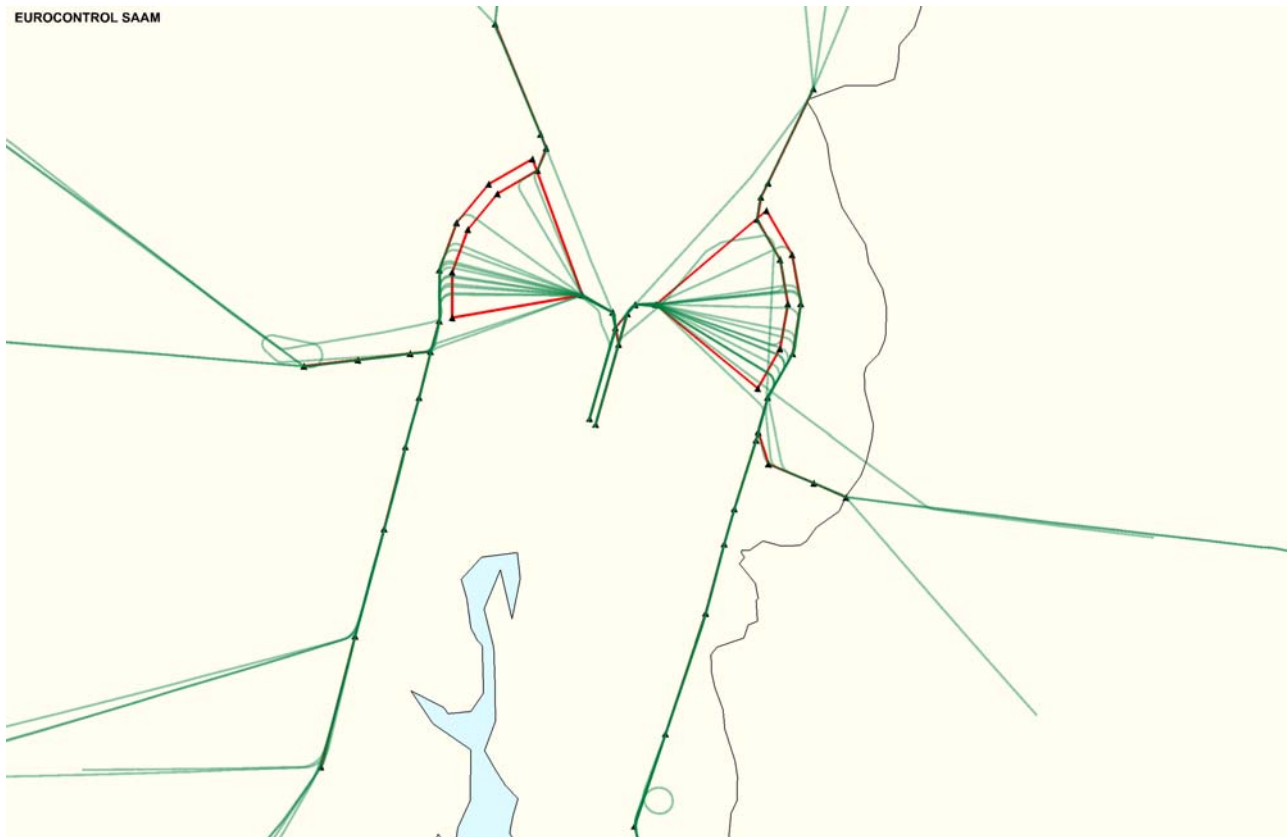
EX 5-1 – ENGM RWY 19 (MPO)



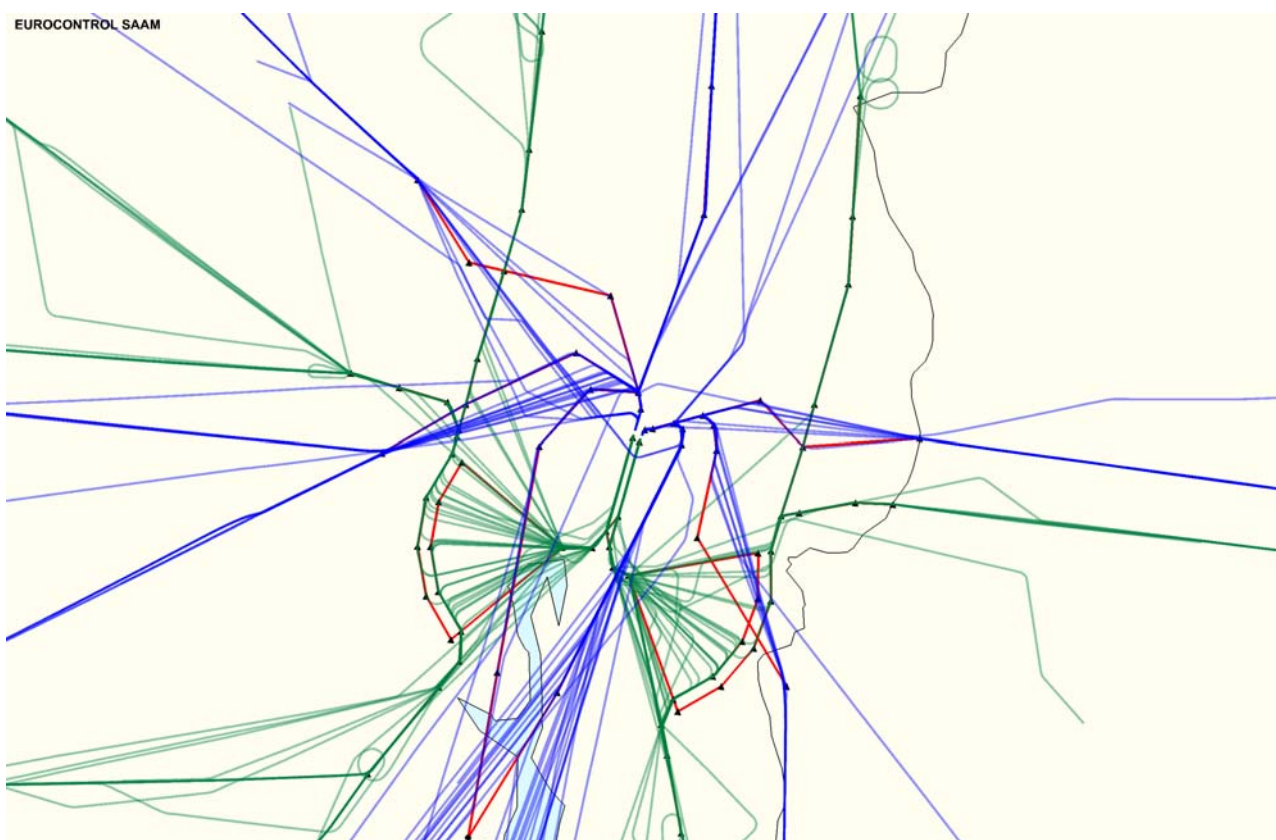
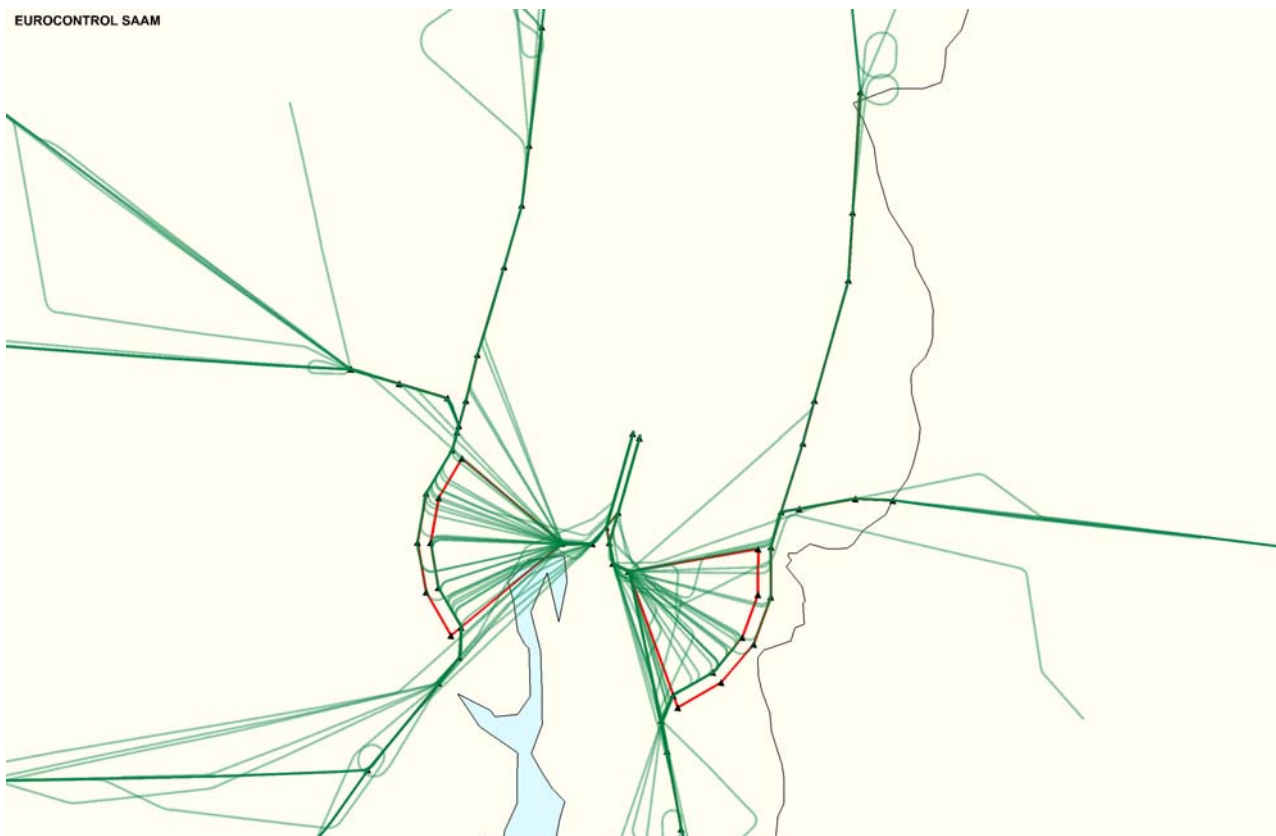
EXE 6 – ENGM RWY 19 (MPO)



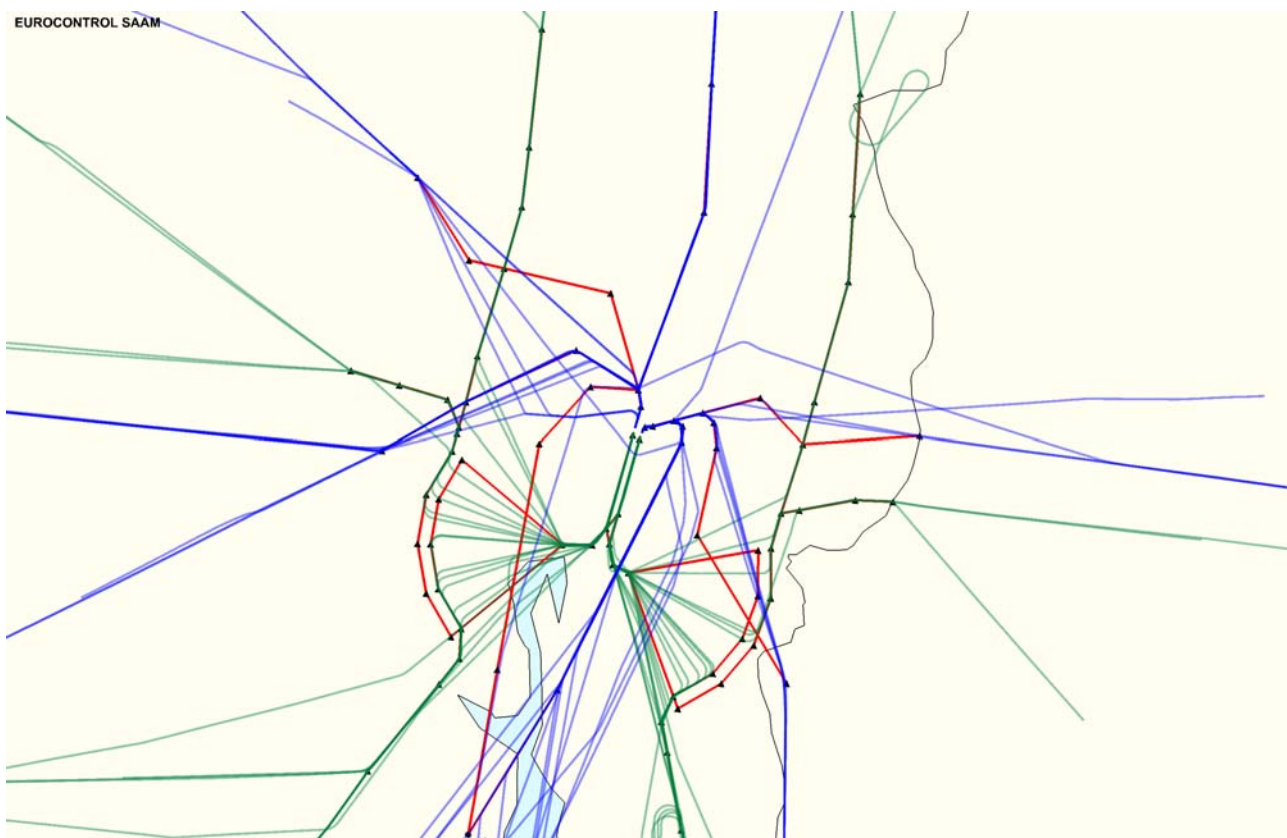
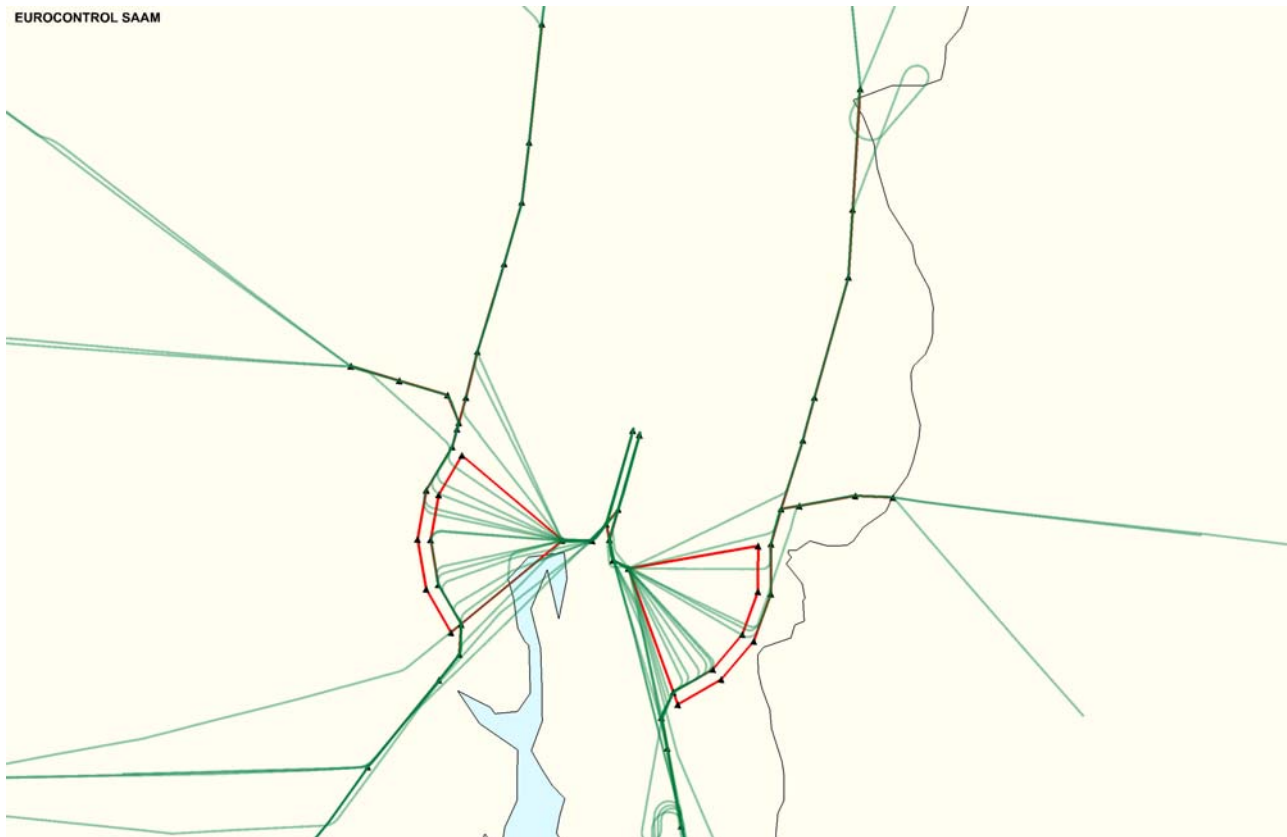
EXE 6-1 – ENGM RWY 19 (MPO)



EXE 7 – ENGM RWY 01 (MPO)



EXE 7-1 – ENGM RWY 01 (MPO)

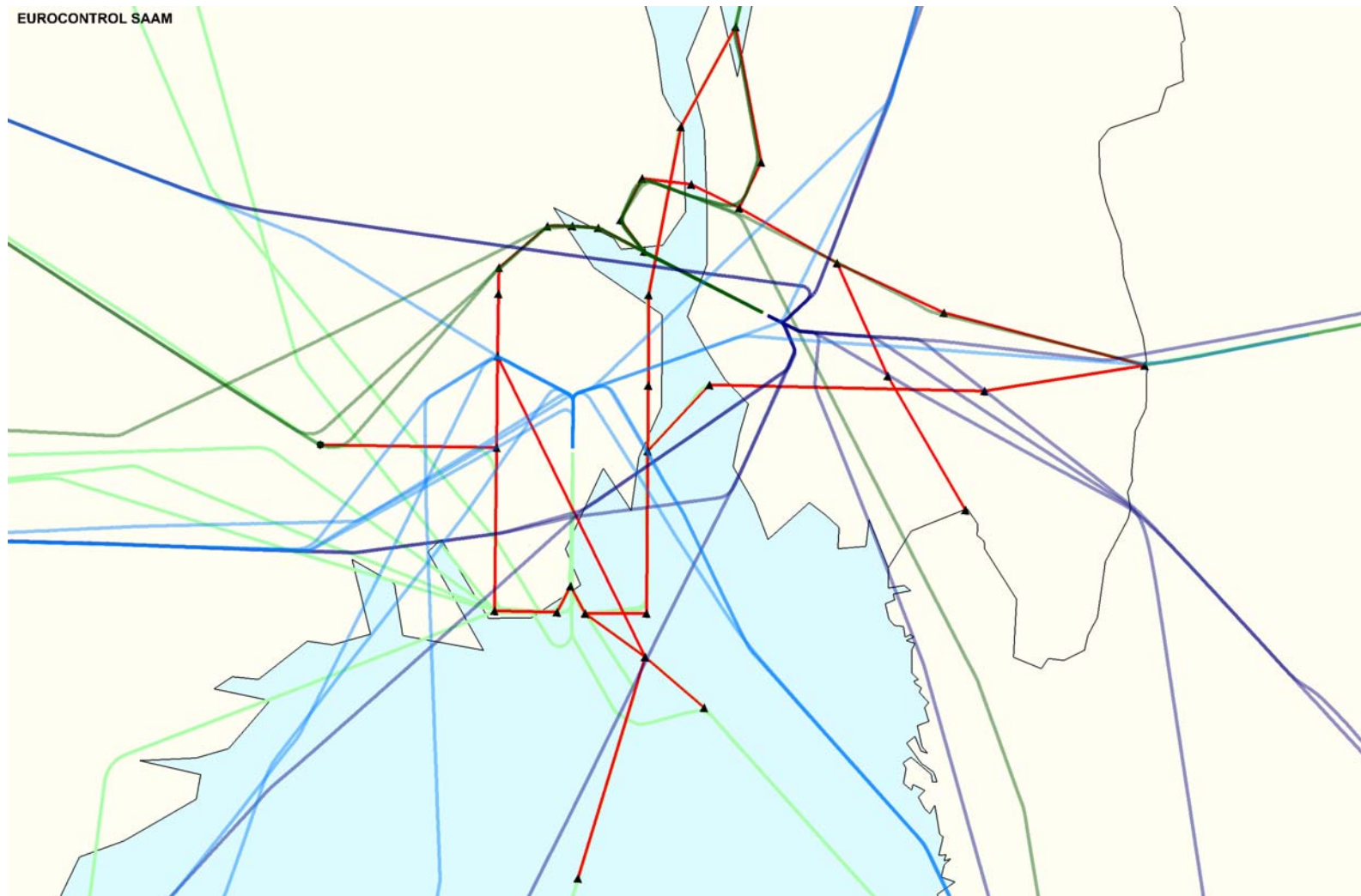


ANNEX N: FARRIS TMA - 2D FLOWN TRAJECTORIES

EXE 1 – ENTO RWY18 and ENRY RWY12



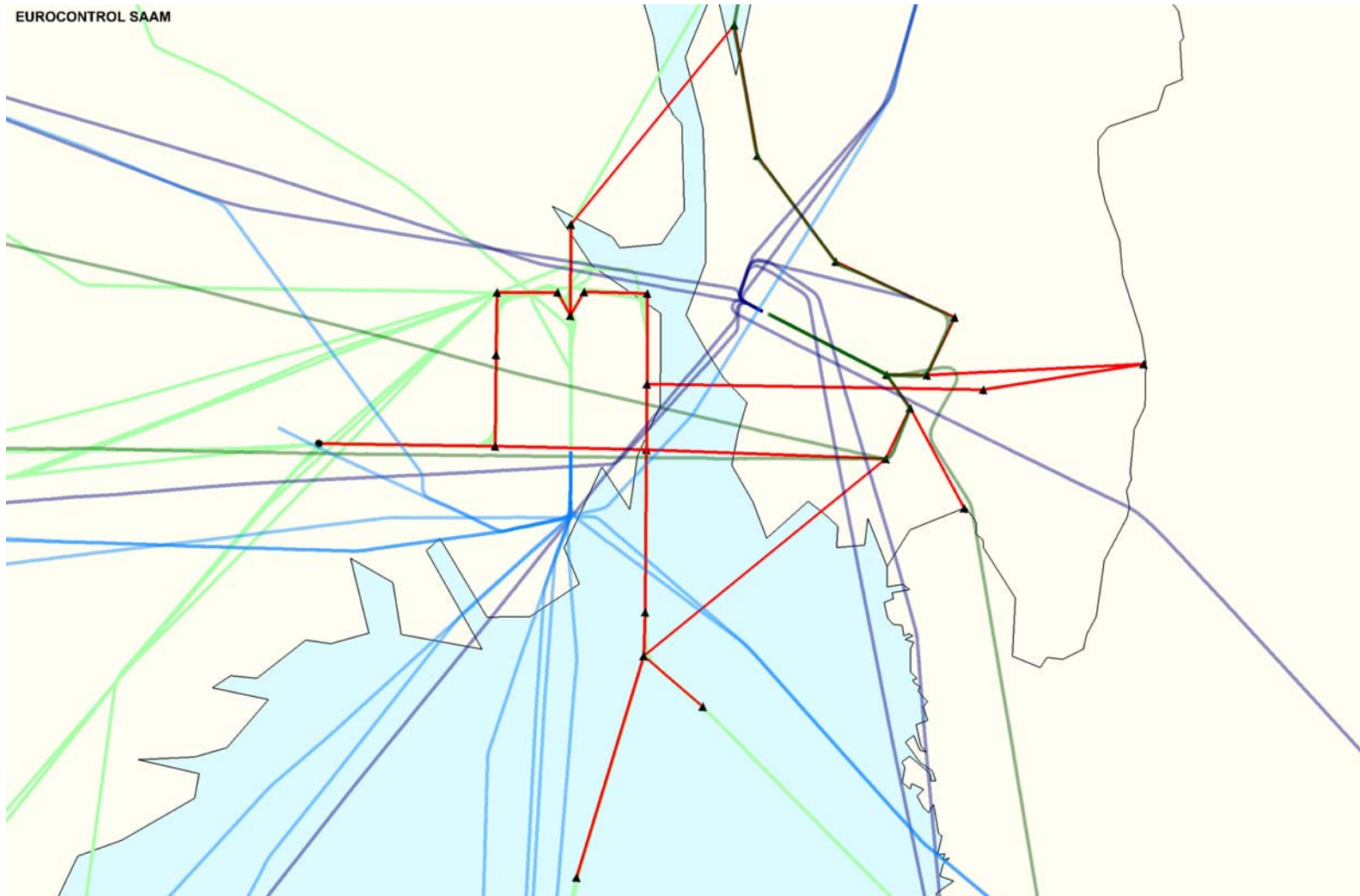
EXE 2 – ENTO RWY36 and ENRY RWY12



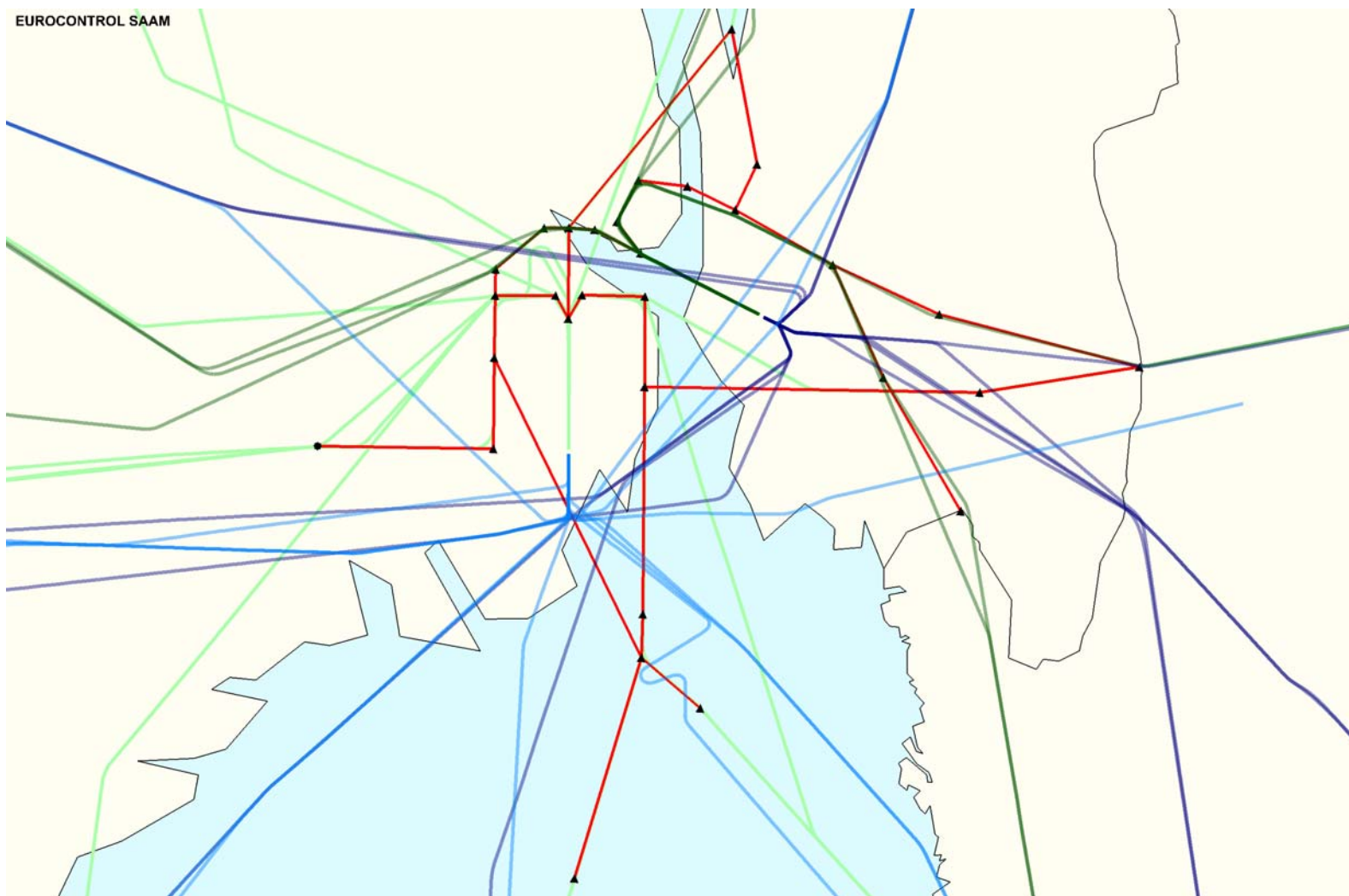
EXE 3 – ENTO RWY36 and ENRY RWY30



EXE 4 (12th March) – ENTO RWY18 and ENRY RWY30



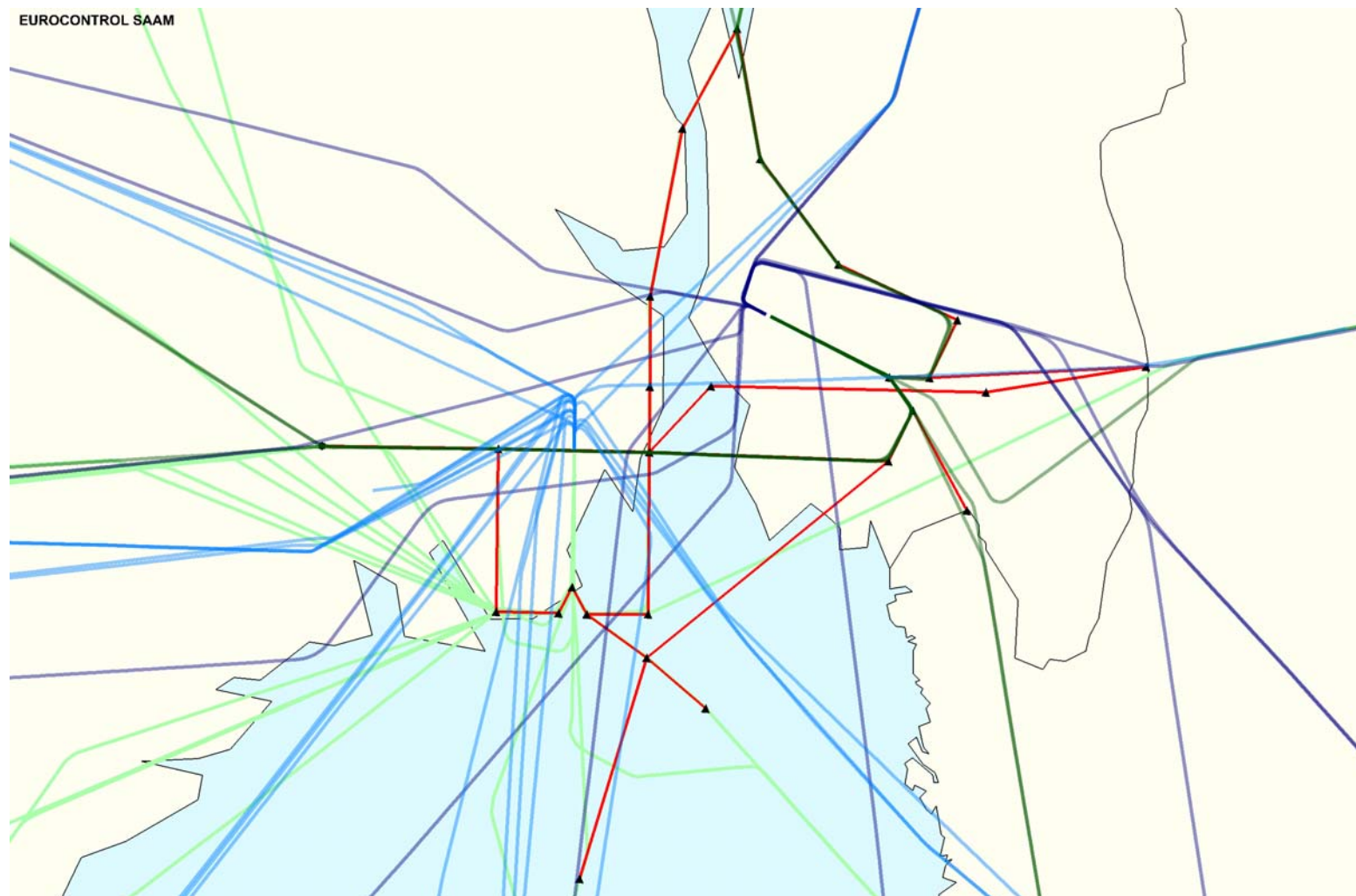
EXE 5 – ENTO RWY18 and ENRY RWY12



EXE 6 – ENTO RWY18 and ENRY RWY30



EXE 7 – ENTO RWY36 and ENRY RWY30



Page intentionally left blank